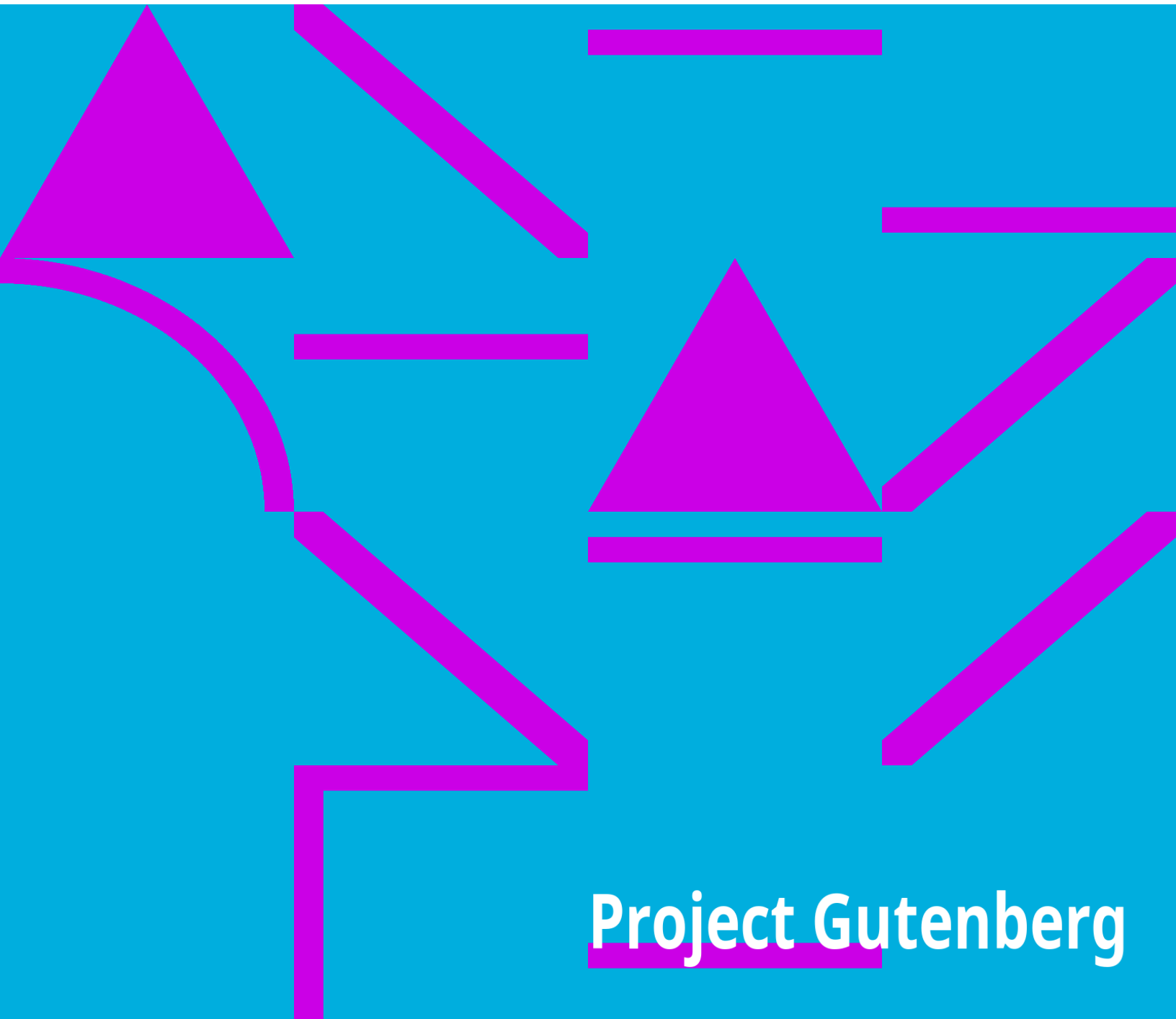


American Men of Mind

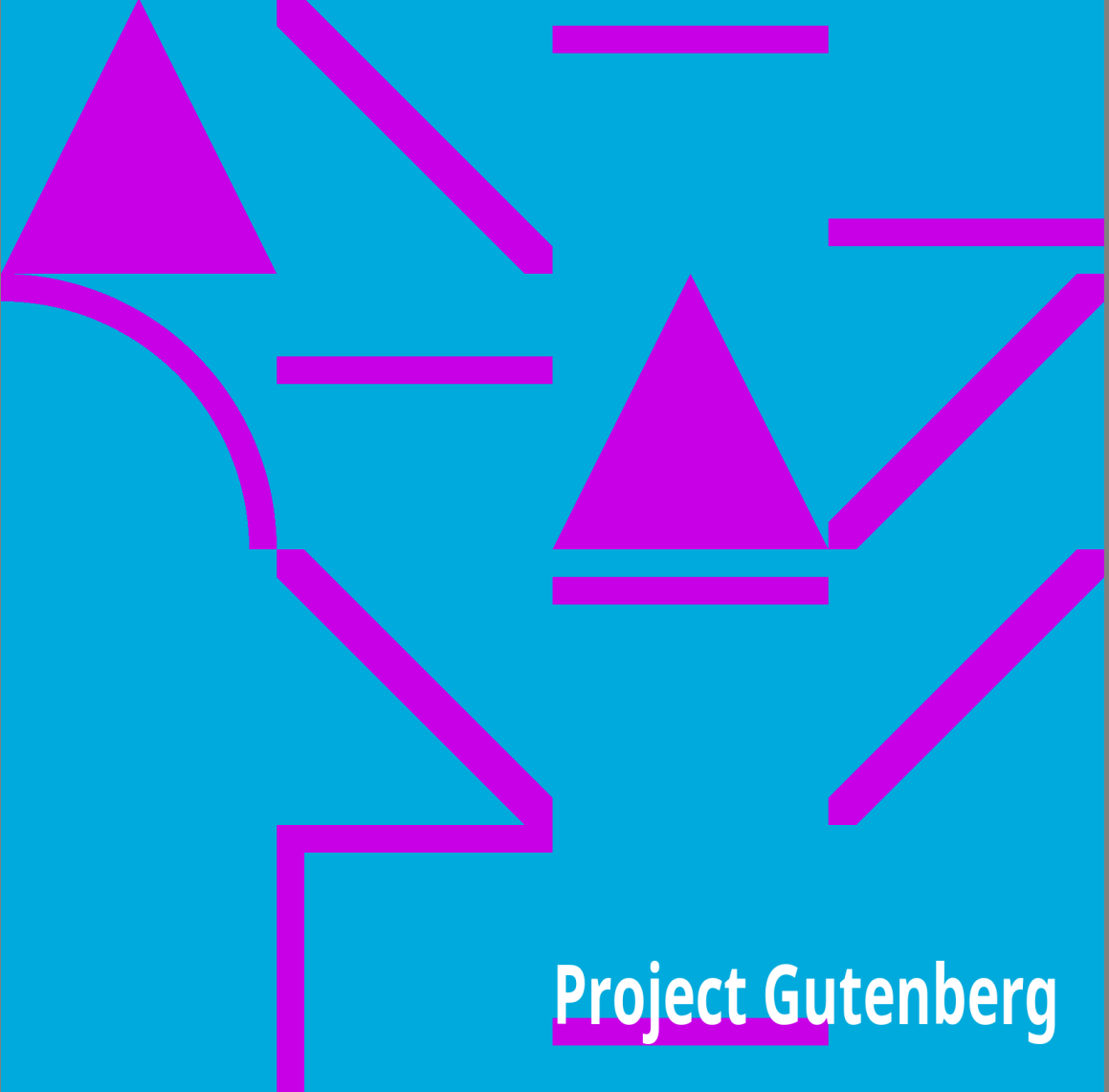
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American Men of Mind

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MEN OF MIND ***

AMERICAN MEN OF MIND

BY

BURTON E. STEVENSON

AUTHOR OF "A GUIDE TO BIOGRAPHY—MEN OF ACTION,"
"A SOLDIER OF VIRGINIA," ETC.; COMPILER OF "DAYS AND
DEEDS—POETRY," "DAYS AND DEEDS—PROSE," ETC.

GARDEN CITY NEW YORK

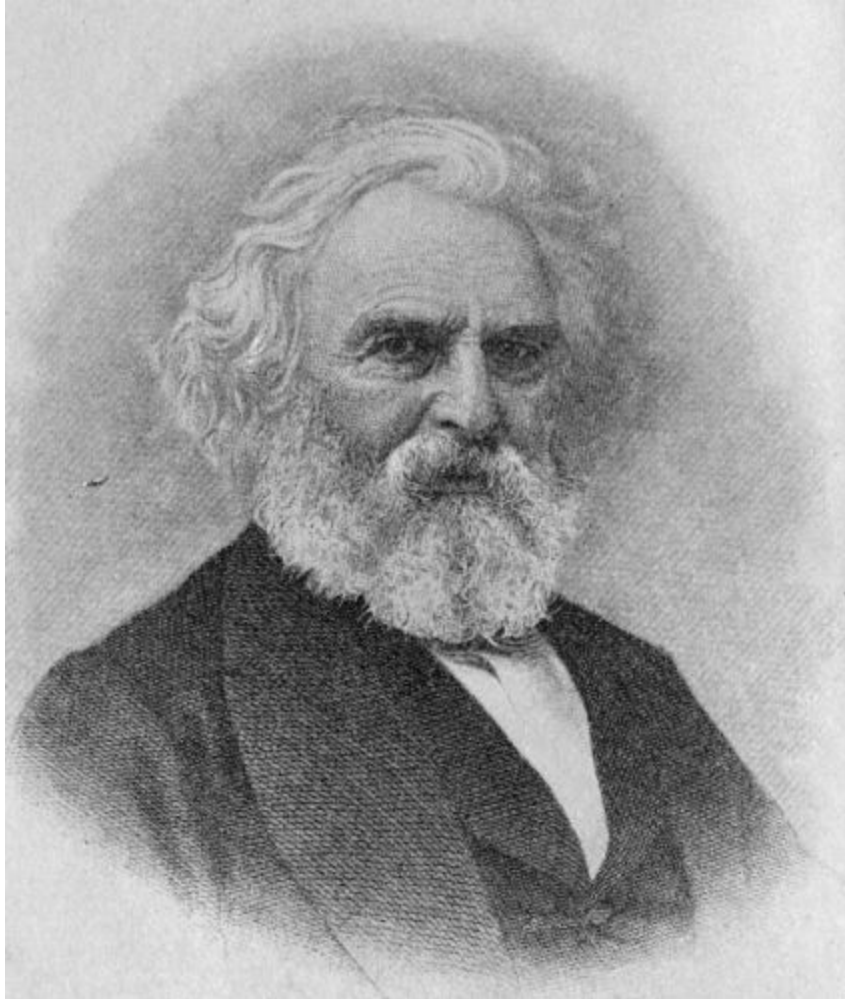
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LONGFELLOW

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CHAPTER I

"MEN OF MIND"

In the companion volume of this series, "Men of Action," the attempt was made to give the essential facts of American history by sketching in broad outline the men who made that history—the discoverers, pioneers, presidents, statesmen, soldiers, and sailors—and describing the part which each of them played.

It was almost like watching a great building grow under the hands of the workmen, this one adding a stone and that one adding another; but there was one great difference. For a building, the plans are made carefully beforehand, worked out to the smallest detail, and followed to the letter, so that every stone goes exactly where it belongs, and the work of all the men fits together into a complete and perfect whole. But when America was started, no one had more than the vaguest idea of what the finished result was to be; indeed, many questioned whether any enduring structure could be reared on a foundation such as ours. So there was much useless labor, one workman tearing down what another had built, and only a few of them working with any clear vision of the future.

The convention which adopted the Constitution of the United States may fairly be said to have furnished the first plan, and George Washington was the master-builder who laid the foundations in accordance with it. He did more than that, for the plan was only a mere outline; so Washington added such details as he found necessary, taking care always that they accorded with the plan of the founders. He lived long enough to see the building complete in all essential details, and to be assured that the foundation was a firm one and that the structure, which is called a Republic, *would* endure.

All that has been done since his time has been to build on an addition now and then, as need arose, and to change the ornamentation to suit the taste of the day. At one time, it seemed that the whole structure might be rent

asunder and topple into ruins; but again there came a master-builder named Abraham Lincoln, and with the aid of a million devoted workmen who rallied to his call, he saved it.

There have been men, and there are men to-day, who would attack the foundation were they permitted; but never yet have they got within effective striking distance. Others there are who have marred the simple and classic beauty of the building with strange excrescences. But these are only temporary, and the hand of time will sweep them all away. For the work of tearing down and building up is going forward to-day just as it has always done; and the changes are sometimes for the better and sometimes for the worse; but, on the whole, the building grows more stately and more beautiful as the generations pass.

It was the work of the principal laborers on this mighty edifice which we attempted to judge in "Men of Action," and this was a comparatively easy task, because the work stands out concretely for all to see, and, as far as essentials go, at least, we are all agreed as to what is good work and what is bad. But the task which is attempted in the present volume is a much more difficult one, for here we are called upon to judge not deeds but thoughts—thoughts, that is, as translated into a novel, or a poem, or a statue, or a painting, or a theory of the universe.

Nobody has ever yet been able to devise a universal scale by which thoughts may be measured, nor any acid test to distinguish gold from dross in art and literature. So each person has to devise a scale of his own and do his measuring for himself; he has to apply to the things he sees and reads the acid test of his own intellect. And however imperfect this measuring and testing may be, it is the only sort which has any value for that particular person. In other words, unless you yourself find a poem or a painting great, it isn't great for you, however critics may extol it. So all the books about art and literature and music are of value only as they improve the scale and perfect the acid test of the individual, so that the former measures more and more correctly, and the latter bites more and more surely through the glittering veneer which seeks to disguise the dross beneath.

It follows from all this that, since there are nearly as many scales as there are individuals, very few of them will agree exactly. Time, however, has a

wonderful way of testing thoughts, of preserving those that are worthy, and of discarding those that are unworthy. Just how this is done nobody has ever been able to explain; but the fact remains that, somehow, a really great poem or painting or statue or theory lives on from age to age, long after the other products of its time have been forgotten. And if it is really great, the older it grows, the greater it seems. Shakespeare, to his contemporaries, was merely an actor and playwright like any one of a score of others; but, with the passing of years, he has become the most wonderful figure in the world's literature. Rembrandt could scarcely make a living with his brush, industriously as he used it, and passed his days in misery, haunted by his creditors and neglected by the public; to-day we recognize in him one of the greatest artists who ever lived. Such instances are common enough, for genius often goes unrecognized until its possessor is dead; just as many men are hailed as geniuses by their contemporaries, and promptly forgotten by the succeeding generation. The touchstone of time infallibly separates the false and the true.

Unfortunately, to American literature and art no such test can be applied, for they are less than a century old—scarcely out of swaddling clothes. The greater portion of the product of our early years has long since been forgotten; but whether any of that which remains is really immortal will take another century or two to determine. So the only tests we can apply at present are those of taste and judgment, and these are anything but infallible.

Especially is this true of literature. Somebody announced, not long ago, that "the foremost poet of a nation is that poet most widely read and truly loved by it," and added that, in this respect, Longfellow was easily first in America. No doubt many people will agree with this dictum; and, indeed, the test of popularity is difficult to disregard. But it is not at all a true test, as we can see easily enough if we attempt to apply it to art, or to music, or to public affairs. Popularity is no more a test of genius in a poet than in a statesman, and when we remember how far astray the popular will has sometimes led us in regard to politics, we may be inclined to regard with suspicion its judgments in regard to literature.

The test of merit in literature is not so much wide appeal as intelligent appeal; the literature which satisfies the taste and judgment of cultured

people is pretty certain to rank higher than that which is current among the uncultured. And so with art. Consequently, for want of something better, the general verdict of cultured people upon our literature and art has been followed in these pages.

Two or three other classes of achievers have been grouped, for convenience, in this volume—scientists and educators, philanthropists and reformers, men of affairs, actors and inventors—and it may be truly argued concerning some of them that they were more "men of action," and less "men of mind" than many who were included in the former volume. But all distinctions and divisions and classifications are more or less arbitrary; and there is no intention, in this one, to intimate that the "men of action" were not also "men of mind," or vice versa. The division has been made simply for convenience.

These thumb-nail sketches are in no sense the result of original research. The material needed has been gathered from such sources as are available in any well-equipped public library. An attempt has been made, however, to color the narrative with human interest, and to give it consecutiveness, though this has sometimes been very hard to do. But, even at the best, this is only a first book in the study of American art and letters, and is designed to serve only as a stepping-stone to more elaborate and comprehensive ones.

There are several short histories of American literature which will prove profitable and pleasant reading. Mr. W. P. Trent's is written with a refreshing humor and insight. The "American Men of Letters" series gives carefully written biographies of about twenty-five of our most famous authors—all that anyone need know about in detail. There is a great mass of other material on the shelves of every public library, which will take one as far as one may care to go.

But the important thing in literature is to know the man's work rather than his life. If his work is sound and helpful and inspiring, his life needn't bother us, however hopeless it may have been. The striking example of this, in American literature, is Edgar Allan Poe, whose fame, in this country, is just emerging from the cloud which his unfortunate career cast over it. The life of the man is of importance only as it helps you to understand his work.

Most important of all is to create within yourself a liking for good books and a power of telling good from bad. This is one of the most important things in life, indeed; and Mr. John Macy points the way to it in his "Child's Guide to Reading."

Only second to the power to appreciate good literature is the power to appreciate good art. For the material in this volume the author is indebted largely to the excellent monographs by Mr. Samuel Isham and Mr. Lorado Taft on "American Painting," and "American Sculpture." There are many, guides to the study of art, among the best of them being Mr. Charles C. Caffin's "Child's Guide to Pictures," "American Masters of Painting," "American Masters of Sculpture," and "How to Study Pictures"; Mr. John C. VanDyke's "How to Judge of a Picture," and "The Meaning of Pictures," and Mr. John LaFarge's "Great Masters." In the study of art, as of literature, you will soon find that America's place is as yet comparatively unimportant.

For the chapter on "The Stage," Mr. William Winter's various volumes of biography and criticism have been drawn upon, more especially with reference to the actors of the "old school," which Mr. Winter admires so deeply. There are a number of books, besides these, which make capital reading—Clara Morris's "Life on the Stage," Joseph Jefferson's autobiography, Stoddart's "Recollections of a Player," and Henry Austin Clapp's "Reminiscences of a Dramatic Critic," among them.

The material for the other chapters has been gathered from many sources, none of which is important enough to be mentioned here. Appleton's "Cyclopedia of American Biography" is a mine from which most of the facts concerning any American, prominent twenty years or more ago, may be dug; but it gives only the dry bones, so to speak. For more than that you must go to the individual biographies in your public library.

If you live in a small town, the librarian will very probably be glad to permit you to look over the shelves yourself, as well as to give you such advice and direction as you may need. In the larger cities, this is, of course, impossible, to say nothing of the fact that you would be lost among the thousands of books on the shelves. But you will find a children's librarian whose business and pleasure it is to help children to the right books. If this

book helps you to form the library habit, and gives you an incentive to the further study of art and literature, it will more than fulfill its mission.

CHAPTER II

WRITERS OF PROSE

It is true of American literature that it can boast no name of commanding genius—no dramatist to rank with Shakespeare, no poet to rank with Keats, no novelist to rank with Thackeray, to take names only from our cousins oversea—and yet it displays a high level of talent and a notable richness of achievement. Literature requires a background of history and tradition; more than that, it requires leisure. A new nation spends its energies in the struggle for existence, and not until that existence is assured do its finer minds need to turn to literature for self-expression. As Poor Richard put it, "Well done is better than well said," and so long as great things are pressing to be done, great men will do their writing on the page of history, and not on papyrus, or parchment, or paper.

So, in the early history of America, the settlers in the new country were too busily employed in fighting for a foothold, in getting food and clothing, in keeping body and soul together, to have any time for the fine arts. Most of the New England divines tried their hands at limping and hob-nail verse, but prior to the Revolution, American literature is remarkable only for its aridity, its lack of inspiration and its portentous dulness. In these respects it may proudly claim never to have been surpassed in the history of mankind. In fact, American literature, as such, may be said to date from 1809, when Washington Irving gave to the world his inimitable "History of New York." It struck a new and wholly original note, with a sureness bespeaking a master's touch.

Where did Irving get that touch? That is a question which one asks vainly concerning any master of literature, for genius is a thing which no theory can explain. It appears in the most unexpected places. An obscure Corsican lieutenant becomes Emperor of France, arbiter of Europe, and one of the three or four really great commanders of history; a tinker in Bedford County jail writes the greatest allegory in literature; and the son of two

mediocre players develops into the first figure in American letters. Conversely, genius seldom appears where one would naturally look for it. Seldom indeed does genius beget genius. It expends itself in its work.

Certainly there was no reason to suppose that any child of William Irving and Sarah Sanders would develop genius even of the second order, more especially since they had already ten who were just average boys and girls. Nor did the eleventh, who was christened Washington, show, in his youth, any glimpse of the eagle's feather.

Born in 1783, in New York City, a delicate child and one whose life was more than once despaired of, Washington Irving received little formal schooling, but was allowed to amuse himself as he pleased by wandering up and down the Hudson and keeping as much as possible in the open air. It was during these years that he gained that intimate knowledge of the Hudson River Valley of which he was to make such good use later on. He still remained delicate, however, and at the age of twenty was sent to Europe. The air of France and Italy proved to be just what he needed, and he soon developed into a fairly robust man.

With health regained, he returned, two years later, to America, and got himself admitted to the bar. Why he should have gone to this trouble is a mystery, for he never really seriously tried to practise law. Instead, he was occupying himself with a serio-comic history of New York, which grew under his pen into as successful an example of true and sustained humor as our literature possesses. The subject was one exactly suited to Irving's genius, and he allowed his fancy to have free play about the picturesque personalities of Wouter Van Twiller, and Wandle Schoonhoven, and General Van Poffenburgh, in whose very names there is a comic suggestion. When it appeared, in 1809, it took the town by storm.

Irving, indeed, had created a legend. The history, supposed to have been written by one Diedrich Knickerbocker, gives to the story of New York just the touch of fancy and symbolism it needed. For all time, New York will remain the Knickerbocker City. The book revealed a genuine master of kindly satire, and established its author's reputation beyond possibility of question. Perhaps the surest proof of its worth is the fact that it is read today as widely and enjoyed as thoroughly as it ever was.

It is strange that Irving did not at once adopt letters as a profession; but instead of that, he entered his brothers' business house, which was in a decaying condition, and to which he devoted nine harassed and anxious years, before it finally failed. That failure decided him, and he cast in his lot finally with the fortunes of literature. He was at that time thirty-five years of age—an age at which most men are settled in life, with an established profession, and a complacent readiness to drift on into middle age.

Rarely has any such choice as Irving's received so prompt and triumphant a vindication, for a year later appeared the "Sketch Book," with its "Rip Van Winkle," its "Legend of Sleepy Hollow" and "The Spectre Bridegroom"—to mention only three of the thirty-three items of its table of contents—which proved the author to be not only a humorist of the first order, but an accomplished critic, essayist and short-story writer. The publication of this book marked the culmination of his literary career. It is his most characteristic and important work, and on it and his "History," his fame rests.

He lived for forty years thereafter, a number of which were spent in Spain, first as secretary of legation, and afterwards as United States minister to that country. It was during these years that he gathered the materials for his "Life of Columbus," his "Conquest of Granada," and his "Alhambra," which has been called with some justice, "The Spanish Sketch Book." A tour of the western portion of the United States resulted also in three books, "The Adventures of Captain Bonneville," "Astoria," and "A Tour on the Prairies." His last years were spent at "Sunnyside," his home at Tarrytown, on the Hudson, where he amused himself by writing biographies of Mahomet, of Goldsmith, and of George Washington.

All of this was, for the most part, what is called "hack work," and his turning to it proves that he himself was aware that his fount of inspiration had run dry. This very fact marks his genius as of the second order, for your real genius—your Shakespeare or Browning or Thackeray or Tolstoi—never runs dry, but finds welling up within him a perpetual and self-renewing stream of inspiration, fed by thought and observation and every-day contact with the world.

Irving's closing years were rich in honor and affection, and found him unspoiled and uncorrupted. He was always a shy man, to whom publicity of any kind was most embarrassing; and yet he managed to be on the most intimate of terms with his time, and to possess a wide circle of friends who were devoted to him.

Such was the career of America's first successful man of letters. For, strangely enough, he had succeeded in making a good living with his pen. More than that, his natural and lambent humor, his charm and grace of style, and a literary power at once broad and genuine, had won him a place, if not among the crowned heads, at least among the princes of literature, side by side with Goldsmith and Addison. Thackeray called him "the first ambassador whom the New World of letters sent to the Old," and from the very first he identified American literature with purity of life and elevation of character, with kindly humor and grace of manner—qualities which it has never lost.

Two years after the appearance of the "Sketch Book," another star suddenly flamed out upon the literary horizon, and for a time quite eclipsed Irving in brilliancy. It waned somewhat in later years, but, though we have come to see that it lacks the purity and gentle beauty of its rival, it has still found a place among the brightest in our literary heaven—where, indeed, only one or two of the first magnitude shine. J. Fenimore Cooper was, like Irving, a product of New York state, his father laying out the site of Cooperstown, on Lake Otsego, and moving there from New Jersey in 1790, when his son was only a year old. James, as the boy was known, was the eleventh of twelve children—another instance of a single swan amid a flock of ducklings.

Cooperstown was at that time a mere outpost of civilization in the wilderness, and it was in this wilderness that Cooper's boyhood was passed. And just as Irving's boyhood left its impress on his work, so did Cooper's in even greater degree. Mighty woods, broken only here and there by tiny clearings, stretched around the little settlement; Indians and frontiersmen, hunters, traders, trappers—all these were a part of the boy's daily life. He grew learned in the lore of the woods, and laid up unconsciously the stores from which he was afterwards to draw.

At the age of eleven, he was sent to a private school at Albany, and three years later entered Yale. But he had the true woodland spirit; he preferred the open air to the lecture-room, and was so careless in his attendance at classes that, in his third year, he was dismissed from college. There is some question whether this was a blessing or the reverse. No doubt a thorough college training would have made Cooper incapable of the loose and turgid style which characterizes all his novels; but, on the other hand, he left college to enter the navy, and there gained that knowledge of seamanship and of the ocean which make his sea stories the best of their kind that have ever been written. His sea career was cut short, just before the opening of the war of 1812, by his marriage into an old Tory family, who insisted that he resign from the service. He did so, and entered upon the quiet life of a well-to-do country gentleman.

For seven or eight years, he showed no desire nor aptitude to be anything else. He had never written anything for publication, had never felt any impulse to do so, and perhaps never would have felt such an impulse but for an odd accident. Tossing aside a dull British novel, one day, he remarked to his wife that he could easily write a better story himself, and she laughingly dared him to try. The result was "Precaution," than which no British novel could be duller. But Cooper, finding the work of writing congenial, kept at it, and the next year saw the publication of "The Spy," the first American novel worthy of the name. By mere accident, Cooper had found his true vein, the story of adventure, and his true field in the scenes with which he was himself familiar. In Harvey Birch, the spy, he added to the world's gallery of fiction the first of his three great characters, the other two being, of course, Long Tom Coffin and Leatherstocking.

The book was an immediate success, and was followed by "The Pioneers" and "The Pilot," both remarkable stories, the former visualizing for the first time the life of the forest, the latter for the first time the life of the sea. Let us not forget that Cooper was himself a pioneer and blazed the trails which so many of his successors have tried to follow. If the trail he made was rough and difficult, it at least possesses the merits of vigor and pristine achievement. "The Spy," "The Pioneers," and "The Pilot" established Cooper's reputation not only in this country, but in England and France. He became a literary lion, with the result that his head, never very firmly set upon his shoulders, was completely turned; he set himself up as a mentor

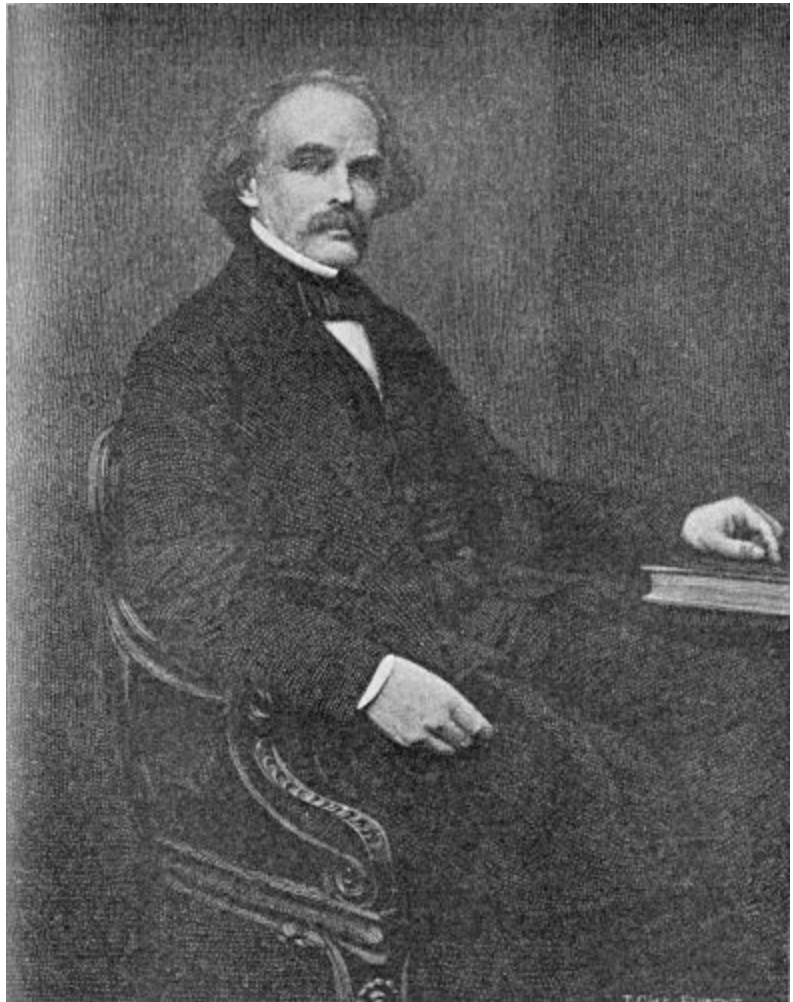
and critic of both continents, and while his successive novels continued to be popular, he himself became involved in numberless personal controversies, which embittered his later years.

The result of these quarrels was apparent in his work, which steadily decreased in merit, so that, of the thirty-three novels that he wrote, not over twelve are, at this day, worth reading. But those twelve paint, as no other novelist has ever painted, life in the forest and on the ocean, and however we may quarrel with his wooden men and women, his faults of taste and dreary wastes of description, there is about them some intangible quality which compels the interest and grips the imagination of school-boy and gray-beard alike. He splashed his paint on a great canvas with a whitewash brush, so to speak; it will not bear minute examination; but at a distance, with the right perspective, it fairly glows with life. No other American novelist has added to fiction three such characters as those we have mentioned; into those he breathed the breath of life—the supreme achievement of the novelist.

For seventeen years after the publication of "The Spy," Cooper had no considerable American rival. Then, in 1837, the publication of a little volume called "Twice-Told Tales" marked the advent of a greater than he. No one to-day seriously questions Nathaniel Hawthorne's right to first place among American novelists, and in the realm of the short story he has only one equal, Edgar Allan Poe.

We shall speak of Poe more at length as a poet; but it is curious and interesting to contrast these two men, contemporaries, and the most significant figures in the literature of their country—Poe, an actor's child, an outcast, fighting in the dark with the balance against him, living a tragic life and dying a tragic death, leaving to America the purest lyrics and most compelling tales ever produced within her borders; Hawthorne, a direct descendant of the Puritans, a recluse and a dreamer, his delicate genius developing gradually, marrying most happily, leading an idyllic family life, winning success and substantial recognition, which grew steadily until the end of his career, and which has, at least, not diminished—could any contrast be more complete?

Nathaniel Hawthorne was a direct descendant of that William Hawthorne who came from England in 1630 with John Winthrop in the "Arabella," and was born at Salem, Massachusetts, the family's ancestral home, in 1804. He was a classmate of Longfellow at Bowdoin College, graduating without especial distinction, and spending the twelve succeeding years at Salem, living a secluded life in accordance with his abnormally shy and sensitive disposition. He was already resolved on the literary life, and spent those years in solitary writing. The



HAWTHORNE

result was a morbid novel, "Fanshawe," and a series of short stories, none of which attracted especial attention or gave indication of more than average talent. Not until 1837 did he win any measure of success, but that year saw the publication of the first series of "Twice-Told Tales," which, by their charm and delicacy, won him many readers.

Even at that, he found the profession of letters so unprofitable that he was glad to accept a position as weigher and gauger at the Boston custom-house, but he lost the place two years later by a change in administration; tried, for a while, living with the Transcendentalists at Brook Farm, and finally, taking a leap into the unknown, married and settled down in the old manse at Concord. It was a most fortunate step; his wife proved a real inspiration,

and in the months that followed, he wrote the second series of "Twice-Told Tales," and "Mosses from an Old Manse," which mark the culmination of his genius as a teller of tales.

Four years later, the political pendulum swung back again, and Hawthorne was offered the surveyor-ship of the custom-house at Salem, accepted it, and moved his family back to his old home. He held the position for four years, completed his first great romance, and in 1850 gave to the world "The Scarlet Letter," perhaps the most significant and vital novel produced by any American. Hawthorne had, at last, "found himself." A year later came "The House of the Seven Gables," and then, in quick succession, "Grandfather's Chair," "The Wonder Book," "The Snow-Image," "The Blithedale Romance," and "Tanglewood Tales."

A queer product of his pen, at this time, was a life of Franklin Pierce, the Democratic candidate for the Presidency; and when Pierce was elected, he showed his gratitude by offering Hawthorne the consulship at Liverpool, a lucrative position which Hawthorne accepted and which he held for four years. Two years on the continent followed, and in 1860, he returned home, his health breaking and his mind unsettled, largely by the prospect of the Civil War into which the country was drifting. He found himself unable to write, failed rapidly, and the end came in the spring of 1864.

Of American novelists, Hawthorne alone shows that sustained power and high artistry belonging to the masters of fiction; and yet his novels have not that universal appeal which belongs to the few really great ones of the world. Hawthorne was supremely the interpreter of old New England, a subject of comparatively little interest to other peoples, since old New England was distinguished principally by a narrow spiritual conflict which other peoples find difficult to understand. The subject of "The Scarlet Letter" is, indeed, one of universal appeal, and is, in some form, the theme of nearly all great novels; but its setting narrowed this appeal, and Hawthorne's treatment of his theme, symbolical rather than simple and concrete, narrowed it still further. Yet with all that, it possesses that individual charm and subtlety which is apparent, in greater or less degree, in all of his imaginative work.

Contemporary with Hawthorne, and surviving him by a few years, was another novelist who had, in his day, a tremendous reputation, but who is now almost forgotten, William Gilmore Simms. We shall consider him—for he was also a maker of verse—in the next chapter, in connection with his fellow-townsmen, Henry Timrod and Paul Hamilton Hayne. So we pause here only to remark that the obscurity which enfolds him is more dense than he deserves, and that anyone who likes frontier fiction, somewhat in the manner of Cooper, will enjoy reading "The Yemassee," the best of Simms's books.

Hawthorne stands so far above the novelists who come after him that one rather hesitates to mention them at all. With one, or possibly two, exceptions, the work of none of them gives promise of permanency—so far as can be judged, at least, in looking at work so near that it has no perspective. Prophesying has always been a risky business, and will not be attempted here. But, whether immortal or not, there are some five or six novelists whose work is in some degree significant, and who deserve at least passing study.

Harriet Beecher Stowe is one of these. Born in 1811, the daughter of Lyman Beecher, and perhaps the most brilliant member of a brilliant family, beginning to write while still a child, and continuing to do so until the end of her long life, Mrs. Stowe's name is nevertheless connected in the public mind with a single book, "Uncle Tom's Cabin," a book which has probably been read by more people than any other ever written by an American author. Mrs. Stowe had lived for some years in Cincinnati and had visited in Kentucky, so that she had some surface knowledge of slavery; she was, of course, by birth and breeding, an abolitionist, and so when, early in 1851, an anti-slavery paper called the "National Era" was started at Washington, she agreed to furnish a "continued story."

The first chapter appeared in April, and the story ran through the year, attracting little attention. But its publication in book form marked the beginning of an immense popularity and an influence probably greater than that of any other novel ever written. It crystallized anti-slavery sentiment, it was read all over the world, it was dramatized and gave countless thousands their first visualization of the slave traffic. That her presentation of it was in many respects untrue has long since been admitted, but she was writing a

tract and naturally made her case as strong as she could. From a literary standpoint, too, the book is full of faults; but it is alive with an emotional sincerity which sweeps everything before it. She wrote other books, but none of them is read to-day, except as a matter of duty or curiosity.

And let us pause here to point out that the underlying principle of every great work of art, whether a novel or poem or painting or statue, is sincerity. Without sincerity it cannot be great, no matter how well it is done, with what care and fidelity; and with sincerity it may often attain greatness without perfection of form, just as "Uncle Tom's Cabin" did. But to lack sincerity is to lack soul; it is a body without a spirit.

We must refer, too, to the most distinctive American humorist of the last half century, Samuel Langhorne Clemens—"Mark Twain." Born in Missouri, knocking about from pillar to post in his early years, serving as pilot's boy and afterwards as pilot on a Mississippi steamboat, as printer, editor, and what not, but finally "finding himself" and making an immense reputation by the publication of a burlesque book of European travel, "Innocents Abroad," he followed it up with such widely popular stories as "Tom Sawyer," "Huckleberry Finn," "The Prince and the Pauper," and many others, in some of which, at least, there seems to be an element of permanency. "Huckleberry Finn," indeed, has been hailed as the most distinctive work produced in America—an estimate which must be accepted with reservations.

Three living novelists have contributed to American letters books of insight and dignity—William Dean Howells, George W. Cable and Henry James. Mr. Howells has devoted himself to careful and painstaking studies of American life, and has occasionally struck a note so true that it has found wide appreciation. The same thing may be said of Mr. Cable's stories of the South, and especially of the Creoles of Louisiana; while Mr. James, perhaps as the result of his long residence abroad, has ranged over a wider field, and has chosen to depict the evolution of character by thought rather than by deed, in his early work showing a rare insight. Of the three, he seems most certain of a lasting reputation.

Others of less importance have made some special corner of the country theirs, and possess a sort of squatter-right over it. To Bret Harte belongs

mid-century California; to Mary Noailles Murfree, the Tennessee mountains; to James Lane Allen and John Fox, present-day Kentucky; to Mary Johnston, colonial Virginia; to Ellen Glasgow, present-day Virginia; to Stewart Edward White, the great northwest. Others cultivate a field peculiar to themselves. Frank R. Stockton is whimsically humorous, Edith Wharton cynically dissective; Mary Wilkins Freeman is most at home with rural New England character; and Thomas Nelson Page has done his best work in the South of reconstruction days.

But of the great mass of fiction being written in America to-day, little is of value as literature. It is designed for the most part as an amusing occupation for idle hours. Read some of it, by all means, if you enjoy it, since "all work and no play makes Jack a dull boy"; but remember that it is only the sweetmeat that comes at the end of the meal, and for sustenance, for the bread and butter of the literary diet, you must read the older books that are worth while.

It may be questioned whether America has produced any poet or novelist or essayist of the very first rank, but, in another branch of letters, four names appear, which stand as high as any on the scroll. The writing of history is not, of course, pure literature; it is semi-creative rather than creative; and yet, at its best, it demands a high degree of imaginative insight. It appears at its best in the works of Prescott, Motley, Bancroft and Parkman.

George Bancroft was, of this quartette, the most widely known half a century ago, because he chose as his theme the history of America, and because he was himself for many years prominent in the political life of the country. Born in Massachusetts in 1800, graduating from Harvard, and, after a course of study in Germany, resolving to be a historian, he returned to America and began work on his history, the first volume of which appeared in 1834. Three years later, came the second volume, and in 1840, the third.

Glowing with national spirit as they did, they attracted public attention to him, and he was soon drawn into politics. During the next twelve years he

held several government positions, among them Secretary of the Navy and Minister to England, which gave him access to great masses of historical documents. It was not until 1852 that his fourth volume appeared, then five more followed at comparatively frequent intervals. Again politics interrupted. He was sent as Minister to Prussia and later to the German Empire, again largely increasing his store of original documents, with which, toward the last, he seems to have been fairly overburdened. In 1874, he published his tenth volume, bringing his narrative through the Revolution, and eight years later, the last two dealing with the adoption of the Constitution. His last years were spent in revising and correcting this monumental work.

It is an inspiring record—a life devoted consistently to one great work, and that work the service of one's country, for such Bancroft's really was. Every student of colonial and revolutionary America must turn to him, and while his history has long since ceased to be generally read, it maintains an honored place among every collection of books dealing with America. It is easily first among the old-school histories as produced by such men as Hildreth, Tucker, Palfrey and Sparks.

At the head of the other school, which has been called cosmopolitan because it sought its subjects abroad rather than at home, stands William Hickling Prescott. Of this school, Washington Irving may fairly be said to have been the pioneer. We have seen how his residence in Spain turned his attention to the history of that country and resulted in three notable works. Prescott, however, was a historian by forethought and not by accident. Before his graduation from Harvard, he had determined to lead a literary life modelled upon that of Edward Gibbon. His career was almost wrecked at the outset by an unfortunate accident which so impaired his sight that he was unable to read or to write except with the assistance of a cumbrous machine. That any man, laboring under such a disability, should yet persevere in pursuing the rocky road of the historian seems almost unbelievable; yet that is just what Prescott did.

Let us tell the story of that accident. It was while he was at Harvard, in his junior year. One day after dinner, in the Commons Hall, some of the boys started a rude frolic. Prescott took no part in it, but just as he was leaving, a great commotion behind him caused him to turn quickly, and a hard piece

of bread, thrown undoubtedly at random, struck him squarely and with great force in the left eye. He fell unconscious, and never saw out of that eye again. Worse than that, his other eye soon grew inflamed, and became almost useless to him, besides causing him, from time to time, the most acute suffering. But in spite of all this, he persisted in his determination to be a historian.

After careful thought, he chose for his theme that period of Spanish history dominated by Ferdinand and Isabella, and went to work. Documents were collected, an assistant read to him for hours at a time, notes were taken, and the history painfully pushed forward. The result was a picturesque narrative which was at once successful both in Europe and America; and, thus encouraged, Prescott selected another romantic theme, the conquest of Mexico, for his next work. Following this came the history of the conquest of Peru, and finally a history of the reign of Philip II, upon which he was at work, when a paralytic stroke ended his career.

Prescott was fortunate not only in his choice of subjects, but in the possession of a picturesque and fascinating style, which has given his histories a remarkable vogue. Fault has been found with him on the ground of historical inaccuracy, but such criticism is, for the most part, unjustified. His thoroughness, his judgment, and his critical faculty stand unimpeached, and place him very near the head of American historians.

Prescott's successor, in more than one sense, was John Lothrop Motley. A Bostonian and Harvard man, well-trained, after one or two unsuccessful ventures in fiction, he turned his attention to history, and in 1856 completed his "Rise of the Dutch Republic," for which he could not find a publisher. He finally issued it at his own expense, with no little inward trembling, but it was at once successful and seventeen thousand copies of it were sold in England alone during the first year. It received unstinted praise, and Motley at once proceeded with his "History of the United Netherlands." The opening of the Civil War, however, recalled his attention to his native land, he was drawn into politics, and did not complete his history until 1868. Six years later appeared his "John of Barneveldt"; but his health was giving way and the end came in 1877.

In brilliancy, dramatic instinct and power of picturesque narration, Motley was Prescott's equal, if not his superior. The glow and fervor of his narrative have never been surpassed; his characters live and breathe; he was thoroughly in sympathy with his subject and found a personal pleasure in exalting his heroes and unmasking his villains. But there was his weakness; for often, instead of the impartial historian, he became a partisan of this cause or that, and painted his heroes whiter and his villains blacker than they really were. In spite of that, or perhaps because of it—because of the individual and intensely earnest personal point of view—his histories are as absorbing and fascinating as any in the world.

The last of this noteworthy group of historians, Francis Parkman, is also, in many respects, the greatest. He combined the virtues of all of them, and added for himself methods of research which have never been surpassed. Through it all, too, he battled against a persistent ill-health, which unfitted him for work for months on end, and, even at the best, would permit his reading or writing only a few minutes at a time.

Like the others, Parkman was born in Boston, and, as a boy, was so delicate that he was allowed to run wild in the country, acquiring a love of nature which is apparent in all his books. In search of health, he journeyed westward from St. Louis, in 1846, living with Indians and trappers and gaining a minute knowledge of their ways. The results of this journey were embodied in a modest little volume called "The Oregon Trail," which remains the classic source of information concerning the far West at that period.

Upon his return to the East, he settled down in earnest to the task which he had set himself—a history, in every phase, of the struggle between France and England for the possession of the North American continent. Years were spent in the collection of material—and in 1865 appeared his "Pioneers of France in the New World," followed at periods of a few years by the other books completing the series, which ends with the story of Montcalm and Wolfe.

The series is a masterpiece of interpretative history. Every phase of the struggle for the continent is described in minute detail and with the intimate touch of perfect knowledge; every actor in the great drama is presented with

incomparable vividness, and its scenes are painted with a color and atmosphere worthy of Prescott or Motley, and with absolute accuracy. His work satisfies at once the student and the lover of literature, standing almost unique in this regard. His flexible and charming style is a constant joy; his power of analysis and presentment a constant wonder; and throughout his work there is a freshness of feeling, an air of the open, at once delightful and stimulating. He said the last word concerning the period which his histories cover, and has lent to it a fascination and absorbing interest which no historian has surpassed. The boy or girl who has not read Parkman's histories has missed one of the greatest treats which literature has to offer.

Other historians there are who have done good service to American letters and whose work is outranked only by the men we have already mentioned—John Bach McMaster, whose "History of the People of the United States" is still uncompleted; James Ford Rhodes, who has portrayed the Civil War period with admirable exhaustiveness and accuracy; Justin Winsor, Woodrow Wilson, William M. Sloane, and John Fiske. John Fiske's work, which deals wholly with the different periods of American history, is especially suited to young people because of its simplicity and directness, and because, while accurate, it is not overburdened with detail.

We have said that, during the Colonial period of American history, most of the New England divines devoted a certain amount of attention to the composition of creaking verse. More than that, they composed histories, biographies and numberless works of a theological character, which probably constitute the dullest mass of reading ever produced upon this earth. The Revolution stopped this flood—if anything so dry can be called a flood—and when the Revolution ended, public thought was for many years occupied with the formation of the new nation. But in the second quarter of the nineteenth century there arose in New England a group of writers who are known as Transcendentalists, and who produced one of the most important sections of American literature.

Transcendentalism is a long word, and it is rather difficult to define, but, to put it as briefly as possible, it was a protest against narrowness in intellectual life, a movement for broader culture and for a freer spiritual life. It took a tremendous grip on New England, beginning about 1830, and kept it for nearly forty years; for New England has always been more or less

provincial—provincialism being the habit of measuring everything by one inadequate standard.

The high priest of the Transcendental movement was Amos Bronson Alcott, born on a Connecticut farm in 1799, successively in youth a clockmaker, peddler and book-agent, and finally driven by dire necessity to teaching school. But there could be no success at school-teaching for a man the most eccentric of his day—a mystic, a follower of Oriental philosophy, a non-resistant, an advocate of woman suffrage, an abolitionist, a vegetarian, and heaven knows what besides. So in the end, he was sold out, and removed with his family to Concord, where he developed into a sort of impractical idealist, holding Orphic conversations and writing scraps of speculation and criticism, and living in the clouds generally.

Life would have been far less easy for him but for the development of an unexpected talent in one of his daughters, Louisa May Alcott. From her sixteenth year, Louisa Alcott had been writing for publication, but with little success, although every dollar she earned was welcome to a family so poor that the girls sometimes thought of selling their hair to get a little money. She also tried to teach, and finally, in 1862, went to Washington as a volunteer nurse and labored for many months in the military hospitals. The letters she wrote to her mother and sisters were afterwards collected in a book called "Hospital Sketches." At last, at the suggestion of her publishers, she undertook to write a girls' story. The result was "Little Women," which sprang almost instantly into a tremendous popularity, and which at once put its author out of reach of want.

Other children's stories, scarcely less famous, followed in quick succession, forming a series which has never been equalled for long-continued vogue. Few children who read at all have failed to read "Little Men," "Little Women," "An Old-Fashioned Girl," "Eight Cousins," and "Rose in Bloom," to mention only five of them, and edition after edition has been necessary to supply a demand which shows no sign of lessening. The stories are, one and all, sweet and sincere and helpful, and while they are not in any sense literature, they are, at least, an interesting contribution to American letters.

But to return to the Transcendentalists.

The most picturesque figure of the group was Margaret Fuller. Starting as a morbid and sentimental girl, her father's death seems suddenly to have changed her, at the age of twenty-five, into a talented and thoughtful woman. Her career need not be considered in detail here, since it was significant more from the inspiration she gave others than from any achievement of her own. She proved herself a sympathetic critic, if not a catholic and authoritative one, and a pleasing and suggestive essayist.

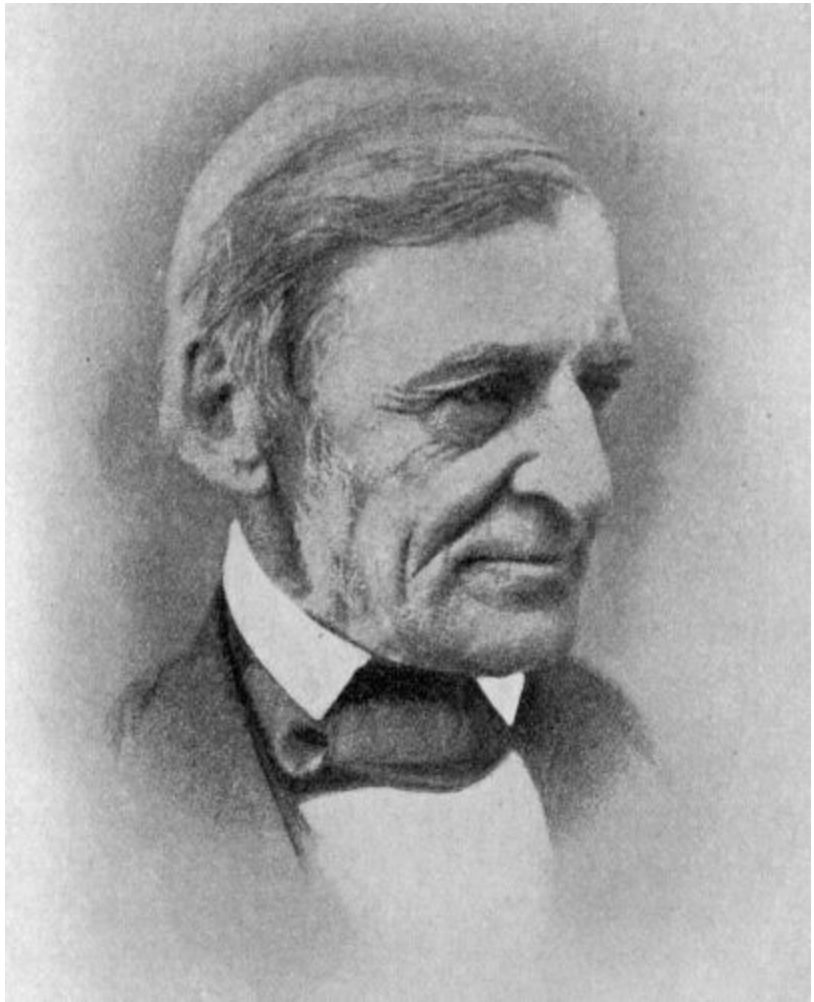
What she might have become no one can tell, for her life was cut short at the fortieth year. She had spent some years in Italy, in an epoch of revolutions, into which she entered heart and soul. A romantic marriage, in 1847, with the Marquis Ossoli, served further to identify her with the revolutionary cause, and when it tumbled into ruins, she and her husband escaped from Rome and started for America. Their ship encountered a terrific storm off Long Island, was driven ashore, broken to pieces by the waves, and both she and her husband were drowned.

By far the greatest of the Transcendental group and one of the most original figures in American literature was Ralph Waldo Emerson—a figure, indeed, in many ways unique in all literature. Born in Boston in 1803, the son of a Unitarian clergyman and a member of a large and sickly family, he followed the predestined path through Harvard College, graduating with no especial honors, entered the ministry, and served as pastor of the Second Church of Boston until 1832. Then, finding himself ill at ease in the position, he resigned, and, settling at Concord, turned to lecturing, first on scientific subjects and then on manners and morals. His reputation grew steadily, and, especially in the generation younger than himself, he awakened the deepest enthusiasm.

In 1836, the publication of a little volume called "Nature" gave conclusive evidence of his talent, and, followed as it was by his "Essays," "Representative Men," and "Conduct of Life," established his reputation as seer, interpreter of nature, poet and moralist—a reputation which has held its own against the assaults of time.

And yet no personality could be more puzzling or elusive. He was at once attractive and repulsive—there was a certain line which no one crossed, a charmed circle in which he dwelt alone. There was about him a certain

coldness and detachment, a self-sufficiency, and a prudence which held him back from giving himself unreservedly to any cause. He lacked heart and temperament. He was a homely, shrewd and cold-blooded Yankee, to put it plainly. Yet, with all that, he was a serene and benignant figure, of an inspiring optimism, a fine patriotism, and profound intellect—a stimulator of the best in man. Upon this basis, probably, his final claim to memory will rest.



EMERSON

Another
Transcendental

eccentric with more than a touch of genius was Henry David Thoreau, and it is noteworthy that his fame, which burned dimly enough during his life, has flamed ever brighter and brighter since his death. This increase of reputation is no doubt due, in some degree, to the "return to nature," which has recently been so prominent in American life and which has gained a wide hearing for so noteworthy a "poet-naturalist"; but it is also due in part to a growing recognition of the fact that as a writer of delightful, suggestive and inspiring prose he has had few equals.

Thoreau is easily our most extraordinary man of letters. Born in Concord of a poor family, but managing to work his way through Harvard, he spent some years teaching; but an innate love of nature and of freedom led him to

seek some form of livelihood which would leave him as much his own master as it was possible for a poor man to be. To earn money for any other purpose than to provide for one's bare necessities was to Thoreau a grievous waste of time, so it came about that for many years he was a sort of itinerant tinker, a doer of odd jobs. Another characteristic, partly innate and partly cultivated, was a distrust of society and a dislike of cities. "I find it as ever very unprofitable to have much to do with men," he wrote; and finally, in pursuance of this idea, he built himself a little cabin on the shore of Walden pond, where he lived for some two years and a half.

It was there that his best work was done, for, at bottom, Thoreau was a man of letters rather than a naturalist, with the most seeing eye man ever had. "Walden, or Life in the Woods," and "A Week on the Concord and Merrimac Rivers" contain the best of Thoreau, and any boy or girl who is interested in the great outdoors, as every boy and girl ought to be, will enjoy reading them.

The last of the Transcendental group worthy of mention here is George William Curtis, a versatile and charming personality, not a genius in any sense, but a writer of pleasant and amusing prose, an orator of no small ability, and one of the truest patriots who ever loved and labored for his country. It is in this latter aspect, rather than as the author of "Nile Notes" and "The Potiphar Papers," that Curtis is best remembered to-day. The books that he produced have, to a large extent, lost their appeal; but the work he did during the dark days of reconstruction and after entitles him to admiring and grateful remembrance.

It is scarcely possible to close a chapter upon American prose writers without referring to at least one of the great editors who have done so much to mould American public opinion. To James Gordon Bennett and Charles A. Dana only passing reference need be made; but Horace Greeley deserves more extended treatment.

Early in the last century, on a rocky little farm in New Hampshire, lived a man by the name of Zaccheus Greeley, a good neighbor, but a bad manager

—so bad that, in 1820, when his son Horace was nine years old, the farm was seized by the sheriff and sold for debt. The proceeds of the sale did not pay the debt, and so, in order to escape arrest, for they imprisoned people for debt in those days, Zaccheus Greeley fled across the border into Vermont, where his family soon joined him. He managed to make a precarious living by working at odd jobs, in which, of course, the boy joined him whenever he could be of any use.



GREELEY

He was a rather remarkable boy, with a great fondness for books, and when he was eleven years old, he tried to get a position in a printing office, but was rejected because he was too young. Four years later, he heard that a boy was wanted in an office at East Poultney, and he hastened to apply for the position. He was a lank, ungainly and dull-appearing boy, and the owner of the office did not think he could ever learn to be a printer, but finally put him to work, with the understanding that he was to receive nothing but his board and clothes for the first six months, and after that forty dollars a year additional.

The boy soon showed an unusual aptitude for the business, and finally decided that the little village was too restricted a field for his talents. With youth's sublime confidence, he decided to go to New York City. He managed to get a position in a printing office there, and two years later, at the age of twenty-two, he and a partner established the first one-cent daily newspaper in the United States. It was ahead of the times, however, and had to be abandoned after a few months.

But he had discovered his peculiar field, and in 1840 he established another paper which he called the "Log Cabin," in which he supported William Henry Harrison through the famous "log cabin and hard cider" campaign. The paper was a success, and in the year following he established the New York "Tribune," which was destined to make him both rich and famous. For more than thirty years he conducted the "Tribune," making it the most influential paper in the country. He became the most powerful political writer in the United States, and in every village groups gathered regularly to receive their papers and to see what "Old Horace" had to say. He was to his readers a strong and vivid personality—they had faith in his intelligence and honesty, and they believed that he would say what he believed to be right, regardless of whose toes were pinched. It was as different as possible to the anonymous journalism of to-day, when not one in a hundred of a newspaper's readers knows anything about the personality of the editor.

We have already referred to the fact that, at the beginning of secession, Greeley doubted the right of the North to compel the seceding states to remain in the Union. Indeed, he counselled peaceful separation rather than war, as did many others, but he was later a staunch supporter of President Lincoln's policy.

We have also spoken of the fact that, when Grant was re-nominated for President in 1872, a large section of the party, believing him incompetent, broke away from the party and named a candidate of their own. The party they formed was called the Liberal Republican, and their candidate was Horace Greeley. They managed to secure for him the support of the Democratic convention, which placed him at the head of the Democratic ticket, but they could not secure the support of the Democrats themselves, who could not forget that Greeley had been fighting them all his life; and the result was that he was overwhelmingly defeated. He had not expected

such a result, his health had been undermined by the labors and anxieties of the campaign, and before the rejoicing of the Republicans was over, Greeley himself lay dead.

SUMMARY

IRVING, WASHINGTON. Born at New York City, April 3, 1783; went abroad for health, 1804; returned to America, 1806; published "Knickerbocker's History of New York," 1809; attaché of legation at Madrid, 1826-29; secretary of legation at London, 1829-32; minister to Spain, 1842-46; died at Sunnyside, near Tarrytown, New York, November 28, 1859.

COOPER, JAMES FENIMORE. Born at Burlington, New Jersey, September 15, 1789; entered Yale, 1802, but left after three years; midshipman in United States navy, 1808-11, when he resigned his commission; published first novel, "Precaution," anonymously, 1820, and followed it with many others; died at Cooperstown, New York, September 14, 1851.

HAWTHORNE, NATHANIEL. Born at Salem, Massachusetts, July 4, 1804; graduated at Bowdoin College, 1825; served in Custom House at Boston, 1838-41; at Brook Farm, 1841; settled at Concord, Massachusetts, 1843; surveyor of the port of Salem, 1846-49; United States consul at Liverpool, 1853-57; published "Twice-Told Tales," 1837; "Mosses from an Old Manse," 1846; "The Scarlet Letter," 1850; "The House of the Seven Gables," 1851; and a number of other novels and collections of tales; died at Plymouth, New Hampshire, May 19, 1864.

STOWE, HARRIET BEECHER. Born at Litchfield, Connecticut, June 14, 1812; educated at Hartford, Connecticut; taught school there and at Cincinnati; published "Uncle Tom's Cabin," 1852; "Dred," 1856; and a number of other novels; died at Hartford, Connecticut, July 1, 1896.

CLEMENS, SAMUEL LANGHORNE. Born at Florida, Missouri, November 30, 1835; apprenticed to printer, 1847; alternated between mining and newspaper work, until the publication of "Innocents Abroad," 1869, made him famous as a humorist; died at Redding, Connecticut, April 22, 1910; published many collections of short stories and several novels.

BANCROFT, GEORGE. Born at Worcester, Massachusetts, October 3, 1800; graduated at Harvard, 1817; collector of the port of Boston, 1838-41;

Democratic candidate for governor of Massachusetts, 1844; secretary of the navy, 1845-46; minister to Great Britain, 1846-49; minister to Berlin, 1867-74; published first volume of his "History of the United States," 1834, last volume, 1874; died at Washington, Jan. 17, 1891.

PRESCOTT, WILLIAM HICKLING. Born at Salem, Massachusetts, May 4, 1796; published "History of the Reign of Ferdinand and Isabella," 1838; "Conquest of Mexico," 1843; "Conquest of Peru," 1847; "History of the Reign of Philip II," 1858; died at Boston, January 28, 1859.

MOTLEY, JOHN LOTHROP. Born at Dorchester (now part of Boston), Massachusetts, April 15, 1814; graduated at Harvard, 1831; studied abroad, 1831-34; United States minister to Austria, 1861-67, and to Great Britain, 1869-70; published "Rise of the Dutch Republic," 1856; "History of the United Netherlands," 1868; "Life and Death of John of Barneveld," 1874; died in Dorset, England, May 29, 1877.

PARKMAN, FRANCIS. Born at Boston, September 16, 1823; graduated at Harvard, 1844; published "The Conspiracy of Pontiac," 1851, and continued series of histories dealing with the French in America to "A Half Century of Conflict," 1892; died at Jamaica Plain, near Boston, November 8, 1893.

ALCOTT, AMOS BRONSON. Born at Wolcott, Connecticut, November 29, 1799; a book-peddler and school-teacher, conducting a school in Boston, 1834-37; removed to Concord, 1840; published "Orphic Sayings," 1840; "Tablets," 1868; "Concord Days," 1872; "Table-Talk," 1877; "Sonnets and Canzonets," 1882; died at Boston, March 4, 1888.

ALCOTT, LOUISA MAY. Born at Germantown, Pennsylvania, November 29, 1832; teacher in early life and army nurse during Civil War; published "Little Women," 1868; "Old-Fashioned Girl," 1869; "Little Men," 1871, and many other children's stories; died at Boston, March 6, 1888.

FULLER, SARAH MARGARET, MARCHIONESS OSSOLI. Born at Cambridgeport, Massachusetts, May 23, 1810; edited *Boston Dial*, 1840-42; literary critic *New York Tribune*, 1844-46; published "Summer on the Lakes," 1843; "Woman in the Nineteenth Century," 1845; "Papers on Art and Literature,"

1846; went to Europe, 1846; married Marquis Ossoli, 1847; drowned off Fire Island, July 16, 1850.

EMERSON, RALPH WALDO. Born at Boston, Massachusetts, May 25, 1803; graduated at Harvard, 1821; Unitarian clergyman at Boston, 1829-32; commenced career as lecturer, 1833, and continued for nearly forty years; edited the *Dial*, 1842-44; published "Nature," 1836; "Essays," 1841; "Poems," 1846; "Representative Men," 1850; and other books of essays and poems; died at Concord, Massachusetts, April 27, 1882.

THOREAU, HENRY DAVID. Born at Concord, Massachusetts, July 12, 1817; graduated at Harvard, 1837; lived alone at Walden Pond, 1845-47; published "A Week on the Concord and Merrimac Rivers," 1849; "Walden, or Life in the Woods," 1854; died at Concord, May 6, 1862. Several collections of his essays and letters were published after his death.

CURTIS, GEORGE WILLIAM. Born at Providence, Rhode Island, February 24, 1824; joined the Brook Farm Community, 1842, and afterwards spent some years in travel; published "Nile Notes of a Howadji," "The Howadji in Syria," "The Potiphar Papers," and other books; prominent as an anti-slavery orator and as the editor of "Harper's Weekly"; died at West New Brighton, Staten Island, August 31, 1892.

GREELEY, HORACE. Born at Amherst, New Hampshire, February 3, 1811; founded *New York Tribune*, 1841; member of Congress from New York, 1848-49; candidate of Liberal-Republican and Democratic parties for President, 1872; died at Pleasantville, Westchester County, New York, November 29, 1872.

CHAPTER III

WRITERS OF VERSE

"Poetry," says the Century dictionary, "is that one of the fine arts which addresses itself to the feelings and the imagination by the instrumentality of musical and moving words"; and that is probably as concise a definition of poetry as can be evolved. For poetry is difficult to define. Verse we can describe, because it is mechanical; but poetry is verse with a soul added.

It is for this very reason that there is so wide a variance in the critical estimates of the work of individual poets. The feelings and imagination of no two persons are exactly the same, and what will appeal to one will fail to appeal to the other; so that it follows that what is poetry for one is merely verse for the other. Tastes vary in poetry, just as they do in food. Indeed, poetry is a good deal like food. We all of us like bread and butter, and we eat it every day and get good, solid nourishment from it; but only the educated palate can appreciate the refinements of caviar, or Gorgonzola cheese, or some rare and special vintage. So most of us derive a mild enjoyment from the works of such poets as Longfellow and Tennyson and Whittier; but it requires a trained taste to appreciate the subtle delights of Browning or Edgar Allan Poe.

Now the taste for the simple and obvious is a natural taste—the child's taste, healthy, and, some will add, unspoiled; but poetry must be judged by the nicer and more exacting standard, just as all other of the fine arts must. I wonder if you have ever read what is probably the most perfect lyric ever written by an American? I am going to set it down here as an example of what poetry can be, and I want you to compare your favorite poems, whatever they may be, with it. It is by Edgar Allan Poe and is called

TO HELEN

Helen, thy beauty is to me

Like those Nicæan barks of yore;
That gently, o'er a perfumed sea,
The weary, wayworn wanderer bore
To his own native shore.

On desperate seas long wont to roam;
Thy hyacinth hair, thy classic face,
Thy Naiad airs, have brought me home
To the glory that was Greece
And the grandeur that was Rome.

Lo! in yon brilliant window-niche
How statue-like I see thee stand,
The agate lamp within thy hand!
Ah, Psyche, from the regions which
Are Holy Land!

In 1821—the same year which saw the publication of *The Spy*, the first significant American novel—there appeared at Boston a little pamphlet of forty-four pages, bound modestly in brown paper boards, and containing eight poems. Two of them were "To a Waterfowl" and "Thanatopsis," and that little volume marked the advent of the first American poet—William Cullen Bryant. Out of the great mass of verse produced on our continent for two centuries after the Pilgrim Fathers landed on Plymouth Rock, his was the first which displayed those qualities which make for immortality.

Before him our greatest poets had been Philip Freneau, the "Poet of the Revolution"; Francis Scott Key, whose supreme achievement was "The Star-Spangled Banner"; Fitz-Greene Halleck, known to every school-boy by his "Marco Bozzaris," but chiefly memorable for a beautiful little lyric, "On the Death of Joseph Rodman Drake"; and Drake himself, perhaps the greatest of the four, but dying at the age of twenty-five with nothing better to his credit than the well-known "The American Flag," and the fanciful and ambitious "The Culprit Fay." But these men were, at best, only graceful versifiers, and Bryant loomed so far above them and the other verse-makers of his time that he was hailed as a miracle of genius, a sort of Parnassan giant whose like had never before existed. We estimate him more correctly to-day as a poet of the second rank, whose powers were limited but

genuine. Indeed, even in his own day, Bryant's reputation waned somewhat, for he never fulfilled the promise of that first volume, and "To a Waterfowl" and "Thanatopsis" remain the best poems he ever wrote.

William Cullen Bryant was born at Cummington, Massachusetts, in 1794, the son of a physician, from whom he received practically all his early training, and who was himself a writer of verse. The boy's talent for versification was encouraged, and some of his productions were recited at school and published in the poet's corner of the local newspaper. In 1808, when Bryant was fourteen years old, the first volume of his poems was printed at Boston, with an advertisement certifying the extreme youth of the author. It contained nothing of any importance, and why anyone should care to read dull verse because it was written by a child is incomprehensible, but the book had some success, and Bryant's father was a very proud man.

Three years later, Bryant entered Williams College, but soon left, and, not having the means to pay his way through Yale, gave up the thought of college altogether, and began the study of law. He also read widely in English literature, and while in his seventeenth year produced what may fairly be called the first real poem written in America, "Thanatopsis," a wonderful achievement for a youth of that age. Six months later came the beautiful lines, "To a Waterfowl," and Bryant's career as a poet was fairly begun. In 1821 came the thin volume in which these and other poems were collected, and its success finally decided its author to relinquish a career at the bar and to turn to literature.

In the years that followed, Bryant produced a few other noteworthy poems, yet it is significant of the thinness of his inspiration that, though he began writing in early youth and lived to the age of eighty-four, his total product was scant in the extreme when compared with that of any of the acknowledged masters. His earnings from this source were never great, and, removing to New York, he secured, in 1828, the editorship of the *Evening Post*, with which he remained associated until his death.

In his later years, he became an imposing national figure. But his poetry never regained the wide acceptance which it once enjoyed, largely because taste in verse has changed, and we have come to lay more stress upon beauty than upon ethical teaching.

America has never lacked for versifiers, and Bryant's success encouraged a greater throng than ever to "lisp in numbers"; but few of them grew beyond the lisping stage, and it was not until the middle of the century that any emerged from this throng to take their stand definitely beside the author of "Thanatopsis." Then, almost simultaneously, six others disengaged themselves—Longfellow, Whittier, Poe, Lowell, Holmes and Emerson—and remain to this day the truest poets in our history.

Of Emerson we have already spoken. His poetry has been, and still is, the subject of controversy. To some, it is the best in our literature; to others, it is not poetry at all, but merely rhythmic prose. It is lacking in passion, in poetic glow—for how can fire come out of an iceberg?—but about some of it there is the clean-cut beauty of the cameo. You know, of course, his immortal quatrain,

Rhodora! if the sages ask thee why
This charm is wasted on the earth and sky,
Tell them, dear, that if eyes were made for seeing,
Then Beauty is its own excuse for being.

More than once he hit the bull's-eye, so to speak, in just that splendid way.

Of the others, Henry Wadsworth Longfellow is easily first in popular reputation, if not in actual achievement. Born at Portland, Maine, in 1807, of a good family, he developed into an attractive and promising boy; was a classmate at Bowdoin College of Nathaniel Hawthorne, and after three years' study abroad, was given the chair of modern languages there. For five years he held this position, filling it so well that in 1834 he was called to Harvard. He entered upon his duties there after another year abroad, and continued with them for eighteen years. The remainder of his life was spent quietly amid a congenial circle of friends at Cambridge. He was essentially home-loving, and took no strenuous interest in public affairs; for this reason, perhaps, he won a warmer place in public affection than has been accorded to any other American man-of-letters, for the American people is a home-loving people, and especially admires that quality in its great men.

From his earliest youth, Longfellow had written verses of somewhat unusual merit for a boy, though remarkable rather for smoothness of rhythm

than for depth or originality of thought. His modern language studies involved much translation, but his first book, "Hyperion," was not published until 1839. It attained a considerable vogue, but as nothing to the wide popularity of "Voices of the Night," which appeared the same year. Two years later appeared "Ballads and Other Poems," and the two collections established their author in the popular heart beyond possibility of assault. They contained "A Psalm of Life," "The Reaper and the Flowers," "The Village Blacksmith," and "Excelsior," which, however we may dispute their claims as poetry, have taken their place among the treasured household verse of the nation.

Four years later, in "The Belfry of Bruges and Other Poems," he added two more to this collection, "The Day is Done" and "The Bridge." The publication, in 1847, of "Evangeline" raised him to the zenith of his reputation. His subsequent work confirmed him in popular estimation as the greatest of American poets—"Hiawatha," "The Courtship of Miles Standish," and such shorter poems as "Resignation," "The Children's Hour," "Paul Revere's Ride," and "The Old Clock on the Stairs."

But, after all, Longfellow was not a really great poet. He lacked the strength of imagination, the sureness of insight and the delicacy of fancy necessary to great poetry. He was rather a sentimentalist to whom study and practice had given an exceptional command of rhythm. The prevailing note of his best-known lyrics is one of sentimental sorrow—the note which is of the very widest appeal. His public is largely the same public which weeps over the death of little Nell and loves to look at Landseer's "The Old Shepherd's Chief Mourner." Longfellow and Dickens and Landseer were all great artists and did admirable work, but scarcely the very highest work. But Longfellow's ballads "found an echo in the universal human heart," and won him an affection such as has been accorded no other modern poet. His place is by the hearth-side rather than on the mountain-top—by far the more comfortable and cheerful position of the two.

The year of Longfellow's birth witnessed that of another American poet, more virile, but of a narrower appeal—John Greenleaf Whittier. Whittier's birthplace was the old house at East Haverhill, Massachusetts, where many generations of his Quaker ancestors had dwelt. The family was poor, and the boy's life was a hard and cramped one, with few opportunities for

schooling or culture; yet its very rigor made for character, and developed that courage and simplicity which were Whittier's noblest attributes.

What there was in the boy that moved him to write verse it would be difficult to say—some bent, some crotchet, which defies explanation. Certain it is that he did write; his sister sent some of his verses to a neighboring paper, and the result was a visit from its editor, William Lloyd Garrison, who encouraged the boy to get some further schooling, and afterwards helped him to secure a newspaper position in Boston. But his health failed him, and he returned to Haverhill, removing, in 1836, to Amesbury, where the remainder of his life was spent.

He had already become interested in politics, had joined the abolitionists, and was soon the most influential of the protestants against slavery. Into this battle he threw himself heart and soul. It is amusing to reflect that, though a Quaker and advocate of non-resistance, he probably did more to render the Civil War inevitable than any other one man. During the war, his lyrics aided the Northern cause; and as soon as it was over, he labored unceasingly to allay the evil passions which the contest had aroused. He lived to the ripe age of eighty-five, simply and bravely, and his career was from first to last consistent and inspiring, one of the sweetest and gentlest in history.

Although Whittier was endowed with a brighter spark of the divine fire than Longfellow, he himself was conscious that he did not possess

The seerlike power to show
The secrets of the heart and mind.

He was lacking, too, in intellectual equipment—in culture, in mastery of rhythm and diction, in felicitous phrasing. And yet, on at least two occasions, he rang sublimely true—in his denunciation of Webster, "Ichabod," and in his idyll of New England rural life, "Snow-Bound."

The third of these New England poets, and also the least important, is Oliver Wendell Holmes. Born at Cambridge, in the inner circle of New England aristocracy, educated at Harvard, and studying medicine in Boston and Paris, he practiced his profession for twelve years, until, in 1847, he was called to the chair of anatomy and physiology at Harvard, continuing in

that position until 1882. He lived until 1894, the last survivor of the seven poets whom we have mentioned.

During his student days, Holmes had gained considerable reputation as a writer of humorous and sentimental society verse, and during his whole life he wrote practically no other kind. Long practice gave him an easy command of rhythm, and a careful training added delicacy to his diction. He became remarkably dexterous in rhyme, and grew to be the recognized celebrant of class reunions and public dinners. Urbane, felicitous and possessing an unflagging humor, he was the prince of after-dinner poets—not a lofty position, be it observed, nor one making for immortal fame. His highwater mark was reached in three poems, "The Chambered Nautilus," "The Deacon's Masterpiece," and that faultless piece of familiar verse, "The Last Leaf," all of which are widely and affectionately known. He lacked power and depth of imagination, the field in which he was really at home was a narrow one, and the verdict of time will probably be that he was a pleasant versifier rather than a true poet.

His claim to the attention of posterity is likely to rest, not on his verses, but upon a sprightly hodgepodge of imaginary table-talk, called "The Autocrat of the Breakfast-Table"—a warm-hearted, kindly book, which still retains its savor.

And this brings us to our most versatile man-of-letters—James Russell Lowell. Born at Cambridge, in the old house called "Elmwood," so dear to his readers, spending an ideal boyhood in the midst of a cultured circle, treading the predestined path through Harvard, studying law and gaining admission to the bar—such was the story of his life for the first twenty-five years. As a student at Harvard, he had written a great deal of prose and verse of considerable merit, and he continued this work after graduation, gaining a livelihood somewhat precarious, indeed, yet sufficient to render it unnecessary for him to attempt to practice law. But it was not until 1848 that he really "struck his gait."

Certainly, then, he struck it to good purpose by the publication of the "Biglow Papers" and "A Fable for Critics," and stood revealed as one of the wisest, wittiest, most fearless and most patriotic of moralists and satirists. For the "Biglow Papers" mark a culmination of American humorous and

satiric poetry which has never since been rivalled; and the "Fable for Critics" displays a satiric power unequalled since the days when Byron laid his lash along the backs of "Scotch Reviewers."

Both were real contributions to American letters, but as pure poetry both were surpassed later in the same year by his "Vision of Sir Launfal." These three productions, indeed, promised more for the future than Lowell was able to perform. He had gone up like a balloon; but, instead of mounting higher, he drifted along at the same level, and at last came back to earth.

The succeeding seven years saw no production of the first importance from his pen, although a series of lectures on poetry, which he delivered before the Lowell Institute, brought him the offer of the chair at Harvard which Longfellow had just relinquished. Two years later, he became editor of the *Atlantic Monthly*, holding the position until 1861. During this time, he wrote little, but the opening of the Civil War gave a fresh impetus to his muse, his most noteworthy contribution to letters being the "Commemoration Ode" with which he marked its close—a poem which has risen steadily in public estimation, and which is, without doubt, the most notable of its kind ever delivered in America. The poems which he published during the next twenty years did little to enhance his reputation, which, as a poet, must rest upon his "Biglow Papers," his odes, and his "Vision of Sir Launfal."

Yet poetry was but one of his modes of expression, and, some think, the less important one. Immediately following the Civil War, he turned his attention to criticism, and when these essays were collected under the titles "Among My Books" and "My Study Windows," they proved their author to be the ablest critic, the most accomplished scholar, the most cultured writer—in a word, the greatest all-around man-of-letters, in America.

This prominence brought him the offer of the Spanish mission, which he accepted, going from Madrid to London, in 1880, as Ambassador to Great Britain, and remaining there for five years. The service he did there is incalculable; as the spokesman for America and the representative of American culture, he took his place with dignity and honor among England's greatest; his addresses charmed and impressed them, and he may be fairly said to have laid the foundations of that cordial friendship between

America and Great Britain which exists to-day. "I am a bookman," was Lowell's proudest boast—not only a writer of books, but a mighty reader of books; and he is one of the most significant figures in American letters.

So we come to the man who measures up more nearly to the stature of a great poet than any other American—Edgar Allan Poe. Outside of America, there has never been any hesitancy in pronouncing Poe the first poet of his country; but, at home, it is only recently his real merit has come to be at all generally acknowledged.

Edgar Allan Poe was born in Boston in 1809—a stroke of purest irony on the part of fate, for he was in no respect a Bostonian, and it was to Bostonians especially that he was anathema. His parents were actors, travelling from place to place, and his birth at Boston was purely accidental. They had no home and no fortune, but lived from hand to mouth, in the most precarious way, and both of them were dead before their son was two years old. He had an elder brother and a younger sister, and these three babies were left stranded at Richmond, Virginia, entirely without money. Luckily they were too young to realize how very dark their future was, and the Providence which looks after the sparrows also looked after them. The wife of a well-to-do tobacco merchant, named John Allan, took a fancy to the dark-eyed, dark-haired boy of two, and, having no children of her own, adopted him.

It was better fortune than he could have hoped for, for he was brought up in comfort in a good home, and his foster-parents seem to have loved him and to have been ambitious for his future. He was an erratic boy, and was soon to get into the first of those difficulties which ended by wrecking his life. For, entering the University of Virginia, he made the mistake of associating with a fast set, with whom he had no business, and ended by losing heavy sums of money, which he was, of course, unable to pay, and which his foster-father very properly refused to pay for him. Instead, he removed the boy from college and put him to work in his office at Richmond.

Edgar felt that, in refusing to pay his debts, his foster-father had besmirched his honor. The thought rankled in his soul, and he ended by running away from home. He got to Boston, somehow, and enlisted in the army, serving for three years as a private. At the end of that time, there was a

reconciliation between him and his foster-father, and the latter provided a substitute for him in the army, and secured him an appointment to the military academy at West Point.

Why Poe should have felt that he was fitted for army life is difficult to understand, since he had always been impatient of discipline; but to West Point he went and very promptly got into trouble there, which culminated, at the end of the year, in court-martial and dismissal. He knew that his foster-father's patience was exhausted, and that he could expect nothing more from him, and he soon proved himself incapable of self-support.

He drifted from New York to Baltimore, often without knowing where his next meal was coming from, and finally, at Baltimore, his father's widowed sister gave him a home, and he soon married her fragile daughter, Virginia Clemm. But he had long been a prey to intemperance, and his habits in consequence were so irregular that he was unable to retain any permanent position. The truth seems to be that Poe was of a temperament so intensely nervous and sensitive that the smallest amount of alcoholic stimulant excited him beyond control, and he lacked the will-power to leave it alone altogether, which was his only chance of safety.

Yet he had gained a certain reputation with discerning people by the publication of a few poems of surprising merit, as well as a number of tales as remarkable and compelling as have ever been written in any language. That is a broad statement, and yet it is literally true. Not only is Poe America's greatest poet, but he is still more decidedly her greatest short-story writer—so much the greatest, that with the exception of Nathaniel Hawthorne, she has never produced another to rival him.

If further testimony to his genius were needed, it might be found in the fact that he was still unable to make a living with his pen, and was forced to see his wife growing daily weaker without the means to provide her proper nourishment. His sufferings were frightful; he was compelled to bend his pride to an appeal for public charity, and the death of his wife wrecked such moral self-control as he had remaining.

The rest is soon told. There was a rapid deterioration, and on October 3, 1849, he was found unconscious in a saloon at Baltimore, where an election had been in progress and where Poe had been made drunk and then used as

an illegal voter. He was taken to a hospital, treated for delirium tremens, and died three days later, a miserable outcast, at an age where he should have been at the very zenith of his powers. The pages of the world's history show no death more pathetically tragic.

Such a death naturally offended right-thinking people. Especially did it offend the New England conscience, which has never been able to divorce art from morals; and as the literary dominance of New England was at that time absolute, Poe was buried under a mass of uncharitable criticism. It should not be forgotten that he had struck the poisoned barb of his satire deep into many a New England sage, and it was, perhaps, only human nature to strike back. So it came to pass that Poe was pointed out, not as a man of genius, but as a horrible example and degrading influence to be sedulously avoided.

With foreign readers, all this counted for nothing. They were concerned not with the life of the man, but with the work of the artist, and they found that work consummately good. They were charmed and thrilled by the haunting melody of his verse and the weird horror of his tales. In his own country, recognition of his genius has grown rapidly of recent years. Within his own sphere, he is unquestionably the greatest artist America can boast—he climbed Parnassus higher than any of his countrymen, and if he did not quite attain a seat among the immortals, he at least caught some portion of their radiance.

After Poe, the man whom foreign critics consider America's most representative poet is another who has been without honor in his own country, and about whom, even yet, there is the widest difference of opinion—Walt Whitman. Whitman was ostracized for many years not because of his life, which was regular and admirable enough, but because of his verse, which is exceedingly irregular in more than one respect.

Whitman was by birth and training a man of the people. His father was a carpenter, and, after receiving a common-school education, the boy entered a printer's office at the age of thirteen. A printer's office is, in itself, a source of education, and Whitman soon began to write for the papers, finally going to New York City, where, for twelve years, he worked on Newspaper Row, as reporter or compositor, making friends with all sorts and conditions of

men and entering heart and soul into the busy life of the great city. The people, the seething masses on the streets, had a compelling fascination for him.

Tiring of New York, at last, he started on a tramp trip to the southwest, worked in New Orleans and other towns, swung around through the northwest, and so back to Brooklyn, where he became, strangely enough, a contractor—a builder and seller of houses. He had been reading a great deal, all these years, but as yet had given no indication of what was to be his literary life-work.

And yet, fermenting inside the man and at last demanding expression, was a strange new philosophy of democracy, all-tolerant, holding the individual to be of the first importance, male and female equal, the body to be revered no less than the soul. For the promulgation of this philosophy, some worthy literary form was needed—poetry, since that was the noblest form, but poetry stripped of conventions and stock phrases, as "fluent and free as the people and the land and the great system of democracy which it was to celebrate." With some such idea as this, not outlined in words, nor, perhaps, very clearly understood even by himself, Whitman set to work, and the result was the now famous "Leaves of Grass," a collection of twelve poems, printed by the author in Brooklyn in 1855.

Like most other philosophies and prophecies, it fell on heedless ears. Few people read it, and those who did were exasperated by its far-fetched diction or scandalized by its free treatment of delicate topics. In the next year, a second edition appeared, containing thirty-two poems; but the book had practically no sale.

Then came the Civil War, and Whitman, volunteering not for the field, but for work in the hospitals, proved that the doctrine of brotherly love, so basic to his poems, was basic also to his character. "Not till the sun excludes you, neither will I exclude you," he had declared; and now he devoted himself to nursing, on battlefield, in camp and hospital, doing what he could to cheer and lighten the worst side of war, an attractive and inspiring figure.

Lincoln, looking out of a window of the White House, saw him go past one day; a majestic person with snow-white beard and hair, his cotton shirt open at the throat, six feet tall and perfectly proportioned; and the President,

without knowing who he was, but mistaking him probably for a common laborer, turned to a friend who stood beside him and remarked, "There goes a man!" And Whitman was a man. Up to that time, he had never been ill a day; but two years later, at the age of fifty-three, his health gave way, under the strain of nursing, and from that time until his death he was, physically, "a man in ruins." Mentally, he was as alert and virile as ever.

He was given a clerical position in one of the departments at Washington after that, remaining there until, in 1873, an attack of paralysis incapacitated him even for clerical labor. Meanwhile he had issued his poems of the war, under the title "Drum-Taps," and had softened some hostile hearts by the two noble tributes to Lincoln there included, "O Captain, my Captain!" and "When Lilacs last in the Dooryard Bloom'd." But his poetry brought him no income and, for a time, after his removal to Camden, New Jersey, where the remainder of his life was to be passed, he was in absolute want. Friends increased, however; his poems were re-issued, and his last years were spent in the midst of a circle of disciples, who hailed Whitman as a seer and prophet and were guilty of other fatuities which made the judicious grieve and did much to keep them alienated from the poet's work.

Since his death, his fame has become established on a firmer basis than hysterical adulation; but it is yet too soon to attempt to judge him, to say what his ultimate rank will be. It seems probable that it will be a high one, and it is possible that, centuries hence, the historian of American letters will start with Whitman as the first exponent of an original and democratic literature, disregarding all that has gone before as merely imitative of Europe.

Of our lesser poets, only a few need be mentioned here. Bayard Taylor, born in Pennsylvania in 1825, of Quaker stock and reared in the tenets of that sect, at one time loomed large in American letters, but it is doubtful whether anything of his has the quality of permanency. His personality was a picturesque and fascinating one and his life interesting and romantic.

A poor boy, burning with the itch to write and especially to travel; at the age of nineteen making his way to England, and from there to Germany; spending two years in Europe, enduring hardships, living with the common people; and finally returning home to find that his letters to the newspapers

had been read with interest and had won a considerable audience—these were the first steps in his struggle for recognition. He collected his letters into a book called "Views Afoot," which at once became widely popular, and his reputation was made.

But it was a reputation as a reporter and traveller, and Taylor, much as he despised it, was never able to get away from it. He became, perforce, a sort of official traveller for the American people, journeyed in California, in the Orient, in Russia, Lapland—in most of the out-of-the-way corners of the world—and his books of travel were uniformly interesting and successful. They do not attract to-day, not, as Park Benjamin put it, because Taylor travelled more and saw less than any other man who ever lived, but because they lack the charm of style, depth of thought, and keenness of observation which the present generation has come to expect.

During all this time, Taylor was struggling with pathetic earnestness for recognition as a novelist and poet, but with poor measure of success. His novels were crude and amateurish, and have long since become negligible; but his verse is somewhat more important. His travels in the East furnished him material for his "Poems of the Orient," which represent him at his best.

His ambition, however, was to write a great epic; but for this he lacked both intellectual and emotional equipment, and his attempts in this field were virtual failures. These failures were to him most tragic; not only that, but he found himself financially embarrassed, and was forced to turn to such hack work as the writing of school histories in order to gain a livelihood. But his friends, of whom he had always a wide circle, secured him the mission to Germany, and he entered on his duties in high spirits—only to die suddenly one morning while sitting in his library at Berlin. A generous, impulsive and warm-hearted man, Bayard Taylor will be remembered for what he was, rather than for what he did.

Two other poets, whose deaths occurred not many months ago, have made noteworthy contributions to American letters—Edmund Clarence Stedman and Thomas Bailey Aldrich. Of the two, Aldrich was by far the better craftsman, his verse possessing a wit, a daintiness and perfection of finish which sets it apart in a class almost by itself. In prose, too, Aldrich wrote attractively, but always rather with the air of a dilettante, and without the

depth and passion of genius. Stedman also possessed wit and polish, though in less degree, and the verse of both these men is delightful reading.

More recent still has been the death of a man whose verse ranks with that of either Stedman or Aldrich—Richard Watson Gilder. Some of his lyrics are very beautiful, but they appeal to the intellect rather than to the heart. Perhaps for this reason, as well as for a certain lack of substance and virility, his verse has never had a wide appeal.

Two men whose names have become household words because of their delightful verses for and about children are Eugene Field and James Whitcomb Riley. Field is the greater of the two, for he possessed a depth of feeling and insight which is lacking in Riley. Few lyrics have been more widely popular than his "Little Boy Blue" and "Dutch Lullaby"; while Riley's "Little Orphant Annie" and "The Raggedy Man" are equally well known.

Alice and Phoebe Cary are remembered for a few simply-written lyrics; Julia Ward Howe's "Battle-Hymn of the Republic" lives as the worthiest piece of verse evoked by the Civil War; and Joaquin Miller is known for a certain rude power in song; but none of them is of sufficient importance to demand extended study.

It will be noted that, among all the poets who have been mentioned here, not one was distinctively of the South. Poe's youth was spent in Richmond, but he was in no sense Southern. Indeed, the South has only three names to offer of even minor importance—Sidney Lanier, Henry Timrod, and Paul Hamilton Hayne. None of these men produced anything of the first order, and much of their verse is marred by amateurishness and want of finish—the result, in the first place, of defective training, and, in the second place, of an incapacity for taking pains, of a habit which relied too much on "inspiration" and too little on intellectual effort.

For verse, to be perfect, must be polished like a diamond, slowly and carefully, until every facet sparkles. This means that the right word or phrase must be searched for until it is found. Perhaps you have read Mr.

Barrie's inimitable story "Sentimental Tommy," and you will remember how Tommy failed to write the prize essay because he couldn't think of the right word, and would be satisfied with no other. Well, that is the spirit. Somebody has said that "easy writing makes hard reading," and this is especially true of poetry. Inspiration doesn't extend to technic—that must be acquired, like any art, with infinite pains.

Of the three poets, Lanier, Timrod, and Hayne, Lanier was by far the greatest, and has even become, in a small way, the centre of a cult; but his voice, while often pure and sweet, lacks the strength needed to carry it down the ages. He is like a little brook making beautiful some meadow or strip of woodland; but only mighty rivers reach the ocean. Lanier is memorable not so much for his work as for the gallant fight he made against the consumption which he had contracted as the result of exposure in the Confederate army during the Civil War. The war also played a disastrous part in the lives of both Hayne and Timrod, for it impoverished both of them, and did much to hasten the latter's death.

Timrod, too, rose occasionally to noble utterance, but his voice is fainter and his talent more slender than Lanier's. His life was a painful one, marred by poverty and disease, and he died at the age of thirty-eight. Hayne's work is even less important, for he did not, like Timrod and Lanier, touch an occasional height of inspired utterance. His name is cherished in his native state of South Carolina, and in Georgia, where his last years were spent; but his poems are little read elsewhere.

Timrod and Hayne were both born at Charleston, South Carolina, as was a third poet and novelist, who, in his day, loomed far larger than either of them, but who is now almost forgotten, except by students of American literature—William Gilmore Simms. Few American writers have produced so much—eighteen volumes of verse, three dramas, thirty-five novels and volumes of short stories, and about as many more books of history, biography and miscellany—and none, of like prominence in his day, has dropped more completely out of sight. In common with the other Southern writers we have mentioned, Simms lacked self-restraint and the power of self-criticism.

Genius has been defined as the capacity for taking pains; and perhaps it is because Southern writers have lacked this capacity that none of them has proved to be a genius. Elbert Hubbard says that Simms "courted oblivion—and won her" by returning to the South after having achieved some success in the North; but it is doubtful if this had anything to do with it. The truth is that Simms's work has lost its appeal because of its inherent defects, and there is no chance that its popularity will ever be regained. And yet, while his verse is negligible—although he always thought himself a greater poet than novelist—some of his tales of the Carolinas and the Southwest possess a rude power and interest deserving of a better fate. Certainly Simms seems to have been the best imaginative writer the antebellum South produced.

American imaginative literature to-day resembles a lofty plateau rather than a mountain range. It shows a high level of achievement, but no mighty peaks. Novelists and poets alike have learned how to use their tools; they work with conviction—but in clay rather than in marble. In other words, they work without what we call inspiration; they have talent, but not genius. This is, perhaps, partly the fault of the age, which has come to place so high a value upon literary form that the quality of the material is often lost sight of. Let us hope that some day a genius will arise who will be great enough to disregard form and to strike out his own path across the domain of letters.

Meanwhile, it is safe to advise boys and girls to spend their time over the old things rather than over the new ones. There is so much good literature in the world that there is really no excuse for reading bad, and the latest novel will not give half the solid entertainment to be got from scores of the older ones. One of the most valuable and delightful things in the world is the power to appreciate good literature. To have worthy "friends on the shelf," in the shape of great books, is to insure oneself against loneliness and ennui.

SUMMARY

BRYANT, WILLIAM CULLEN. Born at Cummington, Massachusetts, November 3, 1794; studied at Williams College, 1810-11; admitted to the bar, 1815; published "Thanatopsis," 1816; editor-in-chief *New York Evening Post*, 1829; published first collection of poems, 1821, and others from time to time until his death, at New York City, June 12, 1878.

LONGFELLOW, HENRY WADSWORTH. Born at Portland, Maine, February 27, 1807; graduated at Bowdoin College, 1825; travelled in Europe, 1826-29; professor of modern languages at Bowdoin, 1829-35; professor of modern languages and *belles lettres* at Harvard, 1836-54; published "Voices of the Night," 1839; "Ballads and Other Poems," 1841; "Poems on Slavery," 1842; and many other collections of his poems, until his death at Cambridge, Massachusetts, March 24, 1882.

WHITTIER, JOHN GREENLEAF. Born at Haverhill, Massachusetts, December 17, 1807; attended Haverhill Academy; edited "American Manufacturer," at Boston, 1829; edited the *Haverhill Gazette*, 1830; became secretary of the American Anti-Slavery Society, 1836; member of Massachusetts legislature, 1835-36; settled at Amesbury, Massachusetts, 1840; published "Legends of New England," 1831; "Moll Pitcher," 1832; and many other collections of his poems until his death at Hampton Falls, New Hampshire, September 7, 1892.

HOLMES, OLIVER WENDELL. Born at Cambridge, Massachusetts, August 29, 1809; professor of anatomy and physiology, Harvard Medical School, 1847-82; published "Autocrat of the Breakfast Table," 1858; "Elsie Venner," 1861; "Songs in Many Keys," 1861; and other collections of poems and essays; died at Cambridge, October 7, 1894.

LOWELL, JAMES RUSSELL. Born at Cambridge, Massachusetts, February 22, 1819; graduated at Harvard, 1838; professor of *belles lettres* at Harvard, 1855; editor *Atlantic Monthly*, 1857-62; editor *North American Review*, 1863-72; minister to Spain, 1877-80; minister to Great Britain, 1880-85;

published "A Year's Life," 1841; "Vision of Sir Launfal," 1845; "A Fable for Critics," 1848; "The Biglow Papers," 1848; and many other collections of essays, criticisms, and poems; died at Cambridge, August 12, 1891.

POE, EDGAR ALLAN. Born at Boston, January 19, 1809; entered University of Virginia, 1826; ran away from home, 1827; published "Tamerlane and Other Poems, by a Bostonian," 1827; enlisted in the army as Edgar A. Perry, rising to rank of sergeant-major, 1829; entered West Point, July 1, 1830; dismissed, March 6, 1831; married Virginia Clemm, 1835, who died in 1847; published "Poems," 1831; "Tales of the Grotesque and Arabesque," 1840; died at Baltimore, October 7, 1849.

WHITMAN, WALT OR WALTER. Born at West Hills, Long Island, May 31, 1819; a printer, carpenter, and journalist in early life; volunteered as army nurse, 1861; seized with hospital malaria, 1864; held government position at Washington, 1864-73; disabled by paralysis and removed to Camden, New Jersey, where he died, March 26, 1892. "Leaves of Grass," published originally in 1855, was many times revised, a final edition appearing in 1892.

TAYLOR, BAYARD. Born at Kennett Square, Chester County, Pennsylvania, January 11, 1825; apprenticed to a printer, 1842; travelled on foot through Europe, 1844-46; in Egypt, Asia Minor, and Syria, 1851-52; in India, China, and Japan, 1852-53; secretary of legation at St. Petersburg, 1862-63; minister to Berlin, 1878; died at Berlin, December 19, 1878. He published collections of poems and travel letters.

STEDMAN, EDMUND CLARENCE. Born at Hartford, Connecticut, October 8, 1833; entered Yale, 1839, leaving in junior year; was correspondent *New York World*, 1861-63; later became stockbroker in New York City, retiring only a short time before his death in New York, January 18, 1908. Published several collections of poems.

ALDRICH, THOMAS BAILEY. Born at Portsmouth, New Hampshire, November 11, 1836; editor of *Every Saturday*, 1870-74; editor of *The Atlantic Monthly*, 1881-90; published "Bells," 1855; "Ballad of Baby Bell," 1856; and many other collections of poetry, together with several novels and collections of short stories; died March 19, 1907.

FIELD, EUGENE. Born at St. Louis, Missouri, September 2, 1850; began newspaper work at age of twenty-three, and ten years later became associated with the *Chicago Daily News*, where most of his work appeared; his first book of verse, "A Little Book of Western Verse," was published in 1889, and a number of others followed; died at Chicago, Illinois, November 4, 1895.

RILEY, JAMES WHITCOMB. Born at Greenfield, Indiana, 1853; entered journalism at Indianapolis, 1873; wrote first verses, 1875; first book of verse, "The Old Swimmin'-Hole and 'Leven More Poems," published in 1883; numerous volumes since then.

LANIER, SIDNEY. Born at Macon, Georgia, February 3, 1842; served in Confederate Army, and suffered exposure which resulted in consumption; studied and practised law till 1873; then decided to devote life to music and poetry; played first flute in the Peabody Symphony Orchestra at Baltimore; lecturer on English literature at Johns Hopkins University, 1879-81; complete poems published 1881; died at Lynn, North Carolina, September 7, 1881.

TIMROD, HENRY. Born at Charleston, South Carolina, December 8, 1829; educated at the University of Georgia, studied law and supported himself as a private tutor until the Civil War; war correspondent and then assistant editor of *The South Carolinian*, at Columbia, until Sherman burned the town; died at Columbia, South Carolina, October 6, 1867; his poems, edited by Paul Hamilton Hayne, published 1873.

HAYNE, PAUL HAMILTON. Born at Charleston, South Carolina, January 1, 1830; graduated at the University of South Carolina, edited *Russell's Magazine* and the *Literary Gazette*, and served for a time in the Confederate Army; first poems published 1855; complete edition, 1882; died near Augusta, Georgia, July 6, 1886.

SIMMS, WILLIAM GILMORE. Born at Charleston, South Carolina, April 17, 1806; admitted to bar, 1827, but abandoned law for literature and journalism; first poems published 1827; resided at Hingham, Massachusetts, 1832-33, where longest poem, "Atalantis," was written; first novel, "Martin Faber," published 1833, and followed by many others; returned to South Carolina, 1833, and died at Charleston, June 11, 1870.

CHAPTER IV

PAINTERS

If background and tradition are needed for literature, they are even more needed for art, and it is curiously worth noting that the background and traditions of England did not serve for her child across the sea. In both literature and art, so far as vital and significant achievement is concerned, the young nation had to find itself, and, starting from a rude and rough beginning, work its way upward of its own strength. Perhaps in no other way may the youth of America be so completely realized as by the thought that all of real importance in both literature and art which she can boast has been produced within the past ninety years—little more than the three score years and ten which the Psalmist assigned as the span of a single life.

We do not mean to say that European influence is not plainly to be traced in both our art and literature. There is a family resemblance, so to speak, as between a child and its parents, and yet the child has an individuality of its own. In literature, Cooper, Poe, Hawthorne, Longfellow, Whitman are distinctively American; and, as we shall find, so are our masters of painting and sculpture.

American art begins with John Singleton Copley. There had been daubers before him, as there were after, but Copley was the first man born in America who produced paintings which the world still contemplates with pleasure. Copley was born in Boston in 1737, his father dying shortly afterwards, and his mother supporting herself by keeping a tobacco shop. About 1746 she married again, most fortunately for her son, for her second husband was Peter Pelham, a mezzotint engraver of considerable merit, who gave the boy lessons in drawing. He proved an apt and precocious pupil, and by the time he had reached seventeen had executed a number of portraits.

His reputation steadily increased, and his income from his work was so satisfactory that he hesitated to try his fortunes in the larger field of London. Finally, in 1774, he sailed for England, and in the next year sent for his family to join him there. The opening of the Revolution persuaded him to stay in England, as there would be no demand for his work in America in so tumultuous a time. In London his talents brought him ample patronage, his income enabled him to live the stately and dignified life he loved, so that, when the Revolution ended, there seemed no reason why he should abandon it for the crudities of Boston. He therefore continued in London until the end of his life, which came in 1815.

Copley was a laborious and painstaking craftsman, setting down what he saw upon canvas with uncompromising sincerity. He worked very slowly and many stories are told of how he tried the patience of his sitters. The result was a series of portraits which preserve the very spirit of the age—serious, self-reliant and capable, pompous and lacking humor. His later work has an atmosphere and repose which his early work lacks, but it is less important to America. His early portraits, which hang on the walls of so many Boston homes, and which Oliver Wendell Holmes called the titles of nobility of the old Boston families, are priceless documents of history.

Copley was an artist from choice rather than necessity; he followed painting because it assured him a good livelihood, and he was a patient and painstaking craftsman. His life was serene and happy; he was without the tribulations, as he seems to have been without the enthusiasms of the great artist. Not so with his most famous contemporary, Benjamin West, whose life was filled to overflowing with the contrast and picturesqueness which Copley's lacked.

West was born in 1738 at a little Pennsylvania frontier settlement. His parents were Quakers, and to the rigor and simplicity of frontier life were added those of that sect. But even these handicaps could not turn the boy aside from his vocation, for he was a born painter, if there ever was one. At the age of six he tried to draw, with red and black ink, a likeness of a baby he had been set to watch; a year later, a party of friendly Indians, amused by some sketches of birds and leaves he showed them, taught him how to prepare the red and yellow colors which they used on their ornaments. His mother furnished some indigo, brushes were secured by clipping the family

cat—no doubt greatly to its disgust—and with these crude materials he set to work.

His success won him the present of a box of paints from a relative in Philadelphia. With that treasure the boy lived and slept, and his mother, finally discovering that he was running away from school, found him in the garret with a picture before him which she refused to let him finish lest he should spoil it. That painting was preserved to be exhibited sixty-six years later.

The boy's talent was so evident, and his determination to be a painter so fixed, that his parents finally overcame their scruples against an occupation which they considered vain and useless, and sent him to Philadelphia. There he lived as frugally as possible, saving his money for a trip to Italy, and finally, at the age of twenty-two, set sail for Europe.

His success there was immediate. He gained friends in the most influential circles, spent three years in study in Italy, and going to London in 1764, received so many commissions that he decided to live there permanently. He wrote home for his father to join him, and to bring with him a Miss Shewell, to whom West was betrothed. He also wrote to the young lady, stating that his father would sail at a certain time, and asking her to join him. The letter fell into the hands of Miss Shewell's brother, who objected to West for some reason, and who promptly locked the girl in her room. Three friends of West's concluded that this outrage upon true love was not to be endured, smuggled a rope-ladder to her, and got her out of the house and safely on board the vessel. These three friends were Benjamin Franklin, Francis Hopkinson and William White, the latter the first Bishop of the American Episcopal Church, and the exploit was one which they were always proud to remember. Miss Shewell reached London safely and the lovers were happily married.

Meanwhile West's success had been given a sudden impetus by his introduction to King George III. The two men became lifelong friends, and the King gave him commission after commission, culminating in a command to decorate the Royal Chapel at Windsor. His first reverse came when the King's mind began to fail. His commissions were cancelled and his pensions stopped. He was deposed from the Presidency of the Royal

Academy, which he had founded, and was for a time in needy circumstances; but the tide soon turned, and his last years were marked by the production of a number of great paintings. He died at the age of eighty-two, and was buried in St. Paul's Cathedral with splendid ceremonies. So ended one of the most remarkable careers in history.

West was, perhaps, more notable as a man than as an artist, for his fame as a painter has steadily declined. His greatest service to art was the example he set of painting historical groups in the costume of the period instead of in the vestments of the early Romans, as had been the custom. This innovation was made by him in his picture of the death of General Wolfe, and created no little disturbance. His friends, including Reynolds, protested against such a desecration of tradition; even the King questioned him, and West replied that the painter should be bound by truth as well as the historian, and to represent a group of English soldiers in the year 1758 as dressed in classic costume was absurd. After the picture was completed, Reynolds was the first to declare that West had won, and that his picture would occasion a revolution in art—as, indeed, it did.

It is difficult to understand the habit of thought which insisted on clothing great men in garments they could never by any possibility have worn, yet it persisted until a comparatively late day. The most famous example in this country is Greenough's statue of Washington, just outside the Capitol. One looks at it with a certain sense of shock, for the Father of His Country is sitting half-naked, in a great arm chair, with some drapery over his legs, and a fold hanging over one shoulder. We shall have occasion in the next chapter to speak of it and of its maker.

Another of West's services to art was the wholehearted way in which he extended a helping hand to any who needed it. He was always willing to give such instruction as he could, and among his pupils were at least four men who added not a little to American art—Charles Willson Peale, Gilbert Stuart, John Trumbull, and Thomas Sully.

Peale was born in Maryland in 1741, and was, among other things, a saddler, a coach-maker, a clock-maker and a silversmith. He finally decided to add painting to his other accomplishments, so he secured some painting materials and a book of instructions and set to work. In 1770, a number of

gentlemen of Annapolis furnished him with enough money to go to England, a loan which he promised to repay with pictures upon his return. West received him kindly, and when Peale's money gave out, as it soon did, welcomed him into his own house. Peale remained in London for four years, returning to America in time to join Washington as a captain of volunteers, and to take part in the battles of Trenton and Germantown.

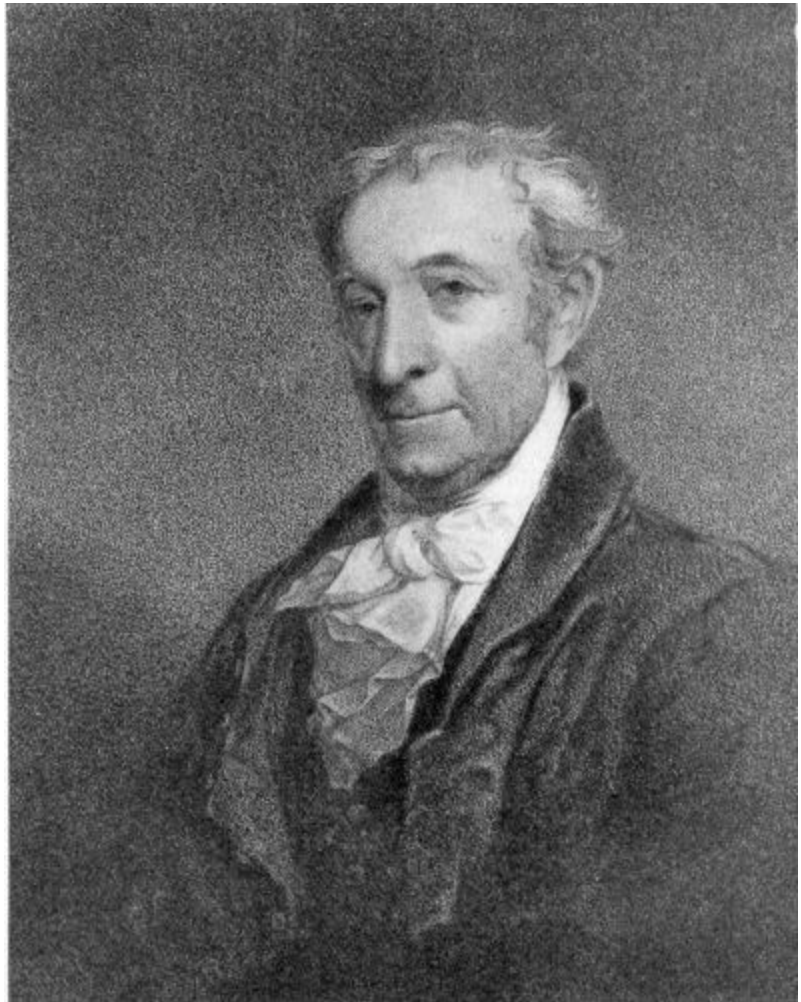
After the war he continued painting, but, in 1801, his mind, always alert for new experiences, was led away in a strange direction. The bones of a mammoth were discovered in Ulster County, New York, and Peale secured possession of them, had them taken to Philadelphia, and started a museum. It rapidly increased in size, for all sorts of curiosities poured in upon him, and he began a series of lectures on natural history, which, whether learned or not, proved so interesting that large and distinguished audiences gathered to hear him. In 1805, he founded the Pennsylvania Academy of the Fine Arts, the oldest and most flourishing institution of the kind in the country. He lived to a hale old age, never having known sickness, and dying as the result of incautious exposure. Like West, his life is more interesting than his work, for while he painted fairly good portraits, they were the work rather of a skilled craftsman than of an artist.

The second of West's pupils whom we have mentioned, Gilbert Stuart, was by far the greatest of the earlier artists. He was born near Newport, R. I., in 1755, his father being a Jacobite refugee from Scotland. He began to paint at an early age, worked faithfully at drawing, and finally, at the age of nineteen, began portrait painting in earnest. One of his first pictures was a striking example of a remarkable characteristic, the power of visual memory, which he retained through his whole life. His grandmother had died five or six years before, but he painted a portrait of her, producing so striking a likeness that it immediately brought him orders for others. But Newport had grown distasteful to him, and in 1775, he started for London.

How he got there is not certainly known, but get there he did, without money or friends, or much hope of making either, and for three years lived a precarious life, earning a little money, borrowing what he could, twice imprisoned for debt, and with it all so gay and brilliant and talented that those he wronged most loved him most. Finally, he was introduced to Benjamin West, and found in him an invaluable friend and patron. For

nearly four years, Stuart worked as West's student and assistant, steadily improving in drawing, developing a technique of astonishing merit, and, more than that, one that was all his own.

His portraits soon attracted attention, and at the end of a few years, he was earning a large income. But he squandered it so recklessly that he was finally forced to flee to Ireland to escape his creditors. They pursued him, threw him into prison, and the legend is that he painted most of the Irish aristocracy in his cell in the Dublin jail.



STUART

At last, in 1792, he returned to America, animated by a desire to paint a portrait of Washington. Arrangements for a sitting were made, but it is related that Stuart, although he had painted many famous men and was at ease in most society, found himself strangely embarrassed in Washington's presence. The President was kindly and courteous, but the portrait was a failure. He tried again, and produced the portrait which remains to this day the accepted likeness of the First American. You will find it as the frontispiece to "Men of Action," and it is worth examining closely, for it is an example of art rarely surpassed, as well as a remarkable portrait of our most remarkable citizen.

Gilbert Stuart still holds his place among the greatest of American portrait painters. His heads, painted simply and without artifice, and yet with high imagination, are unsurpassed; they possess insight, they accomplish that greatest of all tasks, the delineation of character. Stuart's portraits—as every portrait must, to be truly great—show not only how his sitters looked but *what they were*. Art can accomplish no more than that.

The anecdotes which are told of him are innumerable, and most of them have to do with his hot temper, which grew hotter and hotter as his years increased and he became more and more a public character. One day, a loving husband, whose wife Stuart had put on canvas in an unusually uncompromising way, complained that the portrait did not do her justice.

"What an infernal business is this of a portrait painter," Stuart cried, at last, his patience giving way. "You bring him a potato and expect him to paint you a peach!"

But look at his portrait at the beginning of this chapter, and you will see a witty and kindly old gentleman, as well as an irascible one.

John Trumbull was a student of West's at the same time that Stuart was. He was a year younger, and was a son of that Jonathan Trumbull, afterwards governor of Connecticut, whose title of Brother Jonathan, given him by Washington, became afterwards a sort of national nickname. He was an infant prodigy, graduating from Harvard at an age when most boys were entering, and afterwards going to Boston to take lessons from Copley. The outbreak of the Revolution stopped his studies; he enlisted in the army, won rapid promotion, and finally resigned in a huff because he thought his commission as colonel incorrectly dated.

In 1780, he sailed for France, on his way to London, met Benjamin Franklin in Paris and from him secured a letter of introduction to Benjamin West, who welcomed him with his unfailing cordiality; but he had scarcely commenced his studies when he was arrested and thrown into prison. The reason was the arrest and execution at New York of Major André, who was captured with Benedict Arnold's treasonable correspondence hidden in his boot, and who was hanged as a spy. Knowing that Trumbull had been an officer in the American army, and anxious to avenge André's death, the

King ordered his arrest, but West interceded for him and secured his release several weeks later.

Warned that England was unsafe for him, Trumbull returned to America and remained there until after the close of the Revolution. The beginning of 1784 saw him again in London, at work on his two famous paintings, "The Battle of Bunker Hill" and "The Death of General Montgomery," and from that time until his death he was occupied almost exclusively with the painting of pictures illustrating events in American history—"The Surrender of Cornwallis," "The Battle of Princeton," "The Capture of the Hessians at Trenton," to mention only three. In 1816 he received a commission to paint four of the eight commemorative pictures in the Capitol at Washington, and completed the last one eight years later, this being his last important work.

Trumbull is in no respect to be compared with Gilbert Stuart, but his work was done with a painstaking accuracy which makes it valuable as a historical document. For the personages of his pictures he painted a great number of miniatures from life, which, in many cases, are the only surviving presentments of some of the most prominent men of the time.

After Gilbert Stuart, Thomas Sully was by far the greatest of the men who studied in West's studio. Stuart aside, there was no American painter of the day to equal him. He was born in England in 1783, but was brought to this country by his parents at the age of nine. The Sullys were actors of some talent and secured an engagement at Charleston, South Carolina, and there the boy was placed first in school, and then in the office of an insurance broker. He spent so much time making sketches that his employer decided he was destined for art and not for business, and secured another clerk.

Young Sully thoroughly agreed with this and started out to be an artist. He had no money, nor means of earning any, but he managed to secure some desultory instruction, and this, added to his native talent, enabled him to begin to paint portraits for which uncritical persons were willing to pay. But it was a hard road, and none was more conscious of his deficiencies than himself. He knew that he needed training, and finally started for England with a purse of four hundred dollars in his pocket, which had been subscribed by friends, who were each to be repaid by a copy of an old master.

Arrived at London, Sully at once got himself introduced to Benjamin West, who received him "like a father," admitted him to his studio, and aided him in many ways. He remained there, painting by day, drawing by night, studying anatomy in every spare moment, and living on bread and potatoes and water in order to make his money last as long as possible. At the end of nine months it was gone, and he was forced to return to America.

But those nine months of study had given him just what he needed, and his talent soon gained recognition. Orders poured in upon him at good prices; and though his prosperity afterwards dwindled somewhat, he never again experienced the pangs of poverty. He made Philadelphia his home, and for nearly half a century occupied a house on Chestnut Street which had been built for him by Stephen Girard. His work is in every way worthy of respect—firm and serious and rich with a warm and mellow color.

Benjamin West had many other pupils—indeed, his studio was a sort of incubator for American artists—but none of them won any permanent fame. One, Washington Allston, achieved considerable contemporary reputation, but it seems to have resulted more from his own winning personality than from his work. He possessed a charm which fairly dazzled all who met him, notably Coleridge and Washington Irving. His smaller canvasses, graceful figures or heads, to which he attached little importance, are more admired to-day than his more ambitious ones.

Another pupil was John Vanderlyn, of Dutch stock, as his name shows, a protégé of Aaron Burr, and the painter of the best known portrait of his daughter, Theodosia, as well as of Burr himself. When Burr, an outcast in fortune and men's eyes, fled to Paris, Vanderlyn, who had made some reputation there, was able to repay, to some extent, the kindness which Burr had shown him. His work shows care and serious thought, but his last years were embittered by the indifference of the public, and he died in want.

That versatile genius and hale old man, Charles Willson Peale, to whom we have already referred, had many children, and he christened them with most distinguished names, so that, in the end, he could boast himself the father of

Raphael, Rembrandt, Rubens and Titian. Alas that the name does not make the man! Only one of them, Rembrandt, achieved any distinction in art, and that but a faint and far-off reflection of the master whose name he bore.

Like his father, he was interested in many things besides his art; he conducted a museum at Baltimore, introduced illuminating gas there, wrote voluminous memoirs, and, living until 1860, became a sort of dean of the profession. An example of his work will be found in "Men of Action," the likeness of Thomas Jefferson given there being a reproduction from a portrait painted by him. His portraits are not held in high estimation at the present day, for, while correct enough in drawing, they show little insight. We have come to demand something more than mechanical skill, and that "something more," which makes the artist and divides him from the artisan, is exactly what Rembrandt Peale did not possess.

It is interesting, too, to note that one of the most promising painters of the time was S. F. B. Morse. In the Yale School of Fine Arts hangs a portrait of Mrs. De Forest, and in the New York City Hall one of Lafayette, both of them from his brush, and both not unworthy the best traditions of American art. But a chance conversation about electricity turned his thoughts in that direction, and he abandoned painting for invention—the result being the electric telegraph. We shall speak of him further in the chapter on inventors.

The passing of Washington Allston and his group marked the end of Benjamin West's influence, and, in a way, of English influence, on American painting. It marked, too, a lapse in interest, for it was a long time before it found for itself an adequate mode of expression. There are, however, two or three men of the period whom we must mention, not so much because of their achievements, which had little significance, as because of their remarkable and inspiring lives.

Chester Harding, reared on the New York frontier, a typical backwoodsman, by turns a peddler, a tavern-keeper, and house-painter, and a failure at all of them, got so deeply in debt that he ran away to Pittsburgh to escape his creditors, and there, to his amazement, one day saw an itinerant

painter painting a portrait. Before that, he had secured work of some sort, and his wife had joined him. Filled with admiration for the artist's work, he procured a board and some paint, and sat down to paint a portrait of his wife. He actually did produce a likeness, and, delighted at the result, practiced a while longer, and then, proceeding to Paris, Kentucky—perhaps through some association of the name with the great art centre of Europe—boldly announced himself as a portrait painter, and got about a hundred people to pay him twenty-five dollars apiece to paint them.

He spent some time at Cincinnati, and got as far west as St. Louis, where he journeyed nearly a hundred miles to find Daniel Boone living in his log cabin on his Missouri land, and painted the portrait of that old pioneer which is reproduced in "Men of Action." Boone was at that time ninety years of age, and Harding found him living almost alone, roasting a piece of venison on the end of his ramrod, as had been his custom all his life.

One of the most surprising things in the history of American art is the facility with which men of all trades turned to portrait painting, apparently as a last resort, and managed to make a living at it. During the first half of the last century, the country seems to have been overrun with wandering portrait painters, whose only equipment for the art was some paint and a bundle of brushes. They had, for the most part, no training, and that anyone, in a time when money was scarce and hardly earned, should have paid it out for the wretched daubs these men produced is a great mystery. But they did pay it out, and, as we have seen, Harding earned no less than twenty-five hundred dollars in a comparatively short time.

With such of this money as he had been able to save, he went to Philadelphia and spent two months in study there; then he returned to his old home, and astonished his neighbors by paying his debts. He astonished them still more when they found he was making money by painting portraits, for which he now charged forty dollars each, and his aged grandfather felt obliged to protest.

"Chester," he said, having called him aside so that none could overhear, "I want to speak to you about your present mode of life. I think it no better than swindling to charge forty dollars for one of those effigies. Now I want

you to give up this way of living and settle down on a farm and become a respectable man."

However excellent this advice may have been, Chester had gone too far to heed it. He had decided to go to England, but he stayed in America long enough to earn money to buy a farm for his parents and to settle his own family at Northampton. This duty accomplished, he set sail for London, and his success there was immediate, due as much to his remarkable personality as to his work. He returned to America in 1826, and spent the rest of his life here, painting most of the political leaders of the country. It has been said of his portraits that his heads are as solid as iron and his coats as uncompromising as tin, while his faces shine like burnished platters.

Remarkable as Harding's story is, it is no more so than that of many of his contemporaries. Francis Alexander, for instance, born in Connecticut in 1800, a farm boy and afterwards a school teacher, never attempted painting until he was over twenty. Then one day, having caught a pickerel, its beauty reminded him of a box of water-colors a boy had left him, and he attempted to paint the fish, with such success that he was filled with amazement and delight. He practiced a while longer, decorating the white-washed walls of a room with rude landscapes filled with cattle, horses, sheep, hogs and chickens. All the neighbors came to see his work and marvelled at it, though none of them cared to have his house similarly decorated; but finally one of them offered Alexander five dollars if he would paint a full-length portrait of a child.

Other orders followed, and finally with sixty dollars in his pocket, he started for New York. Some years later, he sought Gilbert Stuart, at Boston, got some systematic instruction and ended by painting very passable portraits.

Some amusing stories are told of the persistency with which he hunted for orders. In 1842, Charles Dickens visited America for the first time, and while his ship was yet out of sight of land, the pilot clambered on board, and after him Alexander, who begged the great novelist for the privilege of painting his portrait. Dickens, amused at his enterprise, consented, and Alexander's studio, during the sittings, became the centre of literary Boston. It is a curious commentary upon Alexander's development that, after a trip

or two abroad, he professed to find the crudities of his native land unbearable, and spent his last years in Italy.

A third self-made artist was John Neagle, whose portrait of Gilbert Stuart, which heads this chapter, is the best that exists. Neagle was apprenticed, when a boy, to a coach-painter, and soon was spending his spare time practicing a more ambitious branch of the painting profession. As soon as he was through his apprenticeship he set up as a portrait painter, and travelled over the mountains to Lexington, Kentucky, hoping to fare as well as Harding had. But he found the field already pre-empted by two other painters, one of whom, Matthew Jouett, was an artist of considerable skill.

Neagle had a hard time getting back home again, but he finally reached Philadelphia, and spent most of the remainder of his life there. Practice and study gave him a certain skill; he visited Boston and had the advantage of some instruction from Gilbert Stuart, but his work remained to the end inferior to either Harding's or Alexander's.

Henry Inman had a more varied talent than any of these men, for besides portraits he painted genre scenes and landscapes, and excelled in all of them. At the age of fourteen, he had been apprenticed to a painter by the name of John Wesley Jarvis, a picturesque character, better remembered by his anecdotes than by his work; and when his apprenticeship was over he began painting on his own account in New York and afterwards in Philadelphia. For a time his popularity was very great and his income large; but reverses came, ill health followed, and he died in poverty at the age of forty-five.

It is worth noting that, up to this time, practically no landscapes had been produced by American artists. A few of them had tried their hands at landscape work, but soon abandoned it for the more profitable field of portraiture. The first of the American school of landscapists may be fairly said to be Asher Brown Durand. Durand was the eighth of eleven children, and his father, who managed a small farm on the slope of Orange Mountain, in New Jersey, was renowned throughout the neighborhood for his mechanical ingenuity. Much of this ingenuity his son inherited, and his first artistic effort was an attempt to reproduce the woodcuts in his school books by engraving them on little plates which he had beaten out of copper cents.

This led to his being apprenticed to an engraver, and after his apprenticeship was over, he devoted three years to engraving the plate of Trumbull's "Signing of the Declaration of Independence." The work was excellently done and established Durand's reputation.

But he was not satisfied with engraving, and soon abandoned it for the more creative work of painting. He tried his hand first at portraiture, in which he had considerable success; but he turned more and more to landscape work as the years went on. He practiced it continuously until his eighty-third year. Then he laid down his brush forever, saying, "My hand will no longer do my bidding," and the remaining seven years of his life were passed peacefully on the farm where he was born.

Durand's work is marked throughout by sincerity and skill, if not by genius. His portraits were in a style especially his own, thorough in workmanship, delicately modelled and strongly painted. His landscapes, too, are his own, clearly and definitely finished, and with a bewitching silvery gray tone, which could have come only by painting direct from his subject in the open air, a practice exceptional at the time. His pictures are not "compositions," in the artistic sense of the term—that is, he did not combine detail into a balanced whole; they are rather studies or sketches from nature, with a central point of interest. But the work is done so truly and with such patience and enthusiasm that it deserves the sincerest admiration.

Joined with Durand as the earliest of the landscapists is Thomas Cole. Cole was born in England and did not come to America until he had reached his nineteenth year, but he afterwards became so good an American that he declared he would give his left hand to have been identified with America by birth instead of adoption. He found employment in Philadelphia as an engraver. Then, after some practice, he got together a kit of painting materials, and started to tramp about the country as a portraitist. He found the woods full of them, and competition so fierce that he was unable to make a living; but, determining to be an artist at any cost, he returned to Philadelphia and passed a fearful winter there, living on bread and water, half frozen by the cold, with only a cloth table-cover for overcoat and bed, and suffering tortures from inflammatory rheumatism. A second trying winter followed, but in the spring of 1825 he removed to New York, and his privations were at an end.

For in those years of suffering he had developed a delicate art as a landscapist, and he found a ready sale for his pictures, at first at low prices, it is true; but his fame spread rapidly, and he was able, in 1829, to go abroad and spend three years in Italy and England. He lived only to the age of forty-seven, his last years being passed principally in his studio in the Catskills, where some of his most famous pictures were painted.

Cole was widely known for many years for the various series of moral and didactic pictures which he was fond of painting. Perhaps the most famous of these was his "Voyage of Life," showing infancy, youth, manhood, and old age floating down the stream of time. The taste of the period approved them, and they were especially popular for schoolrooms, lecture-halls and other places where youth would have a chance to gaze upon and gather edification from them. It has since come to be recognized that the proper way to tell a story is by words and not by pictures, and "The Voyage of Life," and "Course of Empire," and "The Cross and the World" have, for the most part, been relegated to the attic.

Durand and Cole were the founders of the famous Hudson River, or White Mountain school, which loomed so large in American art half a century ago. Its members, now rather regarded in the light of primitives, gloried in the views of the Hudson, especially as seen from the Catskills, and journeyed into the wilds of the Rockies and the Yellowstone in search of sublime subjects—too sublime to be transferred to canvas. They loved nature—loved to copy her minutely and literally, loved to live in her hills and woods. Some of them came afterwards to see that, after all, this was not art, or only one of her lower forms—that to achieve a great result, a picture must express an idea.

Cole had a pupil and disciple, who did some admirable work, in Frederick Edwin Church. Church was born in 1826, and lived with Cole in his house in the Catskills until the latter's death. He then established himself in New York, and proceeded to visit the four corners of the earth in search for grandiose scenes. For he made the mistake of thinking that the greatness of a landscape lay in its subject rather than in its execution; so he painted views of the Andes, and Niagara, and Cotopaxi, and Chimborazo, and the Parthenon, throwing in rainbows and sunsets and mists for good measure. These pictures were welcomed with the wildest enthusiasm—just as Clarke

Mills's statue of General Jackson had been, fifteen years before. Strange to say, they were not absurd, as that amazing figure is, but were really fine examples of clever handling and of a true, if untrained, feeling.

Two men attempted to duplicate Church's success, but with very indifferent result. They were Albert Bierstadt and Thomas Moran. The former sought the Rocky Mountains for his subjects; the latter, the Yosemite and the Yellowstone; but neither of them succeeded in transferring to canvas more than a pale and unconvincing presentment of the wonders of those regions.

Durand also had a disciple, more famous than Cole's, in Frederick Kensett, the best known of the so-called Hudson River school. He was a close follower of Durand in believing that nature should be literally rendered, but he missed the truth of the older man by working in his studio from drawings and sketches, instead of in the open air direct from his subject. So he got into the habit of painting all shadows a transparent brown, and of making his rocks and trees brilliant by touching in high-lights where he thought they ought to be instead of where they actually should have been. He surpassed Durand, however, in his range of subject, for all hours and seasons had their charm for him, while Durand was really at home only in the full light of a summer day.

On this foundation a loftier structure was soon built and the builders were George Inness, Alexander Wyant and Homer D. Martin. Inness was the oldest of the three, having been born in 1825, and was contemporary with some of the most arbitrary and hide-bound of the nature copyists. But he felt the weakness of the method and himself attained a much fuller and completer art. He seems to have dabbled with paint and brushes from his youth, but had little regular instruction, studying, for the most part, from prints of old pictures, and finally, in 1847, getting a chance to see the original when a friend offered to send him to Europe. He passed fifteen months in Rome, and afterwards a year at Paris.

A long period of assimilation followed, in which he developed a theory of art and struggled to transfer it to canvas. It was a sound and true theory, and is worth setting down here for its own sake. "The purpose of the painter," Inness held, "is to reproduce in other minds the impression which a scene had made upon him. A work of art does not appeal to the intellect or to the

moral sense. Its aim is not to instruct, not to edify, but to awaken an emotion. It must be a single emotion, if the work has unity, as every such work should have, and the true beauty of the work consists in the beauty of the sentiment or emotion which it inspires. Its real greatness consists in the quality and force of this emotion."

To the very last, Inness's work was changing and developing to fit this theory. He steadily gained mastery of tone and breadth of handling, of true harmony, and it is his crowning merit that he does to some extent succeed in "reproducing in other minds the impression which the scene made upon him."

Alexander H. Wyant was a pupil of Inness, journeying from the little Ohio town where he was born to see him and to ask for advice and aid, which Inness freely gave. Wyant's boyhood had been the American artist's usual one—an early fondness for drawing, a little practice, and then setting up as a painter. In 1873 he joined an expedition to Arizona and New Mexico. The hardships which he endured resulted in a stroke of paralysis and he was never again able to use his right hand. With an inspiring patience, he set to work to learn to use his left hand, and grew to be more skillful with it than he had been with his right.

But even at his best, Wyant's appeal is more limited than Inness's. He learned to paint a typical picture, a glimpse of rolling country seen between the trunks of tall and slender birches or maples, and was content to paint variations of it over and over. That he sometimes did it superbly cannot be denied, and he possessed a certain delicate refinement, an ability to throw upon his pictures the silvery shimmer of summer sunshine, in which no other American artist has ever surpassed him.

The third, and in some respects the most interesting member of the group is Homer D. Martin. Born in Albany in 1838, he turned naturally to painting and began to produce pictures after only two weeks' instruction. At first, he was a disciple of Kensett, with brown shadows and artificial high-lights, but study of nature soon cured these mannerisms, and he grew steadily in skill and power, until he succeeded in imparting to his pictures the deep, grave and sobering sentiment, which is the keynote of his work. His coast views,

with their swirl and almost audible thunder of billow, are considered his crowning achievements.

This culmination of the Hudson River school brings us fairly to our own times and to the work of men still living, for the period just preceding and following the Civil War was marked by no new impulse in American art and by no work which demands attention. But in the early seventies, there were a number of Americans studying at home or in Europe who have since won a wide reputation for inspiring achievement.

Foremost among these is Elihu Vedder, born in New York City in 1836, and following, in his manhood, the manifest bent of his childish years. He went to Paris before he was of age, and from there to Rome, where he spent five years. The five succeeding years were spent in America, and finally, in 1866, he settled in Rome and has since made it his home. He represents a revival of the classical quality of Raphaël or Michael Angelo, though he belongs to no school, and his work has from the very first possessed a distinct originality. He has held to the old simplicity, which minimized detail and exalted the subject. General recognition came to him in 1884, when he published his illustrations to the Rubaiyat of Omar Khayyam—the most sympathetic and beautiful pictorial comment which has ever been given any book of poetry. Since then he has executed much decorative work of a high order, though the mastery in this branch of the art is held by another.

That other is John LaFarge, admittedly the greatest mural painter the world has seen in recent years. His life was a fortunate one. His father, an officer of the French marine, came to this country in 1806, married, and purchased a great plantation in Louisiana, from which he derived a large revenue. His son, born in 1835, grew up in an artistic atmosphere of books and pictures, and was early taught to draw. When, after some study of law, he visited Paris, his father advised him to take up the study of art as an accomplishment, and he entered one of the studios, merely as an amateur, at the same time gaining admittance, through his family connections, to the inner artistic circles of the capital. For some years he studied art, not to become a painter, but because he wished to understand and appreciate great work, and at the end of that time, he returned to New York and entered a lawyer's office.

But he was ill at ease there, and finally definitely decided upon an artistic career, went to Newport and worked under the guidance of William Morris Hunt, painting everything, but turning in the end to decorative work, and afterwards to stained glass. In these he has had no equal, and his high achievement, as well as the wide appreciation his work has won, is peculiarly grateful to Americans, since LaFarge's career has been characteristically American. He had little actual study in Europe, and yet possesses certain great traditions of the masters to a degree unequalled by any compatriot.

Of his work as a whole, it is difficult to speak adequately. Perhaps its most striking characteristic is the thought that is lavished upon it, so that the artist gives us the very spirit of his subjects. In inspiration, in handling, in drawing, and in color, LaFarge stands alone. No man of his generation has equalled him in the power to lift the spectator out of himself and into an enchanted world by the consummate harmony of strong, pure color. This feeling for color culminated in his stained-glass work—probably the richest color creations that have ever been fashioned on this earth. In all his varied mass of production there is nothing that lacks interest and charm.

We have referred to LaFarge's study under William Morris Hunt, and we must pause for a moment to speak of the older artist. His artistic career was in some respects an accident, for, developing a tendency to consumption in his late boyhood, his mother took him to Rome and remained there long enough to enable him to imbibe some of the artistic traditions of the Eternal City and to begin work with H. K. Brown, the sculptor. He found the work so congenial that he persuaded his mother to omit the course at Harvard which had been expected of him, and to permit him to devote his life to art.

For five or six years thereafter, he studied at Rome and Paris, then for three years he was with Millet at Barbizon. Finally, in 1855, he returned to America, settling first at Newport and afterwards at Boston. He painted many portraits and figure pieces, and was an active social and artistic influence to the day of his death. As an artist, he lacked training, and remained to the end an amateur of great promise, which was never quite fulfilled.

And this brings us to the most eccentric, the most striking, and in some respects the greatest artist of his time—James Abbott McNeill Whistler. Whistler was born at Lowell, Massachusetts, in 1834. His grandfather, of an English family long settled in Ireland, had been a member of Burgoyne's invading army, but afterwards joined the American service, and, after the close of the Revolution, settled at Lowell. His father was a distinguished engineer, and major in the army, and after his death in 1849, it was natural that young Whistler should turn to the army as a career. He entered West Point in 1851, remained there three years, and was finally dropped for deficiency in chemistry.

There was one study, however, in which he had distinguished himself, and that was drawing; and after his dismissal he went to Paris, where he studied for two or three years. Then he removed to London, where most of the remainder of his life was spent. His work, striking and original, was at first utterly misunderstood by the public. The most famous piece of hostile criticism to which he was subjected was Ruskin's remark, after looking at "The Falling Rocket" in 1877, that here was a fellow with the effrontery to charge a hundred guineas for flinging a pot of paint in the public's face. Some further years of abuse followed, and then the pendulum swung the other way, and the eccentric artist became a sort of cult. In the end, he won a wide reputation, and before his death was recognized as one of the leading painters of his time.

And this reputation was deserved, for his work possesses a rare and delicate beauty, individual to it. His portraits of his mother and of Thomas Carlyle are admirable in their simplicity and quiet dignity; and many of his "harmonies," as he liked to call them, are so complete and flawless that they are works of pure delight. Whistler always declared that he had no desire to reproduce external nature, but only beautiful combinations of pattern, and tone; what he meant, probably, was that he sought, not external realities, but the spirit which underlies them. That, of course, has been the quest of every great painter.

If Whistler was a law unto himself, so, in another sense, is Winslow Homer, who has worked out for himself an individual point of view and method of expression. Born in Boston in 1836, and early developing a taste for drawing, he entered a lithographer's shop at the age of nineteen and two

years later set up for himself. During the Civil War he acted as correspondent and artist for *Harper's Weekly*, and, when peace came, began his paintings with a series of army scenes. After that he tried his hand at landscape, and finally found his real vocation as a painter of the sea. From the first, his pictures possessed obvious sincerity. More than that, they convince by their absolute veracity, as a reproduction of the thing seen—seen, be it understood, by the eyes of the artist—and so they have lived and been remembered where more ambitious work would have been forgotten. Again, he chooses his subjects with a fine disregard of what other men have done or decided that it was impossible to do, and painted them in a manner wholly independent and original. No other artist has so conveyed on canvas the weight and buoyancy and enormous force of water; no one else approaches him as an interpreter of the power of the sea.

Lineal successor of Inness is Dwight William Tryon, not that his work resembles the older man's, but because both paint the American landscape with a deep personal feeling and with a superb technique. Tryon has not yet developed into so commanding a figure as Inness, but there is no telling what the future holds for him, for his work seems as full of poetry and emotion as the older man's, with a spirit more delicate and a foundation more firm.

The work of Francis D. Millet has attracted wide attention and is also full of promise and inspiration. Millet has the American versatility—he has been a war-correspondent, an illustrator, has written travels, criticism, and even fiction, has acted as an expert on old pictures, raised carnations, and even, in time of need, performed surgical operations on wounded soldiers—all of it, not as an amateur, but as a professional asking no odds of anyone. In addition to which, he has been a painter, and a painter whose work has shown no sign of haste or distraction. The quiet, human side of English life in the seventeenth and eighteenth centuries is what has most appealed to him, the country parlors and white-washed kitchens, peopled with travellers and buxom serving-maids, and these groups are unusually attractive and well executed.

Allied with Millet in taste and viewpoint, and with a much wider popularity, is Edwin A. Abbey. Beginning his career as an illustrator, he soon reached the front rank in that profession, especially with his illustrations of classic

English poems, into whose spirit he has entered so completely that he might better be called their interpreter than their illustrator. From pen-and-ink work, he progressed naturally to oil, and here, too, he has achieved some notable triumphs—so notable, indeed, that, though American, he was chosen by the English government to paint the official picture of the coronation of King Edward VII. It is a curious coincidence that the official picture of the coronation of Queen Victoria was also painted by an American, C. R. Leslie.

More important than Abbey, and perhaps the greatest American artist alive to-day is John Singer Sargent, whose nationality has occasioned no little controversy. Born in Florence of American parents, receiving his artistic training in Paris, residing since in England, though with much travelling through Europe and only two or three trips to the land of his allegiance, he may still be held an American, if descent counts for anything. His paintings have been shown wherever pictures are to be seen and he has received for them all honors that a painter can receive.

Before the freedom and certainty of Sargent's art criticism stands abashed. His portraits have a wonderful effect of vitality, and a purity and brilliancy of color which have never been surpassed; but most noteworthy of all, he achieves the supreme triumph of the portrait painter by comprehending and displaying character. He shows the very soul of his sitter, without malice but also without mercy. Only towards children does he show tenderness, and then he paints with a wonderful and varied charm. Not only of people but of places does he give the character—a room takes on personality; silks, velvets, furniture, bric-à-brac are all eloquent. On the whole, his qualities are such that he may rightly be considered the greatest portrait painter since Reynolds and Gainsborough. The portrait of Edwin Booth, at the beginning of the chapter dealing with the stage, is an excellent specimen of his work.

Sargent's portraits have placed him among the masters of all time, but perhaps he is most widely known by his remarkable decorations in the Boston Public Library, which in the original and in photographic reproductions, have given the keenest delight to thousands and thousands of persons. It is impossible to give any detailed description here of these masterpieces of decorative art, so perfect technically that they might almost serve as a canon to decorative painters.

American painting may be said to have reached its culmination in Sargent, yet there are two other painters, who, if they fall below him in sheer genius, possess a charm and originality all their own. One of these is George de Forest Brush, who, somewhat after the fashion of Holbein, looks for a beauty of spirit independent of form or feature. He paints mothers and children not as young goddesses rollicking with cherubs, but as grave and tender women, who have sacrificed without regret something of their health and youthful freshness to the children they hold in their arms. In such groups there is a note of penetrating peace, a delicate distinction, which give Brush a position by himself.

The other is John W. Alexander, whose work is interesting as introducing a certain new element into art—a concentration of energy on the originality of the first general effect, including nothing that does not interest, and yet giving the effect of completeness. In Alexander's portraits there is nothing to distract the interest from the personality of the sitter, and he usually achieves a delineation of character direct and truthful.

Here this short review of the great personalities of American art must end. There are many other painters alive to-day whose work is full of promise, and who may yet achieve great places in the world's Pantheon. Indeed, it would almost seem that a renaissance of American art is at hand. The country has emerged from the crudities of its first years, and from the mediocre conventionality of its middle period, without having lost the freshness and enthusiasm conducive to high achievement. Its face is toward the sunrise.

SUMMARY

COPLEY, JOHN SINGLETON. Born at Boston, July 3, 1737; went to Europe, 1771, and spent the remainder of his life there, principally in London; associate of Royal Academy, 1771; full member, 1773; died at London, September 9, 1815.

WEST, BENJAMIN. Born at Springfield, Chester County, Pennsylvania, October 10, 1738; studied in Italy, 1760-63; settled in London, 1763; became court historical painter, 1772; president of the Royal Academy for many years; died at London, March 11, 1820.

PEALE, CHARLES WILLSON. Born at Chestertown, Maryland, April 16, 1741; with Copley at Boston, 1768-69; went to London, 1770; and studied under Benjamin West; returned to America, 1774; served in Revolution, 1776-77; opened "Peale's Museum," 1802; died at Philadelphia, February 22, 1827.

STUART, GILBERT. Born at Narragansett, Rhode Island, December 3, 1755; went to London and became pupil of West, 1775; returned to United States, 1792; died at Boston, July 27, 1828.

TRUMBULL, JOHN. Born at Lebanon, Connecticut, June 6, 1756; served in Revolution, attaining rank of colonel; studied under West in London, and returned to America, 1804; died at New York City, November 10, 1843.

SULLY, THOMAS. Born at Horncastle, Lincolnshire, England, June 8, 1783; brought to America at the age of nine; went to London, 1809, and studied under West; settled in Philadelphia in 1810, and spent the remainder of his life there, dying November 5, 1872.

ALLSTON, WASHINGTON. Born at Naccamaw, South Carolina, November 5, 1779; graduated at Harvard, 1800; studied at Royal Academy and at Rome, returning to America, 1809; died at Cambridge, Massachusetts, July 9, 1843.

VANDERLYN, JOHN. Born at Kingston, New York, October 15, 1775; studied art abroad, 1796-1801; and spent subsequent years in Europe, returning to

America in 1815; died at Kingston, September 24, 1852.

PEALE, REMBRANDT. Born in Bucks County, Pennsylvania, February 22, 1778; went to London and studied under West, 1801-03; died at Philadelphia, October 3, 1860.

HARDING, CHESTER. Born at Conway, Massachusetts, September 1, 1792; studied in London, 1823-26; died at Boston, April 1, 1866.

ALEXANDER, FRANCIS. Born in Connecticut, 1800; went to Europe in 1831, finally taking up his residence in Florence, where he died.

NEAGLE, JOHN. Born at Boston, November 4, 1796; died at Philadelphia, September 17, 1865.

INMAN, HENRY. Born at Utica, New York, October 20, 1801; served seven years' apprenticeship with John Wesley Jarvis; died at New York City, January 17, 1846.

DURAND, ASHER BROWN. Born at Jefferson, New Jersey, August 21, 1796; apprenticed to Peter Maverick, an engraver, 1812; president of National Academy of Design, 1845-61; died at South Orange, New Jersey, September 17, 1886.

COLE, THOMAS. Born at Bolton-le-Moors, Lancashire, England, February 1, 1801; came to America, 1819; settled in New York, 1825; died at Catskill, New York, February 11, 1848.

CHURCH, FREDERIC EDWIN. Born at Hartford, Connecticut, May 4, 1826; pupil of Thomas Cole; National Academician, 1849; died at New York City, April 7, 1900.

BIERSTADT, ALBERT. Born at Düsseldorf, Germany, January 7, 1830; brought to America, 1831; early developed a taste for art, and studied at Düsseldorf, 1853-57; returned to America and remained here, except for brief visits to Europe; died at New York City, February 18, 1902.

MORAN, THOMAS. Born at Bolton, England, January 12, 1837; came to America, 1844; National Academician, 1884; still living in New York City.

KENSETT, JOHN FREDERICK. Born at Chester, Connecticut, March 22, 1818; in Europe, 1840-44; National Academician, 1849; died at New York City, December 16, 1872.

INNESS, GEORGE. Born at Newburgh, New York, May 1, 1825; National Academician, 1868; died at Bridge of Allan, Scotland, August 3, 1894.

WYANT, ALEXANDER H. Born at Port Washington, Ohio, January 11, 1836; studied in Germany and settled in New York, 1864; suffered paralytic stroke, 1877, and afterwards painted with left hand; died at New York City, November 29, 1892.

MARTIN, HOMER DODGE. Born at Albany, New York, October 28, 1836; opened New York studio, 1862; National Academician, 1875; died at St. Paul, Minnesota, February 12, 1897.

VEDDER, ELIHU. Born at New York City, February 26, 1836; in Paris and Italy, 1856-61; and, after a year or two in America, returned to Italy, where he has since resided; National Academician, 1865.

LA FARGE, JOHN. Born at New York City, March 31, 1835; studied under Couture and Hunt; National Academician, 1869; president Society of American Artists and Society of Mural Painters.

HUNT, WILLIAM MORRIS. Born at Brattleboro, Vermont, March 31, 1824; studied under Couture and Millet, 1846-55; opened Boston studio, 1856; died at Appledore, Isles of Shoals, New Hampshire, September 8, 1879.

WHISTLER, JAMES ABBOTT MCNEILL. Born at Lowell, Massachusetts, 1834; entered West Point Academy, 1851, but soon left; settled in Paris, 1856, and studied art two years, and then settled in London, where the remainder of his life was passed; died there, July 17, 1903.

HOMER, WINSLOW. Born at Boston, February 24, 1836; accompanied Army of Potomac in its campaigns, 1861-62; National Academician, 1865.

TRYON, DWIGHT WILLIAM. Born at Hartford, Connecticut, August 13, 1849; National Academician, 1891.

MILLET, FRANCIS DAVIS. Born at Mattapoisett, Massachusetts, November 3, 1846; drummer 60th Massachusetts Volunteers, 1864; graduated at Harvard,

1869; studied at Antwerp, 1871-72; correspondent Russo-Turkish war, 1877-78; director of decorations World's Columbian Exposition, 1892-93.

ABBAY, EDWIN AUSTIN. Born at Philadelphia, April 1, 1852; educated at Philadelphia Academy of Fine Arts; went to England, 1878, and has since made that his home.

SARGENT, JOHN SINGER. Born at Florence, Italy, 1856; studied under Carolus Duran; has made England his home; Royal Academician, 1891; National Academician, 1897.

CHAPTER V

SCULPTORS

If background and tradition are needed for painting, how much more are they needed for sculpture! America was settled by a people entirely without sculptural tradition, for, in the early seventeenth century, British sculpture did not exist. More than that, to most of the settlers, art, in whatever form, was an invention of the devil, to be avoided and discouraged. So it is not surprising that two centuries elapsed before the first American statue made its shy and awkward appearance.

In considering the achievements of American sculpture, we must remember that it is still an infant. That it is a lusty infant none will deny, though some may find it lacking in that grace and charm which come only with maturity.

The first man born in America who was foolhardy enough deliberately to choose sculpture as a profession was Horatio Greenough, born in 1805, of well-to-do parents, and carefully educated. It is difficult to say just what it was that turned the boy to this difficult and exacting art—an unknown art, too, so far as America was concerned. But he seems to have begun woodcarving at an early age, and to have progressed from that to chalk and on to plaster of Paris. The American national habit of whittling was perhaps responsible for the development of more than one sculptor.

At any rate, by the time he was twelve years old, Horatio Greenough had produced some portrait busts in chalk, and, after having tried unsuccessfully to learn clay-modelling from directions in an old encyclopedia, took some lessons from an artist who chanced to be in Boston, and from a maker of tombstones, got a little insight into the method of carving marble.

These lessons, elementary as they must have been, were very valuable to the boy, and his work showed such promise that his father finally consented to his adopting this strange profession, insisting only that he first graduate

from Harvard, on the ground that a college education would be of value, whatever his vocation. So he entered college at the age of sixteen, devoting all his spare time to reading works of art, to drawing and modelling, and the study of anatomy. He had also the good fortune to meet and win the friendship of Washington Allston, who advised him as to plans of study.

Immediately upon graduation, he sailed for Italy, which was, sadly enough, to be the Mecca of American sculptors for many years to come. For Italian sculpture was bound hand and foot by the traditions of classicism, to which our early sculptors soon fell captive. Greenough was no exception, and some years of study in the Italian studios rivetted the chains.

His first commission was given him by J. Fenimore Cooper. It was a group called the "Chanting Cherubs," and when it was sent home for exhibition, it awakened a tempest of the first magnitude. Puritan ideas were outraged at sight of the little naked bodies, the group was declared indecent, and the bitter controversy was not stilled until it was withdrawn from view. Greenough wrote of Cooper, "he saved me from despair; he employed me as I wished to be employed; and has, up to this moment, been a father to me in kindness"—a singularly interesting addition to the portrait of the great novelist, famous for his enmities rather than for his friendships.

The tragedy of Greenough's life was the fate of his great statue of Washington, of which we have already spoken. He conceived the work on a high plane, "as a majestic, god-like figure, enthroned beneath the dome of the Capitol at Washington, gilded by the filtered rays of the far-falling sunlight." Perhaps it was too high, but on its execution Greenough labored faithfully for eight years. "It is the birth of my thought," he wrote. "I have sacrificed to it the flower of my days, and the freshness of my strength; its every lineament has been moistened by the sweat of my toil and the tears of my exile. I would not barter away its association with my name for the proudest fortune that avarice ever dreamed."

It will be seen from the above that Greenough's epistolary style was florid and grandiose in the extreme, but no doubt there was a foundation of sincerity beneath it. A bitter disappointment awaited him. The ponderous figure reached Washington safely in 1843, and was conveyed to the Capitol, where, beneath the rotunda, its predestined pedestal awaited it. But the

statue was found too large to pass the door, and when the door was widened and the great stone rolled inside, the floor settled so ominously that it was hastily withdrawn.

It does not seem to have occurred to anyone that the floor might be braced; instead, the pedestal was set up outside, facing the building, and the statue hoisted into place. It speedily became the butt of public ridicule. Once the fashion started, no one looked at it without a smile.

Greenough was in despair. "Had I been ordered to make a statue for any square or similar situation at the metropolis," he wrote, still in his inflated style, "I should have represented Washington on horseback and in his actual dress. I would have made my subject purely a historical one. I have treated my subject poetically, and confess I would feel pain in seeing it placed in direct flagrant contrast with every-day life."

But that is exactly how it was placed, and it is the incongruity of this contrast which strikes the beholder and blinds him to the merits of the work. For Greenough has represented Washington seated in a massive armchair, naked except for a drapery over the legs and right shoulder, one hand pointing dramatically at the heavens, the other extended holding a reversed sword. It shows sincerity and faithful work, and had it been placed within the rotunda, would no doubt have been impressive and majestic. Where it stands, it is a hopeless anachronism.

This was the first colossal marble carved by an American. Fronting it on one of the buttresses of the main entrance of the Capitol, is the second, also by Greenough. It is a group called "The Rescue," and shows a pioneer saving his wife and child from being tomahawked by an Indian, while his dog watches the struggle with a strange apathy—almost with a smile. Like most of his other work, it is stilted and unconvincing; but let us remember that Greenough was the pathfinder, the trail-blazer, and as such to be honored and admired.

Greenough's fame, such as it was, was soon to be eclipsed by that of a man born in the same year, but later in development because he had a harder road to travel. Hiram Powers was born into a large and poverty-stricken family. While he was still a boy, his father removed from the sterile hills of Vermont to the almost frontier town of Cincinnati, Ohio. He seems to have

had little schooling, but was put to work as soon as he was old enough to contribute something toward the family exchequer. He did all sorts of odd jobs, and soon developed an unusual talent, that of modelling faces.

Those were the halcyon days of the dime museum, and there was one at Cincinnati. Its proprietor chanced to hear of the boy's gift for modelling, and offered him employment as a modeller of wax figures. Of course Powers accepted, for this was work after his own heart, and he succeeded not only in producing some figures which resembled definite human beings, but "breathed the breath of life into them" by means of clock-work devices, which enabled them to move their heads and arms in a manner sufficiently jerky, but at the same time astonishing to the simple people who visited the museum to behold its wonders.

Emboldened by this success, the young genius produced an "Inferno," or "Chamber of Horrors," which, when completed, was an immense success—too immense, indeed, for it had to be closed because of the fearful impression it made upon the ladies, who fainted in their escorts' arms whenever they gazed upon its terrors. One is inclined to suspect that the ladies might have withstood the horrors of the sight, but for a desire to prove their extreme sensibility. Fainting was more fashionable eighty years ago than it is to-day.

Powers soon developed from this work a talent for catching likenesses, and, searching for a wider field, proceeded finally to Washington, where he modelled busts in wax of Andrew Jackson, Daniel Webster, John C. Calhoun, John Marshall, and other celebrities of the period. From wax, he naturally wished to graduate into marble, and in 1837, left America for Italy, never to return. Greenough, then laboring away at his Washington, assisted him in various ways; and Hawthorne met him in Italy and was much impressed by him, as his "Italian Note-Book" shows.

In 1843, he completed the figure which was destined to make him famous, the "Greek Slave." The statue was supposed to represent a maiden captured by the Turks, "stripped and manacled and offered for sale in the market place," and so had a sentimental appeal which went straight to the heart of a sentimental people, and overcame any antagonism which her nudity might have produced. It inspired Elizabeth Barrett Browning to a not very

noteworthy sonnet, clergymen gave it certificates of character, so to speak, and "it made a sensation wherever shown, and was fondly believed to be the greatest work of sculpture known to history." Let us say at once that it is an engaging and creditable piece of work, and worthy, in the main, of the enthusiasm which it excited.

The "Greek Slave" was only the beginning. Powers turned out one statue after another with considerable rapidity, but his reputation rests mainly to-day on his portrait busts of men. It is characteristic of artists that the things they do best and easiest they value least, and this was so with Powers. His portrait busts were, in a sense, mere pot-boilers; he lavished himself upon his ideal figures. But these are now ranked as unimaginative and commonplace.

Third among our early sculptors of importance was Thomas Crawford, born eight years later than Greenough and Powers, and preceding the latter to the grave by many years, yet leaving behind him a mass of work which, if it shows no great imagination, displays considerable poetic refinement. Driven to Italy because it was only there that marble work could be well and economically done, he lived there for some years, earning a bare subsistence by the production of second-rate portrait busts and copies of antique statuary. Then he attracted the attention of Charles Sumner, and with his help, was enabled, in 1839, to produce his first important work, the "Orpheus," now in the Boston Museum. Many others followed, but they were of that ideal and sentimental type, very foreign to modern taste.

Crawford was an indefatigable workman, and few American museums are without one or more examples of his product. In the public square at Richmond, Virginia, stands one of his most important monuments, crowned by an astonishing equestrian figure of Washington, which he himself executed. Two of the subordinate statues are also his—those of Patrick Henry and Thomas Jefferson—and represent the best work he ever did.

Another of his productions is the great figure of Freedom which crowns the dome of the Capitol at Washington, not unworthily. By a fortunate chance, which the sculptor could hardly have foreseen, the bulky and roughly modelled figure gains airiness and majesty from its lofty position, where its

sickly-sweet countenance and clumsy adornment are refined by distance. It has become, in a way, a national ideal, a part of the Republic.

The success of these three men and the immense reputation which they attained naturally attracted others to a profession whose rewards were so exalted. The first to achieve anything like an enduring reputation was Henry Kirke Brown, born in Massachusetts in 1814. He early displayed some talent for portrait painting, and went to Boston to study under Chester Harding. Chance led him to model the head of a friend, and the result was so interesting that he then and there renounced painting for sculpture.

Naturally, his eyes turned to Italy, but he had no money to take him there, so perforce remained at home, getting such instruction as he could. In 1837, at the age of twenty-three, he produced his first marble bust, and within the next four years, had carved at least forty more, besides four or five figures. From all this work, he managed to save the money needed for the trip to Italy, but after four years in the Italian studios, he sailed for home again. On July 4, 1856, the second equestrian statue to be set up in the United States was unveiled in Union Square, New York City, and gave Brown a reputation which still endures.

It is a statue of Washington, and, in some amazing fashion, Brown succeeded in producing a work of art, which, in some respects, has never been surpassed in America, and which has served as a pattern and guide to other sculptors from that day to this. It is a sincere, honest and dignified embodiment of the First American. Brown did some notable work after that, but none of it possesses the high inspiration which produced the noble and commanding figure which dominates Union Square.

We have said that it was the second equestrian statue produced in America. The first may still be seen by all who, on entering or leaving the White House, glance across the street at the public square beyond. One glance is certain to be followed by others, for that statue is not only the first, it is the most amazing ever set up in a public place in this country. It has divided with Greenough's "Washington," at the other end of Pennsylvania Avenue, the horrors of being a national joke. Its author was Clarke Mills, and its inception is probably unparalleled in the history of sculpture.

Mills was born in New York State in 1815, lost his father while still a child, and at the age of thirteen was driven by harsh treatment to run away from the uncle with whom he had made his home. Thenceforward he supported himself in any way he could—as farm-hand, teamster, canal-hand, post-cutter, and finally as cabinet maker. He drifted about the country; to New Orleans, and finally to Charleston, South Carolina, where he learned to do stucco work, and whiled away his leisure hours by modelling busts in clay.

With Yankee ingenuity, he invented a process of taking a cast from the living face, and this simple method of getting a likeness enabled him to turn out busts so rapidly and cheaply that he had all the work he could do. He was, of course, anxious to try his hand at marble, and procuring a block of native Carolina stone, hewed out, with infinite labor, a bust of that South Carolina idol, John C. Calhoun. It was the best bust ever made of that celebrated statesman, and was the beginning of Mills's good fortune, and of the sequence of events which resulted in his statue of the hero of New Orleans.

For his Calhoun attracted much attention and secured him other commissions—among them, one for the busts of Webster and Crittenden. To get these, he was forced to go to Washington, and there he met the Hon. Cave Johnson, President of the Jackson Monument Commission, which had got together the funds for an equestrian statue of that old hero. Johnson suggested to Mills that he submit a design for this statue. As Mills had never seen either General Jackson or an equestrian statue, and had only the vaguest idea of what either was like, he naturally felt some doubt of his ability to execute such a work; but Johnson pointed out that this was only modesty, and so Mills finally evolved a design, which the commission accepted.

Then he went to work on his model, and executed it on an entirely new principle, which was to secure a balanced figure by bringing the hind legs of the horse under the centre of its body. Congress donated for the bronze of the statue the British cannon which Jackson had captured at New Orleans, and after many trials and disheartening failures, it was finally cast, hoisted into place, and dedicated on the eighth of January, 1853.

The whole country gazed at it in wonder and admiration, for surely never had another work of art so unique and original been unveiled in any land. Mills had balanced his horse adroitly on his hind legs, and represented the rider as clinging calmly to this perilous perch and doffing his chapeau to the admiring multitude. A delighted Congress added \$20,000 to the price already paid, while New Orleans ordered a replica at an even higher figure. Absurd as the statue is, it yet must command from us a certain respect for the enthusiast who designed it. Remember, he had never seen an equestrian statue, because there was none in the country for him to see; he had no notion of dignified sculptural treatment; but he did what he could, as well as he was able.

Mills was the last of the primitives, for following him came Erasmus D. Palmer and Thomas Ball, the two men who, more than any others, shaped the course and guided the development of American sculpture.

Erasmus Palmer was born in 1817, and followed the trade of a carpenter. But in the odd moments of 1845, he made a cameo portrait of his wife, which was a rather unusual likeness. Encouraged by this success, he practised further, and ended by abandoning his saws and planes to devote his whole time to carving portraits. But the constant strain so weakened his eyes, that he was about to return to carpentering, when a friend suggested that he try his hand at modelling in clay. The result was the "Infant Ceres," modelled from one of his own children, which, reproduced in marble, created a sensation at the exhibitions in 1850.

From that moment, Palmer's career was steadily upwards. It culminated eight years later in his delightful figure, the "White Captive," reminiscent in a way of the "Greek Slave," but a better work of art, and one which stands among the most charming achievements of American sculpture. One of its wonders, too—wonder that an untrained hand and an unschooled brain should have been able to create a work of art at once so tender and so firm. Following it came some admirable portrait busts; and finally, in 1862, his "Peace in Bondage." No doubt the sculptor's beautiful and adequate conception sprang from the tragic period which gave it birth; for "Peace in Bondage" shows a winged female figure leaning wearily against a tree-trunk, and gazing hopelessly into space. It is carved in high relief, with great skill and insight. In fact, nothing finer had been produced in America.

With this work, American art may be said to have found itself. It not only raised the standard of achievement, but it put an end at once and forever to the idea that study in Italy was necessary to artistic success. For only once did Palmer visit Europe, and then it was to stay but a short time. In fact, Italy was artistic poison for many men; its art lacked originality and vigor, and it sapped the native strength of many of the Americans who worked in its studios.

Thomas Ball was an exception to this; for, in spite of many years abroad, he remained always characteristically American. He comes next to Palmer in strength and rightness of achievement; his work, like his life, was earnest and noble.

Thomas Ball's father was a house and sign painter of Boston, with some artistic skill, which he passed on to his son. That was the boy's only inheritance, and when his father died, he undertook the support of the family, first as a boy-of-all-work in the New England Museum, and then as a cameo-cutter. From that he graduated naturally to engraving, miniature painting, and finally to portraiture.

His first attempt at modelling resulted in a bust of Jenny Lind, done entirely from photographs, which had a wide vogue, for the Swedish Nightingale was then at the height of her popularity. Other more ambitious work followed, and finally, at the age of thirty-five, he was able to realize his ambition to study in the studios of Florence. But he found the Italian environment less inspiring than he had hoped, and two years later he was back in Boston, working on an equestrian statue of Washington—the first equestrian group in New England and the fourth in the United States. He built his plaster model with his own hands, and was three years getting it ready. The result was a work which ranks among the first equestrian statues of the country. Other works of importance followed, among them the well-known emancipation group showing Lincoln blessing a kneeling slave, which was unveiled at Washington in 1875.

The years touched Ball lightly, and at seventy years of age, he undertook his greatest work, an elaborate Washington monument for the town of Methuen, Massachusetts. The principal figure, a gigantic Washington in bronze, was exhibited at the Columbian Exposition of 1893, and received

the highest honors of the exposition—a distinction it richly merited by its nobility of a conception and execution. Thomas Ball, indeed, set a new standard in public statuary, and one which no successor has dared to disregard. The far-reaching effects of his influence and that of Erasmus Palmer can hardly be over-estimated.

One of the most engaging and versatile personalities in the whole range of American art was that of William Wetmore Story. Born at Salem, Massachusetts, in 1819, graduated at Harvard, admitted to the bar, the author of a volume of graceful verse and of a valuable life of his father, Chief Justice Story, he yet, in 1851, put all this work aside, adopted sculpture as a profession, and, proceeding to Rome, opened a studio there.

It was from the first an extraordinary studio, attracting the most brilliant people of Rome in literature as well as art; and if Story did not quite practise the perfection he was somewhat fond of preaching, it was because of his very versatility, which absorbed his talent in so many directions that it could not be concentrated in any. His imagination outran his achievement, and the most famous of his works, his statue of Cleopatra, owes its reputation not so much to its own merit, which is far from overwhelming, as to the ecstatic description of it which Nathaniel Hawthorne included in "The Marble Faun." A master of literature is not necessarily an inspired critic of art, and it is to be suspected that Hawthorne permitted some of the fire of his imagination to play about the cold and uninspired marble.

"Cleopatra" marked Story's culmination. He fell away from it year by year, producing a long line of figures whose only impressive features were the names he gave them—"The Libyan Sibyl," "Semiramis," "Salome," "Medea," and so on. However, he did much to increase the popularity of sculpture, for the stories he attempted to tell in stone by means of heavy-browed, frowning women in classic costume and with classic names, were exactly suited to the child-like intelligence of his public. He gave art, too—as William Penn gave the Quakers—a sort of social sanction because of his own social position. If the son of Chief Justice Story could turn sculptor, surely that profession was not so irregular, after all!

Another sculptor who shared with Story the admiration of the public was Randolph Rogers, born at Waterloo, New York, in 1825. Until the age of

twenty-three such modelling as he did was done in the spare moments of a business life; but when he gave an exhibition of the results of this labor, his employers were so impressed that they provided the money needed to send him to Italy, where he was to spend the remainder of his life, with the exception of five years' residence in New York. Two of his earlier figures are his most famous, his "Nydia" and his "Lost Pleiad." Scores of replicas in marble of these two figures were made during their author's life time, and they still retain for many people a simple and pathetic charm. Nearly every one, of course, has made the acquaintance of Nydia, the blind girl, in Bulwer-Lytton's "The Last Days of Pompeii," and so gaze at Rogers's fleeing figure with eyes too sympathetic to see its faults.

Far more important is the work of William H. Rinehart, of the same age as Rogers, and resembling him somewhat in development. Born on a Maryland farm, his early years were those of the average farmer's boy, but at last some blind instinct led him to abandon farming for stonecutting, and he became assistant to a mason and stonecutter of the neighborhood. As soon as he had learned his trade, at the age of twenty-one, he went to Baltimore, where there was work in plenty, and where he could, at the same time, attend the night schools of the Maryland Institute. This sounds much easier than it really was. To devote the evenings to study, after ten and often twelve hours of the hardest of all manual labor, required grit and moral courage such as few possess.

He was soon trying his hand at modelling, and convinced, at last, that sculpture was his vocation, he managed, by the time he was thirty, to save enough money for a short period of study at Rome. Three years of work at Baltimore, after that, gave him some reputation, and he then returned to Rome, to spend the remainder of his life there.

If you have ever visited the Metropolitan Museum of Art in New York City, you have seen, in the hall of statuary, one of Rinehart's most characteristic groups, "Latona and Her Children." The mother half seated, half lying upon the ground, gazes tenderly down at the two sleeping children, sheltered in the folds of her mantle. The whole work possesses a serene poetic charm and dignity very noteworthy; and this and other groups are among the most beautiful that any American ever turned out of an Italian studio.

Rinehart was one of the last American disciples of the classic school. Certainly no art could have been more opposed to his than the frank and vivid realism of his immediate successor, John Rogers. Born in Salem, Massachusetts, the son of a family of merchants, he was educated in the common schools, worked for a time in a store, and then entered a machine shop as an apprentice, working up through all the grades, until finally he was in charge of a railroad repair shop.

During all these years he had no suspicion of artistic talent within himself, but one day in Boston he happened to see a man modelling some images in clay. In that instant, the artist instinct clutched him, and procuring some clay and modelling tools, he spent all his leisure in practice. This leisure was scant enough, for his trade kept him employed fourteen hours of every day; but at the age of twenty-nine he was able to secure an eight months' vacation, which he spent in Europe, principally at Paris and Rome. He returned to America greatly discouraged, for the only thing he saw in Europe was classic sculpture, with which he had no sympathy and which, indeed, he could not understand.

So, abandoning all thought of making sculpture a profession, he went to work as a draughtsman in Chicago, amusing himself, at odd hours, by the construction of a group of small figures, which he called "The Checker Players." It was exhibited at a charity fair, and awakened so much interest and delight that Rogers burned his bridges behind him by resigning his position, and proceeded to New York, and rented a studio, determined to be a sculptor in spite of classicism.

The outbreak of the Civil War furnished him a host of subjects which he treated with a patriotic fervor that went straight to the heart of an overwrought people. "The Returned Volunteer," "The Picket-Guard," "The Sharp-shooters," "The Camp-fire," "One More Shot," and many others, came from his studio in rapid succession. They were all thoroughly American, and some were even admirably sculptural. They, at least, stood for an original idea, and deserve better treatment than the silent contempt which, in these days, is about all that has been accorded them.

At about this time, there came upon the scene the first and only really famous woman sculptor in the history of American art, Harriet Hosmer. She

had had an unusual childhood, and had grown into an original and engaging woman. Born in 1830, at Watertown, Massachusetts, the daughter of a physician, she inherited her mother's delicate constitution, and her father encouraged her in an outdoor life of physical exercise such as only boys, at that time, were accustomed to. She became expert in rowing, riding, skating and shooting, developed great endurance, filled her room with snakes and insects and birds' nests, and in a clay pit at the end of her father's garden modelled rude figures of animals.

A few years of schooling followed this wild girlhood; then she was sent to Boston to study drawing and modelling; but finding that no woman would be admitted to the Boston Medical School, whose course in anatomy she was anxious to take, she went to St. Louis and entered the medical college there. Finally, in 1852, accompanied by her father and Charlotte Cushman, she set sail for Italy.

She remained there for eight years, turning out a number of very creditable figures, which, if not great, at least possess some measure of grace and charm. Nathaniel Hawthorne, in his "Italian Note-Book," has left a vivid impression of Miss Hosmer, whose eccentricity of dress and manner impressed him deeply, as did also the work which she showed him. But she never reached any high development.

Which brings us to the present of American art, for the sculptors we have yet to consider are either yet alive or have died so recently that they belong to the present rather than the past.

The first and one of the most important of these is John Quincy Adams Ward, born in 1830 on an Ohio farm. An accident showed the possession of latent talent, for some good pottery clay happened to be discovered on his father's farm, and his guardian angel inspired the boy to take a handful of it and model the grotesque countenance of a negro servant. The result was striking, and no doubt he felt within himself some of the stirrings of genius, but not until 1849 did he realize his vocation. Then, while on a visit to a sister in Brooklyn, he happened to pass the open door of H. K. Brown's studio. The glimpse he caught of the scene within fascinated him; he returned again and again, and ended by entering the studio as a pupil.

He could have found no better master, and for seven years he remained there, assisting Brown in every detail of his work. His first group, modelled after long study, was his "Indian Hunter," now placed in Central Park, New York—a group instinct with vitality—a glimpse of a forgotten past, evoked with the skill of a master. It was the first of a long line of statues, many of them portraits of contemporaries, a field in which Ward has no superior. It is perhaps the highest tribute which could be paid the man to say that, with all his great production, he has never done bad work, never produced anything trifling or unworthy.

A fellow student with Ward in Henry Kirke Brown's studio was Larkin G. Meade, the first indication of whose talent was a unique one. One winter morning, about the middle of the century, the good people of Brattleboro, Vermont, were astonished to find set up in one of the public squares of the town a colossal snow image, in the form of a majestic angel—crude, no doubt, in execution, but singularly effective. Inquiry developed that it was the work of young Meade, then only fifteen years of age. The incident got into the newspapers, magnified considerably, and attracted the attention of old Nicholas Longworth, of Cincinnati, who, on more than one occasion, had himself appeared as angel to struggling artists.

It was so in this case. Mr. Longworth wrote to Brattleboro, making some inquiries as to the essential truth of the story, and having satisfied himself on that point, offered to help the boy to get an artistic education. The offer was accepted, and young Meade was placed in Brown's studio, going afterwards to Italy. While there, he heard of the assassination of President Lincoln, and prepared an elaborate design in plaster for a national monument to the martyred President's memory. As soon as this was completed, he started for home with it, arriving at precisely the right moment. The rage for monument building was sweeping up and down the land. Councils, legislatures, all sorts of public and private bodies, were making appropriations to commemorate some particular hero of the Civil War, which was just ended; Meade's design appealed to the popular imagination, and the commission was awarded him.

The monument, which was destined to cost a quarter of a million dollars, was by far the most important that had ever been erected in this country, and the inexperienced young sculptor sailed back to Italy to begin work. Not until 1874 was it sufficiently completed to dedicate, and the last group of statuary was not put in place until ten years later. All this time, the sculptor had spent quietly in his studio at Florence, quite apart from the world of progress or of new ideas in art, and long before his work was finished, public taste had outgrown it and found it uninspired and commonplace.

Much more important to American art is the work of Olin Levi Warner, the son of an itinerant Methodist preacher, whose wanderings prevented the boy getting any regular schooling. During his childhood, he had shown considerable talent for carving statuettes in chalk, and he finally decided to immortalize his father by carving a portrait bust of him. For a stone, he "set" a barrel of plaster in one solid mass and then, breaking off the staves, began hacking away at it with such poor implements as he could command. It was a well-nigh endless task, but "it's dogged that does it," and the boy worked doggedly away until the bust was completed. It was considered such a success that young Warner, convinced of his vocation, set to work to earn enough money to go abroad. For six years he worked as a telegrapher, and it was not until 1869, when he was twenty-five years old, that he had saved the money needed.

Three years later he returned to New York, and opened a studio, but met with a reception so dismal and indifferent that, after a four years' desperate struggle, he was forced to abandon the fight and return to his father's farm. Anxious for any employment, he applied to Henry Plant, President of the Southern Express Company, for work. Mr. Plant was interested, and instead of offering him a job as messenger or teamster, gave him a commission for two portrait busts.

It was the turning point in Warner's career, for the busts he produced were of a craftsmanship so delicate and beautiful that they at once established his position among his fellow-sculptors, though years elapsed before he received any wide public recognition. The truth is that he was too great and sincere an artist to cater to a public taste which he had himself outgrown; so

that, until quite recently, he has remained a sculptor's sculptor. His untimely death, in 1896, from the effects of a fall while riding in Central Park, brought forth a notable tribute from his fellow-craftsmen, and students of sculpture have come to recognize in him one of the most delicate and truly inspired artists in our history.

But the most powerful influence in the recent development of American sculpture has been that great artist, Augustus Saint Gaudens. Born in 1848, at Dublin, Ireland, of a French father and an Irish mother, he was brought to this country while still an infant. Perhaps this mixed ancestry explains to some degree Saint Gaudens's peculiar genius. At the age of thirteen, he was apprenticed to a cameo-cutter in New York City, and worked for six years at this employment, which demands the utmost keenness of vision, delicacy of touch, and refinement of manner. His evenings he spent in studying drawing, first at Cooper Union and then, outgrowing that, at the National Academy of Design. So it happened that, at the age of twenty, when most men were just beginning their special studies, Saint Gaudens was thoroughly grounded in drawing and an expert in low relief.

Another thing he had learned; and let us pause here to lay stress upon it, for it is the thing which must be learned before any great life-work can be done. He had learned the value of systematic industry, of putting in so many hours every day at faithful work. The weak artist, whether in stone or paint or ink, always contends that he must wait for inspiration, and so excuses long periods of unproductive idleness, during which he grows weaker and weaker for lack of exercise. The great artist compels inspiration by whipping himself to his work and setting grimly about it, knowing that the "inspiration," so-called, will come. For inspiration is only seeing a thing clearly, and the one way to see it clearly is to keep the eyes and mind fixed upon it.

At the age of twenty, then, Saint Gaudens was not only a trained artist, but an industrious one. Three years in the inspiring atmosphere of Paris, and three years in Italy, followed; and finally, in 1874, he landed again at New York with such an equipment as few sculptors ever had. And seven years later he proved his mastery when his statue of Admiral Farragut was unveiled in Union Square, New York. That superb work of art made its

author a national figure, and Saint Gaudens took definitely that place at the head of American sculpture which was his until his death.

Six years later Saint Gaudens's "Lincoln" was unveiled in Lincoln Park, Chicago, and was at once recognized as the greatest portrait statue in the United States. It has remained so—a masterpiece of exalted conception and dignified execution. Other statues followed, each memorable in its way; but Saint Gaudens proved himself not only the greatest but the most versatile of our sculptors by his work in other fields—by portraits in high and low relief, by ideal figures, and notably by the memorial to Robert Gould Shaw, a work distinctively American and without a counterpart in the annals of art. It is the spiritual quality of Saint Gaudens's work which sets it apart upon a lofty pinnacle—the largeness of the man behind it, the artist mind and the poet heart.

Saint Gaudens's death in 1907 deprived American art of one of its most commanding figures, but there are other American sculptors alive to-day whose work is noteworthy in a high degree. One of these is Daniel Chester French. Born of a substantial New England family, and showing no especial artistic talent in youth, one day, in his nineteenth year, he surprised his family by showing them the grotesque figure of a frog in clothes which he had carved from a turnip. Modelling tools were secured for him, and he went to work. The schooling which prepared him for his remarkable career was of the slightest. He studied for a month with J. Q. A. Ward, and for the rest, worked out his own salvation as best he could.

His first important commission came to him at the age of twenty-three—the figure of the "Minute Man" for the battle monument at Concord, Massachusetts. It was unveiled on April 19, 1875, and attracted wide attention. For here was a work of strength and originality produced by a young man without schooling or experience—produced, too, without a model, or, at least, from nothing but a large cast of the "Apollo Belvidere," which was the only model the sculptor had. But there was no hint of that famous figure under the clothes of the "Minute Man." It had been entirely concealed by the personality and vigor he had impressed upon his work.

After that Mr. French spent a year in Florence, but he returned to America at the end of that period to remain. He has grown steadily in power and

certainty of touch, rising perhaps to his greatest height in his famous group, "The Angel of Death and the Young Sculptor," intended as a memorial to Martin Milmore, but touching the universal heart by its deep appeal, conveyed with a sure and admirable artistry. Mr. French's great distinction is to have created good sculpture which has touched the public heart, and to have done this with no concession to public taste.

Another sculptor who has gained a wide appreciation is Frederick MacMonnies, who for sheer audacity and dexterity of manipulation is almost without a rival. He was born in Brooklyn in 1863, his father a Scotchman who had come to New York at the age of eighteen, and his mother a niece of Benjamin West. The boy's talent revealed itself early, and was developed in the face of many difficulties. Obligated to leave school while still a child and to earn his living as a clerk in a jewelry store, he still found time to study drawing, and at the age of sixteen had the good fortune to attract the attention of Saint Gaudens, who received him as an apprentice in his studio.

No better fate could have befallen the lad, and the five years spent with Saint Gaudens gave him the best of all training in the fundamentals of his art. Some years in Paris followed, where he replenished his slender purse with such work as he could find to do, until, in 1889, his "Diana" emerged from his studio, radiant and superb. A year later came his statue of "Nathan Hale," and there was never any lack of commissions after that. "Nathan Hale" stands in City Hall Park, New York City, the very embodiment of that devoted young patriot. The artist has shown him at the supreme moment when, facing the scaffold, he uttered the memorable words which still thrill the American heart, and expression and sentiment were never more perfectly in accord. He struck the same high note with his famous fountain at Chicago Exposition, where hundreds of thousands of people suddenly discovered in this young man a national possession to be proud of.

A year later his name was again in every mouth, when the Boston Public Library refused a place to perhaps his greatest work, the dancing "Bacchante," which has since found refuge in the Metropolitan Museum at New York—a composition so original and daring that it astonishes while it delights.

Like MacMonnies, George Gray Barnard began life as a jeweller's apprentice, became an expert engraver and letterer, and finally, urged by a ceaseless longing, deserted that lucrative profession for the extremely uncertain one of sculpture. A year and a half of study in Chicago brought him an order for a portrait bust of a little girl, and with the \$350 he received for this, he set off for Paris. That meagre sum supported him for three years and a half—with what privation and self-denial may be imagined; but he never complained. He lived, indeed, the life of a recluse, shutting himself up in his studio with his work, emerging only at night to walk the streets of Paris, lost in dreams of ambition. That from this period of ordeal came some of the deep emotion which marks his work cannot be doubted.

This quality, which sets Barnard apart, is well illustrated in his famous group, "The Two Natures," suggested by a line of Victor Hugo, "I feel two natures struggling within me." Two male figures are shown, heroic in size and powerfully modelled, a victor half erect bending over a prostrate foe.

Besides these men, who are, in a way, the giants of the American sculptors of to-day, there are, especially in New York, many others whose work is graceful and distinctive. Paul Wayland Bartlett, Herbert Adams, Charles Niehaus, John J. Boyle, Frank Elwell, Frederick Ruckstuhl, to mention only a few of them, are all men of originality and power, whose work is a pleasure and an inspiration, and to whose hands the future of American sculpture may safely be confided.

SUMMARY

GREENOUGH, HORATIO. Born at Boston, September 6, 1805; graduated at Harvard, 1825; went to Italy, 1825, and made his home there, with the exception of short visits to America and France; died at Somerville, Massachusetts, December 18, 1852.

POWERS, HIRAM. Born at Woodstock, Vermont, July 29, 1805; modelled wax figures at Cincinnati, Ohio, for seven years; went to Washington, 1835, and to Florence, 1837; died there, June 27, 1873.

CRAWFORD, THOMAS. Born at New York City, March 22, 1814; went to Italy, 1834, and took up residence at Rome for the remainder of his life; afflicted with sudden blindness in 1856, and died at London, October 16, 1857.

BROWN, HENRY KIRKE. Born at Leyden, Massachusetts, February 24, 1814; studied in Italy, 1842-46; opened Brooklyn studio, 1850; died at Newburgh, New York, July 10, 1886.

MILLS, CLARKE. Born in Onondaga County, New York, December 1, 1815; died at Washington, January 12, 1883.

PALMER, ERASTUS DOW. Born at Pompey, Onondaga County, New York, April 2, 1817; opened studio in Albany, 1849; in Paris, 1873-74; died at Albany, New York, March 9, 1904.

BALL, THOMAS. Born at Charlestown, Massachusetts, June 3, 1819; practised painting, 1840-52; adopted sculpture, 1851; resided in Florence, Italy, 1865-97; opened New York studio, 1898.

STORY, WILLIAM WETMORE. Born at Salem, Massachusetts, February 19, 1819; graduated at Harvard, 1838; admitted to the bar, 1840; published a volume of poems, 1847; went to Italy, 1848, and lived at Florence until his death, October 5, 1895.

ROGERS, RANDOLPH. Born at Waterloo, New York, July 6, 1825; removed to Italy, 1855; died at Rome, January 15, 1892.

RINEHART, WILLIAM HENRY. Born in Maryland, September 13, 1825; removed to Rome, 1858, and died there, October 28, 1874.

ROGERS, JOHN. Born at Salem, Massachusetts, October 30, 1829; visited Europe, 1858-59; died, July 27, 1904.

HOSMER, HARRIET G. Born at Watertown, Massachusetts, October 9, 1830; studied in Rome, 1852-60; opened Boston studio, 1861; died at Cambridge, Massachusetts, February 21, 1908.

WARD, JOHN QUINCY ADAMS. Born at Urbana, Ohio, June 29, 1830; studied under H. K. Brown, 1850-57; studio in New York City since 1861.

MEADE, LARKIN GOLDSMITH. Born at Chesterfield, New Hampshire, January 3, 1835; studied under Brown and in Florence; artist at the front for *Harper's Weekly* during Civil War; afterwards returned to Florence and made his home there.

WARNER, OLIN LEVI. Born at Suffield, Connecticut, April 9, 1844; studied in Paris, 1869-72; opened New York studio, 1873; died there, August 14, 1896.

SAINT GAUDENS, AUGUSTUS. Born at Dublin, Ireland, March 1, 1848; came to America in infancy; learned trade of cameo cutter; studied at Paris, 1867-70; Rome, 1870-72; opened New York studio, 1872; died at Corinth, N. H., August 3, 1907.

FRENCH, DANIEL CHESTER. Born at Exeter, New Hampshire, April 20, 1850; studied in Boston and Florence; studio in Washington, 1876-78; in Boston, 1878-87; in New York since 1887.

MACMONNIES, FREDERICK. Born at Brooklyn, New York, September 20, 1863; studied under Saint Gaudens, 1880-84; also at Paris, and has spent many of the succeeding years in France.

BARNARD, GEORGE GRAY. Born at Bellefonte, Pennsylvania, May 24, 1863; studied at Paris, 1884-87; spent some years in New York, and then returned to France.

CHAPTER VI

THE STAGE

The golden age of American acting was not so very long ago. Most white-haired men remember it, and love to talk of the days of Booth and Forrest and Charlotte Cushman. Joseph Jefferson, the last survivor of the old régime, died just the other day, and to the very end showed the present generation the charm and humor of Bob Acres and Rip Van Winkle.

No doubt that golden age is made to appear more golden than it really was by the mists of time; but undoubtedly the old actors possessed a mellowness, a solidity, a sort of high tradition now almost unknown. These qualities were due in part, perhaps, to the long and arduous stock company training, where, in the old days, every actor must serve his apprenticeship, and in part to the study of the classic drama which had so large a place in stock company repertoire.

Success was infinitely harder to win than it is to-day. There were fewer theatres, so that the great actors were forced to play together, to their mutual advantage and improvement. The multiplication of theatres at the present time, and the vast increase of the theatre-going public, has led to the "star" system—to the placing of an actor at the head of a company, as soon as he has won a certain reputation. And, since care is taken that the "star" shall outshine all his associates, it follows that he has no one to measure himself with, he is no longer on his metal, and his growth usually stops then and there.

But let us be frank about it. The attitude of the public toward the theatre has changed. To-day we would not tolerate the heavy melodramas which enchained our parents and grandparents. The age of rant and fustian has passed away, and Edwin Forrest could never gain a second fortune from such a combination of these qualities as "Metamora." We are more sophisticated; we refuse to be thrilled by Ingomar, no matter how loudly he

bellows. What we ask for principally is to be amused, and consequently the great effort of the theatre is to amuse us, for the theatre must cater to its public. So, if the stage to-day is not what it was fifty years ago, the fault lies principally in front of the footlights and not behind them.

To the student of American acting, one name stands out before all the rest, the name of Booth. No other actors in this country have ever equalled the achievements of Junius Brutus Booth and of his son, Edwin Booth. They possessed the genius of tragedy, if any men ever did, and no one who saw them in their great moments can forget the impression of absolute reality which they conveyed.

Junius Brutus Booth was the son of an eccentric silversmith of London, and was born there in 1796. Let us pause here to remark that, just as the greatest

Frenchman who ever lived was an Italian, and the greatest Russian woman a German, so most of the early American actors were either English or Irish. This sounds rather Irish itself; but it is true. Certainly, in the end Napoleon Bonaparte became as French as any Frenchman and the Empress Catherine II Russian to the core; and the English and Irish actors who came to these shores in search of fame and fortune, and who found them and spent the



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remainder of their lives here, have every right to be considered in any account of the American stage which they did so much to adorn.

Junius Brutus Booth, then, was born in London in 1796. Twenty years before, his father had been so carried away by Republican principles that he had sailed for America to join the ranks of the army of independence, but he was captured and sent back to England. So it will be seen that he was something more than a mere silversmith; but he was very successful at his trade, and was able to give his son a careful classical education, to fit him for the bar. Imagine his chagrin when the boy, after a short experience in amateur theatricals, announced his intention of becoming an actor.

He secured some small parts, made a tour of the provinces, and finally, in London, engaged in a remarkable war with the great tragedian, Edmund Kean, which divided the town into two factions. But Booth tired of the struggle, in which the odds were all against him, and in 1821 sailed for America. He won an instant success, and was a great popular favorite until the day of his death. He was a short, spare, muscular man, with a pale countenance, set off by dark hair and lighted by a pair of piercing blue eyes, and he possessed a voice of wonderful compass and thrilling power. Upon the stage he was formidable and tremendous, giving an impression of overwhelming power, in which his son, perhaps, never quite equalled him.

Shortly after his arrival in America, Booth bought a farm near Baltimore, and there, on November 13, 1833, Edwin Booth was born. There was a great shower of meteors that night, which, if they portended nothing else, may be taken as symbolical of the career of America's greatest tragedian. He was the seventh of ten children, all of whom inherited, in some degree, their father's genius. It was not without a trace of madness, and reached a fearful culmination in John Wilkes Booth, when he shot down Abraham Lincoln at Ford's Theatre in Washington.

From the first, Edwin Booth felt himself destined for the stage. His father did not encourage him, but finally, in 1849, consented to his appearance with him in the unimportant part of Tressel, in "King Richard the Third." From that time on, he accompanied his father in all his wanderings, and partook of the strange and sad adventures of that wayward man of genius. In 1852, he went with his father to California, and was left there by the

elder Booth, who no doubt thought it the best school for the boy's budding talent. There, in the Sandwich Islands, and in Australia, among the rough crowds of the mining camps, he had four years of the most severe training that hardship, discipline, and stern reality can furnish. Amid it all his genius grew and deepened, and when he returned again to the east in 1856 he was no longer a novice, but an accomplished actor.

His last years in California had been shadowed by a great sorrow—the sudden and pitiful death of his father. The elder Booth had for years been subject to attacks of insanity, brought on, or at least intensified, by extreme intemperance. On one occasion he had attempted to commit suicide. On another, he had had his nose broken, an accident which so interfered with his voice that he did not regain complete control of it for nearly two years. On his return from California, where he had left his son, he stopped at New Orleans, and remained there a week, performing to crowded houses. He then started north by way of the Mississippi, and was found dying in his stateroom a few days later. He had been caught in a severe rain as he left New Orleans, a cold developed, complications followed, and for forty-eight hours he lay unattended in his stateroom, without that medical attention which he was unable or unwilling to summon. He died November 30, 1852, and his body was interred at Greenmount Cemetery, Baltimore, in a grave afterwards marked by a monument erected by his son Edwin.

This was only one of many tragedies which darkened the life of Edwin Booth, for, to use the words of William Winter, he was "tried by some of the most terrible afflictions that ever tested the fortitude of a human soul. Over his youth, plainly visible, impended the lowering cloud of insanity. While he was yet a boy, and while literally struggling for life in the semi-barbarous wilds of old California, he lost his beloved father, under circumstances of singular misery. In early manhood he laid in her grave the woman of his first love, the wife who had died in absence from him, herself scarcely past the threshold of youth, lovely as an angel and to all who knew her precious beyond expression. A little later his heart was well nigh broken and his life was well nigh blasted by the crime of a lunatic brother that for a moment seemed to darken the hope of the world. Recovering from that blow, he threw all his resources and powers into the establishment of the grandest theatre in the metropolis of America, and he saw his fortune of more than a million dollars, together with the toil of some of the best years

of his life frittered away. Under all trials he bore bravely up, and kept the even, steadfast tenor of his course; strong, patient, gentle, neither elated by public homage nor embittered by private grief."

It has been said that Booth returned from California a finished actor. He had, besides, the prestige of a great name, and he was welcomed with open arms. He had not yet reached the summit of his skill, but he showed an extraordinary grace and "a spirit ardent with the fire of genius." From that time forward, his career was one of lofty endeavor and of high achievement. In the great characters of Shakespeare, especially in those of Hamlet, Richard the Third, and Iago, he had no rivals, and no one who witnessed him in any of these parts ever outlived the deep impression the performance made. During the last two or three years of his life his health failed gradually, and he was finally compelled to leave the stage. On April 19, 1893, he suffered a stroke of paralysis from which he never rallied, lingering in a semi-conscious state until June 7th, when he sank rapidly and died.

Of his art no words can give an adequate idea. It was essentially poetic, full of a strange and compelling charm. His great moments laid upon his audience the spell of his genius, and rank with the highest achievements of any actor who ever lived. His countenance—

"That face which no man ever saw
And from his memory banished quite,
The eyes in which are Hamlet's awe
And Cardinal Richelieu's subtle light"—

as Thomas Bailey Aldrich wrote of Sargent's portrait, which heads this chapter—was a strange and moving one, and in range of expression unsurpassed. His eyes were especially wonderful, dark brown, but seeming to turn black in moments of passion, and conveying, with electrical effect, the actor's thought. He was unique. He stood apart. The American stage has never produced another like him.

Second only to Edwin Booth in sheer glory of achievement stands Edwin Forrest. He fell far below Booth in grace, in charm, and in poetic insight, but he surpassed him in physical equipment for the great parts of tragedy,

particularly in his voice, magnificent, vibrating, with an extraordinary depth and purity of tone.

Unlike Booth, Forrest came from no family of actors, nor inherited a name famous in the annals of the stage. He was born in Philadelphia in 1806, his father being a Scotchman, employed in Stephen Girard's bank, and making just enough money to keep his family of six children from actual want. He died when Edwin was thirteen years old, and his widow, by opening a little store, managed to support the children. She was a serious and devout woman and decided that Edwin should enter the ministry. But meantime, he must earn a living, so he was apprenticed to a cooper.

How long he stayed with the cooper nobody knows; but it could not have been long, for already he was fired with an ambition to be an actor, and after some experience as an amateur, astonished and grieved his mother by announcing that he was going on the stage. He made his first appearance on the 27th of November, 1820, as Young Norval, in Home's tragedy of "Douglas," and was an immediate success. His youth—remember, he was but fourteen—his handsome face and manly bearing, and, above all, that wonderful and resonant voice, won the audience at once, and his career was begun.

But many hardships awaited him. The theatres of New York and Philadelphia had their companies of well-known and well-trained actors. There was no hope for him in either of those cities; but at last he secured an engagement to play juvenile parts at Pittsburgh, Cincinnati, Lexington, and other towns of the middle west, at a salary of eight dollars a week. This, of course, was scarcely enough to keep body and soul together, but all Forrest wanted was a chance, and he did not murmur at the suffering and hardship which followed.

For business was poor, and Forrest did not always receive even that eight dollars. The end came at Dayton, Ohio, where the company went to pieces. Forrest, without money and almost without clothes, walked the forty miles to Cincinnati, where, after a time, he found another position. Such was the beginning of his career, and this hard novitiate lasted for four years, until, in 1826, at the age of twenty, he was able to return to New York and secure an engagement at the old Bowery Theatre. He was an instant success, and from

year to year his wonderful powers seemed to increase, until he became easily the most famous actor of the day.

But his fame was soon to be dulled by unfortunate personalities. Conceiving a jealousy of Macready, the famous English actor, he hissed him at a performance in Edinburgh, and when Macready came to America in 1849, Forrest's followers broke in upon a performance at the Astor Place opera house, and a riot followed in which twenty-two men were killed. A quarrel with his wife led to the divorce court, and the suit was decided against him.

The end was pathetic. He had been troubled with gout for a long time, and in 1865, it took a malignant turn, paralyzing the sciatic nerve, so that he lost the use of one hand, and could not walk steadily. His power had left him, and in the five years that followed, he played to empty houses and an indifferent public, not content to retire, but hoping against hope that he might in some way regain his lost prestige. A stroke of paralysis finally ended the hopeless struggle.

Forrest's art was of a cruder and more robust sort than Edwin Booth's who, by the way, was named after him. He was greatest in characters demanding a great physique, a commanding presence and—yes, let us say it!—a loud voice. Coriolanus, Spartacus, Virginius—those were his roles, and no man ever looked more imposing in a Roman toga.

Forrest, during his English engagement of 1845, and on other occasions, shared the honors with a remarkable actress, Charlotte Cushman. And perhaps none ever had a more astonishing career. Born in Boston in 1816, her youth was one of poverty, for her father died while she was very young, leaving no property. The girl was remarkably bright, and soon developed a contralto voice of unusual richness and compass. She sang in a choir and assisted to support the family from the age of twelve, securing such musical instruction as she could. In 1834, she made her first appearance in opera and scored a tremendous success. A splendid career seemed opening before her, when suddenly, a few months later, her voice, strained by the soprano parts which had been, assigned her, failed completely.

Her friends advised her to become an actress, and she went diligently to work, not allowing herself to despond over that first great disappointment.

For the next seven years, she worked faithfully learning the new profession from the very bottom. "I became aware," she said, "that one could never sail a ship by entering at the cabin windows; he must serve and learn his trade before the mast." In that way she learned hers, playing minor parts, doing cheerfully the drudgery of her profession, refusing all offers for more important work until she felt herself thoroughly capable of undertaking it. One would wish that her example might be taken to heart by her sisters of the present day.

At last her chance came. In 1842, William C. Macready, the great English tragedian, visited the United States, and in Charlotte Cushman he found a splendid support. Indeed, she divided the honors with him. A year later, she went to London and won immense applause. "Since the first appearance of Edmund Keane, in 1814," said a London journal, in speaking of her first night as "Bianca," "never has there been such a *début* on the stage of an English theatre." For eighty-four nights she appeared with Edwin Forrest. "All my successes put together," she wrote to her mother, "would not come near my success in London."

In the winter of 1845 she tried one of the most daring experiments ever made by an actress, appearing as Romeo to her sister, Susan Cushman's, Juliet. It was a notable success. Her deep contralto voice made it possible for her to give a complete illusion of the young and handsome lover. She played other male characters in after years, notably Hamlet, and created a deep impression in them. Her sister was a lovely girl, and an accomplished actress, and their "Romeo and Juliet" ran for two hundred nights. Susan Cushman would no doubt also have won high fame as an actress, but she soon retired from the stage, marrying the distinguished chemist and author, James Sheridan Muspratt, of Liverpool.

Charlotte Cushman returned to America in the fall of 1849, and was received with acclamation. There was never any question, after that, of her position as the greatest English-speaking actress, and that position she easily maintained until her death. She gathered wealth as well as fame, built a villa at Newport, and in 1863 earned nearly nine thousand dollars for the United States Sanitary Commission by benefit performances. Energetic, resolute, faithful, impatient of any achievement but the highest, she seemed the very embodiment of many of Shakespeare's greatest creations. She

possessed a strange, and weird genius, akin, in some respects, to that of Edwin Booth, and her delineation of the sublime, the beautiful, the terrible has never been surpassed. A noble interpreter of noble minds, Charlotte Cushman stands for the supreme achievement of the actress.

What Booth and Forrest were to tragedy, William J. Florence was to comedy. Indeed, he may be said to have gone farther than either Booth or Forrest, for he founded a school and gave to the stage the chivalrous, light-hearted and lucky Irishman, who has since become so familiar to the drama, however rare he may be outside the theatre.

Florence was born in Albany, New York, in 1831. His family name was Conlin, from which it will be seen that he came naturally by his insight into Irish character; but he changed this name when he went upon the stage to the more romantic and euphonious one of Florence. He gave evidence of possessing unusual dramatic talent while still a boy, and made his début on the regular stage at the age of eighteen. He had the usual hardships of the young actor, playing in various stock companies without attracting especial attention, and finally, in 1853, marrying Malvina Pray, herself an actress of considerable ability.

It was at this time that Florence began to find his field in the delineation of Irish and Yankee characters, his wife appearing with him, and together they won a wide popularity. Florence wrote some plays and a number of sprightly songs, which his wife sang inimitably. He himself improved steadily in his acting, and, especially in the gentle humor and melting pathos with which he clothed his characters, stood quite alone. A tour through England added to his fame, and his songs were soon being sung and whistled in the streets pretty generally wherever the English tongue was spoken. One song in particular, called "Bobbing Around," had immense popularity.

But Florence was more than a mere song-writer Irish comedian. In his later years he proved himself to be an actor of high attainments and no one who ever witnessed a performance of "The Rivals," with Jefferson as Bob Acres, and Florence as Sir Lucius O'Trigger, will ever forget his finished and glowing impersonation.

When Edwin Forrest, heart-broken and discredited, died in 1872, he left his manuscript plays to another great tragedian, whom he regarded as his legitimate successor, John McCullough. In some respects McCullough was a greater actor than Forrest, for he possessed that quality of poetic insight and high imagination which Forrest lacked, while in physical equipment for the great characters of tragedy he was in no whit his inferior.

John McCullough was born in Coleraine, Ireland, in 1837, his parents, who were small farmers, bringing him to this country at the age of sixteen. They settled at Philadelphia and the boy was apprenticed to a chair-maker, but he soon broke away from that hum-drum employment, and in 1855, appeared in a minor part in "The Belle's Strategem." His story, after that, was the usual one of long years of training in various stock companies. He gradually worked his way into prominence, and finally in 1866, became associated with Edwin Forrest, taking the second parts in the latter's plays; and, after Forrest's death, taking his place as the first impersonator of robust tragedy in America.

For ten years his success was tremendous—then came the sad ending. McCullough had always been supremely great in characters requiring the delineation of madness—*Virginius*, *King Lear*, *Othello*. Whether this had anything to do with the final tragedy cannot be said, but in 1884, while playing at Chicago, he broke down in the midst of a performance, and had to be led from the stage. His mind was gone; he never rallied, and ended his days in an asylum for the insane.

One of the most successful engagements McCullough ever had was in 1869 and for some years thereafter, when, with Lawrence Barrett, he appeared at the Bush Street theatre in San Francisco. Barrett's name is also closely associated with that of Edwin Booth, for he played opposite Booth through many seasons—*Othello* to Booth's *Iago*, *Cassius* to Booth's *Brutus*, and so on; and the two formed a combination which for sheer genius has never been surpassed. But Barrett never commanded the adoration of the public as Booth did, because he lacked that power of enchantment which Booth possessed in a supreme degree. His mind was austere, he could win respect but not affection, and, as a result, criticism was more captious, honors came grudgingly or not at all, and the fight for recognition was up-hill all the way.

Lawrence Barrett was born in 1838, and he began his theatrical career at the age of fifteen. After the usual hard stock-company experience, he secured a New York engagement, where, for nearly two years, he supported such actors as Charlotte Cushman and Edwin Booth. From New York he went to Boston for a similar engagement, but at the outbreak of the Civil War he left the stage, accepted a captaincy in the Twenty-eighth Massachusetts Infantry, and served through the war with distinction. Then he returned to the theatre, gaining an ever-increasing reputation until his death.

Clara Morris called him "The Man with the Hungry Eyes," and they were hungry, for life was always a battle to him. From an obscure and humble position, without fortune, friends, or favoring circumstances he had fought his way upward in the face of indifference, disparagement and cold dislike.

Clara Morris has told the story of her own life better than anyone else could tell it, and has shown in doing it the very qualities which made most for her success—a wide sympathy, an impetuous heart, and an invincible optimism. She, too, had a hard struggle at the first—entering the ballet at the age of fifteen to help her mother after her father's death, and working her way up until she secured a New York engagement with Augustin Daly's famous stock company, where she soon was sharing the honors with Ada Rehan. Ill health shortened her acting career, and compelled her retirement from the stage when at the very height of her powers.

Just the other day there died in California another woman who won a great public a generation ago by a genius and charm seldom equalled. Helena Modjeska's story was an unusual one. Born in Cracow, Poland, in 1844, the daughter of a great musician, her early years were passed in an inspiring atmosphere, and almost from the first she felt an impulse toward the stage. But her family refused to permit her to become an actress, and it was not until after her marriage that her chance came. Her husband consented to a few trial appearances, and her success was so great that she was soon engaged as leading lady for the theatre at Cracow.

But her husband incurred the ill-will of the authorities by his political writings, and she herself got into trouble with them by resisting the Russian censorship of the Polish theatre. It was evident that arrest and banishment for either or both of them might come at any moment, and under this

incessant and increasing worry, her health began to fail. So she renounced the theatre, as she thought, forever, came to America, purchased a ranch in California, and settled down to spend the remainder of her life in quiet. But Edwin Booth, John McCullough, and others, encouraged her to study English and appear upon the American stage. She did so, and four months later appeared at San Francisco as Adrienne Lecouvreur. She had an instant success, and for more than thirty years maintained her position as one of the greatest actresses of the day.

Her personal fascination was of an exceedingly rare kind, her figure tall and graceful, her face wonderfully attractive in its intellectual charm and eloquent mobility. Shakespeare was her chief delight, and as Juliet, Rosalind and Ophelia she enchanted thousands.

On the evening of Thursday, November 25, 1875, an audience assembled at one of the theatres of Louisville, Kentucky, to witness "the first appearance upon any stage" of "a young lady of Louisville." The young lady in question had chosen as her vehicle Shakespeare's Juliet, which was certainly beginning at the top; she was only sixteen years of age and had never received any practical stage training; her experience of life was narrow and provincial—and yet, when the curtain rang down for the last time, the discerning ones in that audience knew that, despite the crudity of the performance, a new star had arisen and a great career begun. For that "young lady of Louisville" was Mary Anderson. Her story is unique in the history of the American stage.

Born in California in 1859, but taken to Louisville a year later; her father, Charles Joseph Anderson, dying in 1863, an officer in the Confederate army, Mary Anderson was reared by her mother in the Roman Catholic faith and received her education in a parochial school at Louisville. She left school before she was fourteen, and two years later, as we have seen, was upon the stage. Her first appearance won her an engagement at Louisville, and for thirteen years thereafter she was an actress, never in a stock company, but always a star. Then, at the very meridian of her career, she married and retired forever from the stage.

Mary Anderson's charm was not that of a great actress, for a great actress she never became. She had not the training necessary to finished and rounded work. Her charm was rather that of a sweet and gracious personality, of a beautiful nature and a high sincerity. Sumptuously beautiful, and possessed of a clear and resonant voice, such statuesque characters as Galatea and Hermione attracted her irresistibly, and in these she achieved her greatest triumphs.

Scarcely second to her was Ada Rehan, born a year later, appearing on the stage two years earlier, in other words, at the age of thirteen. Ada Rehan, appropriately enough, was born at Limerick, Ireland, and the roguish and perverse Irish spirit was ever uppermost in her acting. She was brought to America when she was five years old, and lived and went to school in Brooklyn. Two of her elder sisters were upon the stage, but she does not seem to have indicated any especial desire to imitate them, and her first appearance was by accident. An actress playing a small part in "Across the Continent" was taken suddenly ill, and the child, who happened to be at the theatre, was hastily dressed for it and taught her few lines; but she displayed so much readiness and natural talent that, at a family council which followed the performance, it was decided that she should proceed with a stage career, and she was soon regularly embarked.

This meant a long and severe course of training in the stock companies maintained at the various theatres throughout the country to support such wandering stars as Booth and McCullough, and Barrett, and Adelaide Neilson, and she emerged from this training well grounded in all the business of the actress. In 1879, she attracted Augustin Daly's attention, and from that time forward until Daly's death, she was the leading woman at his famous New York house, becoming one of the most admired figures upon the stage. Her art, luminous and sparkling, especially fitted her for high comedy, and it was there that she achieved her greatest distinction.

Ada Rehan's name was closely associated for many years with that of John Drew, also a member of the Daly company, and a son of the famous "Mr. and Mrs. John Drew," two of the most versatile, charming and popular members of the old school. The elder John Drew was born in Ireland in 1825, but came to America at the age of twenty and spent the remainder of his life here, except for a few absences on tour. He was considered the best

Irish comedian on the American stage. His wife, born in London in 1820 of a theatrical family, appeared in child's parts at the age of eight, came to this country at the age of twenty, and made a great success here in high comedy parts. Their son can scarcely be said to have fulfilled the promise of his early years, but seems to be content with an achievement which shows him to be an accomplished and finished, but by no means inspired or imaginative, actor.

Another family as celebrated in American theatrical annals as that of John Drew was E. L. Davenport's. Davenport himself had received his training in the old stock companies, and notably as Junius Brutus Booth's support in a number of plays. He was equally at home in tragedy and comedy. Associated with him after their marriage in 1849 was his wife, Fanny Elizabeth Vining, an actress of considerable ability.

No less than six of their children followed the stage as a career. The most famous of them was Fanny Davenport, whose stage career began when she was a mere baby. Her young girlhood was occupied with soubrette parts, but she soon developed unusual emotional powers, and attracted Augustin Daly's notice. He added her to his stock company in 1869, and she soon won a notable success in such parts as Lady Gay Spanker, Lady Teazle and Rosalind.

Perhaps no American actor ever had a more remarkable career than William Warren. Born in 1812, the son of a player of considerable reputation, his first appearance was at the age of twenty. For twelve years his history was that of most other struggling actors, but in 1846 he became connected with the Howard Athenæum at Boston, where he remained for thirty-five years, retiring permanently from the stage in 1882.

During his career, he had given 13,345 performances and had appeared in 577 characters, a record which has probably never been approached. He was especially notable in his representations of the "fine old English gentleman," and he became to Boston a sort of Conservatory of Acting in himself. That he was appreciated both as man and artist his long residence in Boston proves.

He was a cousin of one of the best loved actors who ever trod the American stage—Joseph Jefferson; but their careers were very different, for Jefferson,

in the last quarter century of his life confined himself to a few parts—practically to four, Bob Acres, Rip Van Winkle, Dr. Pangloss and Cabel Plummer. In these he was inimitable. Something is gained and lost, of course, by either of these methods; one is inclined to think the wiser plan, that making for the greatest achievement, is a wide diversity of parts, and constant creation of new ones. And yet, when one looks back upon Jefferson's delicate and cameo-clear impersonations, one would not have him different.

Joseph Jefferson was the third of his name to challenge American theatre-goers. His grandfather, born in England, in 1774, came to America twenty-three years later and spent the remainder of his life here, gaining some reputation as a comedian. His father is said to have had little ability, and to have been careless and improvident. The third of the name was born in Philadelphia in 1829, and began his stage career at the age of three, appearing as the child in "Pizarro," which must have frightened him nearly to death.

His father died when he was only fourteen, and the lad joined a company of strolling players, who made their way through Texas, and during the war with Mexico, followed the American army into Mexican territory. American drama was in no great demand, so at Matamoras Jefferson opened a stall for the sale of coffee and other refreshments, making enough money to get back to the United States.

For the next ten years he appeared in stock companies in the larger eastern cities, meeting such players as Edwin Forrest, James E. Murdoch, and Edwin Adams; but the one who influenced him most was his own half-brother, Charles Burke, an unusually accomplished serio-comic. William Warren also ranked high in his affections.

The turning point of his career came in 1857 when he became associated with Laura Keane at her theatre in New York. Here his first part was one with which he was afterwards so closely identified, that of Dr. Pangloss, and then came "Our American Cousin," in which he gained a notable success as Asa Trenchard, and in which Edward A. Sothorn laid the foundation of the fantastic character of Lord Dundreary, which was to make him famous. A year later, he created another of his great characters, Caleb

Plummer, in "The Cricket on the Hearth," and soon afterwards, the most famous of all, Rip Van Winkle, which remained to the end his supreme impersonation.

After that time, his career was a golden and happy one. He won the affection of the American public as perhaps no recent player has ever done. His art had a peculiarly wide appeal because it was fine and sweet; he won sympathy and inspired affection; and seemed the very embodiment of the tender, artless and lovable characters it was his joy to represent.

Jefferson's death marked the passing of the last of the "old school"—that mellow, fluent, and accomplished circle of players who seem so different to their successors. But public taste is different too. We care no longer for the rantings and heroics of Virginius and Spartacus and all the rest of those toga-clothed gentlemen who differed from each other only in their names. We demand something more subtle, more—yes, let us say it!—intellectual. The modern who came nearest to answering this demand, to showing us the complex thing which we know human nature to be, was Richard Mansfield. A great artist, whom no difficulty appalled, he gave the American public, season after season, the most significant procession of worthy dramas that one man ever produced.

Mansfield was born in Heligoland in 1857, and studied for the East Indian civil service, but came to Boston and opened a studio, studied art, and then suddenly abandoned it for the stage. Curiously enough, he began with small parts in comic opera, and a few years later, made one of the funniest Kokos who ever appeared in "The Mikado." But he soon changed to straight drama, and the first great success of his career was as Baron Chevrial in "A Parisian Romance," a part which was given him after other actors had refused to take it, and in which he created a real sensation. His reputation was secure after that, and grew steadily until the swift and complete collapse from over-work, which ended his life at the age of fifty-one.

Are there any great players alive in America to-day? E. H. Sothorn, perhaps, comes nearest to greatness, and has at least won respectful attention by a sincerity and earnestness which have accomplished much. He is the son of Edward Askew Sothorn, whose career was a most peculiar one. Intended for the ministry, he chose the stage instead, apparently with no talent for it, and

for six or seven years, only the most unimportant of minor parts were entrusted to him.

One of these was that of Lord Dundreary in "Our American Cousin." It consisted of only a few lines and Sothern accepted it under protest, but he made such a hit in it that it was amplified and became the principal part of the play. In fact, the play became, in the end, a series of monologues for Dundreary. It had some remarkable runs, one, for instance, in London, for four hundred and ninety-six consecutive nights. Sothern continued playing the part until his death. His son is undoubtedly a far greater actor, and may achieve a high and lasting fame.

Associated with him in many of his later and more ambitious productions has been Julia Marlowe, undoubtedly the most finished and accomplished actress in America. She had a thorough training, having been on the stage since her twelfth year, and devoting herself closely to the study of her art. Her sincerity, too, promises much for the future. After Sothern, Otis Skinner is perhaps the most noteworthy, and after him, well, anyone of a dozen, whom it is needless to name here.

It was Joseph Jefferson who remarked that "all the good actors are dead." He meant, of course, that the present seems always of little worth when compared with the past; and this is the case not only with the theatre, but in some degree with all the arts. It is especially true of the theatre, however, because the player lives only in the memories of those who saw him, and memory sees things, as it were, through a golden glow.

SUMMARY

BOOTH, JUNIUS BRUTUS. Born at London, May 1, 1796; first appearance, 1813; came to America, 1821; died on a Mississippi steamboat, November 30, 1852.

BOOTH, EDWIN. Born at Bel Air, Maryland, November 13, 1833; first appearance, 1849; first appearance as "star," as Sir Giles Overreach, 1857; played under management of Lawrence Barrett, 1886-91, in "Hamlet"; founded "The Players' Club," 1888; died at its club-house, in New York City, June 7, 1893.

FORREST, EDWIN. Born at Philadelphia, March 9, 1806; first appearance, 1820; first notable success as Othello, 1826; last appearance in March, 1871; died at Philadelphia, December 12, 1872.

CUSHMAN, CHARLOTTE. Born at Boston, July 23, 1816; first appearance, 1835; played with Macready, 1842-44; in London, 1844-48; died at Boston, February 8, 1876.

FLORENCE, WILLIAM JAMES. Born at Albany, New York, July 26, 1831; first appearance, 1849; died at Philadelphia, November 19, 1891.

MCCULLOUGH, JOHN. Born at Coleraine, Ireland, November 2, 1837; came to America, 1853; first appearance, 1855; broke down mentally and physically, 1884; died in insane asylum at Philadelphia, November 8, 1885.

BARRETT, LAWRENCE. Born at Paterson, New Jersey, April 4, 1838; first appearance, 1853; enlisted in 28th Massachusetts Volunteers, 1861; from 1887 until his death closely associated with Edwin Booth; died at New York City, March 21, 1891.

MORRIS, CLARA. Born at Toronto, Canada, 1849; first appearance, 1861; leading lady, 1869; joined Daly's company, 1870; married Frederick C. Harriott, 1874.

MODJESKA, HELENA. Born at Cracow, Poland, October 12, 1844; first appearance, 1861; first appearance in English at San Francisco, 1877; died in California, April 8, 1909.

ANDERSON, MARY. Born at Sacramento, California, July 28, 1859; first appearance, 1875; married Antonio de Navarro, 1889, and retired from the stage.

REHAN, ADA. Born at Limerick, Ireland, April 22, 1860; came to America in childhood; first appearance, 1874; joined Daly's company, 1879; leading lady there until his death in 1899.

DREW, JOHN. Born at Philadelphia, in 1853; first appearance, 1873; leading man in Daly's company, 1879-99.

DREW, JOHN, SR. Born at Dublin, Ireland, September 3, 1825; first appearance in New York, 1845; died at Philadelphia, May 21, 1862.

DREW, MRS. JOHN, SR. (LOUISA LANE). Born at London, January 10, 1820; first appearance when mere child; came to America, 1828; married John Drew, 1850; died at Larchmont, New York, August 31, 1897.

DAVENPORT, EDWARD LOOMIS. Born at Boston, Massachusetts, November 15, 1814; first appearance, 1836; played in England, 1847-54; died at Canton, Pennsylvania, September 1, 1877.

DAVENPORT, FANNY ELIZABETH VINING. Born at London, July 6, 1829; began playing baby parts at age of three; made first appearance, 1847, as Juliet; married E. L. Davenport, January 8, 1849; first appearance in New York, 1854.

DAVENPORT, FANNY LILY GIPSY. Born in London, April 10, 1850; first American appearance, 1862; died at Danbury, Massachusetts, September 26, 1898.

WARREN, WILLIAM. Born at Philadelphia, November 17, 1812; first appearance, 1832; died at Boston, September 21, 1888.

JEFFERSON, JOSEPH. Born at Philadelphia, February 20, 1829; first appearance on stage as child; first became prominent as Asa Trenchard, in "Our American Cousin," 1858; died at West Palm Beach, Florida, April 23, 1905.

SOTHERN, EDWARD ASKEW. Born at Liverpool, England, April 1, 1826; first appearance, 1849; first American appearance, 1852; made his mark as Lord Dundreary, 1858; died at London, January 20, 1881.

SOTHERN, EDWARD H. Born in London; appeared as child; first took leading part, 1887.

CHAPTER VII

SCIENTISTS AND EDUCATORS

To give even the briefest account, within the limits of a single chapter, of the lives of noteworthy American scientists and educators is, of course, quite beyond the bounds of possibility. All that can be done, even at best, is to mention a few of the greatest names and to indicate in outline the particular achievements with which they are associated. That is all that has been attempted here. There are at least a hundred men, in addition to those mentioned in this chapter, whose work is of consequence in the development of American science and education. The record of their achievements is an inspiring one which, if properly told, would occupy many volumes.

In the annals of American science, two names stand out with peculiar lustre—John James Audubon and Louis Agassiz. Neither was, strictly speaking, American, for Agassiz was born in Switzerland and did not come to this country until he was nearly forty years of age; while Audubon was born in French territory, the son of a French naval officer, and was educated in France. But the work of both men was distinctively American, for Audubon devoted his life to the study of American birds, and Agassiz the latter part of his to the study and classification of American fishes—as well as to services of the most valuable kind in the field of geology and paleontology.

Audubon's story is a curious and interesting one. His father, the son of a Vendean fisherman, after working his way up to the command of a French man-of-war, purchased a plantation in Louisiana, which at that time belonged to France. He married there, and there, in 1780, John James Audubon was born. He was a precocious child, and early developed a love for nature, which his parents encouraged in every way they could. He was especially fond of drawing birds and coloring his drawings. He acquired so much skill in doing this that his father sent him to Paris and placed him in the studio of the celebrated painter, David.

It is related of young Audubon that his drawings for many years fell so far short of his ideal, that on each of his birthdays he regularly made a bonfire of all he had produced during the previous year. He cared for nothing else, however, and after his return to America, his home became a museum of birds' eggs and stuffed birds. He took long tramps through the wilderness, with no companions save dog and gun, all the time adding new drawings to his collection. Some birds he was obliged to shoot, afterwards supporting them in natural positions while he painted them; others which he could not approach, he drew with the aid of a telescope, representing them amid their natural surroundings, and all with painstaking care and exactitude.

This work, occupying years of time, and accompanied by every sort of suffering and exposure, by long trips through the wilderness of the west, in heat and cold, snow and rain, was carried forward from pure love of nature and enthusiasm for the work itself, without thought or hope of reward. Audubon's friends began to consider him a kind of harmless madman, for what sane person would devote his life to a work so laborious and seemingly so useless? He made a little money occasionally by giving drawing lessons; but he was never content except when roaming the plains and forests, hunting for some new specimen. For his ambition was to study and draw every kind of bird which lived in America.

In 1824 he happened to be in Philadelphia, and met there a son of Lucien Bonaparte, to whom he showed his drawings. The Frenchman was at once deeply interested, for he saw their beauty and value, and he urged upon Audubon that some arrangement be made by which they could be published and given to the world. The obstacles in the way of such an enterprise were enormous, for the processes of color reproduction at that time were slow and expensive, and it was estimated that the cost of the entire work would exceed a hundred thousand dollars.

But Audubon had overcome obstacles before that, and three years later he issued the prospectus of his famous "Birds of America." It was to consist of four folio volumes of plates, and the price of each copy was fixed at a thousand dollars. Three years more were spent in securing subscriptions, and then the work of publication began, though Audubon had barely enough money to pay for a single issue. Funds came in, however, after the appearance of the first number, and the work went steadily forward to

completion in 1839. It was called by the great naturalist, Cuvier, "the most magnificent monument that art ever raised to ornithology." It contained 448 beautifully colored plates, showing 1065 species of North American birds, each of them life size.

Before it was completed, Audubon had planned another work on similar lines, to be known as "The Quadrapeds of America," and set to work at once to gather the necessary material, which meant the study from life of each of these animals. He even projected an extensive trip to the Rocky Mountains in search of material, but was persuaded by his friends to give it up, as he was then nearly sixty years of age, and suffering from the effects of his long years of exposure. His sons assisted him in the preparation of the work, the first volume of which appeared in 1846, the last in 1854, three years after his death.

Audubon's life illustrates strikingly the compelling power of devotion to an ideal. Few men have met such discouragements as he, and fewer still have overcome them. For many years, in all climates, in all weathers, pausing at no difficulty or peril, his life frequently endangered by wild beasts or still wilder savages, he trudged the pathless wilderness, quite alone, sleeping under a rude shelter of boughs or in a hollow tree, living on such game as he could shoot, seeking only one thing, new birds, and when he found them, observing their habits and setting them on paper with an infinite patience. On one occasion, rats got into the room where his drawings were stored, and destroyed almost all of them; but he set to work at once re-drawing them, where most men would have given up in despair. His work remains to this day the standard one on American birds—a mighty monument to the ideals of its maker.

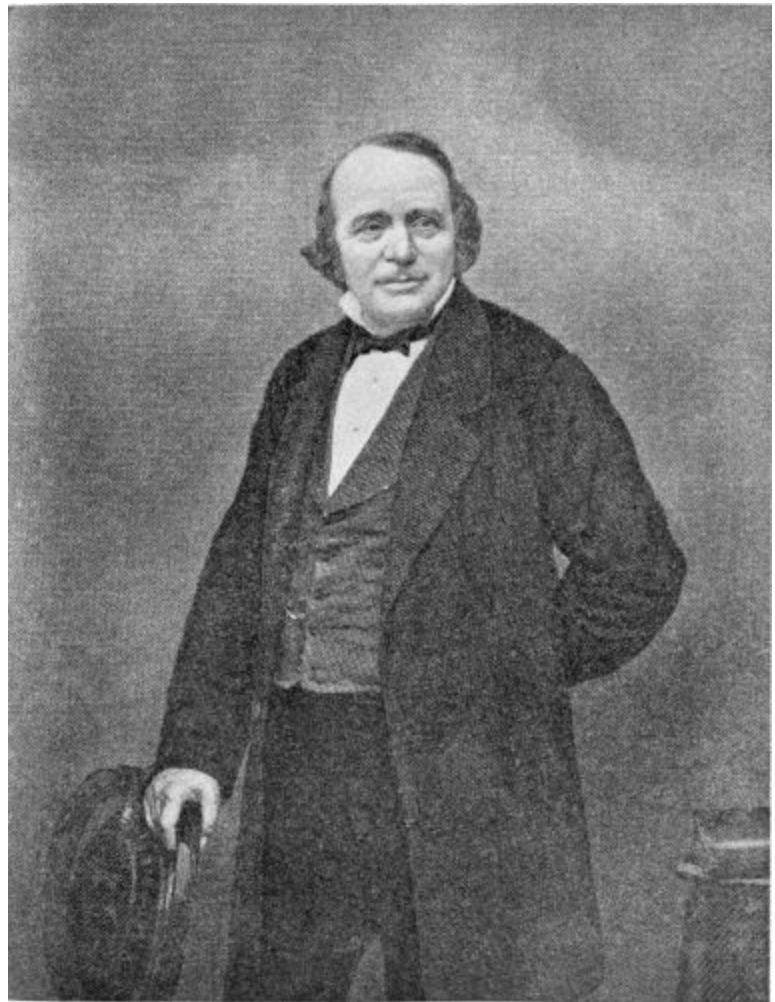
Jean Louis Rudolphe Agassiz was also a born naturalist, but no such obstacles confronted him as Audubon surmounted, nor did he strike out for himself a field so absolutely original. Born in Switzerland in 1807, the descendent of six generations of preachers, but destined for the profession of medicine, he refused to be anything but a naturalist. From his earliest years, he showed a passion for gathering specimens, and his first collection of fishes was made when he was ten years old. He received the very best training to be had in Switzerland, France and Germany, and early attracted attention for original work of the most important description. He came to be

recognized as the greatest authority on fishes in Europe, and his work on fossil fishes, published in 1843, was a contribution to science of the first importance.

In 1846, Agassiz came to the United States, partly to deliver a course of lectures at Boston and partly to make himself familiar with the geology and natural history of this country. His reception was so cordial and he found so much to interest him here, that he accepted the chair of zoology and geology in the Lawrence Scientific School at Cambridge, Massachusetts, and decided to make the United States his home.

He soon made Cambridge a great scientific centre, and proved himself the most inspiring, magnetic and influential teacher of science this country has ever seen.

In succeeding years, he traversed practically the entire country, accumulating vast collections of specimens which formed the foundation of the great natural history museum at Cambridge. He was preparing himself for the publication of a comprehensive work to be called "Contributions to the Natural History of the United States," the first volume of which appeared in 1857. Succeeding years were occupied with a journey to Brazil, another around Cape Horn, and the establishment of the Pekinese Island school of natural history, where he was able to carry out his long



AGASSIZ

contemplated plan of teaching directly from nature. But his labors had impaired his health, and he died in Cambridge in 1873, after a short illness. His grave is marked by a boulder from the glacier of the Aar, and shaded by pine trees brought from his native Switzerland.

Agassiz was one of the most remarkable teachers of science that ever lived. Handsome, enthusiastic, overflowing with vitality, and with a learning broad and deep, his students found in him a real inspiration to intellectual endeavor. His lectures, however technical and abstruse their subjects, were of an incomparable clarity and simplicity. He was one of the first to advocate the teaching of science to women, not in its technical details, but in its broad outlines.

"What I wish for you," he said, one day, addressing a class of girls, "is a culture that is alive and active. My instruction is only intended to show you the thoughts in nature which science reveals.

"A physical fact is as sacred as a moral principle," he used to say. "Our own nature demands from us this double allegiance."

Of the pupils of Agassiz, not the least famous was his son, Alexander, who, after graduating from Harvard, assisted his father in his work, collected many specimens for the museum at Cambridge, and was finally appointed assistant in zoology there. In the following years he put his scientific knowledge to a very practical use. In his geological surveys of the country, he had been impressed with the richness of the copper mines on Lake Superior. For five years, he acted as superintendent of the famous Calumet and Hecla mines, developing them into the most successful copper mines in the world, and himself gaining wealth from them which permitted his making gifts to Harvard aggregating half a million dollars. It was characteristic of him that, after his service with the Calumet and Hecla, he resumed his duties at the museum at Cambridge, and continued as curator until ill health compelled his resignation in 1885.

Among other pupils of Agassiz who won more than ordinary fame as naturalists may be mentioned Albert Smith Bickmore, Alonzo Howard Clark, Charles Frederick Hartt, Alpheus Hyatt, Theodore Lyman, Edward Sylvester Morse, Alpheus Spring Packard, Frederick Ward Putnam, Samuel Hubbard Scudder, Nathaniel Southgate Shaler, William Stimpson, Sanborn

Tenney, Addison Emory Merrill, Burt Green Wilder and Henry Augustus Ward—as brilliant a galaxy of names as American science can boast, bearing remarkable testimony to the inspiring qualities of their great teacher.

What Agassiz did for geology and natural history, Asa Gray to some extent did for botany. Born at Paris, N. Y., in 1810, and at an early age abandoning the study of medicine for that of botany, he accepted, in 1842, a call to the Fisher professorship of natural history at Harvard, a post which he held for over thirty years. Gray's work began at the time when the old artificial system of classification was giving way to the natural system, and he, perhaps more than any other one man, established this system firmly on the basis of affinity.

In 1864, he presented to Harvard his herbarium of more than two hundred thousand specimens, and his botanical library. He remained in charge of the herbarium until his death, adding to it constantly, until it became one of the most complete in the world. His publications upon the subject of botany were numerous and of the highest order of scholarship, and long before his death he was recognized as the foremost botanist of the country.

Scarcely inferior to him in reputation was John Torrey. It was to Torrey that Gray owed his first lessons in botany, and if the pupil afterwards surpassed the master, it was because he was able to build on the foundations which the master laid. John Torrey, born in New York City in 1796, was the son of a Revolutionary soldier, and in early life determined to become a machinist, but afterwards studied medicine and began to practice in New York, taking up the study of botany as an avocation. He found the profession of medicine uncongenial, and finally abandoned it altogether for science, serving for many years as professor of chemistry and botany at the College of Physicians and Surgeons in New York City. The succeeding years brought him many honors, and saw many works of importance issue from his hands.

The progress of the last century in the various branches of science is an interesting study, and America has made no inconsiderable contributions to every one of them. In astronomy, six names are worthy of mention here. The first of these, John William Draper, was noted for his devotion to many other lines of science, especially to photography, and was the first person in

the world to take a photograph of a human being. His service to astronomy was in the application of photography to that science. In 1840, he took the first photograph ever made of the moon, and a few years later published his "Production of Light by Heat," an early and exceedingly important contribution to the subject of spectrum analysis.

His work in astronomy and more especially in physics was carried on most worthily by his son, Henry Draper, who, at his home at Hastings-on-the-Hudson, built himself an observatory, mounting in it a reflecting telescope, which he also made. His description of the processes of grinding, polishing, silvering, testing and mounting it has remained the standard work on the subject. With this telescope he took a photograph of the moon which remains one of the best that has ever been made. Among his other noteworthy achievements were his spectrum photographs of 1872 and 1873, and in 1880 his photograph of the great nebula in Orion, the first photograph of a nebula ever secured. Perhaps the most brilliant discovery ever made in physical science by an American was that by Draper in 1877, when he demonstrated the presence of oxygen in the sun so conclusively that it could not be disputed. It was a sort of *tour de force* that took the scientific world by surprise and gained its author the widest recognition.

The services of Lewis Morris Rutherford to astronomy resembled in many ways those of Draper. Starting in life as a lawyer, he abandoned that profession at the age of thirty-three to devote his whole time to science, principally to the perfection of astronomical photography and spectrum analysis. The service which photography has rendered to astronomy can scarcely be overestimated, and these pioneers in the art were laying the foundations for its recent wonderful developments. He was the first to attempt to classify the stars according to their spectra, and invented a number of instruments of the greatest service in star photography. All in all, it is doubtful if anyone added more to the development of this branch of the science than did he.

Very different from the services of these men were those rendered the science of astronomy by Charles Augustus Young. Called to the chair of astronomy at Princeton University in 1877, he held that important position for thirty years, his courses a source of inspiration to his students. He was a member of many important scientific expeditions, invented an automatic

spectroscope which has never been displaced, measured the velocity of the sun's rotation, and was a large contributor to public knowledge of the science.

Equally important have been the contributions made by Samuel Pierpont Langley, perhaps the greatest authority on the sun alive to-day. He showed a decided fondness for astronomy even as a boy, and at the age of thirty was assistant in the observatory at Harvard. Two years later, he was invited to fill the chair of astronomy in the Western University of Pennsylvania at Pittsburgh, and his work there began with the establishment of a complete time service, the first step toward the present daily time service conducted by the government. In 1870, he began the series of brilliant researches on the sun which have placed him at the head of authorities on that body. His scientific papers are very numerous and his series of magazine articles on "The New Astronomy" did much to acquaint the public with the rapid development of the science. In 1887, he was chosen to the important post of secretary of the Smithsonian Institution, and his recent years have been spent in experimenting with aëronautics.

Simon Newcomb is another who rendered yeoman service to the science. Born in Nova Scotia, the son of the village schoolmaster, he lived to become one of the eight foreign associates of the Institute of France, the first native American since Franklin to be so honored; to win the Huygens medal, given once in twenty years to the astronomer who had done the greatest service to the science in that period, and to receive the highest degree from practically every American college.

In his autobiography he tells how, at the age of five, he began to study arithmetic, at twelve algebra, and at thirteen Euclid. At the age of eighteen, planning to make his way to the United States, he set out on foot, taught school for a year or so, and then attracted the attention of Prof. Joseph Henry, of the Smithsonian Institution, by sending him a problem in algebra. The unusual aptitude for mathematics which the boy possessed so impressed Prof. Henry, that he set him to work as a computer on the Nautical Almanac; but he was soon attracted to "exact," or mathematical astronomy, which became his life work. Some idea of its importance may be gained when it is stated that every astronomer in the world to-day uses his determinations of the movements of the planets and the moon; every

skipper in the world guides his ship by tables which Newcomb devised; and every eclipse is computed according to his tables. He supervised the construction and mounting of the equatorial telescope in the naval observatory at Washington, the Lick telescope, and Russia applied to him, in 1873, for aid in placing her great telescope.

A man of humor, sympathy and anecdote, he found, in the fall of 1908, that he was suffering from cancer, and hastened the work on the moon, which was to be his masterpiece. Ten months later, he was told that his course was nearly run—and his great work was still incomplete.

"Take me to Washington," he said, "I must work while there is time."

And there, lying in agony on his bed, for three weeks he dictated steadily to stenographers on a subject which required the utmost concentration. His indomitable will alone supported him, and a week after the last word had been written, came the end. Verily, there was a man!

The last of the great American astronomers whom we shall mention here is Edward Charles Pickering, whose name is so closely connected with the development of the great observatory at Harvard. Born at Boston, and educated at the Lawrence Scientific School, his first work was in the field of physics, but in 1876, he was appointed professor of astronomy and geodesy, and director of the Harvard observatory, which, under his management, has become of the first importance. His principal work has been the determination of the relative brightness of the stars, and many thousands have been charted. On the death of Henry Draper, the study of the spectra of the stars by means of photography was continued as a memorial to that great scientist, and the results obtained have been of the most important character, including a star map of the entire heavens. Other phases of the science of scarcely less importance have been carefully developed, and the work which has been done under Pickering's direction, is second to none in the history of the science. Not satisfied with the Northern hemisphere, a branch has been established in Peru, in which the observatory's methods of research have been extended to the south celestial pole. So for eighteen years and more, it has kept ceaseless watch of the heavens, with an accuracy of which the world has hardly a conception. For

this great work the scientific world must pay tribute to the genius and perseverance of Edward Charles Pickering.

The second department of science claiming our attention is that of paleontology. Here one of the most eminent of American names is that of Othniel Charles Marsh. A graduate of Yale and firmly grounded in zoology and kindred sciences by a course of study at Heidelberg and Berlin, he returned to the United States in 1866 to accept the chair of paleontology which had been established for him at Yale. The remainder of his life was devoted to the original investigation of extinct vertebrates, especially in the Rocky Mountain regions. In these explorations, more than a thousand new species of extinct vertebrates were brought to light, many of which possess great scientific interest, representing new orders never before discovered in America. So important was this work that the national geological survey undertook the publication of his reports, which formed the most remarkable contributions to the subject ever written in this country, attracting the attention and admiration of the whole scientific world.

Associated with Marsh as paleontologist for the Geological Survey was Edward Drinker Cope, whose work was second only to the older man's in importance. He also devoted much of his attention to the exploration of the Rocky Mountain region, and found that there, in the strata of the ancient lake beds, records of the age of mammals had been made and preserved with a fulness surpassing that of any other known region on earth. The profusion of vertebrate remains brought to light was almost unbelievable. Prof. Marsh, who was first in the field, found three hundred new tertiary species between 1870 and 1876, besides unearthing the remains of two hundred birds with teeth, six hundred flying dragons, and fifteen hundred sea serpents, some of them sixty feet in length. In a single bed of rock not larger than a good sized lecture room, he found the remains of no less than one hundred and sixty mammals.

It was this work which Prof. Cope took up and carried forward. Its importance may be appreciated when it is stated that among these remains are found examples of just such intermediate types of organisms as must have existed if the succession of life on the earth has been an unbroken lineal succession. Here are snakes with wings and legs, and birds with teeth and other snakelike characteristics, bridging the gap between modern birds

and reptiles. The line of descent of the horse, the camel, the hippopotamus and other mammals has been traced to a single ancestor, the result being the proof of the theory of evolution.

The whole work of American paleontology has, of course, been along these lines. Agassiz himself was a living and vital force in it, as were such men as Joseph Leidy and H. F. Osborne.

It is a remarkable fact that one of the few truly original and novel ideas the past century can boast, and the one which has had the deepest influence on geology, had its origin in the brain of an illiterate Swiss chamois hunter named Perraudin. Throughout the Alps, on lofty crags, great boulders were often found, which had no relation to the geology of the region and which were called erratics, because they had evidently come there from a distance. But how? Scientists explained it in many ways, but it remained for the mountaineer to suggest that the boulders had been left in their present positions by glaciers. The scientific world laughed at the idea, but ten years later, it was brought to the attention of Louis Agassiz; he investigated it, became a convert, and saw that its implications extended far beyond the Alps, for these erratic boulders were found on mountains and plains throughout the northern hemisphere. Agassiz found everywhere evidences of glacial action, and became convinced that at one time a great ice cap had covered the globe down to the higher latitudes of the northern hemisphere. So came the conception of a universal Ice Age, now one of the accepted tenets of geology.

The dean of American geologists was Benjamin Silliman, who, at the very beginning of the nineteenth century, took up at Yale University the work which he was to carry on so successfully for more than fifty years. As an inspiring teacher he was scarcely less successful than Agassiz at a later day. His popular lectures began in 1808 and soon attracted to New Haven the brightest young men in the country. Among them was James Dwight Dana, who was to carry on most worthily the work which Prof. Silliman had begun.

James Dwight Dana was attracted to Yale by Prof. Silliman's great reputation and received there the inspiration which started him upon a scientific career. Three years after his graduation, he was appointed assistant to his former instructor, and two years later sailed for the South Seas as mineralogist and geologist of the United States exploring expedition commanded by Charles Wilkes. He was absent for three years and spent thirteen more in studying and classifying the material he had collected. He then resumed his work at Yale, succeeding Prof. Silliman in the chair of geology and mineralogy. His work was recognized throughout the world as most important, and many honors were conferred upon him.

Another famous name in American geology is that of John Strong Newberry. His name is connected principally with the explorations of the Columbia and Colorado rivers. He was afterwards appointed professor of geology and paleontology at the Columbia College School of Mines, and took charge of that department in the autumn of 1866. During his connection with the institution, he created a museum of over one hundred thousand specimens, principally collected by himself, containing the best representation of the mineral resources of the United States to be found anywhere.

Among the pupils of Prof. Silliman who afterwards won a wide reputation was Josiah Dwight Whitney. Graduating from Yale in 1839, he spent five years studying in Europe, and then, returning to America, was connected with the survey of the Lake Superior region, of Iowa, of the upper Missouri, and of California, issuing a number of books giving the results of these investigations, and in 1865, being called to the chair of geology at Harvard.

Still another of Prof. Silliman's pupils was Edward Hitchcock, whose life was an unusually interesting one. His parents were poor and he spent his boyhood working on a farm or as a carpenter, gaining such education as he could by studying at night. Deciding to enter the ministry, he managed to work his way through Yale theological seminary, graduating at the age of twenty-seven. It was here that he came under the influence of Prof. Silliman, and after a laboratory course and much field work, he was chosen professor of chemistry and natural history at Amherst College. He held this position for twenty years, and in 1845 was chosen president of the college, transforming it, before his retirement nine years later, from a poor and

struggling institution into a well-endowed and firmly established one. He had meanwhile served as state geologist of Massachusetts, and completed the first survey of an entire state ever made by authority of a government.

The most important recent contribution to American geology has been the three volume work issued in 1904-5, under the joint editorship of Thomas C. Chamberlain and Rollin D. Salisbury. Both are geologists of wide experience, and their work presents the present status of the science interestingly and simply.

America has had her full share of daring and successful surgeons, and in the science of surgery stands to-day second to no nation on earth, but perhaps the most famous American surgeon who ever lived was Valentine Mott. Dr. Mott was descended from a long line of Quaker ancestors, and was born in 1785. His father was a physician, and Dr. Mott began his medical and surgical studies at the age of nineteen, first in New York City, and afterwards in the hospitals of London, where he made a specialty of the study of practical anatomy by the method of dissection. At that time there was in this country a deep-seated prejudice against the use of the human body for this purpose, and the experience which Dr. Mott secured in London, and which stood him in such good stead in after years, would have been impossible of attainment here. A year was also spent in Edinburgh, and finally, in 1809, Dr. Mott returned to America with an exceptional equipment.

His skill won him a wide reputation and he was soon recognized as one of the first surgeons of the age. His boldness and originality were exceptional, and his success was no doubt due in some degree to his constant practice throughout his life of performing every novel and important operation upon a cadaver before operating upon the living subject. To describe in detail the operations which he originated would be too technical for such a book as this, but many of them were of the first importance. Sir Astley Cooper said of him: "Dr. Mott has performed more of the great operations than any man living, or that ever did live." He possessed all the qualifications of a great operator, extraordinary keenness of sight, steadiness of nerve, and physical

vigor. He could use his left hand as skillfully as his right, and developed a dexterity which has never been surpassed.

It should be remembered that in those days the use of anæsthetics had not yet been discovered, and every operation had to be performed upon the conscious subject, as he lay strapped upon the table shrieking with agony. To perform an operation under such circumstances required an iron nerve. Dr. Mott was one of the first to recognize the value of anæsthetics, and his use of them, immediately following their discovery, greatly facilitated their rapid and general introduction.

It is one of the boasts of American medicine that the first man in the world to conceive the idea that the administration of a definite drug might render a surgical operation painless was an American—Crawford W. Long. Dr. Long graduated from the medical department of the University of Pennsylvania in 1839. When a student, he had once inhaled ether for its intoxicant effects, and while partially under the influence of the drug, had noticed that a chance blow to his shin produced no pain. This gave him the idea that ether might be used in surgical operations, and on March 30, 1842, at Jefferson, Georgia, he used it with entire success. He repeated the experiment several times, but he did not entirely trust the evidence of these experiments. So he delayed announcing the discovery until he had subjected it to further tests, and while these experiments were going on, another American, Dr. W. T. G. Morton, of Boston, also hit upon the great discovery and announced it to the world.

Dr. Morton was a dentist who, in 1841, introduced a new kind of solder by which false teeth could be fastened to gold plates. Then, in the endeavor to extract teeth without pain, he tried stimulants, opium and magnetism without success, and finally sulphuric ether. On September 30, 1846, he administered ether to a patient and removed a tooth without pain; the next day he repeated the experiment, and the next. Then, filled with the immense possibilities of his discovery, he went to Dr. J. C. Warren, one of the foremost surgeons of Boston, and asked permission to test it decisively on one of the patients at the Boston hospital during a severe operation. The request was granted, and on October 16, 1846, the test was made in the presence of a large body of surgeons and students. The patient slept quietly while the surgeon's knife was plied, and awoke to an astonished

comprehension that the dreadful ordeal was over. The impossible, the miraculous, had been accomplished; suffering mankind had received such a blessing as it had never received before, and American surgery had scored its greatest triumph. Swiftly as steam could carry it, the splendid news was heralded to all the world, and its truth was soon established by repeated experiments.

To tell of the work of the men who came after these pioneers in the field of surgery and medicine is a task quite beyond the compass of this little volume. There are at least a score whose achievements are of the first importance, and nowhere in the world has this great science, which has for its aim the alleviation of human suffering, reached a higher development.

Among the physicists of the country, Joseph Henry takes a high place. His boyhood and youth were passed in a struggle for existence. He was placed in a store at the age of ten, and remained there for five years. At the age of fifteen he was apprenticed to a watchmaker, and had some thought of studying for the stage, but during a brief illness, he started to read Dr. Gregory's "Lectures on Experimental Philosophy, Astronomy and Chemistry," and forthwith decided to become a scientist. He began to study in the evenings, managed to take a course of instruction at the academy at Albany, New York, and finally, in 1826, was made professor of mathematics there.

Almost at once began a series of brilliant experiments in electricity which have linked his name with that of Benjamin Franklin as one of the two most original investigators in that branch of science which this country has ever produced. His first work was the improving of existing forms of apparatus, and his first important discovery was that of the electro-magnet. His development of the "intensity" magnet in 1830 made the electric telegraph a possibility. Two years later he was called to the chair of natural philosophy at Princeton University, where he continued his investigations, many of which have been of permanent value to science. In 1846, he was elected first secretary of the Smithsonian Institution, and removed to Washington, where the last forty years of his life were passed in the development of the great scientific establishment of which he was the head. He steadily refused the most flattering offers of other positions, among them the presidency of Princeton, and like Agassiz, he might have answered, when tempted by

larger salaries, "I cannot afford to waste my time in making money." To his efforts is largely due the establishment of the national lighthouse system, as well as that of the national weather bureau.

Besides his services to American science as instructor at Harvard College, Louis Agassiz rendered another when he persuaded Arnold Guyot, his colleague in the college at Neuchâtel, to accompany him to this country. Guyot was at that time forty years old, and was already widely known as a geologist and naturalist, and the delivery of a series of lectures before the Lowell Institute, established his reputation in this country. He was soon invited to the chair of physical geography and geology at Princeton, which he held until his death. He founded the museum at Princeton, which has since become one of the best of its kind in the United States. Perhaps he is best known for the series of geographies he prepared, and which were at one time widely used in schools throughout the United States.

Perhaps no family has been more closely associated with American science than that of the Huguenot Le Conte, who settled at New Rochelle, New York, about the close of the seventeenth century, moving afterwards to New Jersey. There, in 1782, Lewis Le Conte was born. He was graduated at Columbia at the age of seventeen and started to study medicine, but was soon afterwards called to the management of the family estates of Woodsmanston, in Georgia. There he established a botanical garden and a laboratory in which he tested the discoveries of the chemists of the day. His death resulted from poison that was taken into his system while dressing a wound for a member of his family.

His son, John Le Conte, after studying medicine and beginning the practice of his profession at Savannah, Georgia, was called to the chair of natural philosophy and chemistry at Franklin College, and after some years in educational work, was appointed professor of physics and industrial mechanics in the University of California, which position he held until his death, serving also for some years as president of the University. His scientific work extended over a period of more than half a century, being confined almost exclusively to physical science, in which he was one of the first authorities.

Another son of Lewis, Joseph Le Conte, like his brother, studied medicine and started to practice it; but in 1850, attracted by the great work being done by Louis Agassiz, he entered the Lawrence Scientific School at Harvard, devoting his attention especially to geology. After holding a number of minor positions, he became professor of geology and natural history in the University of California in 1869, and his most important work was done there in the shape of original investigations in geology, which placed him in the front rank of American geologists.

Lewis Le Conte had a brother, John Eathan Le Conte, who was also widely known as a naturalist of unusual attainments. He published many papers upon various branches of botany and zoology, and collected a vast amount of material for a natural history of American insects, only a part of which was published. His son, John Lawrence Le Conte, was a pupil of Agassiz, and conducted extensive explorations of the Lake Superior and upper Mississippi regions, and of the Colorado river. He afterwards made a number of expeditions to Honduras, Panama, Europe, Egypt and Algiers, collecting material for a work on the fauna of the world, which, however, was left uncompleted at his death.

American science recently suffered a heavy loss in the death of Nathaniel Southgate Shaler, one of the most brilliant of the pupils of Agassiz, and from 1864 until the time of his death, connected with the geological department of Harvard University, rising to the full professorship in geology, which he held for over twenty years, and to the position of dean of the Lawrence Scientific School. He did much to increase public interest in and knowledge of the development of the science by frequent popular articles in the leading magazines, in addition to more technical books and memoirs intended especially for scientists.

Of living scientists, we can do no more than mention a few. Perhaps the most famous, and dearest to the popular heart is John Burroughs, a nature philosopher, if there ever was one, a keen observer of the life of field and forest, and the author of a long list of lovable books. One of the leaders in the "return to nature" movement which has reached such wide proportions of recent years, he has held his position as its prophet and interpreter against the assaults of younger, more energetic, but narrower men.

Prominent in the same field is Liberty Hyde Bailey, since 1903 director of the College of Agriculture at Cornell University. His early training took place under Asa Gray, and his attention has been devoted principally to botanical and horticultural subjects. He has written many books, his principal work being his *Cyclopedia of American Horticulture*, which has just been completed. Other recent important contributions to science have been made by Vernon L. Kellogg, whose work has dealt principally with American insects, and whose recent book on that subject has been recognized as a standard authority; by Charles Edward Bessey, professor of botany at the University of Nebraska since 1884, a pupil of Dr. Asa Gray and the author of a number of valued books upon the subject which has been his life work; by George Frederick Barker, now emeritus professor of physics in the University of Pennsylvania, and the recipient of high honors at home and abroad; and by many others whom it is not necessary to mention here.

It will be evident enough from the foregoing that American science can boast no men of commanding genius—no men, that is, to rank with Darwin, or Huxley, or Lord Kelvin, or Sir Isaac Newton, to mention only Englishmen. Its record has been one of respectable achievement rather than of brilliant originality, but is yet one of which we have no reason to be ashamed.

Most of the men mentioned in this chapter have, in the widest sense been educators. Agassiz, Gray, Silliman, Guyot—all were educators in the fullest and truest way. It remains for us to consider a few others who have labored in this country for the spread of knowledge. That the present educational system of the United States is not a spontaneous growth, but has been carefully fostered and directed, goes without saying. It is the result, first, of a wise interest and support on the part of the state, which early recognized the importance of educating its citizens, and, second, of the self-sacrificing efforts of a number of intelligent, earnest, and public-spirited men.

One of the first of these was Horace Mann, born in Massachusetts in 1796, the son of a poor farmer. His struggle to gain an education was a desperate

one, and its story cannot but be inspiring. As a child he earned his school books by braiding straw, and his utmost endeavors, between the ages of ten and twenty, could secure him no more than six weeks' schooling in any one year. Consequently he was twenty-three years of age when he graduated from Brown University, instead of seventeen or eighteen, as would have been the case had he had the usual opportunities. He went to work at once as a tutor in Latin and Greek, studied law, was admitted to the bar, elected to the state legislature and afterwards to the senate, and finally entered upon his real work as secretary to the Massachusetts board of education.

He introduced a thorough reform into the school system of the state, made a trip of inspection through European schools, and by his lectures and writings awakened an interest in the cause of education which had never before been felt. His reports were reprinted in other states, attaining the widest circulation. It is noteworthy that as early as 1847, he advocated the disuse of corporal punishment in school discipline. After a service of some years as member of Congress, during which he threw all his influence against slavery, he accepted the presidency of Antioch College, at Yellow Springs, Ohio, where he continued until his death. It was there that the experiment of co-education was tried, and found to work successfully, and the foundations laid for one of the most characteristic of recent great development of higher school education in America. Oberlin College, also in Ohio, had by a few years preceded Dr. Mann's experiment, but the latter's great reputation as an educator caused his ardent advocacy of co-education to carry great weight with the public. From this time on it became a custom, as state universities opened in the west, to admit women, and the custom gradually spread to the east and even to some of the larger colleges supported by private endowments.

Turning to the three great universities, Harvard, Yale, and Princeton, which have done so much for the intellectual welfare of the country, we find a galaxy of brilliant names. On the list of Harvard presidents, three stand out pre-eminent—Josiah Quincy, Edward Everett, and Charles William Eliot. Josiah Quincy, third of the name of the great Massachusetts Quincys, graduated at Harvard in 1790 at the head of his class, studied law, drifted inevitably into politics, held a number of offices, which do not concern us here, and finally, after a remarkable term as mayor of Boston, was, in 1829, chosen president of Harvard. The work that he did there was important in

the extreme. He introduced the system of marking which continued in use for over forty years; instituted the elective system, which permitted the student to shape his course of study to suit the career which he had chosen; secured large endowments, and, when he retired from the presidency in 1845, left the college in the foremost position among American institutions of learning. Edward Everett, who was president of the college from 1846-49, was more prominent as a statesman than as an educator, and an outline of his career will be found in "Men of Action." The third of the trio, Charles William Eliot, whose term as president of the college covered a period of forty years, is rightly regarded as one of the greatest, if not the greatest educator this country has produced.

Graduating from Harvard in 1853, at the age of nineteen, he devoted his attention principally to chemistry, and, after some years of teaching, and of study in Europe, was, in 1865, appointed professor of chemistry in the Massachusetts Institute of Technology. The same year, a revolution occurred in the government of Harvard, which was transferred from the state legislature to the graduates of the college. The effect of the change was greatly to strengthen the interest of the alumni in the management of the university, and to prepare the way for extensive and thorough reforms. Considerable time was spent in searching for the right



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man for president and finally, in 1869, Prof. Eliot was chosen.

That the right man had been found was evident from the first. "King Log has made room for King Stork," wrote Oliver Wendell Holmes, then professor of anatomy and physiology at Harvard, to John Motley. "Mr. Eliot makes the corporation meet twice a month instead of once. He comes to the meeting of every faculty, ours among the rest, and keeps us up to eleven and twelve o'clock at night discussing new arrangements. I cannot help being amused at some of the scenes we have in our medical faculty—this cool, grave young man proposing in the calmest way to turn everything topsy turvy, taking the reins into his hands and driving as if he were the first man that ever sat on the box.

"'How is it, I should like to ask,' said one of our members, the other day, 'that this faculty has gone on for eighty years managing its own affairs and doing it well, and now within three or four months it is proposed to change all our modes of carrying on the school? It seems very extraordinary, and I should like to know how it happens.'

"'I can answer Dr. ——'s question very easily,' said the bland, grave young man. 'There is a new president.'

"The tranquil assurance of this answer had an effect such as I hardly ever knew produced by the most eloquent sentences I ever heard uttered."

The bland young man's innovations did not seem to do much harm to Harvard, for under his administration, her financial resources have been multiplied by ten, as has the number of her teachers, while the number of her students has been multiplied by five. Dr. Eliot has grown into the real head of the educational system of this country; his influence has wrought vast changes in every department of teaching, from the kindergarten to the university. It was his idea that common school education and college education ought to be flexible, ought to be made to fit the needs of the pupil. The result has been the broad development of the elective system—broader than Josiah Quincy ever dreamed of. The same system has changed the whole aspect of the teaching profession, resulting in the demand for a competent training in some specialty for every teacher.

Dr. Eliot, who is in a sense the first living citizen of America, has not attained that position merely by success in his profession. He has devoted time and thought to the great problems of our government, and has taken an active part in many public movements—the race question, the relations of capital and labor, the movement for universal arbitration. He has been honored by France, by Italy, and by Japan, and resigned from his great office, in 1909, at the age of seventy-five, with mental and physical powers in splendid condition, not to retire from active life, but to devote himself even more wholly to the service of his countrymen. In this age of commercial domination, a career such as Dr. Eliot's is more than usually inspiring.

In the history of the administration of Yale university, the most striking personalities are the two Timothy Dwights and Noah Porter. The first Timothy Dwight, born in 1752, and graduating from Yale at the age of seventeen, began to teach, and at the outbreak of the Revolution, enlisted as Chaplain in Parson's brigade of the Connecticut line. It was at this time he wrote a number of stirring patriotic songs, one of which, "Columbia," still lives. At the close of the war, he continued preaching and also opened an academy, at which women were admitted to the same courses with men, and which soon acquired considerable reputation. In 1795, he was called to the presidency of Yale, a position which he held until his death. His administration marked the beginning of a new era in the history of the college. At his accession, the college had about one hundred students, and the instructors consisted of the president, one professor and three tutors. He established permanent professorships and chose such men to fill them as Jeremiah Day, Benjamin Silliman, and James Kingsley. The result of this policy was a steady growth in the number of students, until, at his death, they had increased to over three hundred.

Noah Porter, who came to the presidency in 1871, had been graduated from the college forty years before, during which time he had studied theology, held a number of important charges, was called to the chair of moral philosophy at Yale, and finally elevated to the presidency. His work was most important, one feature of it being the introduction of elective studies, though he insisted also upon a required course, as opposed to the Harvard system. Some of the University's finest buildings were erected during his

administration, and at its close the student body numbered nearly eleven hundred.

He was succeeded in 1886 by Timothy Dwight, grandson of the elder president Dwight, who, for many years has been closely associated with the University, its financial growth being largely due to his efforts. Under his management the growth of the institution was unprecedented, the number of students increasing nearly fifty per cent within five years. He was also prominently identified with the general educational movement throughout the country, and his "True Ideal of an American University," published in 1872, attracted much attention.

Princeton has also had its share of eminent men, among them Jonathan Edwards, John Witherspoon, and James McCosh. Jonathan Edwards was one of the most remarkable characters in American history. Born in 1703, he was the fifth of eleven children and the only son. As a mere child, he developed uncommon qualities, entered Yale College at the age of twelve and graduated at the age of seventeen. His father was a clergyman, and the boy had been brought up in a household and community intensely religious, so that he very early began to have "a variety of concerns and exercises about his soul." It was inevitable, of course, that he should become a minister, and, at the age of nineteen, was ordained and began to preach at a small church in New York City. Edwards seems to have been afflicted from the first with what is in these days irreverently called an in-growing conscience, and early formulated for himself a set of seventy resolutions of the most exalted nature, which, however praiseworthy in themselves, were too high and good for human nature's daily food, and must have made him a most uncomfortable person to live with. He developed, however, into a powerful preacher, and his services were much sought, especially at revivals. One of his sermons, called "Sinners in the Hands of an Angry God," is said to have created a profound impression wherever delivered.

A difference with his congregation at Northampton caused him to resign his pastorate there, and, declining a number of calls to established parishes, he went as a missionary to the Housatonic Indians, at so small an income that his wife and daughters were forced to labor with the needle to support the family. It was while engaged in this work, that an unexpected call came to him to take the presidency of Princeton. He accepted and was installed as

president early in 1758. At once he began a series of reforms in the college administration, but an epidemic of small-pox broke out in the neighborhood, and Edwards, exposing himself to it fearlessly, contracted the disease and died thirty-four days after his installation.

Jonathan Edwards probably came as near to the old idea of a saint as America ever produced. Self-denying, stern, of an exalted piety, and intensely religious, he lived in a world of his own, and was regarded with no little awe and trembling. That he was a power for good cannot be doubted, and his sermons are still read, where those of his contemporaries have long since been forgotten.

Much more important to Princeton, was John Witherspoon, who came to the presidency in 1768, after a distinguished career in Scotland, one of the incidents of which was being taken a prisoner while incautiously watching the battle of Falkirk. He never wholly recovered from the effects of the imprisonment which followed. He brought with him from Scotland a valuable library which he gave to the college, and, finding the college treasury empty, he undertook a vigorous campaign to replenish it, making a tour of New England, and even extending his quest as far as Jamaica and the West Indies. Through his administrative ability and the changes and additions which he made in the course of study, the college received a great impetus.

The service to his adopted country by which Witherspoon will be longest remembered, was the course he followed at the beginning of the Revolution. From the first, he took the side of the colonies, and by precept and example, held not only the great body of Presbyterians true to that cause, but also the Scotch and Scotch-Irish, who were naturally Tories by sympathy. He was a member of the Continental Congress, urged ceaselessly the passage of the Declaration of Independence, was one of its signers, and as a member of succeeding Congresses, distinguished himself by his services. After the close of the war, he returned to Princeton and devoted the remainder of his life to its administration.

Greatest of the three as an educator was James McCosh. A Scotchman, like Witherspoon, a student of the Universities of Glasgow and Edinburgh, a pupil of Thomas Chalmers, he was ordained to the ministry in 1835, and

was a leading spirit in the movement which culminated in the establishment of the Free Church of Scotland. His publications on philosophical subjects brought him the appointment as professor of logic and metaphysics in Queen's College, Belfast, where he remained for sixteen years, drawing to the college a large body of students, and publishing other philosophical works of the first importance. In 1868, he was chosen president of Princeton, and his administration, lasting for nearly a quarter of a century, was remarkably successful. Under him, the student attendance nearly doubled, the teaching staff was more than doubled, and the resources of the college enormously increased. During these years, too, he continued his philosophical work, publishing a series of volumes which are the most noteworthy of their kind ever produced in America.

The temptation is great to dwell upon other educators connected with the great universities: Ira Remsen, and his contributions to chemistry; David Starr Jordan, and his great work on American fishes; Woodrow Wilson, and his contributions to the study of American history; Jacob Gould Schurman, and his work in the field of ethics;—to mention only a few of them—but there is not space to do so here. However, this chapter cannot be closed without some reference to the career of a remarkable woman, an educator in the truest sense, whose influence for good can hardly be estimated—Jane Addams.

John Burns, the English cabinet minister and labor leader, has called her "the only saint America has produced." Her sainthood is of the modern kind, which devotes itself by practical work to the alleviation of suffering and the uplifting of humanity, as opposed to the old fashioned kind of which we were speaking a moment ago in connection with Jonathan Edwards.

Graduating at Rockford College, in 1881, Miss Addams, then a delicate girl, spent two years in Europe. The sight which impressed her most, and which, to a large extent, determined her future career, was that of Mile End Road, the most crowded and squalid district of London, where she beheld a dirty and destitute mob quarreling over food unfit to eat. This vision of squalor and sin never left her, and the result was the establishment, in 1889, of the Social Settlement of Hull House, in the slums of Chicago. For Miss

Addams had come to the conclusion that the only way to reach the destitute and despairing was to dwell among them.

How right she was has been abundantly proved by the splendid work Hull House has done. Its object, as stated in its charter, is "to provide a center for a higher civic and social life; to institute and maintain educational and philanthropic enterprises, and to investigate and improve the conditions in the industrial districts of Chicago." All that it has done, and much more; for it has been a beacon light of progress, pointing the way for like undertakings elsewhere. But most valuable of all has been Miss Addams's personal influence, the inspiration which her life has been to workers everywhere for social betterment, and the message which, by tongue and pen, she has given to the world. As an example of a useful, devoted and well-rounded life, hers stands unique in America to-day.

SUMMARY

AUDUBON, JOHN JAMES. Born near New Orleans, May 4, 1780; published "Birds of America," 1830-39; "Ornithological Biography," 1831-39; "Quadrupeds of America," 1846-54; died at New York City, January 27, 1851.

AGASSIZ, JEAN LOUIS RUDOLPHE. Born at Motier, canton of Fribourg, Switzerland, May 28, 1807; professor of natural history at Neuchâtel, 1832; studied Aar glacier, 1840-41; came to United States, 1846; professor of zoölogy and geology at Cambridge, 1848; curator of Cambridge Museum of Comparative Zoölogy, 1859; travelled in Brazil, 1865-66; around Cape Horn, 1871-72; died at Cambridge, Massachusetts, December 14, 1873.

AGASSIZ, ALEXANDER. Born at Neuchâtel, Switzerland, December 17, 1835; came to United States, 1849; graduated at Harvard, 1855; developed Lake Superior copper mines, 1865-69; curator of Cambridge Museum of Comparative Zoölogy, 1874-85; died at sea, March 29, 1910.

GRAY, ASA. Born at Paris, Oneida County, New York, November 18, 1810; professor of natural history at Harvard, 1842-88; died at Cambridge, Massachusetts, January 30, 1888.

TORREY, JOHN. Born at New York City, August 15, 1796; professor at Princeton and in College of Physicians and Surgeons, New York; State Geologist of New York; United States assayer; died at New York, March 10, 1873.

DRAPER, JOHN WILLIAM. Born at St. Helena, near Liverpool, England, May 5, 1811; came to America, 1832; professor of chemistry University of New York, 1839; president of the Medical College, 1850-73; died at Hastings-on-the-Hudson, New York, January 4, 1882.

RUTHERFORD, LEWIS MORRIS. Born at Morrisania, New York, November 25, 1816; graduated at Williams College, 1834; admitted to bar, 1839;

abandoned law to devote himself to study of physics, 1849; died at Tranquillity, New Jersey, May 30, 1892.

YOUNG, CHARLES AUGUSTUS. Born at Hanover, New Hampshire, December 15, 1834; graduated at Dartmouth, 1858; professor of astronomy at Princeton, 1877-1905; died at Hanover, New Hampshire, January 4, 1908.

LANGLEY, SAMUEL PIERPONT. Born at Roxbury, Boston, August 22, 1834; secretary Smithsonian Institution, 1887-1908.

NEWCOMB, SIMON. Born at Wallace, Nova Scotia, March 12, 1835; came to United States, 1853; graduated Lawrence Scientific School, 1858; professor of Mathematics, U. S. navy, 1861; director Nautical Almanac office, 1877-97; professor mathematics and astronomy Johns Hopkins University, 1884-94; died at Washington, July 11, 1909.

PICKERING, EDWARD CHARLES. Born at Boston, July 19, 1846; graduated Lawrence Scientific School, 1865; professor of astronomy and director of Harvard Observatory since 1877.

MARSH, OTHNIEL CHARLES. Born at Lockport, New York, October 29, 1831; professor paleontology Yale University, 1866, to death at New Haven, March 18, 1899.

COPE, EDWARD DRINKER. Born at Philadelphia, July 28, 1840; professor of natural sciences, Haverford College, 1864-67; paleontologist to United States Geological Survey, 1868 to death at Philadelphia, April 12, 1897.

SILLIMAN, BENJAMIN. Born at North Stratford, Connecticut, August 8, 1779; graduated at Yale, 1796; tutor there, 1799, and professor, 1802; professor emeritus, 1853; died at New Haven, Connecticut, November 24, 1864.

DANA, JAMES DWIGHT. Born at Utica, New York, February 12, 1813; graduated at Yale, 1833; assistant to Professor Silliman, 1836-38; professor of geology and natural history, 1850-64; died at New Haven, April 14, 1895.

NEWBERRY, JOHN STRONG. Born at Windsor, Connecticut, December 22, 1822; professor of geology at school of mines, Columbia College, 1866-90;

state geologist of Ohio, 1869; died at New Haven, Connecticut, December 7, 1892.

WHITNEY, JOSIAH DWIGHT. Born at Northampton, Massachusetts, November 23, 1819; graduated at Yale, 1839; geologist with New Hampshire survey, 1840-42; Lake Superior, 1847-49; state chemist of Iowa, 1855; state geologist of California, 1860-74; professor of geology at Harvard, 1865 to death at Lake Sunapee, New Hampshire, August 18, 1896.

HITCHCOCK, EDWARD. Born at Deerfield, Massachusetts, May 24, 1793; professor of chemistry, Amherst College, 1825; president of the college, 1845-54; died at Amherst, Massachusetts, February 27, 1864.

MOTT, VALENTINE. Born at Glen Cove, Long Island, August 20, 1785; graduated Columbia College, 1806; professor of surgery at Columbia, 1810-35; died at New York City, April 26, 1865.

LONG, CRAWFORD W. Born at Danielsville, Georgia, November 1, 1815; graduated medical department University of Pennsylvania, 1839; died at Athens, Georgia, June 16, 1878.

MORTON, WILLIAM THOMAS GREEN. Born at Charlton, Massachusetts, August 19, 1819; practised dentistry at Boston, 1841-58; discovered anæsthetic properties of ether, 1864; died in New York City, July 15, 1868.

HENRY, JOSEPH. Born at Albany, New York, December 17, 1797; professor of natural philosophy at Princeton, 1832-46; first secretary of Smithsonian Institution, 1846; died at Washington, May 13, 1878.

GUYOT, ARNOLD HENRY. Born near Neuchâtel, Switzerland, September 28, 1807; came to America, 1847; professor of physical geography and geology at Princeton, 1855; died at Princeton, February 8, 1884.

LE CONTE, JOHN. Born in Liberty County, Georgia, December 4, 1818; professor of physics University of California, 1869, to death at Berkeley, California, April 29, 1891.

LE CONTE, JOSEPH. Born in Liberty County, Georgia, February 26, 1823; professor of geology, University of California, 1869; died in Yosemite Valley, California, July 6, 1901.

LE CONTE, JOHN LAWRENCE. Born at New York City, May 13, 1825; surgeon of volunteers during Civil War, and chief clerk of mint at Philadelphia from 1878 until his death there, November 15, 1883.

SHALER, NATHANIEL SOUTHGATE. Born at Newport, Kentucky, February 22, 1841; graduated Lawrence Scientific School, 1862; professor paleontology at Harvard, 1868-87; professor of geology, 1887, to death, April 11, 1906.

MANN, HORACE. Born at Franklin, Massachusetts, May 7, 1796; admitted to the bar, 1823; secretary of Massachusetts Board of Education, 1837-48; member of Congress, 1848-53; president of Antioch College, 1852-59; died at Yellow Springs, Ohio, August 2, 1859.

QUINCY, JOSIAH. Born at Boston, February 4, 1772; member of Congress, 1805-13; mayor of Boston, 1823-28; president of Harvard, 1829-45; died at Quincy, Massachusetts, July 1, 1864.

ELIOT, CHARLES WILLIAM. Born at Boston, March 20, 1834; graduated from Harvard, 1853; taught mathematics and chemistry in Lawrence Scientific School, 1858-69; president of Harvard, 1869-1909.

DWIGHT, TIMOTHY. Born at Northampton, Massachusetts, May 14, 1752; graduated from Yale, 1769; president of Yale, 1795-1817; died at New Haven, Connecticut, January 11, 1817.

PORTER, NOAH. Born at Farmington, Connecticut, December 14, 1811; graduated at Yale, 1831; tutor at Yale, 1833-35; pastor of Congregational churches at New Milford, Connecticut, and Springfield, Massachusetts, 1836-46; professor of metaphysics at Yale, 1846-71; president of Yale, 1871-86; died at New Haven, March 4, 1892.

DWIGHT, TIMOTHY. Born at Norwich, Connecticut, November 16, 1828; graduated at Yale, 1849; studied divinity, 1851-55; professor of sacred literature, 1858; president of Yale, 1886-98.

EDWARDS, JONATHAN. Born at East Windsor, Connecticut, October 5, 1703; pastor of Congregational Church, Northampton, Massachusetts, 1727-50; missionary to the Indians, 1751-58; president of Princeton College, 1758; died at Princeton, March 22, 1758.

WITHERSPOON, JOHN. Born in Haddingtonshire, Scotland, February 5, 1722; president of Princeton, 1768; delegate to Continental Congress, 1774-75; died near Princeton, September 15, 1794.

McCOSH, JAMES. Born at Carskeoch, Ayrshire, Scotland, April 1, 1811; president of Princeton, 1868-88; died at Princeton, November 16, 1894.

ADDAMS, JANE. Born at Cedarville, Illinois, 1860; graduated Rockford College, 1881; opened Hull House, 1889.

CHAPTER VIII

PHILANTHROPISTS AND REFORMERS

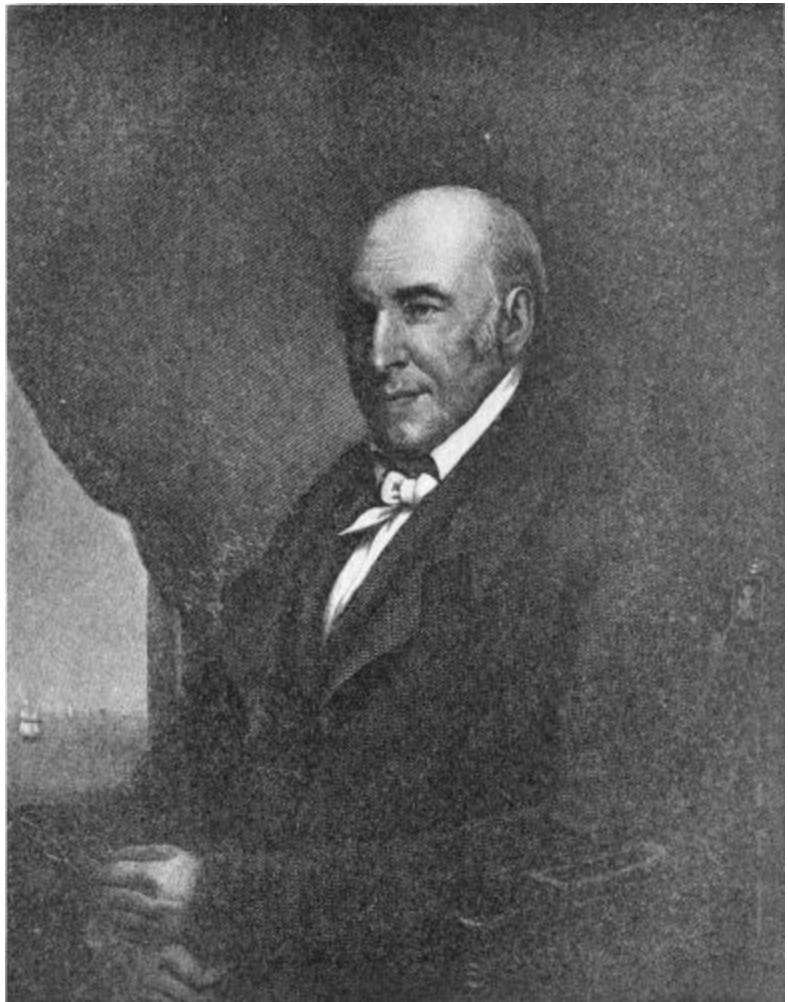
This has been a country celebrated for its great fortunes, and the makers of some of those fortunes will be considered in the chapter dealing with "men of affairs"; but many who have been grouped under that heading might well have been included under this, since, for the most part, the richest men have been the freest in their benefactions. It is worth noting that the recorded public gifts in this country during 1909 amounted to \$135,000,000. The giving of money is, of course, only one kind of benefaction, and not the highest kind, which is the giving of self; but the good which these gifts have rendered possible is beyond calculation.

This kind of philanthropy is no new thing in the United States. It is almost as old as the country itself. Indeed, few of the older institutions of learning but had their origin in some such gift. One of the earliest of such philanthropists was Stephen Girard, whose life-story is unusually interesting and inspiring. The son of a sailor, and with little opportunity for gaining an education, he shipped as cabin-boy, while still a mere child, and after some years of rough knocking around, rose to the position of mate, and finally to a part ownership in the vessel. In 1769, at the age of nineteen, he established himself in the ship business in Philadelphia, but the opening of the Revolution put an end to that business. Not until the close of the war was he able to re-embark in it. The foundation of his fortune was soon laid by his integrity and enterprise, but it was largely augmented in a most peculiar manner.

Two of his vessels happened to be in one of the ports of Hayti, when a slave insurrection broke out there, and a number of the planters hastily removed their treasure to his vessels for safe-keeping. That night, the insurrection reached its height, and the planters, together with their families, were massacred. Heirs to a portion of the treasure were discovered by Mr. Girard,

but he found himself possessed of about \$50,000 to which no heirs could be traced.

With remarkable foresight, Mr. Girard invested largely in the shares of the old Bank of the United States, and in 1812, purchased its building and succeeded to much of its business. He was the financial mainstay of the government during the second war with England—in fact, it was he who made the financing of the war possible. And yet he was, to all outward appearances, a singularly repulsive and hard-fisted old miser. In early youth, an



GIRARD

unfortunate accident had caused the loss of one eye, and his other gradually failed him until he was quite blind; he was also partially deaf, and was sour, crabbed and unapproachable. In small matters he was a miser, ready to avoid paying a just claim if he could in any way do so, living in a miserable fashion and refusing charity to every one, no matter how deserving. He was forbidding in appearance, and drove daily to and from his farm outside of Philadelphia in a shabby old carriage drawn by a single horse. No visitor was ever welcomed at that farm, where its owner dragged out a penurious existence.

Yet in public matters no one could have been more open-handed, and when, after his death in 1831, his will was opened, it created a shock of surprise,

for practically his whole fortune of \$9,000,000 had been bequeathed for charitable purposes. Large sums were given to provide fuel for the poor in winter, for distressed ship-masters, for the blind, the deaf and dumb, and for the public schools. Half a million was given Philadelphia for the improvement of her streets and public buildings; but his principal bequest was one of \$2,000,000, besides real estate, and the residue of his property, for the establishment at Philadelphia of a college for orphans. In 1848, Girard College was opened, and has since then continued its great work, educating as many orphans as the endowment can support. So Girard atoned after his death, for the mistakes of his life.

Almost equally singular was the life of the founder of that splendid government enterprise, the Smithsonian Institution—perhaps the most important scientific center in the world. James Smithson was in no sense an American. Indeed, so far as known, he never even visited the United States, and yet no account of American philanthropy would be complete without him. He was born in France in 1765, and was the illegitimate son of Hugh Smithson, afterwards Duke of Northumberland. He went by his mother's name for the first forty years of his life, being known as James Macie, until, in 1802, he assumed his father's name.

Born under this shadow, the boy soon developed unusual qualities, graduated from Oxford, with high honors in chemistry and mineralogy, and added greatly to his reputation by a series of scientific papers of great importance. A large portion of his life was passed in Europe, where he associated with the greatest scientists of the day, honored by all of them. He died at Genoa at the age of sixty-four, and, when his will was opened, it was seen how the circumstances of his birth had weighed upon him. For, "in order that his name might live in the memory of man when the titles of the Northumberlands are extinct and forgotten," he bequeathed his whole fortune "to the United States of America, to found at Washington, under the name of the Smithsonian Institution, an establishment for the increase and diffusion of knowledge among men." After a suit in chancery, the bequest was paid over to the United States government, amounting to over half a million dollars. In 1846, the Smithsonian Institution was formally established, its first secretary being Joseph Henry, of whose great work there we have already spoken. It has increased in scope and usefulness year by year, and stands to-day without a counterpart in any country.

Peter Cooper also left a portion of his wealth for "the diffusion of knowledge among men," but a different sort of knowledge—the knowledge that would help a man or woman to earn a living. His own career had shown him how necessary such knowledge is. His father was a hatter by trade, and the boy's earliest recollection was of his being employed to pull hair out of rabbit-skins, his head just reaching above the table. But the hat business was unprofitable, and the elder Cooper tried a number of businesses, brewing, brick-making, what not, the boy being required to take part in each of them, so that he had no time for schooling, and had to pick up such odds and ends of knowledge as he could. Finally, in 1808, at the age of seventeen, he was apprenticed to a carriage-maker, and remained with him until he was of age.

After that, the young man himself tried various occupations without great success, until the establishment of a glue factory began to bring him large returns. By the beginning of 1828, he was able to purchase three thousand acres of land within the city of Baltimore and to establish the Canton iron-works, which was the first of his great enterprises tending toward the development of the iron industry in the United States. Other plants were built or purchased, rolling mills and blast furnaces established, and a great impetus given to this branch of manufacture. He practically financed the Atlantic Cable Company, in the face of ridicule, and made the cable possible, and he saved the Baltimore & Ohio Railroad from bankruptcy by designing and building a locomotive—the first ever built in this country—especially adapted to the uneven country over which the track was laid.

The fortune thus acquired he devoted to a well-considered and practical plan of philanthropy. His career had shown him the great value of a trade to any man or woman. The schools taught every kind of knowledge except that which would enable a man to earn a living with his hands, which seemed to him the most important of all. He determined to do what he could remedy this defect, and in 1854, secured a block of land in New York City, at the junction of Third and Fourth Avenues, where, shortly afterwards, the cornerstone was laid of "The Cooper Union for the Advancement of Science and Art." It was completed five years later, and handed over to six trustees; a scheme of education was devised and special emphasis was laid upon "instruction in branches of knowledge by which men and women earn their daily bread; in laws of health and improvement of the sanitary

condition of families as well as individuals; in social and political science, whereby communities and nations advance in virtue, wealth and power; and finally in matters which affect the eye, the ear, and the imagination, and furnish a basis for recreation to the working classes." Free courses of lectures were established, a free reading room, and free instruction was given in various branches of the useful arts. From that day to this, Cooper Union has been an ever-growing force for progress in the life of the great city; it has been a pioneer in the work of industrial education, which has, of recent years, reached such great proportions.

Peter Cooper lived to see the institution which he had founded realize at least some of his hopes for it. He himself lived a most active life, taking a prominent part in many movements looking to the reform of national or civic abuses. In 1876, he was nominated by the national independent party as their candidate for president and received nearly a hundred thousand votes. Since his death, the institution which he founded has grown steadily in importance; other bequests have been added to his, and Cooper Union has come to stand, in a way, for civic righteousness.

The year 1795 saw the birth of two children who were destined to do a great work for their country—George Peabody and Johns Hopkins. Both were the sons of poor parents, with little opportunity for achieving the sort of learning which is taught in schools; but both, by hard experience with the world, gained another sort of learning which is often of more practical value. At the age of eleven, George Peabody was forced to begin to earn his own living, and a place was found for him in a grocery store. His habits were good, he did his work well, and finally, at the age of nineteen, was offered a partnership by another merchant, who had noticed and admired his energy and enthusiasm. The business increased, branch houses were established, and at the age of thirty-five, George Peabody found himself at the head of a great business, his elder partner having retired. He decided to make London his place of residence, and became a sort of guardian angel for Americans visiting the great English capital. He had never married, and it seemed almost as if the whole world were his family. His constant thought was of how he could elevate humanity, and he was not long in putting some of his plans into effect.

In 1852, his native town of Danvers, Massachusetts, celebrated her centennial, and her most distinguished citizen was, of course, invited to be present. He was too busy to attend, but sent a sealed envelope to be opened on the day of the celebration. The seal was broken at the dinner with which the celebration closed, and the envelope was found to contain two slips of paper. On one was written this toast, "Education—a debt due from present to future generations." The other was a check for twenty thousand dollars, afterwards increased to two hundred and fifty thousand, for the purpose of founding an Institute, with a free library and free course of lectures. Four years later, the Peabody Institute was dedicated, its founder being in attendance. Soon afterwards, he decided to build a similar Institute at Baltimore, only on a more elaborate scale, as befitting the greater city, and gave a million dollars for the purpose. It was opened in 1869, twenty thousand school children gathering to meet the donor and forming a guard of honor for him.

Two other great gifts marked his life—the sum of three million dollars for the erection of model tenements for the London poor, and a like sum for the education of the American negro. When, in 1869 the end came in London, a great funeral was held at Westminster Abbey, and the Queen of England sent her noblest man-of-war to bear in state across the Atlantic the body of "her friend," the poor boy of Danvers.

It is a strange coincidence that Baltimore, which had profited so greatly from George Peabody's philanthropy, should also be the object of that of Johns Hopkins. The latter was of Quaker stock, was raised on a farm, and at the age of seventeen became a clerk in his uncle's grocery store at Baltimore. He soon accumulated enough capital to go into business for himself, first as a grocer, then as a banker, and finally as one of the backers of the Baltimore & Ohio Railway. In 1873, he gave property valued at four and a half millions to found in the city of Baltimore a hospital, which, by its charter, is free to all, regardless of race or color; and three and a half millions for the endowment of Johns Hopkins University, which, opened in 1876, has grown to be one of the most famous schools of law, medicine and science in the country.

Another Quaker, Ezra Cornell, is also associated with the name of a great university. Reared among the hills of western New York, helping his father

on his farm and in his little pottery, the boy soon developed considerable mechanical genius, and at the age of seventeen, with the help of only a younger brother, he built a new home for the family, a two-story frame dwelling, the largest and best in the neighborhood. He soon struck out into the world, engaged in businesses of various kinds with varying success, but it was not until he was thirty-six years old that he found his vocation.

It was at that time he became associated with S. F. B. Morse, who engaged him to superintend the erection of the first line of telegraph between Washington and Baltimore. Thereafter he devoted himself entirely to the development of the new invention; succeeded, after many rebuffs and disappointments, in organizing a company to erect a line from New York to Washington, and superintended its construction. It was the first of many, afterwards consolidated into the Western Union Telegraph Company, which, for many years, held a monopoly of the telegraph business of the country, and which made Ezra Cornell a millionaire. He himself was well advanced in years, and finally retired from active life, buying a great estate near Ithaca, New York, where he lived quietly, devising a method for the best disposition of his great fortune.

He at last decided to found an institution "where *any* person can find instruction in *any* study." Work was begun at once, and in 1868, Cornell College was formally opened, over four hundred students entering the first year. The founder's gifts to this institution aggregated over three millions. Many other bequests followed, which have made Cornell one of the most liberally-endowed colleges in the country. Froude, the great English historian, visited it on one occasion, and afterwards said:

"There is something I admire even more than the university, and that is the quiet, unpretending man by whom it was founded. We have had such men in old times, and there are men in England who make great fortunes and who make claim to great munificence; but who manifest their greatness in buying great estates and building castles for the founding of peerages to be handed down from father to son. Mr. Cornell has sought for immortality, and the perpetuity of his name among the people of a free nation. There stands his great university, built upon a rock, to endure while the American nation endures."

The next great benefaction we have to record is, in some respects, unique. John Fox Slater was born in Slatersville, Rhode Island, in 1815. He was the son of Samuel Slater, proprietor of the greatest cotton-mills in New England, and he naturally succeeded to the business upon his father's death. The business prospered, receiving a great impetus from the invention of the cotton-gin, and Slater's wealth increased rapidly.

He had, on more than one occasion, visited the south and seen the negroes at work in the cotton fields. As time went on, the idea grew in his mind that he should do something for these poor laborers to whom, indirectly, his own fortune was due, and in 1882, he set aside the sum of one million dollars for the purpose of "uplifting the lately emancipated population of the Southern States, and their posterity." For this gift he received the thanks of Congress. No part of the gift is spent for grounds or buildings, but the whole income is spent in assisting negroes in industrial education and in preparing them to be the teachers of their own race. By the extraordinary ability of the fund's treasurer, it has been increased to a million and a half, although half a million has been expended along the lines contemplated by the donor. This, with the Peabody fund, comprises a powerful agency in working out the difficult problem of negro education.

The fortunes of such men as Peabody and Cornell and Hopkins and Peter Cooper seem small enough to-day when compared with the gigantic aggregations of money which a few men have succeeded in piling up. Not all of them, by any means, devote their wealth to philanthropy. Here, as in England, there are men concerned only with the idea of building up a family and a great estate; but there are a few who have labored as faithfully to use their wealth wisely as they did to accumulate it.

First of them is Leland Stanford, born in the valley of the Mohawk, studying law, and moving to Wisconsin to practise it, but losing his law library and all his property by fire, and finally joining the rush to the newly-discovered California gold-fields, where he arrived in 1852, being at that time twenty-eight years old. After some experience in the mines, he decided that there were surer ways of getting gold than digging for it, and set up a

mercantile business in San Francisco, which grew rapidly in importance and proved the foundation of a vast fortune. He was the first president of the Central Pacific Railroad, and was in charge of its construction over the mountains, driving the last spike at Promontory Point, Utah, on the tenth of May, 1869. He was prominent in the politics of state and nation, being elected to the United States Senate in 1885.

It is not by his public life, however, that he will be remembered, for he did nothing there that was in any way memorable, but by his gift of twenty million dollars to found a great university at Palo Alto, California, in memory of his only son. On May 14, 1887, the cornerstone of this great institution was laid, and the university was formally opened in 1891. The idea of its founder was that it should teach not only the studies usually taught in college, but also other practical branches of education, such as telegraphy, type-setting, type-writing, book-keeping, and farming. This it has done, and so rapid has been its growth, that it now has over seventeen hundred students enrolled.

After Senator Stanford's death in 1893, the university was further endowed by his widow, Jane Lathrop Stanford, so that the present productive funds of the university, after all of the buildings have been paid for, amount to nearly twenty-five million dollars.

The second of the great givers of recent years is John Davison Rockefeller, whose name is synonymous with the greatest natural monopoly of modern times, the Standard Oil Company. His rise from clerk in a grocery store to one of the greatest capitalists in the history of the world is an interesting one, as well as an important one in the commercial history of America. Born at Richford, New York, in 1839, his parents moved to Cleveland, Ohio, when he was a boy of fourteen, and such education as he had was secured in the Cleveland public schools. He soon left school for business, getting employment first as clerk in a commission house, and at nineteen being junior partner in the firm of Clark & Rockefeller, commission merchants.

At that time the petroleum fields of Pennsylvania were just beginning to be developed, and young Rockefeller's attention was soon attracted to them. He seems to have been one of the first to realize the vast possibilities of the

oil business, and in 1865, he and his brother William built at Cleveland a refinery which they called the Standard Oil Works. They had little money, but unlimited nerve, and very soon began the work of consolidation, which culminated in the formation of the Standard Oil Trust in 1882. They were able to kill competition largely by securing from the railroads lower shipping rates than any competitor, in some cases going so far as to get a rebate on all oil shipped by competitors. That is, if a railroad charged the Standard Oil Company one dollar to carry its oil between two points and charged a competitor a dollar and a quarter for the same service, that extra quarter went, not into the coffers of the railroad, but into the coffers of the Standard Oil Company. Such methods of business have since been made illegal, and the Standard is compelled to do business on the same basis as its competitors, but its vast resources and occupancy of the field give it an advantage which nothing can counteract.

The operations of the Standard Oil Company naturally piled up a great fortune for John D. Rockefeller—how great cannot even be estimated. Not until comparatively recent years, did he turn his attention from making money to spending it, but when he did, it was in a royal fashion. Ten million dollars were given to the University of Chicago, which opened its doors in 1892, and now has an enrollment of over five thousand students; ten million more were given to the General Education Board, organized in 1903, for the purpose of promoting education in the United States, without distinction of race, sex, or creed, and especially to promote and systematize various forms of educational beneficence; a million was given to Yale; the great Rockefeller Institute for Medical Research was founded at New York and liberally endowed; and Mr. Rockefeller's total benefactions probably exceed a total of thirty millions. This will soon be greatly increased, for he has just asked Congress to charter an institution to be known as the Rockefeller Foundation, which he will endow on an enormous scale to carry out various plans of charity, through centuries to come.

He seems recently to have experienced a change of heart, too, toward the public. During his early years, he gained a reputation for coldness and reserve, which made him probably the best-hated man in the United States. Then, suddenly, he changed about. Instead of refusing himself to reporters, he welcomed them; he seemed glad to talk, anxious to show the public that he was by no means such a monster as he was painted; and he has even,

quite recently, written his life story and given it to a great magazine for publication. Seldom before has any public man shown such a sudden and complete change of heart. He still remains, in a sense, an enigma, for it seems possible that the smiling face he has lately turned to the world conceals the real man more effectively than the frowning countenance he wore in former years.

As the dramatist saves his finest effect for the fall of the curtain, so we have saved for the last the most remarkable giver in history—Andrew Carnegie, whose total benefactions amount to at least one hundred millions of dollars. A sum so stupendous would bankrupt many a nation, yet Mr. Carnegie is so far from bankrupt that his gifts show no sign of diminution. The story of how, starting out as a poor boy, on the lowest round of the ladder, he acquired this immense fortune, is a striking one.

Andrew Carnegie was born in Scotland in 1835. His father was a weaver, at one time fairly well-to-do, for he owned four hand looms; but the introduction of steam ruined hand-loom weaving, and after a long struggle, ending in hardship and poverty, the looms were sold at a sacrifice and the family set sail for America. Mrs. Carnegie happened to have two sisters living at Pittsburgh, and there the family settled—by one of those curious chances of fate, the very place in all the world best suited to the development of young Andrew Carnegie's peculiar genius.

At the age of twelve years, he became a wage-earner, his first position being that of bobbin-boy in a cotton mill at Alleghany City, where his salary was \$1.20 a week. Pretty soon he was set to firing a small engine in the cellar of the mill, but he did not like this work, and finally secured a position as messenger boy in the office of the Atlantic & Ohio Telegraph Company, at Pittsburgh. One night, at the end of the month, he did not receive his pay with the rest of the boys, but was told to wait till the others had left the room. He thought that dismissal was coming, and wondered how he could ever go home and tell his father and mother! But he found that he was to be given an increase in salary, from \$11.25 to \$13.50 a month.

"I ran all the way home," said Mr. Carnegie, in telling of the incident, long afterwards. "Talk about your millionaires! All the millions I've made combined, never gave me the happiness of that rise of \$2.25 a month.

Arrived at the cottage where we lived, I handed my mother the usual \$11.25, and that night in bed told brother Tom the great secret. The next morning, Sunday, we were all sitting at the breakfast table, and I said: 'Mother, I have something else for you,' and then I gave her the \$2.25, and told her how I got it. Father and she were delighted to hear of my good fortune, but, motherlike, she said I deserved it, and then came tears of joy."

It was at the dinner given, in 1907, in his honor as "Father of the Corps," by the surviving members of the United States Military Telegraph Corps of the Civil War, that Mr. Carnegie spoke these words, and he continued as follows:

"Comrades, I was born in poverty, and would not exchange its sacred memories with the richest millionaire's son who ever breathed. What does he know about mother or father? They are mere names to him. Give me the life of the boy whose mother is nurse, seamstress, washerwoman, cook, teacher, angel and saint, all in one, and whose father is guide, exemplar, and friend. These are the boys who are born to the best fortune. Some men think that poverty is a dreadful burden, and that wealth leads to happiness. They have lived only one side; they imagine the other. I have lived both, and I know there is very little in wealth that can add to human happiness, beyond the small comforts of life. Millionaires who laugh are rare. My experience is that wealth is apt to take the smiles away."

But we are getting ahead of our story. That small increase in salary meant a good deal to the little family, whose father was working from dawn to dark in the cotton-mill, and whose mother was contributing what she could to the family earnings by binding shoes in the intervals of housework. Meantime the superintendent of the company for which the boy was working happened to meet him while visiting the Pittsburgh office, and it was discovered that both of them had been born near the same town in Scotland. The fact may have had something to do with the boy's subsequent promotion, and it is worth noting that forty years later, he was able to secure for his old employer the United States consulship to the town of their birth. But for the time being, he was busy with his work as messenger-boy. He soon learned the Morse alphabet and practised making the signals early in the morning before the operators arrived. He was soon able to send and receive messages by means of the Morse register—a steel pen which

embossed the dots and dashes of the message on a narrow strip of paper. But young Carnegie soon progressed a step beyond this, and was soon able to read the messages by sound, without need of the register. It was, of course, only a short time after that when he was regularly installed as operator.

He was not to remain long in the telegraph business, however, for Thomas A. Scott, superintendent of the Pittsburgh division of the Pennsylvania Railroad, offered him a position at a salary of \$35 a month. Carnegie promptly accepted, and on February 1, 1853, at the age of seventeen, entered the employ of the road. His promotion was rapid, and he rose to be superintendent of the Pittsburgh division before the success of his other ventures caused him to resign from the service. These ventures were, in the first place, investment in the newly-developed oil-fields of Pennsylvania, which yielded a great profit, and afterwards the establishment of a steel rolling-mill, in the development of which he found his true vocation, building up the most complete system of iron and steel industries ever controlled by an individual. Some idea of the value of the business may be gained from the fact that, when the United States Steel Corporation was organized in 1901 to take over Mr. Carnegie's interests he received for them, first mortgage bonds to the amount of three hundred million dollars.

It is this sum which he has been disposing of for years. Unlike most other philanthropists, he has not used his wealth to endow a great university, but has devoted it mainly to another branch of education, the establishment of free public libraries. He conceived the unique plan of offering a library building, completely equipped, to any community which would agree to maintain it suitably, and, by the beginning of 1909, had, under this plan, given nearly fifty-two millions of dollars for the erection of 1858 buildings, of which 1167 are in this country. Among his other great gifts was one of \$12,000,000, for the founding at Washington of an institution "which shall, in the broadest and most liberal manner, encourage investigation, research, and discovery, show the application of knowledge to the improvement of mankind, and provide such buildings, laboratories, books, and apparatus as may be needed."

The sum of ten millions was given to the great Carnegie Institute, of Pittsburgh; still another ten millions were given to Scottish universities, and

still another for the purpose of providing pensions for college professors in the United States and Canada; and finally five millions for the establishment of a fund to be used for the benefit of the dependants of those losing their lives in heroic effort to save their fellow-men, or for the heroes themselves, if injured only. What great benefaction will next be announced cannot, of course, be foretold, but that some other announcement will some day be forthcoming can scarcely be doubted, since Mr. Carnegie has announced his ambition to die poor.

Although born in Scotland and maintaining a great estate there, he is an American out-and-out. He proved his patriotism during the Civil War by serving as superintendent of military railways and government telegraph lines in the east; and has proved it more than once since by enlisting in the fight for civic betterment and good government. Thousands of benefactions stand to his credit, besides the great ones which have been mentioned above, and it is doubtful if in the history of the world there has ever been another man armed with such power and using it in such a way.

We will end here the story of American benefactions, although scarcely the half of it has been told. During the last forty years, not less than one hundred millions of dollars have been given to American colleges; nearly as much again has been given for the endowment of hospitals, sanitariums and infirmaries; vast sums have been given for other educational or charitable purposes, so that, of the great fortunes which have been accumulated in this country, at least three hundred millions have been returned, in some form or other, to the people. And the end is not yet. Scientific philanthropy is as yet in its infancy. Just the other day, Mrs. Russell Sage set apart the sum of ten million dollars for a fund whose chief and almost sole purpose it is to obtain accurate information concerning social and economic conditions—in other words, to furnish the data upon which the scientific philanthropy of the future will be based. The disposition toward such employment of great fortunes, and away from the selfish piling-up of wealth is one of the most cheering and promising developments of the new century in this great land of ours; the kings of finance are coming to realize that, after all, wealth is useless unless it is used for good, and the next half century will no doubt witness the establishment of philanthropic enterprises on a scale hitherto unknown to history.

We have already said that the highest form of philanthropy is not the giving of money, but the giving of self, and we shall close this chapter with a brief consideration of the careers of a few of the many men and women who, in the course of American history, have devoted their lives to the betterment of humanity, either as ministers of the gospel or as laborers for some great reform.

Among ministers, no name has been more widely known than that of Beecher—first, Lyman Beecher, and afterwards his brilliant son, Henry Ward Beecher.

Lyman Beecher was born in New Haven, Connecticut, in 1775, the son of a blacksmith, and his youth was spent between blacksmithing



BEECHER

and farming. His love of books soon manifested itself, however, and means were found to prepare him for Yale, where he graduated at the age of twenty-two. A further year of study enabled him to enter the ministry. For sixteen years, he was pastor of the Congregational church at Litchfield, Connecticut, and soon took rank as the leading clergyman of his denomination. His eloquence, zeal and courage won a wide reputation, and in 1832, he was offered the presidency of the newly-organized Lane seminary, at Cincinnati. This place he held for twenty years, and his name was continued as president in the seminary catalogue, until his death.

Soon after he assumed this position, the slavery question began to assume the acute phase which ended in the Civil War. Mr. Beecher was, of course,

an Abolitionist, and for a time lived in a turmoil, for many of the seminary students were from the south, while Cincinnati itself was so near the borderline that there was a great pro-slavery sentiment there. But during Mr. Beecher's absence, his trustees tried to allay excitement and, in a way, carry water on both shoulders, by forbidding all further discussion of slavery in the seminary, and succeeded in nearly wrecking the institution, for the students withdrew in a body, and while a few were persuaded to return, the great majority refused to do so and laid the foundation of Oberlin College. For seventeen years, Mr. Beecher labored to restore the seminary's prosperity, but finally abandoned the task in despair. He resigned the presidency in 1852, intending to devote his remaining years to the revision and publication of his works, but a paralytic stroke put an end to his active career.

Mr. Beecher's vigor of mind and body were imparted in a remarkable degree to his children, of whom he had thirteen. Of Harriet Beecher Stowe we have already spoken, but by far the most famous of them was Henry Ward Beecher. Born in 1813, and renouncing an early desire for a sea-faring life in favor of the ministry, he secured his first charge in 1837, and ten years later entered upon the pastorate of Plymouth church, in Brooklyn, where his chief fame was won. The church, one of the largest in the country, soon became inadequate to hold the crowds which flocked to hear his brilliant preaching. As a lecturer and platform orator he soon came to be in such demand that he was at last compelled to decline all such engagements. He took an active part in politics, holding that Christianity was not a series of dogmas, but a rule of everyday life, and did not hesitate to attack the abuses of the day from the pulpit. He was as facile with the pen as with the tongue, and his publications were many and important. All in all, he was one of the most influential and picturesque figures that has ever occupied an American pulpit.

Lyman Beecher was at all times a doughty antagonist, and in 1826 he had been called to Boston to take up the cudgels against the so-called Unitarian movement which had developed there, under the leadership of William Ellery Channing. For six years and a half, he wielded the cudgels of controversy, but with no great effect, for Channing was a foeman in every sense his equal. Channing had graduated at Harvard in 1798, a small man of an almost feminine sensibility, with a singular capacity for winning devoted

attachment from all with whom he came in contact. For two years, he served as tutor in a family at Richmond, Virginia, where he acquired an abhorrence of slavery that lasted through life. Upon his return north, he began the study of theology at Cambridge, and in 1803, became pastor of a church in Boston, where he soon attracted attention by sermons of a rare "fervor, solemnity, and beauty." He was from the first identified with the movement of thought, which came to be known as Unitarian, and gave to the body so-called a consciousness of its position and a clear statement of its convictions with his sermon delivered at Baltimore, in 1819, on the occasion of the ordination of Jared Sparks. For the fifteen years succeeding, Channing was best known to the public as the leader of the Unitarian movement, and his sermons delivered during that period constitute the best body of practical divinity which that movement has produced. In later years, he was identified with many philanthropical and reform movements, and was one of the pillars of the anti-slavery cause, though never adopting the extreme opinions of the abolitionists. Of his rare quality and power as a pulpit orator many traditions remain, and his death at the age of sixty-two removed a great power for righteousness.

Even to give a list of the men and women who have sacrificed their lives in the attempt to carry the gospel of Christianity to heathen nations is beyond the limits of a book like this, but at least mention can be made of two of the earliest, Adoniram Judson and his wife, whose experiences form one of the most thrilling chapters in missionary history.

Adoniram Judson was born in Malden, Massachusetts, in 1788, and after graduating at Brown University, and taking a special course at Andover Theological seminary, became deeply interested in foreign missions, and in 1810, determined to go to Burmah. Securing the support of the London Missionary Society, he sailed for Asia on the nineteenth of February, 1812. Two weeks before, he had married Ann Haseltine, who consented to share his work, and who sailed with him. On that long voyage, they had ample time to discuss and consider the various dogmas of their faith, and they became convinced that the baptism of the New Testament was immersion, and in accordance with this view, both of them were baptized by immersion upon reaching Calcutta. But this change of faith cut them off from the body which had sent them to India, and it was not until 1814 that the Baptists of America took the two missionaries under their care.

Meanwhile, Dr. Judson mastered the Burmese language and began his public preaching. Before long, he baptized his first convert, and pushed forward the work with renewed zeal, translating the gospels into Burmese, publishing tracts in that language, and undertaking the most perilous journeys. The Burmese government had never been friendly, and in 1824, seized the missionaries and threw them into prison. They were confined in the "death hole," reeking with foul air, without light, and were loaded with fetters. Just enough food was given them to keep them alive, and at last, stripped almost naked, they were driven like cattle under the burning sun, to another prison, where it was intended to burn them alive. They were saved by the intercession of Sir Archibald Campbell, but Mrs. Judson's health had been wrecked by the terrible experience. She never recovered, dying two years later. Undaunted by difficulties, Dr. Judson continued his work, completing his translation of the Bible, travelling over India, compiling a Burmese grammar and dictionary, but his labors at last undermined even his constitution and he died at sea in 1850, while on his way to the Isle of France.

Turn we now to Lucretia Mott, one of the most extraordinary women who ever lived in America. Born in Nantucket in 1793, the daughter of a sea-captain named Thomas Coffin, she was raised in the strict Quaker faith, to which her parents belonged. She began teaching while still a girl, and at the age of eighteen, married a fellow teacher, James Mott. It was not long after that, that she developed the "gift" of speaking at the Quaker meetings, simply, earnestly and eloquently. The Quakers had always opposed slavery and Lucretia Mott was soon working heart and soul against it. When the American Anti-Slavery Society was organized in 1833, she was one of four women who joined it, and she proceeded immediately to organize the Female Anti-Slavery Society, the first organization of women in America working for a political purpose. Years of abuse followed, for in those days anti-slavery lecturers were tarred and feathered, their homes burned, and many other indignities heaped upon them. Throughout all this, Mrs. Mott never lost her serenity, and never suffered bodily injury. On one occasion, the annual meeting of the Anti-Slavery Society, in New York, was broken up by a mob, and some of the speakers were roughly handled. Perceiving that some of the women were badly frightened, Mrs. Mott asked her escort to look after them.

"But who will take care of you?" he asked.

"This man will," she said, and smilingly laid her hand upon the arm of one of the leaders of the mob. "He will see me safe through."

The rioter stared down at her for a moment, his conflicting thoughts betraying themselves upon his countenance, then his better nature triumphed and he led her respectfully to a place of safety.

She seems to have possessed the power of charming any audience, and carried her anti-slavery campaign even into Kentucky, where she commanded respectful attention. She was one of the first to take up the question of woman suffrage, and in 1848, with Elizabeth Cady Stanton and a few others, called the first Woman's Suffrage Convention ever held in this country. For fifty years she continued her public work, until she grew to be one of the best known and best loved women in the country. She lived to see the slave freed, and when she died, a great concourse followed her body silently to the grave. As they stood there with bowed heads, a low voice asked, "Will no one say anything?"

"Who can speak?" another voice responded, "The preacher is dead."

In this day of pitying and enlightened treatment of the insane, it is difficult to realize the barbarities which they were called upon to endure a century ago. They were regarded almost as wild beasts, were kept chained in foul and loathsome places, fed with mouldy bread, filthy water, and allowed to die the most miserable death. For everyone used to believe that insanity was a mark of God's displeasure, and the outcast from His heart became equally an outcast from the hearts of men. The insane were regarded with fear and loathing, and it was not until the beginning of the nineteenth century that such men as Dr. Channing began to insist on the presence in human nature, even in its most degraded condition, of grains of good.

It was from Dr. Channing that Dorothea Lynde Dix drank in this theory with passionate faith, and proceeded at once to convert it into action. She was governess of Dr. Channing's children, and had long been interested in

bettering the condition of convicts; but now her attention was turned to the insane and she proceeded at once to master the whole question of insanity, its origin, its development, and its treatment, so far as it was then known. Enlisting the aid of a number of broad-minded men, among them Charles Sumner, she went to work. In one prison, she found two insane women, each confined in a small cage of planks; others were locked in closets, cellars, and stalls; some of them were naked, some were chained, some were regularly beaten and scourged. With all her data at hand, she addressed a memorial to the Massachusetts legislature, setting forth, in page after page, the details of these almost incredible horrors, which she herself had witnessed.

It exploded like a bombshell, for it was a terrific arraignment of the whole state. Her statements were denounced as untrue and slanderous, but a little investigation proved their truth, and with such men behind her as Channing, Horace Mann, and Samuel G. Howe, it was soon apparent that something would be done. The obstructions and delays of politicians were swept away before a steadily rising tide of public indignation, and a large appropriation was made by the legislature to provide proper quarters and proper treatment for insane persons. So Miss Dix won her first great victory, the forerunner of similar ones in almost every state in the union; for she travelled from state to state making the same investigations she had in Massachusetts, arousing public opinion, and compelling legislature after legislature to make adequate provision for the insane. The vastness of this campaign which Miss Dix planned deliberately and which she carried through until she had visited every state east of the Rocky Mountains, gives evidence to her extraordinary character. During the Civil War, she was superintendent of hospital nurses, having the entire control of their appointment and assignment. But the care of the insane was her life work. She resumed it at the close of the war, and carried it forward until her death.

We have already referred more than once, in the course of these chapters, to the anti-slavery agitation which ended in the Civil War. During the second quarter of the nineteenth century, it was the one great political question in America, upon which men were compelled to take one side or the other.

From the first, there existed in the north a band of abolitionists—of men, in other words, who believed that the only solution of the slavery question was to put an end to that institution at once and forever. Of the persecutions which were visited on the abolitionists we have spoken when telling the story of Lucretia Mott. Social ostracism was the least of them.

Perhaps no one person in America did more to crystalize public sentiment against slavery than Lydia Maria Child. An author at the age of seventeen, and writing continuously until her death, coming early under the influence of William Lloyd Garrison, that great leader of the abolitionists, it was inevitable that she should employ her pen to assist the cause. In 1833 appeared her "Appeal for that class of Americans called Africans," the first anti-slavery work printed in America in book form, antedating Mrs. Stowe's "Uncle Tom's Cabin" by nineteen years. It attracted wide attention, enlisting the interest of such men as Dr. Channing, who walked from Boston to Roxbury to thank the author. But it was not without its penalties, for society closed its doors to Mrs. Child, many of her friends deserted her, and she was made the subject of much cruel comment. However, she became more and more interested in the anti-slavery crusade, edited the "National Anti-Slavery Standard," and wrote pamphlet after pamphlet. When John Brown was taken prisoner, she wrote him a letter of sympathy, which drew forth a courteous rebuke from Governor Wise, of Virginia, and a letter from the wife of Senator Mason, the author of the fugitive slave law, threatening her with future damnation. These letters were published and had a circulation of three hundred thousand copies. Wendell Phillips paid an eloquent tribute to her character and influence, at her funeral: "She was the kind of woman," he said, "one would choose to represent woman's entrance into broader life. Modest, womanly, sincere, solid, real, loyal, to be trusted, equal to affairs, and yet above them; a companion with the password of every science and all literature."

But however valuable the services of women like Lucretia Mott and Lydia Maria Child and Harriet Beecher Stowe were in the fight against slavery, the leader and high priest of the movement was William Lloyd Garrison. Born in Newburyport, Massachusetts, in 1805, his was an unhappy boyhood, for his father, a sea-captain of intemperate and adventurous habits, left his family, soon after the boy was born, and was never seen again. The mother, a woman of unusual strength of character, went to work

to earn a living for herself and her son, and it was to her careful training that his development was due. At fourteen years of age, he was apprenticed to a printery and served until he was of age. From the first he was remarkable for his firmness of moral principle and for an inflexible adherence to his convictions, no matter at what cost to himself.

He soon showed, too, that he was destined for something more than a printer—a man who puts in print the ideas of others—that he had ideas of his own. His apprenticeship over, he started a paper of his own, but it was too reformatory for the taste of the day, and proved a failure. The most noteworthy thing in connection with it was the publication of some poems which had been sent in anonymously, and which Garrison, recognizing their merit, discovered to be the work of John G. Whittier, then entirely unknown. He visited the poet, encouraged him to keep on writing, and laid the foundation of a friendship which was broken only by death.

Going to Boston after the failure of his paper, Garrison for a time edited the "National Philanthropist," devoted to prohibition. This paper, too, was a failure, but at Boston Garrison met a man whose influence changed the whole course of his life. His name was Benjamin Bundy. He was a Quaker, and at that time thirty-nine years of age. He was a saddler by trade, but for thirteen years had devoted his life to the anti-slavery cause, forming anti-slavery societies and editing a little monthly paper with a portentous name—"The Genius of Universal Emancipation." Bundy, whose home was in Baltimore, had journeyed to New England in the hope of interesting the clergy in the cause. In this he was bitterly disappointed, but he mightily stirred the heart of young Garrison, who soon became his ally and afterwards his partner in the conduct of the paper. His vigorous editing of it was soon a national sensation. He had seen with dismay the indifference with which the north regarded the great issue—an indifference grounded on the belief that slavery was intrenched by the constitution and that all discussion of it was a menace to the Union. He realized that this indifference could be broken only by heroic measures, and he took the ground that since slavery was wrong, every slave had a right to instant freedom, and that immediate emancipation was the duty of the master and of the state.

Baltimore was at that time one of the centres of the slave trade. There were slave-pens on the principal streets, and Garrison soon witnessed scenes which would have touched a less tender heart. In the first issue of his paper, he denounced this traffic as "domestic piracy," and named some men engaged in it, among them a vessel-owner of his own town of Newburyport. This man immediately had Garrison arrested for "gross and malicious libel," he was found guilty, fined fifty dollars and costs, and as there was no one to pay this, was thrown into prison.

Garrison took his imprisonment calmly enough, but his old friend, John G. Whittier, was deeply distressed and appealed to Henry Clay to secure the release of the "guiltless prisoner." This Clay would probably have done, but he was anticipated by another friend of Garrison's, Arthur Tappan, of New York, who sent the money to pay the fine, and the young agitator was free again, after an imprisonment of forty-nine days. He had not been idle while in prison, but had prepared a series of lectures on slavery, which he proceeded at once to deliver. Then, on the first day of January, 1831, he began in Boston the publication of a weekly paper called the "Liberator," which he continued for thirty-five years, until its fight was won and slavery was abolished.

How well that fight was waged history has shown. In his first number he announced: "I will be as harsh as truth and as uncompromising as justice. On this subject I do not wish to think, to speak, or write with moderation. No! No! Tell the man whose home is on fire to give a moderate alarm; tell the mother to gradually extricate her babe from the fire into which it has fallen; but urge me not to use moderation in a cause like the present. I am in earnest—I will not equivocate—I will not excuse—I will not retreat a single inch—and I will be heard."

And heard he was. The whole land was soon filled with excitement; the apathy of years was broken. From the south came hundreds of letters threatening him with death if he did not desist, and the state of Georgia offered a reward of \$5,000 for his apprehension. In the north, anti-slavery societies were formed everywhere, and the movement grew with great rapidity, in spite of powerful efforts to crush it. There were riots everywhere. Garrison was dragged through the streets of Boston with a rope around his body and his life was saved only by lodging him in jail; Elijah

Lovejoy was slain at Alton, Illinois, while defending his press; Marius Robinson, an anti-slavery lecturer, was tarred and feathered in Mahoning County, Ohio; in the cities of the south, mobs broke into the postoffice and made bonfires of anti-slavery papers and pamphlets found there. Quarrels and dissension in the anti-slavery ranks developed in time, but when the Civil War was over, the leaders of the Republican party united with Garrison's friends in raising for him the sum of \$30,000, and after his death the city of Boston raised a statue to his memory. Perhaps no better estimate of him has ever been made than that of John A. Andrew, war governor of Massachusetts:

"The generation which preceded ours regarded him only as a wild enthusiast, a fanatic, or a public enemy. The present generation sees in him the bold and honest reformer, the man of original, self-poised, heroic will, inspired by a vision of universal justice, made actual in the practice of nations; who, daring to attack without reserve the worst and most powerful oppression of his country and his time, has outlived the giant wrong he assailed, and has triumphed over the sophistries by which it was maintained."

Closely second to Garrison in the awakening of the public conscience to the enormities of slavery was Theodore Parker, one of the purest, most self-sacrificing and interesting of personalities. He came of good stock. His grandfather, John Parker, commanded the little company of minute-men who held the bridge at Lexington on that fateful nineteenth of April, 1775; his father a farmer, and Theodore himself the youngest of eleven children. The family was poor and the boy was brought up to hard labor, with short intervals of schooling now and then. But his thirst for knowledge seems to have been insatiable, and he read everything he could lay his hands on, even to translations of Homer and Plutarch and Rollin's "Ancient History." A century ago, a book was a far greater treasure than it is to-day, when their very number has made us in a way contemptuous of them; and the few which young Parker could secure were read and re-read and learned through and through. His memory was amazing, and at the age of twenty he walked from his home in Lexington to Cambridge, took the entrance examination for Harvard College, passed with honors, and, walking home again, told his unsuspecting father, then in bed, of his success. He could not be spared from the farm, however, nor was there any money to pay for his

maintenance at Cambridge, so he continued working on the farm, keeping up with his class by studying in the evenings and going to Cambridge only to take the examinations.

He undertook teaching after that, and gradually worked his way toward the ministry, to which he was admitted in 1837. He was soon called to Boston, to a congregation independent of sectarian bonds, and here he reached the culmination of his fame, attracting the most cultured people of the city by his breadth of knowledge, warmth of feeling and intensity of conviction. His interest in slavery began early, and by 1845, his share in the anti-slavery struggle had become engrossing. He threw himself into it heart and soul, and no one did more to awaken the conscience of the north. His speeches, letters, sermons, tracts and lectures had an immense influence; he took an active part in aiding runaway slaves to get to Canada, and his labors were incessant and prodigious. His health at last gave way, and the end came in 1860, at Florence, Italy, where he lies buried.

Parker's immense influence was due to the brain rather than to the heart. He possessed no grace of person, music of voice, or charm of manner, none of that fascination which is a part of the great orator. He was a white-hot flame which scorched and seared, an intellect pure and piercing, a self-made instrument to expose the shams of society.

Closely associated with Garrison and Parker in the fight against slavery, and in some ways more famous than either, was Wendell Phillips. The very opposite of Parker, handsome in person, cultivated in manner, with a charm of personality seldom equalled,—the two yet worked hand in hand for a common cause, the one, as it were, supplementing the other.

Wendell Phillips was the son of John Phillips, the first mayor of Boston, and was a year younger than Theodore Parker. He went the way of all well-to-do Boston youth through Harvard, graduating there in 1831, without distinguishing himself particularly, except by his skill in debate and his finished elocution. During one of the revivals of religion which followed the settlement of Dr. Lyman Beecher at Boston, he became a convert, and this marked the beginning of his interest in the great moral question of the day, slavery. It soon became overwhelming, and was given point and passion by a spectacle which he witnessed on October 21, 1835.

He had studied for the law, been admitted to the bar, and opened an office, and looking from his office window on that October day, he saw a mob break up an anti-slavery meeting on the street below, pull William Lloyd Garrison off the platform, tear his clothes from his back, throw a rope around him and drag him through the streets, ready to hang him, and prevented from doing so only by a ruse of the mayor, who got Garrison into the jail and locked him up for safety. That spectacle moved the young lawyer through and through, and from that moment he was an avowed Abolitionist.

"If clients do not come," he had said to a friend a short time before, "I will throw myself heart and soul into some good cause and devote my life to it."

Clients would have come, no doubt, but the good cause came first. His opportunity came in 1837, when Elijah Lovejoy was murdered by a mob at Alton, Illinois, for publishing an anti-slavery paper. Phillips, stirred with indignation, arranged for a public meeting at Faneuil Hall, and was of course present, but with no expectation of speaking. Dr. Channing made an impressive address, and one or two others followed, when James T. Austin, attorney-general of the state, and bitterly opposed to the anti-slavery agitation, arose. He eulogized the Alton murderers, comparing them with the patriots of the Revolution, and declared that Lovejoy had "died as the fool dieth." Some instinct led the chair to call upon Wendell Phillips to reply. He consented, and as he stepped upon the platform won instant admiration by his dignity, his self-possession, and his manly beauty.

"Mr. Chairman," he began, "when I heard the gentleman who has just spoken lay down principles which placed the rioters, incendiaries, and murderers of Alton side by side with Otis and Hancock, with Quincy and Adams, I thought those pictured lips [pointing to the portraits in the hall] would have broken into voice, to rebuke the recreant American, the slanderer of the dead. Sir, for the sentiments he has uttered on soil consecrated by the prayers of Puritans and the blood of patriots, the earth should have yawned and swallowed him up."

The effect of the whole speech was tremendous. At last the abolitionists had found a champion equal to the best, and from that hour to the end of the anti-slavery conflict, he was foremost in the fight. He accepted without

reservation the doctrines which Garrison had formulated: that slavery was under all circumstances a sin and that immediate emancipation was a fundamental right and duty. Up and down the land, obeying every call so far as his strength would permit, he travelled, lecturing against slavery, asking no pecuniary reward. He was soon a great popular favorite—the greatest, perhaps, who ever mounted a lecture platform in America,—and gained a hearing in quarters where, before, abolitionists had been hated and derided. His tact in winning over a turbulent audience was extraordinary; the strongest opponents of the anti-slavery cause felt the spell of his power, and often confessed the justice of his arguments.

When that fight was won and the negro had gained his freedom, Wendell Phillips remained the foremost critic of public men and measures in America, and year after year, he devoted his great gifts to guiding popular opinion. A champion of temperance, of the rights of labor, of the Indians, of equal suffrage, he stood forth until his death an inspiring and august figure—a man who devoted his life wholly to the welfare of his country.

One of the reforms which Wendell Phillips advocated was that of woman suffrage, but this movement has come to be particularly associated with the name of Susan B. Anthony. Like her great predecessor in that cause, Lucretia Mott, Miss Anthony was a Quaker, and the Quakers, it should be remembered, made no distinction of sex when it came to speaking in their meeting-houses. Her father was well-to-do, and she received a careful education, and in 1847, first spoke in public. The temperance movement absorbed her energies at first; then the Abolitionist cause; and finally the work of securing equal civil rights for women. During the winter of 1854, she held woman suffrage meetings in every county in New York State, and the remainder of her life was devoted to this cause.

Her most prominent co-worker was Elizabeth Cady Stanton, whose inspiration came directly from Lucretia Mott, whom she met in 1840, and with whom she joined, eight years later, in issuing a call for the first woman's suffrage convention. The convention was held at Mrs. Stanton's home at Seneca Falls, New York, and from that time forward, she devoted herself entirely to lecturing and writing upon the subject. That the cause of woman suffrage has made so little headway is certainly not because of a lack of devoted and accomplished advocates; it seems rather to be due to

the fact that it has not yet succeeded in winning over the great body of women, who have held aloof and viewed the movement with indifference, if not with suspicion.

We cannot close this consideration of the anti-slavery movement without some reference to that strange fanatic, John Brown, who headed a forlorn hope and gave up his life for an idea. It was the custom at one time to consider John Brown a saint, at the north, and a very emissary of Satan, at the south. One estimate was as untrue as the other. He was merely a misguided old man, grown a little mad, perhaps, from long brooding over one subject.

He was born at Torrington, Connecticut, in 1800, his father being a shoemaker and tanner, who, five years later, moved to Hudson, Ohio, then a mere outpost in the wilderness. He was soon expert in woodcraft, and he relates how, when he was six years old, an Indian boy gave him a yellow marble, the first he had ever seen, and which he treasured for a long time. He had little or no schooling, and a project to educate him for the ministry was cut short by an inflammation of the eyes. He grew up into a tall, handsome man, headstrong, but humane and kind, and easily moved to tears. He married young and had many children, for some of whom a tragic fate was waiting.

He soon became interested in the anti-slavery movement, and, by 1837, was so absorbed by it that he made his family take a solemn oath of active opposition to slavery. Ten years later, he unfolded to Frederick Douglass a plan for a negro insurrection in the Virginia mountains, but nothing came of it. From that time forward, the project seems to have slumbered at the back of his mind, and he grew more and more certain that the only way to end slavery was to arm the blacks and encourage them to fight for freedom. In 1854, his sons emigrated to Kansas, then in the throes of civil war over the slavery question, and their father busied himself raising money to send arms and ammunition into the troubled state. Finally, in September, 1855, he himself removed to Kansas, became the captain of a band of Free State Rangers, took part in the fight at Lawrence, and in some other affairs, and

then, proceeding to the shores of Pottawatomie creek, where several pro-slavery men lived, seized five of them and put them to death.

For this deed he never experienced any compunction; he believed that he was directed by Providence in these "executions," as he called them, and after they were over, he held divine services. His fearful deed sent a thrill of horror through the country, and Brown and his sons became marked men. Their houses were burned, and one of the sons went insane from brooding over the father's deed. Brown himself was charged with murder, treason and conspiracy, and a price put on his head, but no one attempted to arrest him. Another of his sons was soon afterwards shot and killed by pro-slavery men and Brown, hastily collecting a small force, attacked the marauders, and killed or wounded many of them, himself being injured by a spent rifle ball. The fight was known as "the battle of Osawatomie," and Brown was thereafter known as "Osawatomie" Brown.

But the fight in Kansas was about won, and Brown again took up the idea of a slave insurrection. He went to Boston to raise the necessary money, and succeeded in getting it without much trouble, though most of the people who gave it to him had only the haziest kind of an idea of what it was he proposed to do. He bought rifles and ammunition, and also had a thousand pikes made with which to arm the negroes, who, of course, would not know how to use the rifle. Then he got together a band of young men, secured a military instructor; and on July 3, 1859, he appeared at Harper's Ferry, Virginia, hired a small farm near there, and quietly assembled his men and munitions.

Harper's Ferry had been selected because there was a well-equipped arsenal there which would furnish the arms and munitions which he had been unable to buy, and would also serve as a base of operations. Brown intended to proceed to the mountains, gathering up the slaves as he went, and establish headquarters in some strong position, where he could drill his forces and prepare for a raid on the rest of the state. He believed the slaves would flock to him, and that he would soon be at the head of a great army. He tried to get Frederick Douglass to join him, but Douglass refused, and, at last, on the night of Sunday, October 16, 1859, at the head of a little band of twenty-two men, whites and negroes, he moved on the arsenal. They reached the covered bridge over the Potomac without adventure, crossed

until they were near the Virginia side, seized the solitary sentinel who challenged them, broke down the armory gate with a sledge hammer, seized the remainder of the guard, and a few citizens, who attempted to interfere, and were soon firmly in possession of not only the arsenal, but also the little town.

Meanwhile, the country round about was arming, and by noon, of Monday, Brown was so surrounded that he could not escape. Why he had not got away to the mountains in the morning, as he had intended doing, no one knows. The Virginia militia gathered, and in the early evening, a company of United States marines arrived from Washington, under command of Colonel Robert E. Lee and Lieutenant J. E. B. Stuart. They soon found out how small Brown's force was, carried the arsenal by assault, and took Brown and the survivors of his little band prisoners. Brown's two sons were dead, as were seven others of his followers, and seven more had succeeded in escaping, though two were afterwards captured.

The rest is soon told. Brown was swiftly tried and convicted of "treason and conspiring and advising with slaves and others to rebel, and of murder in the first degree," was sentenced to death, and was hanged on December 2, 1859. The affair made the South wild with rage and apprehension, for a slave insurrection was a thing to be trembled at, and Brown's execution similarly affected his friends at the North. He had once remarked, "I am worth a good deal more to hang than for any other purpose," and this was, in a sense, true, for in the words of the great marching song of the Northern armies during the war which followed, "his soul was marching on."

Another branch of philanthropy with which the name of a woman is closely identified is that of caring for the wounded and destitute in time of war or disaster, and the woman is Clara Barton. Born in Massachusetts about 1830, she started in life as a school-teacher, but in 1854 secured a position in the patent office at Washington, where she remained until the opening of the Civil War. The sight of the suffering in the Washington hospitals revealed to her her real vocation, and she determined to devote herself to the care of wounded soldiers on the battlefield. This work of mercy was one that carried with it a wide appeal, and she soon secured influential backing and support.

Her work was so effective that in 1864, she was appointed "lady in charge" of the hospitals at the front of the Army of the James, and in the following year was sent to Andersonville, Georgia, to identify and mark the graves of the Union soldiers buried there. Soon afterwards she was placed by President Lincoln in charge of the search for missing men of the Union armies—a work of the first importance, to which she devoted all her energies, and which she carried on for some years after the war closed, raising the necessary money by lectures and appeals for donations. Thousands of families at the North have reason to thank her for definite knowledge as to the fate of their loved ones.

Her health broke down under the strain, at last, and she went for a rest to Switzerland, but the outbreak of the Franco-German war, in 1870, called her again to duty, assisting the grand duchess of Baden in the preparation of military hospitals, and giving the Red Cross Society the benefit of her experience. In 1871, at the request of the German authorities, she superintended the supplying of work to the poor of Strasburg, after that city had been reduced by siege; and after the fall of Paris, she was placed in charge of the distribution of supplies to the destitute of that great city. At the close of the war, she was decorated with the golden cross of Baden and the iron cross of Germany.

Although the Red Cross societies in Europe had been established as early as 1863, and an international organization completed six years later, the society was not officially recognized by the United States until 1882. The American Association of the Red Cross was at once organized, and Miss Barton chosen its president, a position which she held without opposition for many years. Its object as stated by its constitution is "to organize a system of national relief and apply the same in mitigating suffering caused by war, pestilence, famine and other calamities." Since then, every such occasion has found the society in the forefront of relief work, and it has distributed many millions in assuaging human suffering.

Still another great reform, ridiculed at first, but now recognized as one of the most beneficent movements of the age is associated with a single name.

The reform is the protection of dumb animals, and the name is that of Henry Bergh.

Born in New York City in 1823, the son of a wealthy ship-builder and inheriting his father's fortune at the age of twenty, Henry Bergh, after spending some years in Europe, a portion of them in the diplomatic service of the United States, returned to this country, determined to devote the remainder of his life to the interests of animals.

It was a new idea which he presented to the public, met at first with indifference, then with ridicule and opposition. But as a bold worker in the streets of New York, by a relentless activity in carrying cases of ill-treatment of animals to the courts, and an eloquent advocacy of his cause on the floor of the legislature, he soon won friends and support, as every great cause is bound to do, and finally succeeded in so winning over public sentiment that, in 1866, the legislature passed the laws which he had prepared, creating the Society for the Prevention of Cruelty to Animals, with himself as president. He gave not only his time, but his property to the work, and soon had the society in a prosperous condition, with branches forming in other cities. Indeed, the idea which he fostered has spread to the whole country, and nowhere may animals be mistreated with impunity. The idea that man is responsible not only for the happiness of his fellows, but for the well-being of his beasts marks a long stride forward in ethics.

Bergh's influence, indeed, extended beyond this country. Not only did practically every state in the Union enact the laws for the protection of animals which he had procured from the state of New York, but Brazil, the Argentine Republic, and many other foreign countries did likewise. In 1874, Bergh rescued a little girl from inhuman treatment, and this led to the formation of the Society for the Prevention of Cruelty to Children, which has also done a great work.

No doubt before Bergh's time, there were many people who were pained to see either children or animals mistreated and who passed by with averted eyes. Bergh did not pass by. He made it his business, in the first place, to secure adequate laws for the punishment of cruelty, and in the second place, to provide means for the enforcement of those laws.

There are many of us to-day who are shocked at the injustice and suffering in the world, and who would welcome its regeneration. But wishing for a thing never got it. Nor does philanthropy consist merely in wishing men well. It means labor and self-sacrifice, and frequently obloquy and misunderstanding. The reward of the reformer is usually a stone and a sneer, if nothing worse. But when a man's heart is in the work, stones and sneers seem only to spur him on. They are like wind to a flame, fanning it white-hot. And it is a wonderful commentary on the essential goodness of human nature that never yet, in the history of mankind, has a real and needed reform failed, in the end, of success.

Among latter-day clergymen in America, none has achieved a wider reputation or a greater personal popularity than Phillips Brooks. Born in Boston in 1835, a graduate of Harvard, ordained to the Episcopal ministry at the age of twenty-four, and ten years later called to the rectorship of Trinity church, Boston, it was in this latter field, which he would never leave, that he showed himself to be one of the strongest personalities and noblest preachers of his age. No more striking figure ever appeared in a pulpit. Of magnificent physique, with a striking and massive head and handsome countenance, breathing the very spirit of youth, in spite of his grey hair, he had the interest and attention of any audience before he opened his lips.

Phillips Brooks has been compared to Henry Ward Beecher, and in many things they were alike. But the former's culture, while perhaps less varied than Beecher's, was deeper and richer, his sermons were less brilliant but cast in better form, his appeal was narrower but to a far more influential class. He was, in a word, the preacher of the intellectual. No one who heard him preach ever failed to be startled at first by his tremendous rapidity of delivery—averaging two hundred words a minute—or failed to find himself, at first, lagging behind the equal rapidity of thought. But once accustomed to these—once realizing that, in listening to him there could be no inattention or wandering of wits—his sermon became a source of keenest intellectual delight and noblest spiritual inspiration.

Phillips Brooks often said that he had to preach rapidly, or not at all. In youth he had suffered from something resembling an impediment in his speech, and more measured utterance gave it a chance to recur. Certainly,

no one who ever listened to his fluent and limpid utterance would have suspected it. But he was far more than a great preacher. By his broad tolerance, his lofty character and immense personal influence, he became, in a way, a national figure, the common property of the nation which felt itself the richer for possessing him. A gracious and courtly figure, with a heart as wide as the human race, he lives, somehow, as the true type of clergyman, whose concern is humanity and whose field the world.

Which brings us to the life of the last man we shall consider in this chapter, a man the opposite in many ways of the great clergyman whose career we have just noted, and yet, like him, of broadest sympathies and most sincere convictions; a man whose life was more picturesque, whose battle against fate was harder, and whose achievement was even more remarkable—the greatest evangelist the modern world has ever produced, Dwight L. Moody. If ever a man labored for his fellow-men, he did, and the story of his life reads almost like a romance.

He was born at Northfield, Massachusetts, in 1837, the son of a stonemason, who, disheartened and worn out by business reverses, died when the boy was only four years old. There were nine children, the oldest only fifteen, and when the father's creditors came and took every possession they had in the world, the future looked dark indeed. The mother was urged to place the children in various homes, but she managed to keep them together by doing housework for the neighbors and tilling a little garden.

As soon as he was old enough, Dwight was put to work on a farm, but his earnings were small, and finally, when he was seventeen, he started for Boston to look for something better. He managed to get a position in a shoe-store, and there came under the influence of Edward Kimball, who persuaded him to become a Christian and to join a church. But he was not admitted to membership for nearly a year; so poor was his command of language and so awkward his sentences that it was doubted if he understood Christianity at all, and even when he was admitted, the committee stated that they thought him "very unlikely ever to become a Christian of clear and decided views of gospel truth; still less to fill any extended sphere of public usefulness." How blind, indeed, we often are to the possibilities in human nature!

At the age of nineteen, Dwight removed to Chicago, secured another position as shoe-salesman, and offered his services to a mission school as a teacher. His appearance made anything but a favorable impression, but finally he was told that he might teach provided he brought his own scholars. The next Sunday he walked in at the head of a score of ragamuffins he had gathered up along the wharves. The divine fire seems to have been working in him; he was finding words with which to express himself, and burning for a wider field. So he rented a room in the slum districts which had been used as a saloon and opened a Sunday school there. It was an immense success, soon outgrew the little room, and was removed to a large hall, where, every Sunday, a thousand boys and girls attended. For six years, Moody conducted that school, sweeping it out and doing the janitor work himself, attending to his business as salesman throughout the week. But in 1860, at the age of twenty-three, he decided to devote all his time to Christian work.

He had no income, and to keep his expenses as low as possible, he slept at night on a bench in his school, and cooked his own food. Then the Civil War began, and he erected a tent at the camp near Chicago where the recruits were gathered, and labored there all day, sometimes holding eight or ten meetings. He went with the men to the front, and was at the desperate battles of Shiloh, Murfreesboro, and Chattanooga. The war over, he took up again his work in Chicago. The great fire of 1871 swept away his church, but he soon had a temporary structure erected, and labored on.

By this time, his fame had got abroad, and finally in 1873, his great opportunity came. Accompanied by Ira D. Sankey, the famous singer of hymns, he started on an evangelist tour of Great Britain. At his first meeting only four people were present; at his last, thirty thousand crowded to hear him. In Ireland, the crowds sometimes covered six acres, and during the four months he spent in London, over two million people heard him preach. Great Britain had never before experienced such a religious awakening; but it was as nothing to the reception given him when he returned to America two years later. There are many people still living who remember those wonderful revivals in Philadelphia, New York, and Boston, with their great choirs, and Ira Sankey's singing, and Moody's soul-stirring talks. From that time forward he was easily the first evangelist in the world—perhaps the greatest the world had ever seen.

It is doubtful if any man ever faced and preached to so many people. He spoke to thousands night after night, week in and week out. In his themes he kept close to life, and few men were his equal in making scriptural biography vivid and realistic; in reconstructing scriptural scenes and setting them, as it were, bodily before his audience. He was not a cultured man, as we understand the word—not a man of broad learning; perhaps such learning would only have weakened him—nor did he have the presence and voice which go so far toward the equipment of the orator. But he burned with an intense conviction, and his sermons were so free from art, so direct, so persuasive, that they were perfectly adapted to the end he sought—the conversion of human beings.

SUMMARY

GIRARD, STEPHEN. Born near Bordeaux, France, May 24, 1750; sailed as cabin-boy to West Indies, and then to America; established in Philadelphia, 1769; financial mainstay of government in war of 1812; died at Philadelphia, December 26, 1831.

SMITHSON, JAMES LEWIS MACIE. Born in France in 1765; matriculated from Pembroke College, Oxford, England, 1782; Fellow Royal Society, 1786; distinguished as student of mineralogy and chemistry; died at Genoa, Italy, June 27, 1829.

COOPER, PETER. Born at New York City, February 12, 1791; apprenticed to carriage-maker, 1808; engaged in various enterprises and established Canton Iron Works, Canton, Maryland, 1830; Greenback candidate for President, 1876; died at New York, April 4, 1883.

PEABODY, GEORGE. Born at Danvers, Massachusetts, February 18, 1795; settled in London as a banker, 1837; died there, November 4, 1869.

HOPKINS, JOHNS. Born at Waterbury, Connecticut, May 19, 1795; founded house of Hopkins & Brothers, 1822; chairman of finance committee Baltimore & Ohio railroad, 1855; died at Baltimore, December 24, 1873.

CORNELL, EZRA. Born at Westchester Landing, New York, January 11, 1807; mechanic and miller at Ithaca, New York, 1828-41; member of State Assembly, 1862-63; State Senator, 1864-67; died at Ithaca, New York, December 9, 1874.

SLATER, JOHN FOX. Born at Slatersville, Rhode Island, March 4, 1815; established Slater Fund, 1882; died at Norwich, Connecticut, May 7, 1884.

STANFORD, LELAND. Born at Watervliet, New York, March 9, 1824; Republican governor of California, 1861-63; United States Senator, 1885-93; died at Palo Alto, California, June 20, 1893.

ROCKEFELLER, JOHN DAVISON. Born at Richford, New York, July 8, 1839; partner of Clark & Rockefeller, 1858; built Standard Oil Works, Cleveland, Ohio, 1865; organized Standard Oil Company, 1870; Standard Oil Trust, 1882.

CARNEGIE, ANDREW. Born at Dunfermline, Fifeshire, Scotland, November 25, 1837; came to United States, 1848; telegraph messenger boy, 1851; introduced Bessemer steel process to America, 1868; formed Carnegie Steel Company, 1899; merged into United States Steel Corporation, 1901, when he retired from business.

BEECHER, LYMAN. Born at New Haven, Connecticut, October 12, 1775; pastor of various Congregational churches, 1799-1832; president Lane Theological Seminary, 1832-51; died at Brooklyn, New York, January 10, 1863.

BEECHER, HENRY WARD. Born at Litchfield, Connecticut, June 24, 1813; graduated at Amherst, 1834; pastor of Plymouth Congregational Church, Brooklyn, 1847-87; founder of the *Independent* and the *Christian Union*; died at Brooklyn, March 8, 1887.

CHANNING, WILLIAM ELLERY. Born at Newport, Rhode Island, April 7, 1780; graduated at Harvard, 1798; pastor of Federal Street Church, Boston, 1803-42; died at Bennington, Vermont, October 2, 1842.

JUDSON, ADONIRAM. Born at Malden, Massachusetts, August 9, 1788; graduated at Brown, 1807; started as missionary to Burmah, 1812, and remained in far East until his death, April 12, 1850.

MOTT, LUCRETIA. Born at Nantucket, Massachusetts, January 3, 1793; entered ministry of Friends, 1818; assisted at formation of American anti-slavery society, 1833; called first woman suffrage convention, 1848; died near Philadelphia, November 11, 1880.

DIX, DOROTHEA LYNDE. Born at Worcester, Massachusetts, 1805; devoted her whole life to work for paupers, convicts, and insane persons; superintendent of hospital nurses during Civil War; died at Trenton, New Jersey, July 19, 1887.

CHILD, LYDIA MARIA. Born at Medford, Massachusetts, February 11, 1802; editor *National Anti-Slavery Standard*, 1840-43; published a number of novels; died at Wayland, Massachusetts, October 20, 1880.

GARRISON, WILLIAM LLOYD. Born at Newburyport, Massachusetts, December 10, 1805; began publication of the *Liberator*, 1831; president American Anti-Slavery Society, 1843-65; died at New York City, May 24, 1879.

PARKER, THEODORE. Born at Lexington, Massachusetts, August 24, 1810; studied at Cambridge Divinity School, 1834-36; Unitarian clergyman at Roxbury, 1837; head of an independent society at Music Hall, Boston, 1846; died at Florence, Italy, May 10, 1860.

PHILLIPS, WENDELL. Born at Boston, November 29, 1811; educated at Harvard; admitted to the bar, 1834; leading orator of the Abolitionists, 1837-61; president of the Anti-Slavery Society, 1865-70; Prohibitionist candidate for governor of Massachusetts, 1870; died at Boston, February 2, 1884.

ANTHONY, SUSAN BROWNELL. Born at South Adams, Massachusetts, February 15, 1820; became agitator in cause of woman suffrage, organized National American Woman Suffrage Association and was its president for many years; died March 13, 1906.

STANTON, ELIZABETH CADY. Born at Johnstown, New York, November 12, 1815; graduated at Willard Seminary, 1832; met Lucretia Mott, 1840; held first woman's suffrage convention, 1848; associated with Susan B. Anthony; died at New York City, October 26, 1902.

BROWN, JOHN. Born at Torrington, Connecticut, May 9, 1800; removed with parents to Ohio, 1805; emigrated to Kansas, 1855; won battle of Osawatimie, August, 1856; seized arsenal at Harper's Ferry, Virginia, October 16, 1859; captured, October 18; tried by Commonwealth of Virginia, October 27-31; hanged at Charlestown, Virginia, December 2, 1859.

BARTON, CLARA. Born at Oxford, Massachusetts, 1821; superintended relief work on battle-fields during Civil War; laid out grounds of national cemetery at Andersonville, 1865; worked through Franco-Prussian war, 1870; distributed relief in Strasburg, Belfort, Montpelier, Paris, 1871;

secured adoption of Treaty of Geneva, 1882; president American Red Cross Society, 1881-1904.

BERGH, HENRY. Born at New York City, 1823; secretary of legation at St. Petersburg, 1862-64; organized American Society for the Prevention of Cruelty to Animals, 1866; founded Society for the Prevention of Cruelty to Children, 1874; died at New York City, March 12, 1888.

BROOKS, PHILLIPS. Born at Boston, December 13, 1835; graduated at Harvard, 1855; graduated from Episcopal Seminary, Alexandria, Virginia, 1859; rector of Trinity Church, Boston, 1870-93; elected Bishop of Episcopal Diocese of Massachusetts, 1891; died at Boston, January 23, 1893.

MOODY, DWIGHT LYMAN. Born at Northfield, Massachusetts, February 5, 1837; started missionary work at Chicago, 1856; conducted revival meetings in Great Britain, 1873-75; and devoted the remainder of his life to this work; died at Northfield, December 22, 1899.

CHAPTER IX

MEN OF AFFAIRS

Almost from the first years of her existence America has been known chiefly as a commercial nation, as a nation noted for her men of affairs, rather than for her artists and men of letters. Which is to say that the life of the Republic has been practical rather than artistic, and it is only of late years, except for a sporadic instance here and there, that any genuine artistic impulse has made itself felt.

This is not a cause of reproach. Given the circumstances, it was inevitable that America should develop first on her commercial side. Here was a great continent, stretching thousands of miles to the westward, waiting for man's occupancy. Millions of acres of plain and woodland awaited development. There were cities to found and rivers to bridge and roads to make and soil to till and gold to dig before America could think of writing poetry or painting pictures. Think—it is only three centuries since Jamestown was founded; only a century and a quarter since we became a nation—a mere handbreadth of time when compared with the long centuries of English or French or Italian history. We have already said that for art historic background is necessary; a background of achievement and tradition. Such a background we are just achieving. Besides, during our first century, there were such great deeds of conquest and development to be done that they challenged our strongest men. Great fortunes were made, as a matter of course, and Europe witnessed the unique spectacle of men, born in poverty and obscurity, rising to be captains of the world. It is this which has never ceased to shock the European sense of the fitness of things—that the poor boy of yesterday may be the millionaire of to-morrow and take his place with the greatest of the nation. It is the story of a few such boys which will be told in this chapter.

First is the man who financed the Revolution and who to a large extent made possible its successful termination—Robert Morris. Born in

Liverpool, England, in 1734, he came to this country with his father at the age of thirteen, and a place was soon found for him in the counting-house of Charles Willing, a wealthy merchant of Philadelphia. By his diligence and activity, as well as unusual intelligence, he grew in favor and confidence, until, upon the death of the elder Willing, he was taken into partnership by the latter's son, and by the opening of the Revolution, the firm of Willing & Morris was one of the largest and most prosperous in Philadelphia.

Of English birth, and bound to England by the ties of business, Morris was nevertheless opposed to the stamp-act and was one of those who, in 1765, signed an agreement to import nothing further from England until the act was repealed. He was, however, opposed to independence, and, as a member of the Continental Congress, voted on July 1, 1776, against the Declaration. Three days later he declined to vote, but when the Declaration was adopted, he signed it, and threw in his fortunes unreservedly with his new country. His services were more than valuable—they were indispensable. As a member of the Committee of Ways and Means, he backed the government's credit with his own. Without his aid, the last campaigns of the war would have been impossible. It was he who supplied General Green with munitions of war for the great campaign of the south, and shortly afterwards raised a million and a half on his own notes to assist Washington in the movement which resulted in the capture of Cornwallis at Yorktown. A year later, when the financial situation of the government had become desperate, he organized the Bank of North America to assist in financing it. For three years, he acted as superintendent of finance, with complete control of the monetary affairs of the country. He was a member of the Constitutional Convention, and when the new government was organized, Washington asked him to accept the treasury portfolio, but he declined, suggesting instead Alexander Hamilton. That was not the least of his services to America, for Hamilton was preëminently the man for the place.

It was the striking irony of fate that the man who had controlled the finances of a nation and by his personal exertions saved it from bankruptcy should himself die in a debtor's prison; yet such was the case. A series of unfortunate land speculations swept away his wealth and ruined his credit; he found himself unable to meet his obligations and was seized by his

creditors and thrown into prison, where he remained for some years, and where death found him in 1806.

So Robert Morris was not one of the founders of great fortunes. Turn we to the earliest and perhaps most successful of these, John Jacob Astor, the very type of the astute, large-minded, and far-sighted financier. Born at Waldorf, Germany, in 1763, the son of a poor butcher in whose shop he worked until sixteen years of age, there was nothing in his life or circumstances to indicate the future which lay before him. One of his brothers, however, had come to America and settled at New York, and young John Astor resolved to join him in the land of opportunity. At the age of twenty, he was able to do so, bringing with him some musical instruments to sell on commission, but a chance acquaintance which he made on shipboard changed the whole course of his life.

This acquaintance was that of a furrier, who told young Astor of the great profits to be made by buying furs from the Indians and selling them to the large dealers. Perhaps he exaggerated the profits of the business; at any rate, he fired the ambition of his hearer, and the latter decided to enter the fur business without delay. Upon landing in New York, therefore, he at once secured a position in the shop of a Quaker furrier, and after learning all the details of the business, opened a shop of his own.

Perhaps no one ever worked harder in establishing a business than John Jacob Astor did. Early and late he was at his shop, except when absent on long and arduous purchasing expeditions into the wilderness. More than that, he possessed admirable business judgment, so that, after fifteen years of work, he had succeeded in accumulating a fortune of a quarter of a million dollars. With careful and sagacious management, the business prospered so that Astor was soon able to send his furs to Europe in his own vessels, and bring back European goods. And about this time, he began working on a grandiose and picturesque enterprise.

The English Hudson Bay Company, established many years before, with hundreds of trappers and traders and scores of trading-posts, controlled the rich fur business of Canada and the northwest. We have seen how, years after the events which we are now narrating, the agents of the company tried to save Oregon for England and how Marcus Whitman foiled them.

Astor's plan, in outline, was to render American trade independent of the Hudson Bay Company by establishing a chain of trading-posts from the great lakes to the Pacific, to plant a central depot at the mouth of the Columbia river, and to acquire one of the Sandwich Islands and establish a line of vessels between the western coast of America and the ports of Japan, China and India. Surely a man who could conceive a plan like that was something more than a mere trader, and Astor proceeded at once to carry it into effect.

Two expeditions were sent out, one by land and one by sea, to open up intercourse with the Indians of the Pacific coast, and the settlement of Astoria was planted at the mouth of the Columbia river. Whether Astor would have been able to carry out the remainder of his plan is purely problematical, for before he had it fairly under way, the war of 1812 began, and he was forced to abandon the enterprise. The story of this far-reaching project has been told by Washington Irving in his "Astoria." Until his death, he continued to enlarge and increase his business, and left a fortune estimated at twenty millions of dollars.

The Astor plan of investment is one of the safest, most sagacious in the world. Practically all of his profits were invested by John Jacob Astor in real estate outside the compact portion of the city of New York. As the city grew out to his holdings, he would improve them, rent or sell them, and reinvest further out. In this way the growth of the city marked also the growth of his fortune, and this plan of investment has been followed by his descendants to the present day, until they have become by far the most important owners of real estate in New York City. His son, William B. Astor, gave his life to the preservation and growth of the vast property he inherited, and at his death had more than doubled it, dividing an estate of \$45,000,000 between his two sons.

Not that the whole thought of these two men was money-getting, for their public gifts were numerous and important. The most noteworthy was the Astor library, founded by John Jacob Astor at the suggestion of Washington Irving, and largely added to by his son, the total amount of the Astor donations to it exceeding a million dollars. But they stand as two types of sagacious and hard-headed business men, to whom money-making and the

still more difficult art of money-keeping was an instinctive accomplishment.

The second great American fortune was that founded by Cornelius Vanderbilt, as remarkable and picturesque a character as this country ever produced. Born on Staten Island in 1794, the son of a farmer in moderate circumstances, the boy soon developed a remarkable talent for trade. His father owned a sail-boat, in which he conveyed his produce across the bay to the New York markets, and the boy soon learned to manage this and was intrusted with these daily trips. When he was sixteen years old, he bought a boat of his own, in which he ferried passengers across the bay, and two years later he was owner of two boats and captain of a third. This was the beginning of the great fleet of steamers, sloops and schooners which he built up for the navigation of the shores of New York bay and the Hudson river, which won him the title of "Commodore," which clung to him all his life. Before he was forty years old, he had accumulated a fortune of half a million dollars, and was ready for those great financial operations which marked his later life.

The discovery of gold in California led him to establish a passenger line by way of Lake Nicaragua which netted him ten millions in ten years; he established a fast line of passenger steamships between New York and Havre; and finally was attracted to railway development as a field of enterprise destined to win large returns. In the course of a few years he had secured control of both the Hudson River and New York Central roads, and brought both of them to the highest state of efficiency, and after consolidating them, extended the system to Chicago by the purchase of the Lake Shore, the Canada Southern and Michigan Central. He built a great terminal in New York City, and made the system so profitable that, from it, and a series of fortunate speculations, he accumulated a fortune of \$100,000,000, practically all of which he bequeathed to his eldest son, William Henry. One million was also given for the establishment of Vanderbilt University at Nashville, Tennessee.

Cornelius Vanderbilt, for many years, had a very poor opinion of his son's financial ability, and giving him a small farm on Staten Island, left him to shift for himself. Everyone has read of the incident which changed this opinion. William needed some fertilizer for his farm, and asked his father to

give him a load of manure from his stables. His father told him to go ahead and take a load, and William thereupon brought a great scow up to the pier near the stables, proceeded to load it, and when his father protested, pointed out that he had not specified the kind of load, but that he had meant a scow-load. This bit of sharp practice pleased his father, and, shortly afterwards, the great success with which he managed the Staten Island Railroad, as receiver, established him in his father's confidence. He continued and extended his father's policy of railway investment, and added to the great fortune which had been left him, and which still remains one of the greatest in America, though it has been split up among the different branches of the Vanderbilt family. William himself distributed about two millions in various benevolent and public enterprises, one of the queerest of which was the removal of one of "Cleopatra's Needles" from Egypt to Central Park, New York City, at a cost of over a hundred thousand dollars.

In the business world of New York City, half a century ago, no name was more prominent than that of A. T. Stewart, whose success as a merchant was one of the most astonishing features of the time. Born near Belfast, Ireland, in 1803, Stewart was a descendant from one of those hardy and thrifty Scotch-Irish, whom we have had occasion to mention before. His father was a farmer, but died while the son was still at school, and at the age of twenty the latter came to New York, and after looking over the field, opened a small store on lower Broadway, with a sleeping apartment for himself in the rear. Such was the beginning of the greatest dry-goods business this country ever saw. It increased by leaps and bounds, for Stewart seems to have had a sort of instinctive genius for the business. He was continually moving to larger and larger quarters, and in 1862, built on Broadway a store which was at that time the largest in the world, and which, even in this day of mammoth structures, commands attention. Its cost was nearly three millions, a colossal sum for those days; two thousand people were employed in it and it cost a million a year to run. But it brought a tremendous return, and its owner soon became one of the wealthiest men in New York.

He wanted more than wealth—he hungered for political and social honors which were never fully his. He had made a large contribution to the fund of \$100,000 presented by the merchants of New York to General Grant, and in 1869, Grant appointed him secretary of the treasury. The senate refused to

confirm the appointment, on the ground that the law excluded from that office anyone interested in the importation of merchandise. Grant sent to the senate a message recommending that this law be repealed, but the senate refused; and Stewart thereupon offered to place his business in the hands of trustees and devote its entire profits to charity during his term of office; but still the senate refused, and the nomination was withdrawn. It was a bitter blow to Stewart, nor was his fight for social prominence much more fortunate. As his last stake, as it were, he began the erection of a great marble palace on Fifth Avenue, designed to cost a million and to be the finest private residence in the world, but he died before it was completed.

One of the great industries of the country is that of sugar refining, and it is inseparably connected with the name of Havemeyer, for to the Havemeyers is due its development and its formation into a so-called trust, which practically controls the market, and which has won great wealth for its organizers. The ancestor of the Havemeyers was a thrifty German who came to this country in the latter part of the eighteenth century, and, after engaging in various pursuits, opened a little sugar refinery in New York City, which soon brought him a comfortable income. There, in 1804, William Frederick Havemeyer was born, and after a careful education, entered the refinery, gained a thorough knowledge of the business and, in 1828 succeeded to it, having as a partner his cousin, Frederick Christian Havemeyer. These two men developed the business in a wonderful manner, installing new machinery, inventing new processes, which reduced the manufacturing cost, acquiring possession of other plants and securing government support in the shape of a protective tariff, which made a naturally profitable business doubly so, and netted its owners many millions.

William Frederick Havemeyer found time, in the intervals of running his business, to take a prominent part in New York politics. He was mayor of the city from 1845 to 1851, and again in 1873, dying before the last term was finished.

As far as possible removed from Havemeyer's humdrum existence was that of Phineas Taylor Barnum, the greatest showman the world has ever seen, the originator of the great travelling circus, the exploiter of Tom Thumb and Jenny Lind, the owner of Jumbo, the most famous elephant that ever lived, whose name has passed into the English language as a synonym for bigness.

Barnum was born at Bethel, Connecticut, in 1810. His father was an inn-keeper and died when the boy was fifteen years old, leaving no property. He tried his hand at store-keeping, and failed; ran a newspaper, and was imprisoned for libel, and finally reached New York at about the end of his resources and looking around for something to do. That was in 1834, and by accident he hit upon his real vocation.

A man by the name of R. W. Lindsay was exhibiting through the country an old negro woman named Joice Heth, advertising her as being 161 years old, and as having been the nurse of George Washington. Barnum went to see her and found her an extraordinary-looking object. He has himself told how he was impressed by her.

"Joice Heth," he says, "was certainly a remarkable curiosity, and she looked as though she might have been far older than her age as advertised. She was apparently in good health and spirits, but from age or disease, or both, was unable to change her position; she could move one arm at will, but her lower limbs could not be straightened; her left arm lay across her breast and she could not remove it; the fingers of her left hand were drawn down so as nearly to close it, and were fixed; the nails on that hand were almost four inches long and extended above her wrist; her head was covered with a thick bush of gray hair; but she was toothless and totally blind, and her eyes had sunk so deeply in the sockets as to have disappeared altogether. Nevertheless she was pert and sociable and would talk as long as people would converse with her. She was quite garrulous about 'dear little George,' at whose birth she declared she was present, having been at the time a slave of Elizabeth Atwood, a half-sister of Augustine Washington, the father of George Washington. As nurse, she put the first clothes on the infant, and she claimed to have raised him."

Barnum was so impressed by this extraordinary object, that he bought her for a thousand dollars, putting his last cent into the venture and borrowing

what he lacked. He proceeded to advertise her with characteristic energy, and great crowds thronged to see her, so that his receipts sometimes ran as high as \$1,500 a week. However, the old woman died within a year, and a post-mortem examination showed that she was really only about eighty years old.

But Barnum had found his vocation, that of showman, and after a few unsuccessful ventures, bought Scudder's American Museum, in New York City, and started out on a brilliant career. It is interesting to note that the museum which Barnum purchased consisted in part of the curios collected years before by Charles and Rembrandt Peale. Barnum added to it, was indefatigable in securing curiosities, really created the art of modern advertising, and it was his proudest boast that no one ever left the museum without having got his money's worth. He was one of the first to realize that the best possible advertisement is a pleased customer, and he tried honestly to keep his museum supplied with every novelty. The public soon came to appreciate this, and perhaps his greatest asset was public confidence in his promises. People came to believe that when Barnum advertised a thing, he really had it. But the most fortunate day in all his life was that November day of 1842, when he discovered at Bridgeport, Connecticut, the midget whose real name was Charles S. Stratton, but who was to become world-famous as General Tom Thumb.

The story of Tom Thumb's success reads like a romance. He was quite young when Barnum got him, and the showman took great pains with his education and training, for he wanted the midget to appear a finished man of the world. He became a great public favorite, toured America and Europe, was introduced to kings and princes and made a great fortune for himself and his exhibitor. Barnum struck the apogee of his fortunes when he discovered another midget, Lavinia Warren, who achieved a success scarcely less than Tom Thumb's. Indeed, she and the General fell in love with each other and were married at Grace Church, and as General and Mrs. Tom Thumb were perhaps the greatest drawing cards in the world. Another triumph of his career was his engagement of Jenny Lind for a series of one hundred concerts, at a salary of a thousand dollars a night, the receipts of the tour being over seven hundred thousand dollars.

Barnum had many ups and downs, which he met with an invincible optimism. His museum burned down and he rebuilt it, but it soon burned down again. It was then that the idea occurred to him to establish a travelling museum, exhibiting under a tent, and it was this idea which developed into "The Greatest Show on Earth." It really was the greatest and its owner never spared money in his endeavor to keep it so. Large-hearted, benevolent, a true entertainer, he will always occupy a bright place in the memory of the American public.

Perhaps no name in the history of America was ever more closely connected in the public mind with money-making for its own sake than that of Russell Sage. It will be surprising news to many, who knew him only as a money-lender on a large scale, that he started out on a public career, as alderman, county treasurer, and finally as member of congress for two terms, from 1853 to 1857. He was the first person to advocate, on the floor of congress, the purchase of Mount Vernon by the government. His career on Wall street began shortly after that, at first in a small way; but before his death, he had developed into the greatest individual money-lender in the world.

That was his whole life. He took no part in any political or charitable movement; he had no interest in art, and he lived in the simplest manner. He used his wealth, not to procure enjoyment for himself or other people, but to procure more wealth. He was saving to the point of miserliness; he got the utmost he could out of his money; he never took a vacation—and dying, at the age of ninety, left a fortune of many millions. He had no children and the whole fortune went to his wife. She at once proceeded to bestow it in carefully-considered benevolences, so that the Sage millions are to benefit humanity, after all. In fact, it is doubtful if any other fortune, amassed by a single man, will, in the end, do so much good in the world as will this of Russell Sage, for Mrs. Sage is devoting it to what may be called scientific charity, which has for its object the universal betterment of mankind.

Mrs. Sage, who thus becomes one of the world's great philanthropists, was Margaret Olivia Slocum, of Syracuse, New York, and was married to Mr.

Sage in 1869. She was of a family in only moderate circumstances, and was a school teacher previous to her marriage. The turn of the wheel made her the wealthiest woman in the world, and she proceeded without delay to the carrying out of the immense benevolent enterprises which she had doubtless long meditated.

The name of Cyrus West Field is so closely associated with his supreme achievement, the laying of the first Atlantic cable, that we are apt to forget that he was in the beginning a manufacturer and had amassed a considerable fortune before his attention was called to the possibility of linking Europe to America by a telegraph line laid on the bottom of the Atlantic. It was under A. T. Stewart that Field received his mercantile training, having gone to New York in 1834, at the age of fifteen, from his home in Stockbridge, Massachusetts, and entering Stewart's employ as a clerk.

He was an apt pupil, and before he was of age, owned an establishment of his own for the manufacture and sale of paper. In this business, in the course of a dozen years, he had amassed a fortune so considerable that he was able to retire from active charge of it, and to spend his time in travel. It was in 1853 that the project of carrying a telegraph line across the Atlantic ocean suggested itself to him during a conversation with his brother, who was interested in building a line across Newfoundland. The more he considered and investigated the project, the more feasible it seemed, and he proceeded to organize the New York, Newfoundland and London Telegraph Company, himself taking one fourth of the capital stock, and interesting such other capitalists as Peter Cooper, Moses Taylor, Chandler White and Marshall Roberts.

But the project which had appeared simple enough in theory and on paper, proved extremely difficult of execution. If Field could have foreseen the thirteen years of constant anxiety which awaited him, he would no doubt have hesitated to undertake it. It looked, at first, as though success would crown his efforts almost at the outset, for in 1858, the laying of a cable was completed, and for some days, messages were sent from one continent to the other. Then the signals began to grow fainter and fainter, until they became imperceptible, supposedly from the water of the ocean penetrating the cable covering.

At any rate, the work had to be done all over again, with little money on hand, and the coming of the Civil War helped to make further progress impossible. Field visited Europe more than twenty times in the effort to raise money for the enterprise and to keep it before the public, but it was not until 1865 that another effort to lay the cable could be made. The "Great Eastern," the largest ship in the world, was secured, and began paying out the cable; but twelve hundred miles from shore the cable parted and could not be regained, although every effort was made to grapple it. So the vessel had to put back to England, and Field was confronted with the heart-breaking task of raising even more money. He succeeded in doing so, and in 1866, another expedition started out with a new cable. This time, it met with no serious misadventure, and on July 27, telegraphic communication was re-established between England and America, and has never since been interrupted.

That cable was the first of the hundreds which now encircle the globe. Congress presented the bold adventurer with a gold medal and the thanks of the nation; John Bright pronounced him "the Columbus of modern times, who, by his cable, has moored the New World alongside of the Old"; the Paris exposition of 1867 gave him the grand medal, the highest prize it had to bestow; and he received votes of thanks and medals and presents from all parts of the world.

In 1884, two other cables were laid across the Atlantic by John W. Mackay and James Gordon Bennett, whose private property they remained. Mackay had had an adventurous career, and was destined to be the founder of another of those great American fortunes which are the wonder and admiration of Europe. He was born in Dublin, Ireland, in 1831, his father being another of those sturdy Scotch-Irish of whom we have already had occasion to speak. He was brought to New York at the age of nine; but his father died a short time thereafter and the boy was thrown practically upon his own resources.

When gold was discovered in California in 1849, Mackay joined the crowd that rushed to the new El Dorado, and for several years, he lived a typical miner's life, roughing it in the camps, but gaining little except a thorough knowledge of mining. In 1860, some guiding spirit led him eastward to Nevada; his fortunes there steadily improved, until he became one of the

leading men in the settlement, and in 1872, he made one of the most famous and romantic discoveries in mining history, that of the famous Comstock lode, on a ledge of rock high in the Sierras, under which Virginia City now nestles. So rich in silver was this great ledge of rock and its enormous production added so greatly to the world's supply of silver that the market price fell to a point where such countries as India and China, whose currency was on a silver basis, were seriously embarrassed to maintain values. From one mine alone over \$150,000,000 was taken out. Mackay devoted himself personally to the superintendence of the mines, working in the lower levels with his men, who idolized him.

Let us turn for a moment to the career of another great fortune-builder, the man who was, perhaps, the greatest freebooter the American financial world ever saw, who made his money by destroying rather than building up, and whose wealth finally killed him—Jay Gould. Let us see if we can get some sort of idea of the personality of this extraordinary man.

Born in 1836, a farmer's boy, with only such education as he could pick up, he managed to find time to study surveying, and for two or three years was engaged in making surveys of various New York counties. While thus engaged, he fell in with a wealthy and eccentric individual named Zadock Pratt, who sent him to the western part of the state to select a site for a tannery. He was soon doing a large lumbering business, first with Pratt and then in his own name; but he sold out just before the panic of 1857, and soon after entered upon that career of speculation in New York City which, in the end, made him the best-hated man in America.

Picture the man, small, only five feet six inches in height, with sallow skin and jet black whiskers, his eyes dark and piercing, his whole personality, as one observer put it, "reminiscent of the spider." His reputation was that of an unscrupulous and immoral rascal, who would not hesitate to sacrifice his best friends, if need be. His war against Cornelius Vanderbilt for control of the Erie was one of his typical operations—a war which, when he saw he was losing, he won by issuing \$5,000,000 worth of fraudulent stock. There was never any question about the criminality of this proceeding, and Gould

was forced to flee to New Jersey, where he spent millions in corrupting courts and legislatures—millions, not taken from his own pocket, but from the treasury of the Erie, of which he had control. He was ousted, at last, but not until he had added \$62,000,000 to the indebtedness of the road, of which amount it was asserted Gould had pocketed \$12,000,000.

The culminating feature of his career was his attempt to corner gold, which brought about the famous Black Friday panic of 1869. The scheme, one of the most daring ever attempted by any operator, came near success. Gould is said to have bribed the brother-in-law of President Grant and to have persuaded the President himself not to release any of the government supply of gold. He then succeeded in driving the price up to 162½, when suddenly the bubble burst. Gould, himself, had been warned and succeeded in getting away with his immense profits, covering himself at the expense of his associates, an act of treachery unprecedented even in the stock market.

These were only two of the remarkable operations which he engineered, and which need not be given in detail here. The net result was a fortune of some seventy million dollars, and a reputation for duplicity such as perhaps no man in America ever had before. It is only fair to Gould to say, however, that he accomplished merely what most stock gamblers would like to accomplish, if they could, and that outside of finance, he seems to have been an estimable man, faithful to his wife, devoted to his children, and passionately fond of flowers. He made no gifts of any consequence to charity during his life, nor did he make a single benevolent bequest in his will; but one of his children, Helen Miller Gould, has more than atoned for this by practically devoting her life and her fortune to charitable work. It is doubtful if there is a better-loved woman in America to-day than Helen Gould, who has shown so notably how a life may be consecrated to good works.

The great marble palace which A. T. Stewart built on Broadway, in New York City, to house his business, and which was, at the time, the largest building in the world devoted to a retail business, is now occupied by another great merchant, who, starting from a beginning even smaller than

Stewart's, has built up a business many times as great. John Wanamaker, whatever the growth of the country may be hereafter, will always remain one of America's most representative and most successful men of affairs—both representative and successful because his business has rested from the first on the principle of honest dealing, of making satisfied customers—in a word, upon the altogether modern principle of "your money back, if you want it."

John Wanamaker was born in Philadelphia in 1838, a poor boy with his way to make in the world. He received his education in the common schools, and

at the age of fourteen, entered upon his business career as an errand boy in a book store. From that, he got a clerkship in a clothing store, and for some years acted as salesman, until he could save enough money to start a little store of his own. This he was able to do in 1861, in partnership with a man named Nathan Brown, and ten years later, he was sole owner of a prosperous and growing business. It was at about this time that an idea occurred to him which was destined to revolutionize the retail business of the larger cities of the country.

The idea was simply this: In the great cities, most shoppers have to travel a considerable distance to get to the business centre, and must there waste time and energy going from one store to another to make their purchases.



WANAMAKER

Why not, then, combine all the representative retail businesses into one store, so that the shopper could make all purchases under a single roof, pay for them all at once, and have them all delivered at the same time? Moreover, why could not one great business be conducted more cheaply, and so undersell, the small ones, since a single executive staff would do for it, rent, delivery cost, and a hundred other fixed charges would be reduced, to say nothing of the advantages of large buying, and the advertising which every department would get from all the rest? The idea grew into a carefully-formulated plan, and 1876 saw the start of the great Wanamaker department store, perhaps the most famous retail business in the world.

Its tremendous success is an old story now, and it has found hundreds of imitators. Twenty years after the opening of the Philadelphia store, another was opened in New York in the old Stewart building, to which another building, four times as large, has recently been added. Wanamaker from the first firmly believed in P. T. Barnum's old adage that "A satisfied customer is the best advertisement," and he made every effort to see that none left the Wanamaker stores unsatisfied. He also made it a rule that no visitor to his store should ever be urged to buy anything; that every article of merchandise should be exactly as represented, and that any purchase might be returned and the purchase money would be refunded without question. As a result, Wanamaker got a reputation for fair dealing which proved his greatest asset.

One would think that the management of such a business would fully occupy any man, but Wanamaker found time for many public and benevolent interests. He founded, in 1858, the Bethany Sunday School, which has grown into perhaps the largest in the world and of which he has always been superintendent; he has taken part in many movements for civic reform, and from 1889 to 1893 was postmaster general of the United States. He reorganized the service; set in motion the rural delivery system, the greatest single improvement in its service the department has ever made; and tried to secure a postal telegraph, a postal savings-bank, a parcels post and one-cent letter postage. He was the first official to regard the service as a business pure and simple, and if the reforms he suggested had been carried out, the United States postoffice would now be a model for the world.

The greatest banker and financier in America at the present day is undoubtedly J. Pierpont Morgan, who, however, is known not so well for the millions he has accumulated as for the other millions he has spent in collecting rare objects of art, until he has become the possessor of a collection surpassing any ever possessed by another private individual. That much of this will one day be bequeathed by its owner to the public there can be little doubt.

J. Pierpont Morgan is of a family of bankers. His father, Junius Spencer Morgan, was for many years a partner in the great London banking house of George Peabody & Co., and on the retirement of Mr. Peabody, succeeded him as the head of the business. There was never any doubt of the son's vocation. Born in 1837, and carefully educated, he entered the banking house of Duncan, Sherman & Co. at the age of twenty, and from that time, rose steadily, until he became the head of the greatest banking house in the country. He has been largely concerned in the reorganization of railways and the consolidation of industrial properties, and the magnitude of some of his operations is fairly astounding. During the Cleveland administration, he floated a national bond issue of \$62,000,000; he marketed the securities of the United States Steel Corporation, with a capitalization of \$1,100,000,000; he secured American subscriptions aggregating \$50,000,000 for the British war loan of 1901; he controls over fifty thousand miles of railway, and his interests extend into practically every great financial enterprise in America. He has given large sums of money for public enterprises in New York City, among them a million and a half for a great lying-in hospital. He built the "Columbia," which twice defeated the "Shamrock" in the races for the America's cup, and he has made many valuable gifts to the various museums and libraries of New York City. The power he wields is enormous, but he wields it wisely and legitimately, winning the respect, as well as the admiration of men.

The greatest work of American men of affairs during the past half century has been the upbuilding and extension of the railroad system of the country.

The railroad mileage of the United States at the present time is over three hundred and twenty-five thousand; the total cost of the railroad equipment of the country reaches fourteen billion dollars and the yearly earnings average over two and a half billions. They employ over a million and a half men, whose wages average three million dollars a day—and, it may be added, they kill or injure nearly ninety thousand. But that is a detail. With this vast development of the railroad business the names of some half dozen men are so closely connected that the great systems of the country are generally known as the Hill lines, the Harriman lines, the Vanderbilt lines, the Gould lines, and so on. Of these men we shall try to tell something briefly here.

We have already related how Cornelius Vanderbilt secured control of the New York Central and Hudson River roads, and added to these until he had secured an entrance into Chicago; and how his son, William Henry Vanderbilt, added to this system until it became, and still remains, one of the strongest in the country. We have told, too, of Jay Gould's ideas of railroad management, which seem to have been to get the most out of it for Jay Gould. But when Jay Gould died, he was caught, as it were, with thousands of miles of railroads on his hands. He left four sons, George Gould, Edwin Gould, Howard Gould and Frank Gould, of whom George is the only one that really counts. But he, with a real genius for railroad building, has developed the Gould lines into a great system stretching from Buffalo and Pittsburgh southwestward to Chicago, Omaha, Kansas City, Denver, Ogden, St. Louis, New Orleans, Galveston and away out to El Paso. These lines have played a most important part in the development of the great Southwest, and it is said that George Gould is already blazing a way to the Atlantic seaboard, as an outlet for the mighty freight traffic which his lines control.

No man connected with railroad building in this country has had a more interesting or adventurous career than James J. Hill. Born on a little Canadian farm in 1838, descended from the hardy Scotch-Irish of whom we have spoken so often, his father died when he was fifteen years, and he was left to his own resources. He found work as a wood-chopper, and one day, while he was chopping down a tree a traveler stopped at the house to take dinner, hitching his horse to the gate. The boy noticed that it was tired and fagged and carried it a bucket of water. This attention pleased the traveler,

and as he drove away, tossed the boy a Minnesota newspaper, remarking, "Go out there, young man. That country needs youngsters of your spirit."

The boy read the paper with its glowing accounts of the new country, and the next morning, walking to the tree he had been cutting he hit it one last lick for luck, and announced, "I've chopped my last tree." That tree, it is said, bears to-day a great placard with the words, "The last tree chopped by James J. Hill." It was the last one, for a day or two later the boy started for St. Paul. He brought with him to the United States the lusty body, frugal instincts and good principles of his Scotch-Irish ancestry, and, in addition to those, a self-confidence and sureness of judgment destined to take him far.

He got employment as a shipping clerk in a steamboat office in St. Paul, and so took his first lessons in transportation problems. Pretty soon he was agent for a steamboat line, then he established a fuel and transportation business on his own account and managed it so well that by 1873, he had accumulated a fortune of a hundred thousand dollars. There was in Minnesota at the time a little railroad called the St. Paul & Pacific. It started at St. Paul, but it stopped after it had got only a few hundred miles toward the Pacific. Hill decided to buy it. The price was half a million, so he tramped back to Canada and persuaded the bank of Montreal to let him have the \$400,000 he needed. That was surely one of the most wonderful feats of a wonderful career. The directors of the bank were severely criticised; men laughed at his purchase, pointing out that the road had never paid, and prophesying that it never would pay.

Yet that Jim Crow road was the foundation of the Great Northern system, the Hill line, stretching across Dakota and Montana to Puget Sound. Every man who went into the enterprise with Hill now owns his stock in it as a free gift, for in the intervening years, the cost has been returned to him in the shape of dividends and bonuses. It has never failed to pay regular dividends, and has, perhaps, won public confidence more surely than any other in the country. For James J. Hill has kept faith in the smallest detail with every man who ever entrusted a dollar to his hands. The loyalty of the employes of the Great Northern has passed into a proverb, "Once a Hill man, always a Hill man," and it is true. He knows his road as few other men do. Before he bought the St. Paul & Pacific, he traveled over the route in an ox-cart, studying not only the road, but the people along the way—there

weren't many—and the resources of the country. Before he extended his line to the Pacific, he went the whole distance on foot and horseback.

People laughed at him when he announced that he was going to extend his line to the Pacific. No line had ever been built across the continent without a great subsidy from the government—to secure a subsidy was always the first step; besides, it was believed that the country through which the Great Northern was to extend would not even grow wheat, and the new road was promptly dubbed "Hill's Folly." But in 1893, his line reached the Pacific. A few years later, the owners of the great Northern Pacific were begging him to manage that road, too. For he had created business for his road—a great market in the Orient to fill his west-bound freight cars, and a great market in the eastern United States for Puget Sound lumber to fill his east-bound cars. For remember no railroad can make money unless, after it has hauled a loaded car from one end of the line to the other, it can find another load to put in that same car to haul back again. Hill supplied the business and his story is the wonderful story of the development of the Great Northwest.

Which brings us to the Napoleon of the railroad world, E. H. Harriman. America has never seen another quite like him. When the panic of 1901 was at its height and the financial world seemed trembling in ruins about his head, he refused to break the corner, as he might have done, but sat watching the tape, cool, quiet and calculating, while men failed, banks tottered, and his own associates begged him to yield. For the ambition of this man knew no limitation. His kingdom must stretch from sea to sea and from the lakes to the gulf.

His kingdom lay to the south of Hill's, for he ruled the Union Pacific, and between the two men there was ceaseless war. Physically and mentally they were as far apart as two men could be. Hill is a large man, with massive head and brow, and his eyes are steady and cool and brown, his lips full and sensitive, his whole personality bespeaking force and decision. Quite different was Harriman; a small, ordinary looking man, with glasses and a scraggy mustache, giving the impression of nervous force rather than of

power; an irritable man, easily angered; a fighter clear through, but fighting sometimes when peace were wiser—that was Harriman.

Harriman was born at Hempstead, Long Island, the son of a clergyman with a large family and a small income. The boy was renowned chiefly for his daily fights and for his aversion to study. At the age of fourteen, he was put to work in a broker's office in Wall street, at eighteen he had a partnership, at twenty-two he bought a seat on the stock exchange, and pretty soon entered the railroad field by getting control of the Illinois Central. He at once inaugurated a new policy. Before that time, the prevailing idea of railroad management was to run a road as cheaply as possible and pay big dividends. Harriman's idea was that the biggest dividends would be secured in the end by making a good road, and he proceeded to carry the idea out by putting his road in the very pink of condition. And it paid.

That was the beginning. His great coup was the rebuilding of the Union Pacific. A railroad with 7,500 miles of track, a giant crushed by its own weight, it had gone into a receivership in the panic of 1893. For five years it stayed there, despite the utmost efforts of the giants of finance to lift it out. Then Harriman got possession of it, and taking an engine and a car, turned the train backward and, running in the day time only, went over the road mile by mile. He decided that the road must be made a good road, and he told his executive committee that he needed for his immediate necessities one hundred millions of dollars!

Well, he got the money and he got good men and went to work. The result was soon apparent. Earnings grew, business increased, and the company's credit improved. Never before in the history of railroading had there been such daring rebuilding. The line was levelled down to a maximum grade of forty-one feet to a mile; two hundred and forty-seven feet were scaled off the top of the Great Divide; millions of cubic yards of dirt and stone were blasted out and moved; tunnels were drilled; and, finally, when the Southern Pacific, too, was acquired, a trestle twenty-three miles long was built across Great Salt Lake, through water thirty feet deep, taking railroad trains farther from land than they had ever yet been run, and shortening the road forty-four miles. And the result? The gross earnings have risen to over \$170,000,000 a year, and \$28,000,000 a year are distributed in dividends.

Truly a transformation from the old water-logged road which Harriman took over.

He had his reverses—he attempted to get hold of the Northern Pacific, but it slipped through his fingers; the Burlington was cut out from under his guns, and so was the Rock Island. James J. Hill outgeneraled him more than once, and he was never able to "get back" at Hill effectively.

With Harriman we shall close this chapter on men of affairs. Many others might have been noted. In fact, none of the great industries of the country has been built up except by inspired work. Armour and Cudahy and Swift made the packing business; Marshall Field built up a business in Chicago rivalling Wanamaker's; August Belmont, William C. Whitney, Levi Leiter, Robert Goelet, Pierre Lorillard, and a hundred others, amassed great fortunes. Yet there was nothing in their career different to those of the men already considered in this chapter. They had a genius for money-making. Each in some special field; but, beyond that, they did few memorable things. And so we need not pause longer over them here, except to remark, that it is, in the main, to such men as these, that America owes her great material prosperity.

SUMMARY

MORRIS, ROBERT. Born at Liverpool, England, January 20, 1734; came to America, 1747, and settled at Philadelphia; delegate to Continental Congress, 1775-78; gave his credit to assist in financing Revolution and elected superintendent of finance, 1781; organized Bank of North America, 1781; member of Constitutional Convention, 1787; United States senator, 1789-95; died in debtor's prison at Philadelphia, May 8, 1806.

ASTOR, JOHN JACOB. Born at Waldorf, Germany, July 17, 1763; came to America, 1783, and settled at New York City; founded Astoria, at mouth of Columbia River, 1811; died at New York City, March 29, 1848.

VANDERBILT, CORNELIUS. Born near Stapleton, Staten Island, New York, May 27, 1794; became chief owner Harlem railroad, 1863, and of Hudson River and New York Central roads soon afterwards; died at New York City, January 4, 1877.

STEWART, ALEXANDER TURNEY. Born near Belfast, Ireland, October 12, 1803; came to America, 1823, and established drygoods business at New York City; died there April 10, 1876.

BARNUM, PHINEAS TAYLOR. Born at Bethel, Connecticut, July 5, 1810; opened Barnum's Museum in New York City, 1841; managed Jenny Lind's concert tour, 1850-51; established "Greatest Show on Earth," 1871; died at Bridgeport, Connecticut, April 7, 1891.

SAGE, RUSSELL. Born in Oneida County, New York, August 4, 1816; member of Congress, 1853-57; established himself as broker and money-lender in New York City, 1863; died there, July 22, 1906.

FIELD, CYRUS WEST. Born at Stockbridge, Massachusetts, November 30, 1819; in paper business in New York, 1840-53, retiring with a fortune; organized New York, Newfoundland & London Telegraph Company, 1854; Atlantic Telegraph Company, 1856; laid Atlantic cable, 1866; first message over it, July 29; died at New York City, July 12, 1892.

MACKAY, JOHN WILLIAM. Born at Dublin, Ireland, November 28, 1831; came with parents to America, 1840; went to California, 1850; discovered Bonanza mines, 1872; died, July 20, 1902.

GOULD, JAY. Born at Roxbury, New York, May 27, 1836; established himself as broker in New York City, 1859; notorious for manipulations of various railroad and other securities, and for "Black Friday"; died at New York City, December 2, 1892.

WANAMAKER, JOHN. Born at Philadelphia, July 11, 1838; established clothing house of Wanamaker & Brown, 1861; established department store in Philadelphia, 1876, and in New York City, 1896; Postmaster-General, 1889-93; founded Bethany Sunday School, 1858; president Philadelphia Y. M. C. A., 1870-83.

MORGAN, JOHN PIERPONT. Born at Hartford, Connecticut, April 17, 1837; entered banking business, 1857, and developed present firm of J. P. Morgan & Co., largest private bankers of the United States.

HILL, JAMES J. Born near Guelph, Ontario, September 16, 1838; removed to Minnesota, 1856; entered transportation business; general manager St. Paul, Minneapolis & Manitoba Ry. Co., 1879-82; president since 1883; built Great Northern, with steamship connection with Japan and China, 1883-93; president of Great Northern system since 1893.

HARRIMAN, EDWARD HENRY. Born at Hempstead, Long Island; entered Wall Street as clerk at age of fourteen; entered New York Stock Exchange eight years later; was president and chairman of the board of directors of the Union Pacific, Oregon Short Line, Southern Pacific, Texas & New Orleans, and many other great railway systems; died near New York City, September 9, 1909.

CHAPTER X

INVENTORS

It is a curious fact that the men to whom the world owes most generally get the least reward. The genius in art or letters is seldom recognized as such until long after he himself has passed away—his life is usually embittered by derision or neglect. But, in the history of civilization, the lot of no man has been harder or more thankless than that of the inventor. Poverty and want have always been his portion, and even after he had won his triumph, had compelled public recognition of some great invention, it was usually some one else who won the reward.

America has been especially strong in the field of invention. Indeed, practically all the great labor-saving devices of the past century and more have originated here. "Yankee ingenuity" has passed into a proverb, and a true one, for the country which has produced the steamboat, the cotton gin, the sewing machine, the electric telegraph, the phonograph, the telephone, the typewriter, the reaper and binder, to mention only a few of the achievements of American inventors, may surely claim first place in this respect among the nations of the world. There are few stories more inspiring than that of American invention, and as benefactors to their race, the long line of American inventors may rightly rank before even the great philanthropists whose careers are outlined elsewhere in this volume. Indeed, if we judge greatness by the benefits which a man confers upon mankind, such men as Whitney and Howe and Morse and Bell and Edison far surpass most of the great characters of history.

First of the line is Benjamin Franklin, whose many-sided genius gives him a unique place in American history. His career has been considered in the chapter dealing with our statesmen, but let us pause for a moment here to speak of his inventions. One of them, the Franklin stove, is still in use in hundreds of old houses, and as an economizer of fuel has never been surpassed; another was the lightning-rod. He introduced the basket willow,

the water-tight compartment for ships, the culture of silk, the use of white clothing in hot weather, and the use of oil to quiet a tempest-tossed sea. From none of his inventions did he seek to get any return. The Governor of Pennsylvania offered to give him a monopoly of the sale of the Franklin stove for a period of years, but he declined it, saying, "That, as we enjoy great advantages from the inventions of others, we should be glad to serve others by any invention of ours"—a principle characteristic of Franklin's whole philosophy of life.

After Franklin, came Robert Fulton, the first man successfully to apply the power of the steam-engine to the propulsion of boats. Everyone has heard the story of how, years before, the youthful James Watt first got his idea of the power of steam by noticing how it rattled the lid on his mother's boiling teakettle. From that came the stationary engine, and from that the engine as applied to the locomotive. It remained for Fulton to apply it to water navigation.

Born in Lancaster County, Pennsylvania, of Irish parents, in poor circumstances, the boy received only the rudiments of an education, but developed a surprising talent for painting, so that, when he was seventeen, he removed to Philadelphia and set up there as an artist, painting portraits and landscapes. He remained there for some years, and finally, having made enough money to purchase a small farm for his mother, sailed for London, where he introduced himself to that amiable patron of all American painters, Benjamin West. West, who was at that time at the height of his fame, received Fulton with great kindness, and made a place in his house for him, where he remained for several years.

Those years were not devoted exclusively to painting, for Fulton had developed an interest in mechanics, secured a patent for an improvement in canal locks, invented a "plunging" boat, a kind of submarine, a machine for spinning flax, one for making ropes, one for sawing marble, and many others of minor importance. Finally abandoning art altogether, he went to Paris, where he spent seven years with the family of Joel Barlow, conducting with him a number of experiments; one series of which has developed into the modern submarine torpedo. He succeeded in interesting the French government in his submarine experiments and constructed a boat equipped with a small engine, with which, in the harbor of Brest, he seems

actually to have made some progress under water, remaining under on one occasion for more than four hours. But the French government finally withdrew its support, and finding the British government also indifferent, Fulton sailed for New York in December, 1806.

Here, he succeeded in interesting the United States government, which granted him \$5,000 to continue his submarine experiments, but interest in them soon waned, and Fulton turned his whole attention to the subject of steam navigation. He had been experimenting in this direction for a number of years, and, in conjunction with Chancellor Livingston, of New Jersey, had secured from the legislature of New York the exclusive right and privilege of navigating all kinds of boats which might be propelled by the force of fire or steam on all the waters within the territory of New York for a period of twenty years, provided he would, by the end of 1807, produce a boat that would attain a speed of four miles an hour. Fulton went to work at once, the experiments being paid for by Livingston, and after various calculations, discarded the use of paddles or oars, of ducks' feet which open as they are pushed out and close as they are drawn in, and also the idea of forcing water out of the stern of the vessel. He finally decided on the paddle-wheel, and, in August, 1807, the first American steamboat appeared on the East River. A great concourse witnessed the first trial, incredulous at first, but converted into enthusiastic believers before the boat had gone a quarter of a mile.

She was christened the "Clermont," and soon afterwards made a trip up the Hudson to Albany, to the astonishment of the people living along the banks of that mighty river. The distance of 150 miles, against the current of the river, was covered in thirty-two hours, and there could no longer be any question of Fulton's success. A regular schedule between Albany and New York was established, and the "Clermont" began that great river traffic now carried on by the most palatial river steamers in the world.

After that, it was merely a question of development. More boats were built, improvements were made, and every year witnessed an increase of speed and efficiency. In 1814, in the midst of the second war with England, Fulton built the first steam ship-of-war the world had ever seen, designed for the defense of New York harbor. This ancestor of the modern "Dreadnought" was named "Fulton the First" in honor of her designer. She indirectly

caused his death, for, exposing himself for several hours of a bitter winter day, in supervising some changes on her, he developed pneumonia and died a few days later. Could he re-visit the world to-day and see the wonderful and mighty ships which have grown out of his idea, he would no doubt be as astonished as were the people along the Hudson on that fall day in 1807 when they saw the "Clermont" making her way up the stream against wind and tide.

The same year that Robert Fulton was born, another inventive genius first saw the light in the little town of Westborough, Massachusetts. His name was Eli Whitney, and the work he was to do revolutionized the industrial development of the South, paid off its debts, and trebled the value of its lands. It did something else, too, which was to fasten upon the South the system of negro slavery, resulting in the Civil War. But though he added hundreds of millions of dollars to the wealth of his country, his own reward was neglect, indifference, countless lawsuits and endless vexation of body and spirit.

Whitney's father ran a little wood-working shop where he made wheels and chairs, and there the boy spent every possible hour. At the age of twelve, he made himself a violin, and his progress was so steady, that by the time he was sixteen, he had greatly enlarged the business and had gained the reputation of being the best mechanic in all the country round. He soon discovered the value of education, and managed to prepare himself for Yale College, which he entered in 1789, at the age of twenty-four—an age at which most men had long since graduated and settled in life. But Whitney persevered, graduating in 1792, and almost immediately securing a position as private tutor in a Georgia family, which was to change the whole course of his life.

Until he reached the South, he had never seen raw cotton, only a little of which, indeed, had been raised in the United States. It had not been profitable because of the difficulty of picking out the green cottonseed. To separate one pound of the staple from the seed was a day's work, so that cotton was considered rather as a curiosity than as a profitable crop. Whitney was impressed by the possibilities of cotton culture, could this obstacle be overcome, and devoted his spare time to the construction of the machine upon which his fame rests. At last it was done, and did its work so

perfectly that there could be no question of its success. Experiments showed that with it, one man, with the aid of two-horse power, could clean five thousand pounds of cotton a day!

A patent was at once applied for and every effort made to keep the invention a secret until a patent had been secured. But knowledge of it swept through the state, and great crowds of people came to see the machine. Whitney refused to show it, and after much excitement, a mob one night broke into the building where it was, and carried it away. Others were at once made, using it as a model, and by the time Whitney had secured his patent, they were in successful operation in many parts of the state.

That was the beginning of Whitney's trials. He had not enough money to produce machines rapidly enough to meet the tremendous demand for them, and various rivals sprang up, some of them even claiming the honor of the invention. Other gins were put on the market, differing from Whitney's only in some unimportant detail, and plainly an infringement of his patent; but he had not the means to prosecute their manufacturers. The result was, that after two years of disheartening struggle, Whitney was reduced to bankruptcy.

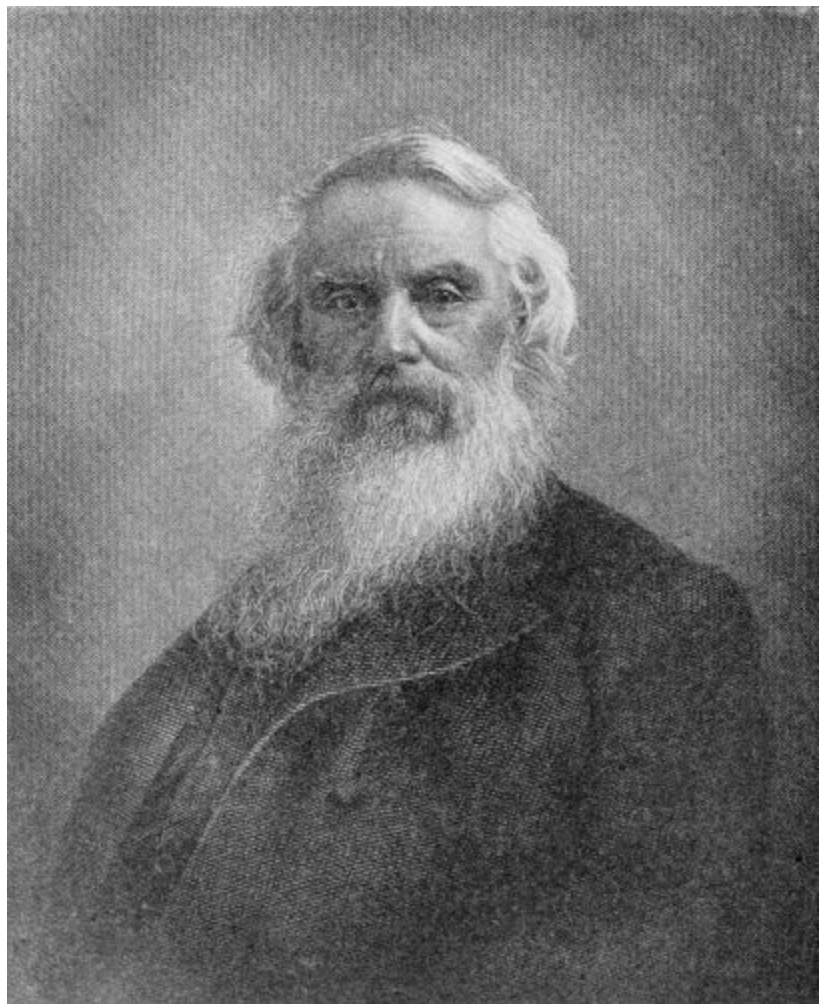
The attitude of the South toward him caused him especial distress. "I have invented a machine," he wrote, "from which the citizens of the South have already realized immense profits, which is worth to them millions, and from which they must continue to derive the most important profits, and in return to be treated as a felon, a swindler, and a villain, has stung me to the very soul. And when I consider that this cruel persecution is inflicted by the very persons who are enjoying these great benefits, and expressly for the purpose of preventing my ever deriving the least advantage from my labors, the acuteness of my feelings is altogether inexpressible."

Finally, the states of North and South Carolina voted him a royalty upon all the machines in use, and this enabled him to pay his debts; but Whitney at last abandoned hope of ever receiving from his invention the returns he had hoped for, and, turning his attention to other business, received, in 1798, a contract from the United States government for 10,000 stand of arms. Eight years were consumed in filling this contract. A contract for 30,000 stand

followed, and so many improvements in design and process of manufacture were made by Whitney that no other manufacturer could compete with him.

The result of all this was that Whitney was enabled to end his life in comparative independence. His last days were his happiest, and he found in the care and affection of a loving family some consolation for the injustice and ingratitude which he had suffered.

Sixteen years after the battle of Bunker Hill, a boy was born in a great frame house at the foot of Breed's Hill, upon which that famous and misnamed battle was really fought. The boy's father was a preacher named Jedediah Morse, and the boy was named Samuel Finley, after his maternal great grandfather, the renowned president of Princeton College, and Breese, after his mother's maiden name, so that he comes down through history as S. F. B. Morse. He received a thorough schooling,



MORSE

graduating from Yale in 1807, and at once turned his attention to art. We have already spoken of his achievements in that respect, which were really

of the first importance. He was an artist, heart and soul, but the whole course of his life was to be changed in a remarkable fashion.

In the autumn of 1832, Morse, being at that time forty-one years of age, sailed from Havre for New York in the ship Sully. It happened that there were on board some scientists who had been interested in electrical development, and the talk one evening turned on electricity. Morse knew little about it, except what he had learned in a few lectures heard at Yale; but when somebody asked how long it took a current of electricity to pass through a wire, and when the answer was that the passage was instantaneous, his interest was aroused.

"If that is the case," he said, "and if the passage of the current can be made visible or audible, there is no reason why intelligence cannot be transmitted instantaneously by electricity."

The company broke up, after a while, but Morse, filled with his great idea, went on deck, and at the end of an hour had jotted down in his notebook the first skeleton of the "Morse alphabet." Before he reached New York, he had made drawings and specifications of his invention, which he seems to have grasped clearly and completely from the first, although its details were worked out only by laborious thought. It was necessary for him to earn a living, and not until three years later was the first rude instrument completed. Two years more, and he had a short line in operation, but it was looked upon as a scientific toy constructed by an unfortunate dreamer. Finally, in 1838, Morse appeared before Congress, exhibited his invention and asked aid to construct an experimental line between Washington and Baltimore. He was laughed at, and for twelve years an extraordinary struggle ensued, Morse laboring to convince the world of the value of his invention, and the world scoffing at him. His own situation was forlorn in the extreme; for his painting was his only means of livelihood, and, absorbed as he was by his great invention, he found painting utterly impossible. His home was a single room in the fifth story of a building at the corner of Nassau and Beekman streets in New York City—a room which served as studio, workshop, parlor, kitchen and bedroom. There he labored and slept, using such money as he could earn for his experiments, and almost starving himself in consequence.

But at last the tide turned. He was appointed to a position in the University of the City of New York, which provided him with better means for experiment, and in 1843, again appeared before Congress. This time, he found some backers, and by a close vote, at the last hour of the session, an appropriation of \$30,000 was made to enable him to construct a line between Washington and Baltimore. Wild with delight and enthusiasm, the inventor went to work, and on the twenty-fourth day of May, 1844, the first message flashed over the wire, "What hath God wrought!"

The wonder and amazement of the public can be better imagined than described. Morse offered to sell his invention to the government for the sum of \$100,000, but the Postmaster General, a thickheaded individual named Cave Johnson, refused the offer, stating that in his opinion, no line would ever pay for the cost of operation!

It was inevitable that rival claimants for the honor of the invention should crop up on every side, but, after years of bitter litigation, Morse succeeded in defending his title, and honors began to pour in upon him. It is worth remarking that the Sultan of Turkey, supposedly the most benighted of all rulers, was the first monarch to acknowledge Morse as a public benefactor. That was in 1848; but the monarchs of Europe soon followed, and in 1858, a special congress was called by the Emperor of the French to devise some suitable testimonial to the great inventor. But perhaps the most fitting testimonial of all were the ceremonies at the unveiling of the Morse monument in New York City in 1871. Delegates were present from every state in the Union, and at the close of the reception, William Orton, president of the Western Union Telegraph Company, announced that the telegraph instrument before the audience was in connection with every other one of the ten thousand instruments in America, and that, beside every instrument an operator was waiting to receive a message. Then a young operator sent this message from the key: "Greeting and thanks to the telegraph fraternity throughout the world. Glory to God in the highest; on earth, peace, good-will to men." Then the venerable inventor, the personification of dignity, simplicity and kindliness, bent above the key, and sent out, "S. F. B. Morse." A storm of enthusiasm swept over the audience, and the scene will never be forgotten by any who took part in it. The proudest boast of many an old operator is that he received that message.

Death came to the inventor a year later, and on the day of his funeral, every telegraph office throughout the land was draped in mourning.

Although to Morse belongs all the credit for the invention of the telegraph, it should, in justice to one man, be pointed out that it would have been impossible but for a discovery which preceded it—that of the electro-magnet. To Joseph Henry, the great physicist, first of Princeton, then of the Smithsonian Institution, this invention is chiefly due. We have already spoken of Professor Henry's work in science, but none of it was more important than his invention, in 1828, of the modern form of electro-magnet—a coil of silk-covered wire wound in a series of crossed layers around a soft iron core, and in 1831, he had used it to produce the ringing of a bell at a distance. It is this magnet which forms the basis of every telegraph instrument—is essential to it, and is the foundation of the entire electrical art. Let it be added to this great scientist's credit that he never sought to patent any of his inventions, giving them, as Franklin had done, free to all the world.

The struggle which Morse made to perfect and secure public recognition of his telegraph and the injustice shown Eli Whitney by the people of the South, were as nothing when compared with the trials of that most unfortunate of all inventors, Charles Goodyear, whose story is one of the most tragic in American annals. No one can read of his struggles without experiencing the deepest admiration for a man who, at the time, was regarded as a hopeless lunatic.

Charles Goodyear was born at New Haven, Connecticut, in 1800. While he was still a child, his father moved to Philadelphia and engaged in the hardware business, in which his son joined him, as soon as he was old enough to do so. But the panic of 1836 wiped the business out of existence, and Goodyear was forced to look around for some other means of livelihood. He had been interested for some time in the wonderful success of some newly-established India-rubber companies, and, out of curiosity, bought an India-rubber life-preserver. Upon examining it, he found a defect in the valve, and inventing an improvement in it, he went to New York with the intention of selling his improvement to the manufacturer. The manufacturer was impressed with the new device, but told Goodyear

frankly that the whole India-rubber business of the country was on the verge of collapse, and indeed, the collapse came a few months later.

The trouble was that the goods which the rubber companies had been turning out were not durable. The use of rubber had begun about fifteen years before, first in France in the manufacture of garters and suspenders, and then in England where a manufacturer named Mackintosh made waterproof coats by spreading a layer of rubber between two layers of cloth. Then, in 1833, the Roxbury India-Rubber Company was organized in the United States, and manufactured an India-rubber cloth from which wagon-covers, caps, coats, and other articles were made. Its success was so great that other companies were organized and seemed on the highroad to fortune, when a sudden reverse came. For the heat of summer melted wagon-covers, caps and coats to sticky masses with an odor so offensive that they had to be buried. So the business collapsed, the various companies went into bankruptcy, and the very name of India-rubber came to be detested by producers and consumers alike.

It was at this time that Charles Goodyear appeared upon the scene—unfortunately enough for himself, but fortunately for humanity—and determined to discover some method by which rubber could be made to withstand the extremes of heat and cold. From that time until the close of his life, he devoted himself wholly to this work, in the face of such hardships and discouragements as few other men have ever experienced. He began his experiments at once, and finally hit upon magnesia as a substance which, mixed with rubber, seemed to give it lasting properties; but a month later, the mixture began to ferment and became as hard and brittle as glass.

His stock of money was soon exhausted, his own valuables, and even the trinkets of his wife were pawned, but Goodyear never for an instant thought of giving up the problem which he had set himself to solve. Again he believed he had discovered the secret by boiling the solution of rubber and magnesia in quicklime and water, when he found to his dismay that a drop of the weakest acid, such as the juice of an apple, would reduce an apparently fine sheet of rubber to a sticky mass. The first real step in the right direction was made by accident, for, in removing some bronzing from a piece of rubber with aqua fortis, he found that the chemical worked a remarkable change in the rubber, which would now stand a degree of heat

that would have melted it before. He called this "curing" India-rubber, and after careful tests, patented the process, secured a partner with capital, rented an old India-rubber works on Staten Island, and set to work, full of hope. But commercial disaster swept away his partner's fortune, and Goodyear could find no one else who would risk his money in so doubtful an enterprise.

Indeed, in all America he seemed to be the only man who had the slightest hope of accomplishing anything with India-rubber. His friends regarded him as a lunatic, and especially when he made himself a suit of clothes out of his India-rubber cloth, and wore it on all occasions. One day a man looking for Goodyear asked one of the latter's friends how he would recognize him if he met him.

"If you see a man with an India-rubber coat on," was the reply, "India-rubber shoes, India-rubber hat, and in his pocket an India-rubber purse with not a cent in it, that's Goodyear."

The description was a good one, for that purse had been without a cent in it for a long time. It was to stay empty for some weary years longer. For he had not yet discovered the secret of making India-rubber permanent, as he found when he tried to fill a contract for a hundred and fifty mail bags ordered by the government. The bags were apparently perfect, but in less than a month began to soften and ferment and were thrown back on his hands. All his property was seized and sold for debt; his family was reduced to the point of starvation, and friends, relatives and even his wife joined in demanding that he abandon this useless quest.

Goodyear was in despair, for he had just made another discovery that seemed to promise success—the discovery that sulphur was the active "curing" agent for India-rubber, and that it was the sulphuric acid in aqua fortis which had wrought the changes in rubber which he had noticed in his experiments. One day, while explaining the properties of a sulphur-cured piece of rubber to an incredulous crowd in a country-store, he happened to let it fall on the red-hot stove. To his amazement it did not melt; it had shrivelled some, but had not softened. And, at last, he had the key, which was that rubber mixed with sulphur and subjected to a certain degree of heat, would be rendered impervious to any extremes of temperature!

But what degree of heat? He experimented in the oven of his wife's cooking-stove, and in every other kind of oven to which he could gain access; he induced a brick-layer to make him an oven, paying him in rubber aprons; he grew yellow and shrivelled, for he and his family were living upon the charity of neighbors; more than once, there was not a morsel of food in the house; his friends thought seriously of shutting him up in an asylum; he tried to get to New York, but was arrested for debt, and thrown into prison. Even in prison, he tried to interest men with capital in his discovery, for he needed delicate and expensive apparatus, and at last two brothers, William and Emory Rider agreed to advance him a certain sum. The laboratory was built, and in 1844, Goodyear astonished the world by producing perfect vulcanized India-rubber with economy and certainty. The long and desperate battle had been won!

Did he reap a fortune? By no means! In one way or another, he was defrauded of his patent rights. In England, for instance, another man who received a copy of the American patent, actually applied for and obtained the English rights in his own name. In 1858, the United States Commissioner of Patents said, "No inventor, probably, has ever been so harassed, so trampled upon, so plundered by that sordid and licentious class of infringers known in the parlance of the world as 'pirates.'" Worn out with work and disappointment, Goodyear died two years later, a bankrupt. But his story should be remembered, and his memory honored, by every American.

Near a little mountain hamlet of central Sweden stands a great pyramid of iron cast from ore dug from the neighboring mountains. It is set up on a base of granite also quarried from those mountains, and bears upon it two names, Nils Ericsson and John Ericsson. The monument marks the place where these two men were born. The life of the former was passed in Sweden and does not concern us, but John Ericsson's name is closely connected with the history of the United States.

He was the son of a poor miner, and one of his earliest recollections was of the sheriff coming to take away all their household goods in payment of a

debt. He was put to work in the iron mines as soon as he was able to earn a few pennies daily, and he soon developed a remarkable aptitude for mechanics. At the age of eleven, he planned a pumping engine to keep the mines free from water, and at the age of twelve, was made a member of the surveying party in charge of the construction of the Gotha ship canal, and was soon himself in charge of a section of the work, with six hundred men under him, one of whom was detailed to follow him with a stool, upon which he stood to use the surveying instruments. It reminds one of Farragut commanding a war ship, at the age of eleven.

In 1826, at the age of twenty-three, he went to England to introduce a flame or gas-engine which he had invented. He remained there for eleven years, and then a fortunate chance won him for the United States. He had been experimenting with a screw or propeller for steamboats, instead of the paddle-wheels as used by Fulton, and finally, equipping a small boat with two propellers, offered the invention to the British admiralty. But the admiralty was skeptical. The United States consul in Liverpool happened to be Francis B. Ogden, a pioneer in steam navigation on the Ohio river. He was impressed with Ericsson's invention, introduced him to Robert F. Stockton, of the United States navy, and on their assurance that the invention would be taken up in the United States, closed up his affairs in England and sailed for this country.

His first experiment was disastrous—though through no fault of his. A ship-of-war called the Princeton was ordered by the government and completed. She embodied, besides screw propellers, many other features which made her a nine days' wonder. A distinguished company boarded her for her trial trip, and it was decided also to test her big guns. But at the first discharge, the gun burst, killing the secretary of state, the secretary of the navy, the captain of the ship, and a number of other well known men. As a consequence, the experiment was stopped and Ericsson was twelve years in securing from the government the \$15,000 he had spent in equipping the Princeton.

However, he was soon to render the country a service which will never be forgotten. In 1861, he appeared before the navy department with a plan for an iron-clad consisting of a revolving turret mounted upon an armored raft. He secured an order for one such vessel, to be paid for only in the event that

it proved successful. The majority of the board which gave the order doubtless laughed in their sleeves as the inventor withdrew, for what chance of success had such a vessel? There were some who even doubted whether she would float—among them her builders, who took the precaution of placing buoys under her before they launched her four months later.

Of the voyage of the little craft from New York to Hampton Roads, and of her epoch-making battle with the Merrimac we have already told. Ericsson had asked that she be named the "Monitor," as a warning to the nations of the world that a new era in naval warfare had begun, and that she was well-named no one could doubt after that momentous ninth of March, 1862. Honors were showered upon the inventor, whose great service to the nation could not be questioned. The following ten years of his life were devoted to the construction of his famous torpedo-boat, the "Destroyer," which, he believed, would annihilate any vessel afloat—the predecessor of all the torpedo-boats, past and present, which have played so important a part in naval warfare. He lived for more than twenty years in a house in Beach street, New York, where he died, in 1889.

The Monitor's attack upon the Merrimac would have been ineffective but for the remarkable guns with which the little craft was armed—two eleven-inch rifled cannon, the invention of John Adolph Dahlgren. Dahlgren had been connected with the ordnance department of the navy at Washington for many years, and his inventions had revolutionized United States gunnery.

Dahlgren was born at Philadelphia, where his father was Swedish consul, a position which he held until his death in 1824. The boy, from his earliest years, had been ambitious to enter the navy, and finally, at the age of seventeen, received his midshipman's warrant. In 1847, he was assigned to ordnance duty at Washington, and began that career of extraordinary energy, which lasted for sixteen years. He saw almost at once the many defects in the cannon which were at that time being manufactured, and soon offered a design of his own, which proved a vast advance over old guns. The Dahlgren gun, as it was called, was of iron, cast solid, with a thick breech adjusted to meet varying pressure strains. The invention of the rifled cannon followed, and it was this weapon which caused even the great armored Merrimac to tremble. Admiral Dahlgren's career was a distinguished one, but no service he rendered his country was more noteworthy than this.

But there are triumphs of peace, as well as of war, and one of the most notable of these was won by Cyrus Hall McCormick when he invented the automatic reaper which bears his name. In 1859, it was estimated that the reaper was worth \$55,000,000 a year to the United States; William H. Seward remarked that, "owing to Mr. McCormick's invention, the line of civilization moves westward thirty miles each year"; and the London Times declared, after it had been tested at the great international exhibition of 1851, that it was "worth to the farmers of England the whole cost of that exhibition." To few men is it given to confer such benefits upon mankind, and the career of this one is well worth dwelling upon.

Cyrus McCormick was born in 1809, in a little house at the hamlet of Walnut Grove, Virginia. His father was a farmer, and was also something of a mechanical genius, and as early as 1816, had tried to build a mechanical reaper. His son inherited this aptitude, and helped his father in mechanical experiments, soon quite outstripping him. As a farmer's boy, his day's work in the fields began at five o'clock in the morning, and in the harvesting season even earlier. But in the harvest field, he found himself unable to keep up with grown men in the hard work of swinging the scythe, and so devised a harvesting-cradle, which made the work so much easier that he was able to do his share. At the age of twenty-two he invented a plough, which threw alternate furrows on either side, and two years later, a self-sharpening plough, which proved a great success.

Then he turned his attention to a mechanical reaper, though his father warned him against wasting time and money on so impracticable a project. But the possibility of making a machine do the hot hand-work of the harvest field fascinated the young man, and he set to work upon the problem. It was not an easy one, for the machine, to be successful, must not only work in fields where the wheat stood straight, but also where it had become tangled and beaten down by wind and rain. In 1831, he produced his first practicable machine, making every part of it himself by hand. Its three essential features have never been changed—a vibrating cutting-blade, a reel to bring the grain within reach of the blade, and a platform to receive the falling grain. The problem had been solved.

Three years, however, were spent in perfecting the minor working parts, then another was built and tested. It worked well, but McCormick was still

not satisfied with it, and not until 1840, was it perfected sufficiently to make him willing to put it on the market. This self-restraint was remarkable, but it had this good effect, that when the machine was finally offered to the public, it was not an experiment. So there were no failures, but a steady increase in demand from the very first, until the great factory, which McCormick early located at Chicago, now turns out nearly two hundred thousand machines a year. The whirl of these machines is heard around the world—everywhere the McCormick reaper is doing its share toward lightening man's labor.

Another of the great victories of peace was won by Elias Howe, when, in 1844, he invented a machine which would sew. Strangely enough, he was at first regarded as an enemy of humanity, rather than as a friend; an enemy, especially, of the poor sewing-women who earned a pitiful living with the needle. Few had the foresight to perceive that it was these very women whose toil he was doing most to lighten!

Elias Howe, born in Spencer, Massachusetts, in 1819, as the son of a poor miller, and was put to work at the age of six to contribute his mite to the support of the family. He was a frail child and slightly lame, so that, after trying in vain to do farm labor, he went to work in the mill, and afterwards in a machine shop, where he learned to be a first-class machinist—knowledge which, at a later day, was to stand him in good stead. He married, at the age of twenty-one, and three children were born to him. Then came a period of illness, during which the young mother supported the family by sewing; and as Howe lay upon his bed, watching his wife at this tedious labor, the thought came to him what a blessing it would be to mankind if a machine could be devised to do that work.

The idea remained with him, and finally led to experiments. Of the many disappointments, the long months of patient labor, the intense thought, the repeated failures, there is not room to tell here; but at last he hit upon the solution of the problem—the use of two threads, making the stitch by means of a shuttle and a needle with the eye near the point. In October, 1844, he produced a rude machine which would actually sew. Another year was spent in perfecting it, while he kept his family from starvation by doing such odd jobs as he could find, and in the winter of 1845, he was ready to introduce his machine to the public.

But here an unforeseen difficulty arose. The public refused to have anything to do with the machine. The tailors declared it would ruin their trade, and refused to try it; nobody could be found who would invest a dollar in it; and Howe, in despair, was forced to put his invention away and to accept a place as railway engineer in order to support his family. Some disastrous years followed, his wife died, and he was left in absolute poverty, but at last came a ray of light. A man named Bliss became interested in Howe's invention, and a few machines were made and marketed in New York. Riots among the workingmen followed, so serious that for a time the use of the machines was stopped; but no human power could stay the wheel of progress, and as the value of the invention came to be recognized, all opposition to it faded away. Howe's royalties grew to enormous proportions, but he had been broken in health by his years of struggle and hardship, and lived only a few years to enjoy them.

George Henry Corliss was another mechanical genius, who, in one respect, anticipated Howe, for about 1842 he actually invented a machine for stitching leather. That was two years before Howe made his discovery. But Corliss was soon attracted to other work, and the development of the sewing machine was left for the other inventor. It was in 1846 that Corliss began to develop those improvements in the steam engine which were to revolutionize its construction. One trouble with the steam engine as then built was that it was not uniform in motion. That is, if the engine was running a lot of machines their speed would vary from moment to moment, as they were started or stopped. For instance, a hundred looms, all running at once, would run at a certain speed, but if some of them were shut off, the speed of the others would increase, so that it was very difficult to regulate them. Again, there was a tremendous waste of power, so that the fuel consumption was out of all proportion to the power actually developed.

It was these defects that Corliss set himself to remedy, and he did it simply by taking a load off the governor, which had always been used to move the throttle-valve. In the Corliss engine, the governor simply indicated to the valves the work to be done, and the saving of fuel was so great that the inventor often installed his engine under a contract to take the saving in coal-bills from a certain period as his pay. One of his great achievements was the construction of a 1400 horse power engine to move all the

machinery at the centennial exposition at Philadelphia, in 1876. The engine, which worked splendidly, was one of the sights of the exposition.

What the sewing-machine is to the needle, the typewriter is to the pen. No other one invention has so revolutionized business, and the credit for the invention of a practicable typewriting machine is due to C. Latham Sholes. Others had tried their hands at the problem before he took it up, but he was the first to hit upon its solution—a number of type-bars carrying the letters of the alphabet operated by levers and striking upon a common centre, past which the paper was carried on a revolving cylinder.

Sholes had a varied and picturesque career. Born in Pennsylvania in 1819, he followed the printer's trade for a number of years, and it was no doubt from the type that he got his idea of engraved dies mounted on type-bars. Finally he removed to Wisconsin, where he edited a paper and soon became prominent in the politics of the state, holding a number of appointive positions. It was in 1866 that he began to experiment with a writing-machine, and his first one, which was patented two years later, was as big as a sewing-machine. Still, it embodied the essential principles of the typewriter as it is made to-day, and after spending five years in perfecting it, Sholes made a contract with E. Remington & Son to manufacture it. It is one of the ironies of fate that the name principally connected with the typewriter in the public mind is that of the manufacturer, the identity of the inventor being completely lost, so far as applied, at least, to the name of any machine.

We have spoken elsewhere of the career of John D. Rockefeller, of the immense fortune he made from petroleum and the manner in which he disposed of a portion of it. It is worth pausing a moment to consider the career of the two men who discovered petroleum, who sunk the first well in search of a larger supply, and who put it on the market. There is scarcely any development of modern life to rank in importance with the introduction of kerosene. It added at once several hours to every day, and who can estimate what these evening hours, spent usually in study or reading, have meant to humanity?

In the early part of the century, whales were so plentiful, especially along the New England coast, that whale, or sperm, oil was used for lighting purposes, and many of the old whale-oil lamps are still in existence. The light they gave was dim and smoky, but it was far better than no light at all. As the years passed, whales became more and more scarce, until sperm oil was selling at over two dollars a gallon. Only the richest people could afford to pay that, and the poor passed their evenings in darkness.

In 1854, a man named James M. Townsend brought to Professor Silliman, of Yale, a bottle of oil, asking him to test it. This was done, and the astonished professor found that here was an oil, whose source he could only guess, which made a splendid illuminant and which also seemed to have some medicinal properties. The oil was from Oil Creek, Pennsylvania, and Townsend, associating with himself a conductor named E. L. Drake, formed the Seneca Oil Company and began gathering the oil by digging trenches. At first it was bottled and sold for medicinal purposes at one dollar a gallon; then Drake suggested that a larger supply might be secured if a well was bored for it. A man familiar with salt well boring was employed, and in 1859 the first well was begun at Titusville.

Most people regarded Drake as a madman, and thought that he was simply throwing money away. The work was costly and slow, and finally, when \$50,000 had been spent without result, the stockholders of the company refused to go further—all except Townsend. That enthusiast managed to rake up another \$500, which he sent to Drake, with instructions to make it go as far as possible. It did not go very far—and yet far enough—for one day the auger, which was down sixty-eight feet, struck a cavity, and up came a flow of oil to within five feet of the surface. Pumping began at the rate of five hundred barrels a day, and fortune seemed in sight. But three months later, the company's works were destroyed by fire, and before they could be rebuilt, scores of other wells had been sunk, many of which were "gushers," requiring no pumping, and the supply was soon so far in excess of the demand that the price of oil tumbled to one dollar a barrel. Discouraged by all this, the Seneca Company sold out its leases and disbanded, leaving Townsend and Drake poorer than they had been before their great discovery.

Years ago, in 1790, to be exact, an Italian scientist named Galvani, experimenting with the legs of a frog, happened to touch the exposed nerves with a piece of metal, while the legs were lying across another piece. He was astonished to see the legs contract violently. Further experiments followed, and the galvanic battery resulted. Years later, our own Professor Henry discovered that if an insulated wire carrying a current of electricity was wrapped around a piece of soft iron, the latter became a magnet. Out of these simple discoveries, came the electric telegraph, and, still more wonderful, the telephone, by which the human voice may be instantly projected hundreds of miles, not only intelligibly, but with every tone and inflection reproduced. In an age of wonders, this is surely one of the greatest.

On February 14, 1876, two applications were made at the patent office at Washington for patents upon the conveyance of sound by electricity. One was filed by Elisha Gray, the other by Alexander Graham Bell. They were practically identical, but it was Bell's good fortune to be the first to make his device practically effective, and so he may fairly be considered the inventor of the telephone.

Alexander Graham Bell was born in Edinburgh, Scotland, in 1847, the son of the famous Alexander Melville Bell, the inventor of the system by which deaf people are enabled to read speech more or less correctly by observing the motion of the lips. The family moved to Canada in 1872, and Alexander Bell came to Boston, where he soon became widely known as an authority in the teaching of the deaf and dumb. The reproduction of the human voice by mechanical means interested him deeply, and his study of the construction of the human ear, with its drum vibrating in response to sound vibrations, gave him the idea of a vibrating piece of iron in front of an electric magnet. He was, however, very poor and had no money to expend in experiments—so poor, indeed, that when attacked by illness, his hospital expenses were paid by his employer, and so friendless that during his illness no one visited him except two or three pupils from his school.

He persevered with his experiments, with such rude apparatus as he could make himself, and the first Bell telephone was brought into existence with an old cigar-box, two hundred feet of wire, and two magnets from a toy fish-pond. In an improved form, it was shown at the Centennial exhibition

of 1876, where Sir William Thomson pronounced it "the greatest marvel hitherto achieved by the electric telegraph." As is always the case, the public was slow to appreciate the importance of the invention, and as late as 1877, Bell was unable to secure \$10,000 for a half interest in the European rights. The rapid growth of the business in this country is almost without a parallel in history, and no invention has added more to the convenience of modern life.

A distinguished scientist one day asked the late Clerk Maxwell what was the greatest scientific discovery of the last half century, and Maxwell answered without an instant's hesitation: "That the Gramme machine is reversible." Probably the whole scientific world will agree with him, for that discovery meant that power will not only produce electricity, but that electricity will produce power. Let us see how that has been applied. Falling water is one of the most powerful agents in the world, and at a great waterfall like Niagara, millions of horsepower go to waste every day. So at the foot of Niagara, great power-houses have been built where the power of the water is converted into electricity. The electricity is conducted along wires for hundreds of miles to the great industrial centres, and there converted back again into power. In other words, the water of Niagara is to-day turning machinery in Buffalo and Albany. The same method of producing power, the cheapest that has ever been discovered, is being installed all over the world, and will, in time, produce a revolution in manufacturing processes.

The vital mechanism in the production of this power is the dynamo, and it is to Charles F. Brush, of Cleveland, Ohio, that its development is principally due. He was interested in electricity from his earliest years, and when he was only thirteen, distinguished himself by making magnetic machines and batteries for the Cleveland high-school, where he was a pupil. During his senior year, the physical apparatus of the school laboratory was placed under his charge, and he constructed an electric motor having its field magnets as well as its armature excited by the electric current. He devised an apparatus for turning on the gas in the street lamps of Cleveland, lighting

it and turning it off again, thus doing away with the expensive process of lighting them and turning them out by hand.

After graduating from the University of Michigan with the degree of mining engineer, he returned to Cleveland, where, in 1875, his attention was drawn to the great need of a more effective dynamo than the clumsy and inefficient types then in use. In two months, Brush had made a dynamo so perfect in every way that it was running until taken to the Chicago Exposition, in 1893. Six months more of experimenting resulted in the Brush arc light, and in 1879 the Brush Electric Company was organized. A year later, the first Brush lights were installed in New York City, and now there is scarcely a town in the country which does not pay tribute to the inventor.

Let us turn for a moment from the field of electricity, in which America has been pre-eminent, to another in which Yankee ingenuity has also led the world—the railroad. It was in this country that the sleeping-car, the diner, the parlor-car were first used; no other country affords such luxury of travel; and no other country has added to railroading any device comparable in importance to the invention of George Westinghouse, the air-brake. Before its introduction, to stop a train brakes must be set painfully by hand, and even then were not always effective. Now, the engineer, by pulling a single lever, sets the brakes instantly all along his train, and so effectively that the passengers sometimes feel as though the train had struck a rock. More than that, should any accident occur, breaking the train in two, the brakes are instantly set automatically. All of which is done by the power of compressed air, working through a series of pipes and air-hose beneath the cars.

George Westinghouse's father was superintendent of the Schenectady Agricultural Works, and it was there that the boy found his vocation. Before he was fifteen, he had modelled and built a steam engine, and followed that with a steel railroad frog, which was so great an improvement over the frogs then in use that it was soon widely adopted, and brought the young inventor both money and reputation. He moved to Pittsburgh, as the centre of the iron and steel business, and began the manufacture of his frogs there.

One day he came across a newspaper account of the successful use of compressed air in the digging of the Mont Cenis tunnel, in Switzerland, and the thought occurred to him that perhaps a railroad train could be controlled by the same agency. He worked over the problem for a time, but when he mentioned his idea to his friends, they were inclined to think it absurd to suppose that a rubber-tube strung along under the cars could work the brakes effectively. However, Westinghouse was not discouraged, but continued to experiment, and the air-brake as we have it to-day was the result.

Which brings us to the most remarkable genius in the field of invention the world has ever known—the man who has made invention, as it were, a business, whose life has been devoted to rendering practical and useful the dreams of other men, who has reduced invention to a science—Thomas Alva Edison. There are some who are inclined to belittle Edison's achievements because some of the greatest of them have been founded upon the ideas of others. He is best known, for instance, as the inventor of the modern incandescent light; but the discovery that light may be obtained from wire heated to incandescence in a glass bulb from which the air has been exhausted, was made when Edison was only two years old. Experiments with this light were made by a dozen scientists, but it remained a mere laboratory curiosity until Edison took hold of it, and with a patience, ingenuity and fertility of resource, in which he stands alone, made it a practicable, efficient and convenient source of light. That the incandescent light, as it is known to-day, is his through and through cannot be questioned.

It is as a scientific inventor that Edison likes to be known. He abhors the word discoverer, as applied to himself. "Discovery is not invention," he once said. "A discovery is more or less in the nature of an accident, while an invention is purely deductive. In my own case, but few, and those the least important, of my inventions, owed anything to accident. Most of them have been hammered out after long and patient labor, and are the result of countless experiments all directed toward attaining some well-defined object."

There is, however, one modern marvel for which Edison is wholly responsible, both for the initial idea and for its practical working-out—the phonograph—but let us tell something of his early life, before we relate the achievements of his manhood.

Born in a little village in Erie County, Ohio, in 1847, Edison was early introduced to the struggle for existence. His father was very poor, being, indeed, the village jack-of-all-trades, and living upon such odd jobs as he was able to procure. The boy, of course, was put to work as soon as he was old enough, and of regular schooling had only two months in all his life. At the age of twelve, he was a train-boy on the Michigan Central Railroad, selling books, papers, candy, and fruit to the passengers. He managed to get some type and an old press and issued a little paper called the "Grand Trunk Herald," containing the news of the railroad. One day, he snatched the little child of the station-master at Port Clements, Michigan, from under the wheels of a train, and in return the grateful father taught the boy telegraphy.

It was the turning-point in his career, for it turned his attention to the study of electricity, with which he was soon fascinated. At eighteen, he was working as an operator at Indianapolis, but he was from the very first, more of an inventor than an operator, and his inventions sometimes got him into trouble. For instance, at one place where he had a night trick, he was required to report the word "six" every half-hour to the manager to show that he was awake and on duty. After a while, he rigged up a wheel to do it for him, and all went well until the manager happened to visit the office one night and found Edison sleeping calmly while his wheel was sending in the word "six." But he nevertheless developed into one of the swiftest operators in the country, all the time devising changes and improvements in the mechanism of telegraphy.

His first great success came with the sale of an improvement in the instruments used to record stock quotations, which enabled these "tickers" to print the quotations legibly on paper tape, and this success enabled him to get some capitalists to finance his experiments with the electric light. The arrangement was that they were to pay the expense of the experiments and to share in such inventions as resulted. For the sake of quiet, he moved out to a little place in New Jersey called Menlo Park, and built himself a shop. Then began that remarkable series of experiments—one of the most

remarkable in history—which resulted in the perfection of the incandescent lamp.

The problem was to find a material for the filament which would give a bright light and which, would, at the same time, be durable, and with this end in view, hundreds and hundreds of different filaments were tried. The difficulties in the way of this experimenting were enormous, since the light only burns when in a vacuum, and the instant the vacuum is impaired, out it goes. At one time, all the lamps he had burning at Menlo Park, about eighty in all, went out, one after another, without apparent cause. The lamps had been equipped with filaments of carbon and had burned for a month. There seemed to be no reason why they should not burn for a year, and Edison was stunned by the catastrophe. He began at once the most exhaustive series of experiments ever undertaken by an American physicist, remaining in his laboratory for five days and nights, dining at his work bench on bread and cheese, and snatching a little sleep occasionally, when one of his assistants was on duty. It was finally discovered that the air had not been sufficiently exhausted from the lamps.

Again success seemed in sight, but soon the lamps began acting queerly again. Worn out with fatigue and disappointment, Edison took to his bed. Ultimate failure was freely predicted, and the price of gas stock rose again. In five months, the inventor had aged five years, but he was not yet ready to give up the fight. And at last it was won, and the incandescent lamp placed on the market. It has not displaced gas, as some people thought it would, but it is the basis of a business which made the inventor sufficiently rich to realize his great ambition of building himself the finest laboratory in the world; where the most expert iron-workers, wood-workers, glass-blowers, metal-spinners, machinists and chemists in the world find employment. Every known metal, every chemical, every kind of glass, stone, earth, wood, fibre, paper, skin, cloth, may be found in its store-rooms, ready for instant use. The library contains one of the finest collections of scientific books and periodicals to be found anywhere. These are the tools, and with them Edison is constantly at work upon a great variety of problems.

The first thing he turned his hand to after his installation in his new laboratory was the phonograph. The patient thought and experiment, extending over many years, lavished on this wonderful invention are almost

unbelievable. The idea had come to him years before, when he had worked out an instrument that would not only record telegrams by indenting a strip of paper with the dots and dashes of the Morse code, but would also repeat the message any number of times by running the indented strip of paper through it.

"Naturally enough," said Edison, in telling the story, "the idea occurred to me that if the indentations on paper could be made to give off again the click of the instrument, why could not the vibrations of a diaphragm be recorded and similarly reproduced? I rigged up an instrument hastily and pulled a strip of paper through it, at the same time shouting 'Hallo!' Then the paper was pulled through again, and listening breathlessly, I heard a distinct sound, which a strong imagination might have translated into the original 'Hallo!' That was enough to lead me to a further experiment. I made a drawing of a model, and took it to Mr. Kruesi, at that time engaged on piece-work for me. I told him it was a talking-machine. He grinned, thinking it a joke; but he set to work and soon had the model ready. I arranged some tin-foil on it and spoke into the machine. Kruesi looked on, still grinning. But when I arranged the machine for transmission and we both heard a distinct sound from it, he nearly fell down in his fright. I must admit that I was a little scared myself." The words which he had spoken into the machine and which were the first ever to be reproduced mechanically, was the old Mother Goose quatrain, starting, "Mary had a Little Lamb."

From that rude beginning came the phonograph, with which Edison has never ceased to experiment. He has made improvements in it from year to year, until it has reached its present high state of efficiency—a state, however, which Edison hopes to improve still further. In addition to the two great inventions of the phonograph and incandescent lamp, which we have dwelt upon here, many more stand to his credit. In fact, he has been the greatest client the patent office ever had, nearly one thousand patents having been issued in his name. At the age of sixty-three, he shows no sign of falling off in either mental or physical energy, and no doubt more than one invention has yet to come from Llewellyn Park before he quits his great laboratory forever.

No one can ever guess at the future of electrical invention. The recent marvelous development of the wireless telegraph, by which the impalpable ether is harnessed to man's service, is an indication of the wonders which may be expected in the future. It was our own Joseph Henry who, in 1842, discovered the electric wave—the "induction" upon which wireless telegraphy depends. He discovered that when he produced an electric spark an inch long in a room at the top of his house, electrical action was instantly set up in another wire circuit in the cellar. After some study, he saw and announced that the electric spark started some sort of action in the ether, which passed through floors and ceilings and all other intervening objects, and caused induction in the wires in the cellar. But wireless telegraphy was made a commercial possibility not by any great scientist, but by a young Italian named Marconi. Already experiments with wireless telephony are going forward, and another half century may see all the labor of the world performed by this wonderful and mysterious force which we call electricity.

From earliest times, man has longed to navigate the air. He has watched with envy the free flight of birds, and has tried to imitate it, usually with disastrous results. The balloon, of course, enabled him to rise in the air, but once there, he was at the mercy of every wind. More recently, balloons fitted with motors and steering gear have been devised, which are to some extent dirigible; but the real problem has been to fly as birds do without any such artificial aid as balloons provide.

Experiments to solve this problem were begun several years ago by Professor S. P. Langley, of the Smithsonian Institution, under government supervision, and pointed the way to other investigators. He proved, theoretically, that air-flight was possible, provided sufficient velocity could be obtained. He showed that a heavier-than-air machine would sustain itself in the air if it could only be driven fast enough. You have all skipped flat stones across the water. Well, that is exactly the principle of the flying machine. As long as the stone went fast enough, it skipped along the top of the water, which sustained it and even threw it up into the air again. When its speed slackened, it sank. So the boy on skates can skim safely across

thin ice which would not bear his weight for an instant if he tried to stand upon it.

So, theoretically, it was possible to fly, but to reduce theory to practice was a very different thing. Professor Langley tried for years and failed. He built a great machine, which plunged beneath the waters of the Potomac a minute after it was launched. All over the world, inventors were struggling with the problem, but nowhere with any great degree of success. It remained for two brothers, in a little workshop at Dayton, Ohio, to produce the first machine which would really fly.

Orville and Wilbur Wright were poor boys, the sons of a clergyman, and apparently in no way distinguished from ordinary boys, except by a taste for mechanics. They had a little workshop, and one day in 1905, they brought out a strange looking machine from it, and announced that it was a flying-machine. The people of Dayton smiled skeptically, and assembled to witness the demonstration with the thought that there would probably soon be need for an ambulance. The gasoline motor with which the machine was equipped, was started, one of the brothers climbed aboard and grasped the levers, the other dropped a weight which started the machine down a long incline. For a moment, it slid along, then its great forward planes caught the air current and it soared gracefully up into the air.

That was a great moment in human history, so great that the crowd looking on scarcely realized its import. They watched the machine with bated breath, and saw it steered around in a circle, showing that it could go against the wind as well as with it. For thirty-eight minutes it remained in the air, making a circular flight of over twenty-four miles. Then it was gently landed and the exhibition was over. Great crowds flocked to Dayton, after that, expecting to see further exhibitions, but they were disappointed. The machine had been taken back to the shop, and the young inventors announced that they were making some changes in it. No one was admitted to the shop, nor were any other flights made.

One day the inventors also disappeared, and months later it was discovered that they had built themselves a little shop on a deserted stretch of the sandy North Carolina coast, and that they were carrying on their experiments there, secure from observation. Enterprising reporters tried to interview

them and failed; but, ambushed afar off, they one day saw the great machine soaring proudly in a wide circle above the sands. A photographer even got a distant photograph of it. There could be no doubt that the Wright brothers had solved the problem of flight.

But not for two years more were they ready for public exhibitions. Then, in 1908, they appeared at Fort Myer, Virginia, ready to take part in the contest set by the United States government. No one who was present on that first day will ever forget his sensations as the great winged creature rose gracefully from the ground and circled about in the air overhead. Again and again flights were made, sometimes with an extra passenger; great speed was attained and the machine was under perfect control. But an unfortunate accident put a stop to the trials, for one day a propellor-blade broke while the machine was in mid-air, and it struck the ground before it could be righted. The passenger, a member of the United States Signal Corps, was instantly killed and Orville Wright was seriously injured.

Meanwhile, the other brother, Wilbur, had gone to Europe, where, first in France, and afterwards in Italy and England, he created a tremendous sensation by his spectacular flights. They were uniformly successful. Not an accident marred them. The governments of Europe were quick to secure the right to manufacture the aeroplane; kings and princes vied with each other in honoring the young inventor, and when he returned to the United States, city, state, and nation combined in a great reception to him and to his brother.

As these lines are being written, in August, 1909, another series of flights has been concluded at Fort Myer. They were successful in every way in fulfilling the government tests, and the Wrights' machine was purchased by the government for \$30,000. Everywhere air-ship flights are being made successfully, and it is only a question of time until the aeroplane becomes a common means of conveyance. Wilbur Wright declares that it is already safer than the automobile, and it would seem that there is in store for man a new and exquisite sensation, that of flight.

Surely, America has cause to be proud of her inventors!

SUMMARY

FULTON, ROBERT. Born at Little Britain, Pennsylvania, 1765; went to London, 1786, to study painting under Benjamin West; abandoned painting, 1793; returned to America, 1806; first successful trip in steamboat, the Clermont, August 11, 1807; died at New York City, February 24, 1815.

WHITNEY, ELLI. Born at Westborough, Massachusetts, December 8, 1765; graduated at Yale, 1792; went to Georgia as teacher and invented cotton-gin, 1792-93; died at New Haven, Connecticut, January 8, 1825.

MORSE, SAMUEL FINLEY BREESE. Born at Charlestown, Massachusetts, April 27, 1791; graduated at Yale, 1810; studied art under Benjamin West in London, and opened studio in New York City, 1823; first president National Academy of Design, 1826-42; designed electric telegraph, 1832; applied for patent, 1837; first line completed between Baltimore and Washington, 1844; died at New York City, April 2, 1872.

GOODYEAR, CHARLES. Born at New Haven, Connecticut, December 29, 1800; began experiments with rubber, 1834; secured patent, 1844; died at New York City, July 1, 1860.

ERICSSON, JOHN. Born in parish of Fernebo, Wermland, Sweden, July 31, 1803; went to England, 1826; came to America, 1839; constructed caloric engine, 1833; applied screw to steam navigation, 1836-41; invented turreted ironclad Monitor, 1862; died at New York City, March 8, 1889.

DAHLGREN, JOHN ADOLPH. Born at Philadelphia, November 13, 1809; lieutenant in navy, 1837; assigned to ordnance duty at Washington, 1847; commander, 1855; rear-admiral, 1863; took important part in naval operations during Civil War; died at Washington, July 12, 1870.

MCCORMICK, CYRUS HALL. Born at Walnut Grove, West Virginia, February 15, 1809; invented mechanical reaper, 1831; died at Chicago, May 13, 1884.

HOWE, ELIAS. Born at Spencer, Massachusetts, July 9, 1819; invented sewing-machine, 1844; died at Brooklyn, New York, October 3, 1867.

CORLISS, GEORGE HENRY. Born at Easton, New York, July 2, 1817; invented Corliss engine, 1849; died at Providence, Rhode Island, February 21, 1888.

SHOLES, CHRISTOPHER LATHAM. Born at Mooresburg, Pennsylvania, February 14, 1819; state senator, Wisconsin, 1848, 1856-58; held many positions of trust in Milwaukee, 1869-78; patented typewriter, 1868.

BELL, ALEXANDER GRAHAM. Born at Edinburgh, Scotland, March 3, 1847; came to Canada, 1870, and to Boston, 1871; invented telephone, 1876; graphophone, 1883.

BRUSH, CHARLES FRANCIS. Born at Euclid, Ohio, March 17, 1849; graduated University of Michigan, 1869; invented modern arc electric lighting; founder Brush Electric Company.

WESTINGHOUSE, GEORGE. Born at Central Bridge, Schoharie County, New York, October 6, 1846; invented rotary engine at age of fifteen; in Union army, 1863-64; invented air brake, 1868; also inventions in railway signals, steam and gas engines, turbines, and electric machinery.

EDISON, THOMAS ALVA. Born at Milan, Ohio, February 11, 1847; established workshop at Menlo Park, New Jersey, 1876; invented megaphone, phonograph, aërophone, incandescent electric lamp, kinetoscope, and many other things.

WRIGHT, ORVILLE. Born at Dayton, Ohio, 1871.

WRIGHT, WILBUR. Born at Dayton, Ohio, 1869.

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Transcriber's Note:

Inconsistent hyphenation in the original document has been preserved.

A reference to the index has been added to the ToC for convenience.

Obvious typographical errors have been corrected. These are:--

[Page 99](#)—ran away to Pittsburg changed to ran away to Pittsburgh

[Page 105](#)—landscapists in Thomas Cole. changed to landscapists is Thomas Cole.

[Page 218](#)—history of adminstration changed to history of administration

[Page 341](#)—rubber and magnesia is quicklime changed to rubber and magnesia in quicklime

[Page 347](#)—played so imporant changed to played so important

[Page 360](#)—power of compresed air, changed to power of compressed air,

[Page 363](#)—The arrrangement changed to The arrangement

[Page 376](#)—Cary, Phœbe, 76. changed (for consistency with main text) to Cary, Phoebe, 76.

[Page 381](#)—Silliman, Benjamin, 202-203-204 changed to Silliman, Benjamin, **202-204**

[Page 382](#)—Warren, William, 177-178-179-184. changed to Warren, William, **177-179, 184.**

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